



INTERMEDIATE STATE PERMIT TO OPERATE

Under the authority of RSMo 643 and the Federal Clean Air Act the applicant is authorized to operate the air contaminant source(s) described below, in accordance with the laws, rules, and conditions set forth herein.

Intermediate Operating Permit Number: OP2020-009

Expiration Date: MAR 16 2025

Installation ID: 087-0016

Project Number: 2014-05-089

Installation Name and Address

Golden Triangle Energy, LLC
15053 Highway 111
Craig, MO 64437
Holt County

Parent Company's Name and Address

Golden Triangle Energy, LLC
15053 Highway 111
Craig, MO 64437

Installation Description:

Golden Triangle Energy is a denatured ethanol, beverage ethanol, and denatured industrial ethanol production facility. Current permits limit production to 21 million gallons of ethanol and denaturant of 1.05 million gallons, for a total denatured ethanol production limit of 22.05 million gallons per consecutive twelve month period. The installation has emission limitations on the following criteria pollutants to qualify for an Intermediate operating permit: PM₁₀, NO_x, VOC, CO, SO_x, and both individual and total HAPs.

MAR 16 2020

Effective Date

A handwritten signature in blue ink, appearing to read "Sarah A. Byrd", written over a horizontal line.

Director or Designee
Department of Natural Resources



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I. Installation Equipment Listing

Emission Units With Limitations

The following list provides a description of the equipment at this installation that emits air pollutants and that are identified as having unit-specific emission limitations. All emission units at this installation are subject to plant wide conditions and some are also subject to unit specific permit conditions. The applicability is detailed in each permit condition.

Grain Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
EU029	FS003	Grain/DDGS Storage Building	Broin	2000	N/A	None
EU001	SV001	Corn Dump Pit with Auger	Broin, C8-11	2000	280 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU002	SV001	Corn Elevator	Schlagel, 36168	2000	280 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU041	SV001	Corn Bin Distribution Conveyor	Schlagel, 1614	2000	280 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU004	SV001	Corn Bin-Surge Bin		2000		Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU005	SV001	Corn Bin #1	Broin, T-841	2000	36,000 Bushel capacity	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU006	SV001	Corn Bin #2	Broin, T-842	2000	150,000 Bushel capacity	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU007	SV001	Corn Bin #3	Broin, T-843	2000	150,000 Bushel capacity	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU042	SV001	Corn Bin Unloading Conveyor	Schlagel, 1214	2000	210 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU003	SV001	Corn Scalper	Schlagel, 1614	2000	280 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU013	SV003	Hammermill and Belt Scale	Broin, G-110	2000	280 tons/hr	Baghouse #2 (C002) Mfr: Broin, Constructed 2000
FS003-1	FS003	Uncaptured Grains	None	2000		None

Fermentation Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
EU023	SV005	Slurry Tank (T203)	Broin	2000	24,000 gal/hr	Distillation Scrubber (C004)
T231		Alpha Amylase Enzyme Tank	Broin	2000	5,000 gallons	None
TA001		Ammonia Tank	Atlas	1975	18,000 gallons	None
EU014	SV004	Fermentation Tank #1 (T301)	Broin	2000	15,420 gal/hr ¹	Fermentation Scrubber (C003)
EU015	SV004	Fermentation Tank #2 (T302)	Broin	2000	15,420 gal/hr ¹	Fermentation Scrubber (C003)
EU016	SV004	Fermentation Tank #3 (T303)	Broin	2000	15,420 gal/hr ¹	Fermentation Scrubber (C003)
EU051	SV004	Fermentation Tank #4 (T314)	Winbco	2016	15,420 gal/hr ¹	Fermentation Scrubber (C003)
EU052	SV004	Fermentation Tank #5 (T315)	Winbco	2016	15,420 gal/hr ¹	Fermentation Scrubber (C003)
EU024	SV005	Yeast Tank (T307)	Broin	2000	24,000 gal/hr	Distillation Scrubber (C004)
T232		Gluc Amylase Enzyme Tank	Broin	2000	5000 gallons	None
EU017	SV004	Beer Well (T304)	Advance	2000	270,000 gallons	Fermentation Scrubber (C003)

Distillation Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
EU018	SV005	Beer Stripper Distillation Column (T401)	Broin	2000	2,400 gal/hr	Distillation Scrubber (C004)
EU019	SV005	Rectifier Distillation Column (T402)	Broin	2000	2,400 gal/hr	Distillation Scrubber (C004)
P407-2		Heat Exchanger Pump	Inomag	2012	600 gallons/hr	None
EU025	SV005	Rundown Tank (T406)	Broin	2000	2,400 gal/hr	Distillation Scrubber (C004)
T801		190 Proof Ethanol Tank	Advance	2000	67,000 gallon capacity	None
EU020	SV005	Side Stripper Distillation Column (T403)	Broin	2000	2,400 gal/hr	Distillation Scrubber (C004)

¹ Fermentation Tanks are bottlenecked by the Distillation process. The bottlenecked maximum hourly design rate is 257 gal/min, 15,420 gal/hr

Distillation Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
EU021	SV005	Molecular Sieve Dehydration Vacuum Pump Vent (T421, T422, T423)	Broin	2000	2,400 gal/hr	Distillation Scrubber (C004)
T802		200 Proof Ethanol Tank	Advance	2000	46,000 gallon capacity	None
EU035	SV005	Industrial Extractive Distillation Column #1	R. Katzen	2006	2,400 gal/hr	Distillation Scrubber (C004)
T830		Impurities Storage Tank		2000		Distillation Scrubber (C004)
EU036	SV005	Industrial Rectifier Distillation Column #2	R. Katzen	2006	2,400 gal/hr	Distillation Scrubber (C004)
EU037	SV005	Industrial Demthylization Distillation Column #3	R. Katzen	2006	2,400 gal/hr	Distillation Scrubber (C004)
EU040	SV005	Industrial Distillation Recycle Rectifier Column #4 (T524)	Katzen	2007	2,400 gal/hr	Distillation Scrubber (C004)
T807		HQ 200 Ethanol Tank	JL Houston	2005	39,725 gallon capacity	None
T808		HQ 190 Ethanol Tank	Winbco	2013	38,500 gallon capacity	None
T809		HQ 190 Ethanol Tank	JL Houston	2005	39,725 gallon capacity	None
EU046		HQ Heat Exchanger				None
T811		HQ 200 Proof Ethanol Storage Tank	Strobel	2007	190,913 gallon capacity	None
EU054	SV005	HQ Molecular Sieve Dehydration	Katzen	2007	1,872 gal/hr	Distillation Scrubber (C004)
T804		HQ 200 Proof Ethanol Storage Tank	Advance	2000	180,000 gallons capacity	None

Distillation Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
T806		HQ 200 Ethanol Tank	JL Houston	2005	39,725 gallon capacity	None
EU028		Boiler #1, Natural gas fired	Johnston, PFTA 1500	2000	60.5 MMBTU/hr	Low NOx Burners
EU034		Boiler #2, Natural gas fired	Johnston, PFTA 1500	2006	60.5 MMBTU/hr	Low NOx Burners
FS005		Cooling Tower #1	Midwest Towers, RS2424	2000	462,000 gal/hr	None
FS006		Cooling Tower #2	EvapTech, C224-424K	2007	576,000 gal/hr	None

Ethanol Product Loading Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
T802		HQ 200 Proof Ethanol Tank	Advance	2000	46,000 gallon capacity	None
T421	SV005	Fuel Mol Sieve #1	Midwest	2000	1,500 gallons capacity	Distillation Scrubber (C004)
T422	SV005	Fuel Mol Sieve #2	Midwest	2000	1,500 gallons capacity	Distillation Scrubber (C004)
T423	SV005	Fuel Mol Sieve #3	Midwest	2000	1,500 gallons capacity	Distillation Scrubber (C004)
T803		Denatured Ethanol Fuel Grade Storage Tank	Advance	2000	180,000 gallon capacity	None
T805		Natural Gasoline Storage Tank	Advance	2000	18,000 gallon capacity	None

Ethanol Product Loading Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
T810		Corrosion Inhibitor Storage Tote	Clawson	1993	390 gallons capacity	None
T815		HQ 190 Ethanol Tank	WINBCO	2010	250,000 gallon capacity	None
T816		HQ 190 Ethanol Tank	WINBCO	2015	250,000 gallon capacity	None
T817		HQ 190 Ethanol Tank	WINBCO	2015	250,000 gallon capacity	None
T804		HQ 200 proof Ethanol storage Tank	Advance	2000	180,000 gallon capacity	None
T811		HQ 200 Storage Tank	Strobel	2007	190,913gallon capacity	None
FS008		Methanol Storage Tank	Clawson tote and Metalcraft tote	2000 and 2005	2 totes at 550 gallons capacity per tote	None
FS009		Tertiary Butyl Alcohol Storage Tank	Plymouth totes	2000	2 totes at 550 gallons capacity per tote	None
FS010		Toluene Storage Barrels	Barrel	NA	55 gallons capacity per barrel	None
FS011		3-Chloropropyl Triethoxysilane, CPTEO, Storage Totes	Schutz totes	NA	1 tote at 275 gallons capacity	None

Ethanol Product Loading Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
FS012		Ethyl Acetate Storage Tote	Plymouth tote, Hoover tote, and Transtore tote	2000	2 totes 550 gallons capacity per tote, Transtore tote is 520 gallons capacity	None
FS013		Isopropyl Alcohol Storage Tote	Hoover totes	2000	2 totes 550 gallons capacity per tote	None
FS014		Ethyl 3-Ethoxypropionate, 3-EEP, Storage Tote	Hoover tote and Clawson totes	1995,1994, and 1998	3 totes 550 gallons capacity per tote	None
FS016		Hexane Storage Barrel	Exxon Mobil, 5120	2009	550 gallon capacity	None
FS007-1		Truck bottom loading of Denatured Ethanol Fuel Grade	Broin, K802	2000	30,000 gal/hr	Open Flare (EU039)
EU039		Open Flare, Natural gas fired	John Zink, LHT-1-12-20-x-1/4X	2005	0.9 MMBTU/hr	Not applicable
FS001-1		HQ Ethanol Rail Loading Rack	SafeRack Top Loading	2015 (replaced original 2000)	30,000 gal/hr	Open Flare (EU039)

Wet Distillers Grains with Syrup (WDGS) and Dried Distillers Grains with Syrup (DDGS) Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
		Centrifuge Deck				None
EU050		Centrifuge #4	Fottweg, Z6E-4/454 DDGS	2015	15,420 gallons/hr	None
EU053		Centrifuge #5	Flottweg, Z6E-4/454 DDGS	2018	15,420 gallons/hr	None
T604	SV016	Concentrate Tank	A&B process, model TS6201	2014	1,000 gallons	None
FS020		Wetcake Conveyor	Broin C-601	2000	37,635 lb wetcake/hr	None
FS021		Wetcake Pad	NA	NA	5,000 ft ² capacity	None
FS022		Wetcake Truck Loading	Broin C-612	2000		None
EU026		Dryer Feed Conveyor	Broin C-610	2000	37,635 lb wetcake/hr	None
T501		Thin Stillage Tank	Advance	2001	200,000 gallon capacity	None
EU022	SV005	Evaporator	Broin	2000	2,400 gal/hr	Distillation Scrubber (C004)
T505		Syrup Storage Tank	Winbco	2015	43,191 gal capacity	None
T505A		Syrup Storage Tank	Winbco	2012	43,666 gal capacity	None
EU026	SV012 or SV006 for RTO bypass	DDGS Dryer, Natural gas fired	ICM, 40136R	2000	36 MMBTU/hr 9 tons DDGS/hr	RTO (C008, EU038) or RTO Bypass (SV006)
EU026	SV012 or SV006 for RTO bypass	DDGS Multclones	Broin S605 A, B, C, D	2000	9 tons DDGS/hr	RTO (C008, EU038) or RTO Bypass (SV006)

Wet Distillers Grains with Syrup (WDGS) and Dried Distillers Grains with Syrup (DDGS) Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
EU027	SV012 or SV006 for RTO bypass	DDGS Cooling/Separating Cyclone	AirLanco, RC-78, #86101116	September 2013	7.4 dry ton/hr	RTO (C008, EU038) or RTO Bypass (SV006)
EU038	SV012	DDGS Dryer Regenerative Thermal Oxidizer, Natural gas fired	Eisemann, VRTO-C	2005	8 MMBTU/hr	Not Applicable
EU026	SV006	DDGS Dryer Regenerative Thermal Oxidizer Bypass	None			
		Dried Solids Conveyor	C607	2000	9 tons/hr	None
EU043	SV001	DDGS Pit Conveyor	C620, Schlagel #1214, C800	2000	9 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU044	SV001	DDGS Loadout Conveyor	Schlagel, 1214, C802	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU008	SV001	DDGS Dump Augers: Screw Augers A, B, and C	C805, C806, C807	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
		DDGS Stockpile				None
EU008	SV001	DDGS Pit A	Broin	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU008	SV001	DDGS Pit B	Broi	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU008	SV001	DDGS Pit C	Broin	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU011	SV001	DDGS Screw Conveyor	Schlagel, 1214, C801	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
EU009	SV001	DDGS Elevator	Schlagel, 36167, E801	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
		Bulk Scale	T802	2000	185 ton/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000

Wet Distillers Grains with Syrup (WDGS) and Dried Distillers Grains with Syrup (DDGS) Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Constructed	MHDR or Capacity	Control Device
EU010	SV001	DDGS Rail Load Spout with flexible shrouds	Broin, XV802-1	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
		Railcar Grate				None
		Railcar Loading Pit				None
EU012	SV001	DDGS Truck Load Spout	Broin, XV8022	2000	185 tons/hr	Baghouse #1 (C001) Mfr: Broin, Constructed 2000
		Truck Loading Grate				None
		Truck Loading Pit				None

Miscellaneous Operations						
EP #	Stack ID	Description	Manufacturer /Model	Date Manufactured	MHDR or Capacity	Control Device
EU045	SV017	Emergency Generator, Natural gas fired, spark ignition, 4 stroke	Cummins, Ford ESG-642	Manufactured January 2005, purchased 2008, installed 2010	74 HP	None
FS002-1		LDAR Monitoring of Fugitive Emissions for units subject to Subpart VV	NA	NA	NA	None
FS002-2		LDAR Monitoring of Fugitive Emissions for units subject to Subpart VVa	NA	NA	NA	None
FS004	FS004	Truck Haul Road	N/A	2000	Length=0.6 miles	Paved Haul Roads

II. Emission Limitations

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the Code of Federal Regulations (CFR) and Code of State Regulations (CSR) for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect on the date of permit issuance. Limitations that apply on both a plant wide basis and emission unit specific basis appear in this section. The permit condition will specify the applicability of the limitation. Plant wide limitations apply to all emission units listed in Section I of this permit.

Permit Condition 072008-001² 10 CSR 10-6.060 Construction Permits Required Construction Permit Permit #072008-001, Issued July 2, 2008 Construction Permit Permit Amendment #072008-001A, Issued June 6, 2009 Construction Permit Permit Amendment #072008-001B, Issued September 12, 2013 Construction Permit #072016-004, Issued July 14, 2016						
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Emission Limitations:

1. Particulate Matter Less than Ten Microns in Diameter (PM₁₀)³

The permittee shall comply with the PM₁₀ emission limitations and compliance demonstrations shown in the table below.

Table 1: Summary of PM₁₀ Emission Limitations

Emission Point	Emission Unit Number	Stack Description	Emission Limitation	Units	Compliance Demonstration	Regulatory Citation
Plant wide			Less than 95.0	Tons per consecutive 12 month period	Recordkeeping	Special Condition #8.A.1.) and Special Condition #8.A.2.
SV001	EU001,EU002, EU003,EU004, EU005,EU006, EU007,EU008, EU009,EU010, EU011,EU012, EU041,EU042, EU043,EU044	Baghouse #1 (C001)	0.272		Performance Testing ⁴	
SV003	EU013	Baghouse #2 (C002)	1.314			

² Unless otherwise noted, Special Conditions are sourced from Construction Permit 072008-001.

³ All references to PM₁₀ in this permit condition include both the filterable and condensable components.

⁴ Baghouse #1 (C001) and Baghouse #2 (C002) have special circumstances for continuous performance testing. The DDGS Cooling/Separating Cyclone (EU027) has continuous performance testing requirements. See the Performance Testing portion of this permit condition for additional details.

Emission Point	Emission Unit Number	Stack Description	Emission Limitation	Units	Compliance Demonstration	Regulatory Citation
SV007	EU027	DDGS Cooling/Separating Cyclone	1.34			
SV008	EU028	Boiler #1	2.01	Tons per consecutive 12 month period	AP42 SCC emission factors or Performance Testing	Special Condition #8.A.1.) and Special Condition #8.A.2.
SV011	EU034	Boiler #2	2.01			
SV012	EU026	DDGS Dryer and RTO	8.67		Performance Testing	
SV012	EU038	RTO	3.4	lbs/hr	Performance Testing	Special Condition #8.A.3.)

2. Nitrogen Oxides (NOx)

The permittee shall comply with the NOx emission limitations and compliance demonstrations shown in the table below.

Table 2: Summary of NOx Emission Limitations

Emission Point	Emission Unit	Stack Description	Emission Limitation	Units	Compliance Demonstration	Regulatory Citation
Plant wide			Less than 95.0	Tons per consecutive 12 month period	Recordkeeping	Special Condition #8.B.2.)
Sum of SV008 and SV012 (dryer outlet)	Sum of EU028 and EU026	Sum of Boiler #1 and DDGS dryer outlet	20.24 ⁵		Performance Testing	Special Condition #8.B.1.)
SV006	EU026	DDGS Dryer	0.04	Lbs/MMBtu	Performance Testing	Special Condition #8.B., Table 4
SV008	EU028	Boiler #1	0.03			
SV011	EU034	Boiler #2	0.032			
SV012	Sum of EU026 and EU038	Sum of DDGS Dryer and RTO	2.69	Lbs/hr		

⁵ In previous permits this limit is referenced as the Group NOx Cap. This emission limitation applies to the sum of the Boiler #1 (EU028) and DDGS dryer (EU026) emissions.

3. Hazardous Air Pollutants (HAPs)

The permittee shall comply with the HAPs emission limitations and compliance demonstrations shown in the table below.

Table 3: Summary of HAP Emission Limitations

Emission Point	Emission Unit	Stack Description	Emission Limitation	Units	Compliance Demonstration	Regulatory Citation
Plant wide			Less than 9.0 any individual HAP; and Less than 24.0 total HAPs	Tons per consecutive 12 month period	Recordkeeping	Special Condition #8.C.1.) and Special Condition #8.C.2.)
SV007	(EU027)	DDGS Cooling/Separating Cyclone (EU027)	0.33 total HAPs	Lbs/hour	Performance Testing	Special Condition #13.B. and Special Condition #13.D.

4. Volatile Organic Compounds (VOCs)

The permittee shall comply with the VOC emission limitations and compliance demonstrations shown in the table below.

Table 4: Summary of VOC Emission Limitations

Emission Point	Emission Unit	Stack Description	Emission Limitation	Units	Compliance Demonstration	Regulatory Citation
Plant wide			Less than 95.0	Tons per consecutive 12 month period	Recordkeeping	Special Condition #8.D.1.)
SV012	EU026	DDGS dryer	95% reduction or less than 10 ppm of VOC		Performance Testing	
SV004	COO3	Fermentation Scrubber	95% reduction or less than 20 ppm VOC			
SV007	EU027	DDGS Cooling/Separating Cyclone (EU027)	2.79	Lbs/hr	Recordkeeping	Special Condition #13.C.

5. Carbon Monoxide (CO)

The permittee shall comply with the CO emission limitations and compliance demonstrations shown in the table below.

Table 5: Summary of CO Emission Limitations

Emission Point	Emission Unit	Stack Description	Emission Limitation	Units	Compliance Demonstration	Regulatory Citation
Plant wide			Less than 95.0	Tons per consecutive 12 month period	Recordkeeping	Special Condition #8.E.1.)
SV012	EU026	DDGS Dryer/RTO	90% reduction or less than 100 ppm		Performance Testing	Special Condition #8.E.3.)

6. Sulfur Oxides (SOx)

The permittee shall comply with the SOx emission limitations and compliance demonstrations shown in the table below.

Table 6: Summary of SOx Emission Limitations

Emission Point	Emission Unit	Stack Description	Emission Limitation	Units	Compliance Demonstration	Regulatory Citation
Plant wide			Less than 95.0	Tons per consecutive 12 month period	Recordkeeping	Special Condition #8.F.1.)

Operational Limitations:

1. Throughput Limitations

The permittee shall comply with the throughput limitations and compliance demonstrations shown in the table below.

Table 7: Summary of Process Throughput Limitations

Operational Parameter	Throughput Limitation	Units	Compliance Demonstration	Regulatory Citations
Denatured Ethanol Production Rate	22,050,000	Gallons per consecutive 12 month period	Recordkeeping	Special Condition #7.A.
Total Ethanol Production Rate	21,000,000			
Maximum Denaturant Usage	1,050,000			
Truck and Rail Grain Receiving	217,778	tons of grain per consecutive 12 month period		Special Condition #5.A.
Truck and Rail DDGS Shipping	78,840	tons of DDGS shipped per consecutive 12 month period		Special Condition #6.A.

2. Combustion Fuel Type
The permittee shall only burn natural gas, combustion of which will produce less than 8 pounds per million Btu of SO_x. (Special Condition #8.F.2.)
3. Paved Haul Roads
 - A. The haul road is 0.6 miles long and 0.24 miles are paved. The permittee shall control fugitive emissions from the 0.24 miles of paved haul roads by paving and washing/cleaning. The permittee shall pave with materials such as asphalt, concrete, and/or other material(s). If materials other than asphalt or concrete are used, the permittee must receive approval from the Air Pollution Control Program. The permittee shall apply the pavement in accordance with industry standards for such pavement so as to achieve control of fugitive emissions while the installation is operating. (Special Condition #3.A.)
 - B. The permittee shall conduct maintenance and/or repair of the road surface as necessary to ensure that the physical integrity of the pavement is adequate to achieve control of fugitive emissions from these areas. (Special Condition #3.B.)
4. Unpaved Haul Roads
 - A. The haul road is 0.6 miles long and 0.36 miles are not paved. The permittee shall inspect the roadway on a weekly basis for wear, frost boils, etc and will observe truck traffic for signs of visible emissions. (Special Condition #4.A.)
 - B. The permittee shall apply dust suppressant or water within 48 hours of observation of fugitive dust emissions caused by car/truck traffic on the road. (Special Condition #4.B.)
 - C. The permittee shall record roadway inspection observation data including but not limited to: (Special Condition #4.C.)
 - i. Date and time of inspection, name of inspector, map or site plan showing location, and site line of visible emission observations and location of road surface problem areas, corrective actions taken to eliminate visible emissions or problem surface conditions. (Special Condition #4.C.1.)
 - ii. The permittee shall report any deviation of monitoring frequency and range. (Special Condition #4.C.2.)
5. Cooling Tower #1 (FS005) and Cooling Tower #2 (FS006) Requirements
 - A. The permittee shall operate and maintain both cooling towers in accordance with the manufacturer's specifications. (Special Condition #17.A.)
 - B. The permittee shall ensure the cooling water design circulation rate is 7,700 gpm for Cooling Tower #1 (FS005) and 9,600 gpm for Cooling Tower #2 (FS006). (Special Condition #17.B.)
 - C. The permittee shall not exceed a drift loss of 548 gallons per minute or 1.4% percent of the water circulation rate. (Special Condition #17.D.)
 - D. The permittee shall not exceed a total dissolved solids (TDS) concentration in the circulated cooling water of 3,500 parts per million (ppm). (Special Condition #17.E.)

Operational Limitations-Add On Control Devices:

1. Grain Storage Building (EU029), Baghouse #1 (C001), and Baghouse #2 (C002)
 - A. Capture Efficiency of the Grain Storage Building (EU029), Baghouse #1 (C001), and Baghouse #2 (C002)
 - i. The permittee shall enclose by ductwork, maintain under negative pressure, and exhaust to Baghouse #1 (C001) the grain handling and process equipment (EU001 to EU012) and to Baghouse #2 (C002) the Hammermill/Belt Scale (EU013). (Special Condition #9.A.)

- ii. The Grain Storage Building (EU029) is a multi purpose building. The DDGS Storage Pile, DDGS truck loading (EU012), and Corn Dump Pit/Auger (EU001) are all located in this building. The building is divided into halves by a low concrete wall separating the DDGS storage from the DDGS truck loading (EU012) and Corn Dump Pit/Auger (EU001) area. The cooling cyclone underflow conveyor (C-620) discharges into this building supplying the DDGS and creating the DDGS Storage Pile. (Special Condition #9.E.)
 - a. The permittee shall control emissions from the DDGS Dump Pits/Augers (EU008), the DDGS Elevator (EU009), the DDGS Rail Load Spout (EU010), and the DDGS Truck Load Spout (EU012) by maintaining negative pressure and routing emissions Baghouse #1 (C001). The equipment is expected to be maintained at 100 percent capture efficiency. No visible emissions are expected from the transferring equipment. (Special Condition #9.E.1.))
 - b. A capture efficiency of 80% is provided for emissions from the Corn Dump Pit/Auger (EU001). Visible emission created from the Corn Dump Pit/Auger (EU001) are expected from this portion of the building. (Special Condition #9.E.2.))
 - c. A wind flow of 2 miles per hour is provided for the DDGS Storage Pile. Visible emissions are expected from the storage pile portion of the building. (Special Condition #9.E.3.))
 - d. No capture efficiency is provided for the Grain Storage Building (EU029). All building openings are not required to indicate the presence of negative pressure for compliance. (Special Condition #9.E.4.))
- B. Operations of Baghouse #1 (C001), and Baghouse #2 (C002)
 - i. The permittee must use the baghouses at all times when the equipment listed in the following table is operating: (Special Condition #10.A.)

Table 8: Equipment Controlled by Baghouses

Emission Unit	Control ID	Type of Control
EU001 through EU012, and EU041 through EU044	C001	Baghouse #1
EU013	C002	Baghouse #2

- ii. The permittee shall operate and maintain the baghouses and any related instrumentation or equipment in accordance with the manufacturer's specifications. (Special Condition #10.B.)
 - iii. The permittee shall equip the baghouses with gauges or meters, which indicate the pressure drop across the baghouses. These gauges or meters shall be located such that Department of Natural Resources' employees may easily observe them. (Special Condition #10.B.)
 - iv. The permittee shall keep replacement bags for the baghouses on hand at all times. The bags shall be made of fibers appropriate for operating conditions expected to occur (i.e. temperature limits, acidic and alkali resistance, and abrasion resistance). (Special Condition #10.C.)
2. Fermentation Scrubber (C003) and Distillation Scrubber (C004)
- A. The permittee must use the corresponding wet scrubber at all times when any of the equipment listed in the table below is in operation: (Special Condition #11.A.)

Table 9: Equipment Controlled by Wet Scrubbers

Equipment Controlled by Wet Scrubbers	Wet Scrubber Description
Fermentation Tank #1 (EU014), Fermentation Tank #2 (EU015), Fermentation Tank #3 (EU016), Fermentation Tank #4 (EU051), and Fermentation Tank #5 (EU052)	Fermentation Scrubber (C003)
Beer Stripper Distillation Column (EU018), Rectifier Distillation Column (EU019), Side Stripper Distillation Column (EU020), Molecular Sieve Dehydration Vacuum Pump Vent (EU021), Evaporator (EU022), Slurry Tank (EU023), Yeast Tank (EU024), Production Rundown Storage Tank (EU025), Industrial Extractive Distillation Column #1 (EU035), Industrial Rectifier Distillation Column #2 (EU036), Industrial Demethylization Column #3 (EU037), and Industrial Distillation Recycle Rectifier Column #4 (EU040)	Distillation Scrubber (C004)

- B. The permittee shall operate and maintain the Fermentation Scrubber (C003) and Distillation Scrubber (C004), and any related instrumentation or equipment, in accordance with the manufacturer's specifications and the operating parameters established by the most recent performance test that the Air Program has determined demonstrates compliance with applicable regulations. (Special Condition #11.B.)
 - C. The permittee shall equip the Fermentation Scrubber (C003) and Distillation Scrubber (C004) with a gauge or meter that indicates the pressure drop across each scrubber, and a water flow meter that indicates the flow to each scrubber. These gauges and meters shall be located in such a way they may be easily observed by Department of Natural Resources' personnel. (Special Condition #11.B.)
 - D. The permittee shall operate the Fermentation Scrubber (C003) and Distillation Scrubber (C004) within the parameters demonstrated by the most recent performance test that the Air Program has determined demonstrates compliance with applicable regulations. Unless otherwise documented via testing, Fermentation Scrubber (C003) has a calculated pressure drop based on Flexiring packing curves and variable speed fan curves ranges from 1.0 to 14 inches water column (in WC) and a water flow rate of 28 gallons per minute with a bisulfite addition rate of 1 gallon/hour of 36% sodium bisulfite. The Distillation Scrubber (C004) has a calculated pressure drop based on Flexiring packing curves and fan curves ranges from 0.4 to 8 in WC, a water flow rate of 4 gallons per minute and no sodium bisulfite. (Special Condition #11.C.)
 - E. The permittee is required to add bisulfite at all times Fermentation Scrubber (C003) is in operation. To change this condition, to operate without bisulfite addition in C003, the permittee must retest the fermentation scrubber without the bisulfite addition. Bisulfite addition can be discontinued only for authorized stack testing. Using the data from the test, the permittee shall calculate the plant wide potential to emit for acetaldehyde. If the potential emissions are less than the current SMAL value, bisulfite addition can be discontinued upon approval and concurrence of calculation method and testing procedures by the Permitting and Compliance/Enforcement Sections of the Air Pollution Control Program. Bisulfite addition must be maintained until approval to discontinue is obtained. (Special Condition #11.D.)
3. Flare (EU039)
 - A. The permittee must use the Flare (EU039) at all times when the Truck Loading Rack (FS007) or Rail Loading Rack (FS001) is in operation loading denatured ethanol. Denatured ethanol means

ethanol product that contains denaturants before or after ethanol loading operations. (Special Condition #12.A.)

- B. The permittee shall operate and maintain the Flare (EU039) in accordance with the manufacturer's specifications. The permittee must operate the Flare (EU039) in accordance with 40 CFR Part 60.18, General Control Device Requirements. (Special Condition #12.B.)
- 4. DDGS Cooling/Separating Cyclone (EU027)
 - A. The permittee shall use the DDGS Cooling/Separating Cyclone (EU027) at all times when the DDGS dryer (EU026) is in operation. (Special Condition #13.A.)
 - B. The permittee shall operate and maintain the DDGS Cooling/Separating Cyclone (EU027) in accordance with the manufacturer's specifications. (Special Condition #13.A.)
- 5. Regenerative Thermal Oxidizer (C008/EU038)
 - A. The permittee must use the Regenerative Thermal Oxidizer (C008/EU038) at all times when the DDGS dryer (EU026) is in operation or any time that regulated PM₁₀, VOC, or HAP emissions are possible; except as detailed in the Alternative Operating Scenario. (Special Condition #14.A.)
 - B. The permittee shall operate and maintain the Regenerative Thermal Oxidizer (C008/EU038) in accordance with the manufacturer's specifications. (Special Condition #14.A.)

Alternative Operating Scenario-Regenerative Thermal Oxidizer

- 1. The permittee may use this alternative operating scenario only during times when the Regenerative Thermal Oxidizer (C008/EU038) requires regeneration. When using this alternative operating scenario, the permittee may bypass the Regenerative Thermal Oxidizer (C008/EU038), Stack SV012, and route emissions from the Multiclones to the RTO Bypass Stack, SV006, and emit directly to the atmosphere in accordance with the following conditions:
 - A. The permittee shall not exceed 50 dryer operating hours per consecutive 12 month period of Regenerative Thermal Oxidizer (C008/EU038) regeneration. (Special Condition #8.B.6.))
 - B. The permittee shall not exceed 12 hours for each Regenerative Thermal Oxidizer (C008/EU038) regeneration event. (Special Condition #8.B.7.))
 - C. The permittee shall include all emissions from the RTO Bypass Stack, SV006, in the compliance demonstrations for all plant wide emission limitations. (Modified Special Condition #8.B.8.))

Performance Testing:

- 1. Frequency of tests
 - A. The permittee shall conduct performance tests to verify the emission rates as indicated this permit condition once every five years from the date of the most recent performance tests for each unit detailed in Table 10 below; except for Baghouse #1 (C001), and Baghouse #2 (C002). (Special Condition #15.I.).
 - a. Baghouse #1 (C001) and Baghouse #2 (C002) have one time testing requirements that have been satisfied. Baghouse #1 (C001) and Baghouse #2 (C002) are not required to undergo testing every five years. Baghouse #1 (C001) and Baghouse #2 (C002) are required to undergo additional testing if:
 - i. The units do not demonstrate compliance with the limitations in this permit or
 - ii. The installation modifies the equipment associated with Baghouse #1 (C001) or Baghouse #2 (C002), the baghouses themselves, or the stack release parameters in a manner that could cause a change in emission rates from Baghouse #1 (C001) or Baghouse #2 (C002).

2. Proposed Test Plan and Final Test Report

- A. The permittee shall submit one electronic copy of the completed proposed test plan form to the Air Pollution Control Program at least thirty days prior to the proposed test date so that the Air Pollution Control Program may arrange a pretest meeting, if necessary, and assure that the test date is acceptable for an observer to be present. The proposed test plant may serve the purpose of notification and must be approved by the Director prior to conducting the required emission testing. (Special Condition #16.A.)
- B. The permittee shall submit one electronic copy of a written report of the performance test results to the Director within thirty days of completion of any required testing. The report must include legible copies of the raw data sheets, analytical instrument laboratory data, and complete sample calculations from the required U. S. EPA Method for at least one sample run. (Special Condition #16.B.)
- C. The permittee shall ensure the test report fully accounts for all operational and emission parameters addressed both in this permit condition as well as in any other applicable state or federal rules or regulations. (Special Condition #16.C.)
- D. The permittee shall submit the electronic copies of the completed proposed test plan form and the written report of the performance test results to the Air Pollution Control Program at stacktesting@dnr.mo.gov.
- E. If the performance testing required by this permit condition indicates that any of the emission limits specified in this permit are being exceeded, the permittee must propose a remedial plan to the Air Pollution Control Program within thirty days of submitting the performance test results. This plan must demonstrate how the permittee will reduce emission rates to show compliance with specified limits in this permit. The permittee must begin to implement any such plan immediately upon its approval by the Director and conduct any additional performance testing . If necessary, additional controls must be started-up within twelve months after the performance test results report. (Special Condition #16.D.)

3. Testing Requirements:

- A. The permittee shall conduct performance tests as specified in Table 10 and: (Special Condition #15.A.)
 - i. The permittee shall conduct performance testing for the control devices shown in the following table when all equipment controlled by these devices are in operation.
- B. The permittee shall use the operating parameters (water flowrate, amount and type of additives, etc.) at which the performance tests are conducted to set the appropriate values used in actual operations of the Fermentation Scrubber (C003), Distillation Scrubber (C004), and the Regenerative Thermal Oxidizer (C008/EU038). Inlet air temperature, pH and pressure may vary under normal operating conditions and can be operated at different values than the value used during stack testing. All operating parameters shall be determined and agreed upon by the Air Pollution Control Program's Compliance/Enforcement Section and the permittee before the start of the performance tests. (Special Condition #15.B. and Special Condition #15.C.)
- C. The permittee shall record all operating parameters on record keeping sheet(s) and be made available to Department of Natural Resources personnel upon request. The frequency of the record keeping is dependent upon the parameters being kept and shall be determined and agreed upon by the Air Pollution Control Program's Compliance/Enforcement Section and the permittee before the start of the performance tests. (Special Condition #15.D.)
- D. The permittee shall conduct the performance tests for the Fermentation Scrubber (C003) for one of the following time periods: (Special Condition #15.E.)

- i. A complete cycle, defined as the time period between transferring the contents of one fermenter to the beer well and transferring the contents of the next fermenter; or (Special Condition #15.E.1.)
 - ii. During period(s) of representative emissions. The permittee shall submit, in the proposed test plan, sufficient data to determine the point(s) of representative emissions. The representative emissions are the average of 3 points identified as highest airflow, lowest airflow, and mid-range airflow going up or down the pressure curve. Testing will consist of three (3) 1-hour runs at each of the 3 points. These points must be approved by the Air Pollution Control Program's Compliance/Enforcement section prior to conducting the tests. Weighted averages can be used to determine the average emission rate. If sufficient data is not supplied supporting these representative emission points, the permittee must conduct testing for the complete cycle as defined above. (Special Condition #15.E.3.)
- E. The testing required may be limited to conducting tests on a representative piece(s) of each type of equipment upon approval by the Director. In addition, an alternate method(s) of quantifying the emission rates of criteria air pollutants from these sources may be used in place of the above testing requirement if requested by the permittee and approved by the Director. (Special Condition #15.F.)

Table 10: Summary of Continuous Testing Requirements

Stack ID	Emission Unit ID	Stack Description	Tested Pollutant	Measured Parameters	To Demonstrate Compliance with the Emission Limit of...		Testing Regulatory Citation
					Emission Limit	Units	
SV007	EU027	DDGS Cooling/ Separating Cyclone	VOC ⁶	1. Pound pollutant/hr, 2. Tons DDGS/hr as tested, and 3. Pound pollutant/ton of DDGS	2.79	Lbs VOC/hr	Special Condition #15.A.1.)
					95.0	Tons VOC /consecutive 12 month period ⁷	
			Total HAP		0.33	Lbs HAPS/hr	
					24.0	Tons total HAPs/consecutive 12 month period ⁷	Special Condition #15.A.2.)
			Speciated HAPS: Acetaldehyde, acrolein, formaldehyde, and methanol		9.0	Tons any individual HAP/consecutive 12 month period ⁷	
			PM ₁₀ ⁸		1.34	Tons PM ₁₀ /consecutive 12 month period	Special Condition #15.A.5.)
					95.0	Tons PM ₁₀ /consecutive 12 month period ⁷	

⁶ Testing shall include all VOC compounds identified in the Midwest Scaling Protocol and all VOC compounds identified in previous tests conducted by the installation.

⁷ This emissions limitation applies plant wide. The emission units indicated in this table are required to perform testing to establish emission factors for this pollutant.

⁸ All references to PM₁₀ in this permit condition include both filterables and condensibles.

Stack ID	Emission Unit ID	Stack Description	Tested Pollutant	Measured Parameters	To Demonstrate Compliance with the Emission Limit of...		Testing Regulatory Citation
					Emission Limit	Units	
SV008	EU028	Boiler #1 ⁹	CO	1. Pound pollutant/hr,	95.0	Tons CO /consecutive 12 month period ⁷	Special Condition #15.A.4.)
			NO _x	2. MMBtu/hr as tested, and	20.24	Tons NO _x /consecutive 12 month period ¹⁰	Special Condition #15.A.6.)
				3. Pound pollutant /MMBtu	95.0	Tons NO _x /consecutive 12 month period ⁷	Special Condition #15.A.
					0.030	Lbs/MMBtu	
SV011	EU034	Boiler #2	CO	1. Pound pollutant/hr,	95.0	Tons CO /consecutive 12 month period ⁷	Special Condition #15.A.4.)
			NO _x	2. MMBtu/hr as tested, and	95.0	Tons NO _x /consecutive 12 month period ⁷	Special Condition #15.A.
				3. Pound pollutant /MMBtu	0.032	Lbs/MMBtu	

⁹ For Boiler #1 (EU028) or Boiler #2 (EU034) the required testing is detailed in the table. The permittee may choose to conduct performance testing or demonstrate compliance using AP-42 emission factors for the following pollutants: VOC, HAP, PM₁₀, and SO_x. (Special Condition #15.A.8.)

¹⁰ The NO_x emission limitation applies to the sum of NO_x emissions of Boiler #1 (EU028), Stack SV008, and the DDGS Dryer Outlet (EU026), Stack SV012. Emissions from the RTO are not included in the emission limit.

Stack ID	Emission Unit ID	Stack Description	Tested Pollutant	Measured Parameters	To Demonstrate Compliance with the Emission Limit of...		Testing Regulatory Citation
					Emission Limit	Units	
SV004	C003	Fermentation scrubber	VOC ⁶	1. Pound pollutant/hr,	95.0% reduction or less than 10.0 ppm of VOC		Special Condition #15.A.1.)
				2. 1000 gallon ethanol/hr as tested, and	95.0	Tons VOC /consecutive 12 month period ⁷	
			Total HAP		24.0	Tons total HAPs/consecutive 12 month period ⁷	
			Speciated HAPS: Acetaldehyde, acrolein, formaldehyde, and methanol	3. Pound pollutant /1000 gallon ethanol	9.0	Tons any individual HAP/consecutive 12 month period ⁷	Special Condition #15.A.2.)
			PM ₁₀		95.0	Tons PM ₁₀ /consecutive 12 month period ⁷	Special Condition #15.A.
			SO _x		95.0	Tons SO _x /consecutive 12 month period ⁷	
SV005	C004	Distillation Scrubber	VOC ⁶	1. Pound pollutant/hr,	95.0	Tons VOC /consecutive 12 month period ⁷	Special Condition #15.A.1.)
			Total HAP	2. 1000 gallon ethanol/hr as tested, and	24.0	Tons total HAPs/consecutive 12 month period ⁷	
			Speciated HAPS: Acetaldehyde, acrolein, formaldehyde, and methanol	3. Pound pollutant /1000 gallon ethanol	9.0	Tons any individual HAP/consecutive 12 month period ⁷	Special Condition #15.A.2.)
			PM ₁₀		95.0	Tons PM ₁₀ /consecutive 12 month period ⁷	Special Condition #15.A.
			SO _x		95.0	Tons SO _x /consecutive 12 month period ⁷	

Stack ID	Emission Unit ID	Stack Description	Tested Pollutant	Measured Parameters	To Demonstrate Compliance with the Emission Limit of...		Testing Regulatory Citation
					Emission Limit	Units	
SV012	C008/ EU038 and EU026	Regenerative Thermal Oxidizer and DDGS dryer	VOC ⁶	1. Pound pollutant/hr,	95.0% reduction or less than 10.0 ppm of VOC		Special Condition #15.A.1.)
			Total HAP	2. MMBtu/hr as tested, and from the RTO and DDGS Dryer	95.0	Tons VOC /consecutive 12 month period ⁷	
				Speciated HAPS: Acetaldehyde, acrolein, formaldehyde, and methanol	24.0	Tons total HAPs/consecutive 12 month period ⁷	
			CO	3. Tons DDGS/hr in the DDGS Dryer as tested	9.0	Tons any individual HAP/consecutive 12 month period ⁷	Special Condition #15.A.2.)
				PM ₁₀	4. Pound pollutant /ton DDGS	90.0% reduction or less than 100.0 ppm of VOC	
			95.0			Tons CO /consecutive 12 month period ⁷	Special Condition #15.A.4.)
			NO _x	8.67		Tons PM ₁₀ /consecutive 12 month period	Special Condition #15.A.5.)
				95.0		Tons PM ₁₀ /consecutive 12 month period ⁷	Special Condition #15.A.
			SO _x	20.24		Tons NO _x /consecutive 12 month period ¹⁰	Special Condition #15.A.6.)
				95.0		Tons NO _x /consecutive 12 month period ⁷	Special Condition #15.A.
				2.69	Lbs/hr ¹¹	Special Condition #15.A.	
				95.0	Tons SO _x /consecutive 12 month period ⁷	Construction Permit Amendment 072008-001B, Special Condition #2.	

¹¹ This emission limitation applies to the sum of NO_x emissions from the DDGS Dryer (EU026) and the RTO (EU038). Both the DDGS Dryer and the RTO exhaust from Stack SV012.

Stack ID	Emission Unit ID	Stack Description	Tested Pollutant	Measured Parameters	To Demonstrate Compliance with the Emission Limit of...		Testing Regulatory Citation
					Emission Limit	Units	
SV006	EU026	DDGS Dryer	NO _x	1. Pound pollutant/hr,	0.040	Lbs/MMBtu	Special Condition #15.A.
				2. MMBtu/hr as tested, and from the DDGS Dryer	95.0	Tons NO _x /consecutive 12 month period ⁷	Special Condition #15.A.
				3. Tons DDGS/hr in the DDGS Dryer as tested			
				4. Pound pollutant /ton DDGS			
SV012	EU038	RTO	PM ₁₀	1. Pound pollutant/hr,	3.4	Lbs/hr	Special Condition #15.A.
				2. MMBtu/hr as tested,	95.0	Tons PM ₁₀ /consecutive 12 month period ⁷	Special Condition #15.A.
				3. Tons DDGS/hr in the DDGS Dryer as tested			
				4. Pound pollutant /ton DDGS			

Monitoring/Recordkeeping:

1. The permittee shall comply with the following requirements to demonstrate compliance with the emission limitations.

Table 11: Monitoring and Recordkeeping Requirements to Demonstrate Compliance with Emission Limitations

Pollutants for which there is an Emission Limitation	The permittee shall monitor and record the....	The permittee shall use the attachments listed, or approved alternative recordkeeping
PM ₁₀	Individual monthly and consecutive 12 month emissions totals	Attachment PM ₁₀
NO _x		Attachment NO _x
HAPs		Attachment Individual HAPs and Attachment Total HAPs
VOC		Attachment VOC
CO		Attachment CO
SO _x		Attachment SO _x

2. The permittee shall comply with the following requirements to demonstrate compliance with the operational limitations.
 - A. Throughput Limitations

Table 12: Monitoring and Recordkeeping Requirements to Demonstrate Compliance with Throughput Limitations

Parameter for which there is a Throughput Limitation	The permittee shall monitor and record the....	The permittee shall use the attachments listed, or approved alternative recordkeeping
Denatured Ethanol Production Rate	Individual monthly and consecutive 12 month throughput totals	Attachment Denatured Ethanol Production Rate
Total Ethanol Production Rate		Attachment Total Ethanol Production Rate
Maximum Denaturant Usage		Attachment Maximum Denaturant Usage
Truck and Rail Grain Receiving	Individual daily, monthly, and consecutive 12 month throughput totals	Attachment Truck and Rail Grain Receiving
Truck and Rail DDGS Shipping		Attachment Truck and Rail DDGS Shipping

- B. Combustion Fuel Type
 - i. The permittee shall obtain supplier certification from all suppliers of natural gas. (Special Condition #8.F.2.)
- C. Paved and Unpaved Haul Roads
 - i. The permittee shall maintain an inspection and maintenance log for the paved and unpaved haul roads. The log shall include the following:

- a. Date, time, and methods of all haul road washing/cleaning and application of dust suppression activities,
 - b. Date, time, and results of all pavement inspections for the paved haul road,
 - c. Date, time, and activities of all maintenance or repair of the paved surface for the paved haul road.
- ii. The permittee shall use Attachment Operating and Maintenance Log, or an approved alternative recordkeeping method, to demonstrate compliance.

D. Cooling Towers

- i. The permittee shall keep records of the monthly and consecutive 12-month totals of the hours of cooling tower operation. (Special Condition #17.C.)
 - ii. The permittee shall collect monthly TDS samples for analysis. The permittee shall record the results of the monthly TDS analysis on a monthly basis to verify the TDS concentration. (Special Condition #17.E.)
 - iii. The permittee may petition the Air Pollution Control Program for a reduced sampling frequency, or removal of the sampling requirement, if TDS sampling results demonstrate compliance for 24 consecutive months. The permittee shall not cease sampling until the petition is approved by the Air Pollution Control Program. (Special Condition #17.F.)
 - iv. The permittee shall verify the drift loss by keeping records of the manufacturer's documentation of the guaranteed drift loss rate. (Special Condition #17.D.)
 - v. The permittee shall keep all manufacturer's documentation onsite and make it readily available to Department of Natural Resources' employees upon request. (Special Condition #17.A.)
 - vi. The permittee shall use Attachment Cooling Towers, or an approved alternative recordkeeping method, to demonstrate compliance.
3. The permittee shall comply with the following requirements to demonstrate compliance with the operational limitations for add on control devices.
- A. Grain Storage Building (EU029), Baghouse #1 (C001), and Baghouse #2 (C002)
- i. The permittee shall demonstrate negative pressure on openings of the ductwork from the grain handling and process equipment (EU001 to EU012) ductwork to Baghouse (C001) and the openings of the ductwork from the Hammermill/Belt Scale (EU013) ductwork to Baghouse (C002) by using visual indicators such as streamers, talc puff test, negative pressure gauges, flags, etc. at openings that are not closed during normal operations. All openings must indicate the presence of negative pressure for compliance. (Special Condition #9.B.)
 - ii. The permittee shall perform a visual indicator check on the visual indicators detailed above for each ductwork opening at least once in every 24-hour period while any operations within the Grain Storage Building (EU029) are occurring. (Special Condition #9.C.)
 - iii. The permittee shall conduct visible emissions observation on the DDGS Dump Pits/Augers (EU008), the DDGS Elevator (EU009), the DDGS Rail Load Spout (EU010), and the DDGS Truck Load Spout (EU012) using the procedures contained in U.S. EPA Test Method 22. Observations are required during a time when the emission unit under observation is operating. Each Method 22 observation shall be conducted for a minimum of six-minutes. The permittee shall comply with the following monitoring schedule:

- a. The permittee shall conduct weekly observations of each emission unit for a minimum of eight consecutive weeks after permit issuance. If no visible emissions are observed during this period then,
 - b. The permittee shall conduct observations of each emission unit once every two weeks for a period of eight weeks. If no visible emissions are observed during this period then,
 - c. The permittee shall conduct observations of each emission unit once per month.
 - d. If at any time the permittee observes visible emissions from any emission unit listed above, the permittee shall submit a report to the Air Pollution Control Program within 10 days of the date the observation occurred. The report shall note the date of the observation, the emission unit(s) from which visible emissions were observed, and corrective actions taken to return the emission unit(s) to a state of no visible emissions.
 - iv. The permittee shall maintain an operating and maintenance log for the Grain Storage Building (EU029), and all material handling and processing equipment located inside the building, which shall include the following: (Special Condition #9.D.)
 - a. Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions. (Special Condition #9.D.1.))
 - b. Maintenance activities, with inspection schedule, repair actions, and replacements, etc. (Special Condition #9.D.2.))
 - c. A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections. Either paper copy or electronic formats are acceptable. (Special Condition #9.D.3.))
- B. Baghouse #1 (C001) and Baghouse #2 (C002)
 - i. The permittee shall monitor and record the operating pressure drop across each baghouse at least once every 24 hours. The operating pressure drop shall be maintained within the design conditions specified by the manufacturer's performance warranty. If the pressure drop reading fall outside of this normal operating range, then the associated equipment shall be shut down as quickly as is feasible and corrective action taken to address the cause of the pressure drop problem. The problem shall be corrected and the baghouse shall be operational before restarting the equipment. (Special Condition #10.D.)
 - ii. The permittee shall maintain an operating and maintenance log for each baghouse which shall include the following: (Special Condition #10.E.)
 - a. Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions (Special Condition #10.E.1.))
 - b. Maintenance activities, with inspection schedule, repair actions, and replacements, etc. (Special Condition #10.E.2.))
 - c. A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections. Either paper copy or electronic formats are acceptable. (Special Condition #10.E.3.))
- C. Fermentation Scrubber (C003) and Distillation Scrubber (C004)
 - i. The permittee shall monitor and record the operating pressure drop across each scrubber at least once every 24 hours. The operating pressure drop shall be maintained within the design conditions specified by the manufacturer's performance warranty and the operational parameters established by the most recent performance test which the Air

- Pollution Control Program accepted as compliant with applicable regulations. (Special Condition #11.E.)
- ii. The permittee shall maintain an operating and maintenance log for each scrubber which shall include the following: (Special Condition #11.F.)
 - a. Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions. (Special Condition #11.F.1.))
 - b. Maintenance activities, with inspection schedule, repair actions, and replacements, etc. (Special Condition #11.F.2.))
 - c. A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspection. Either paper copy or electronic formats are acceptable. (Special Condition #11.F.3.))
- D. Flare (EU039)
- i. The permittee shall maintain records that sufficiently indicate compliance with 40 CFR Part 60.18. (Special Condition #12.B.)
 - ii. The permittee shall maintain an operating and maintenance log for the flare, which shall include the following. (Special Condition #12.C.)
 - a. Incidents of malfunction, with impact on emissions, duration of events, probable cause, and corrective actions taken. (Special Condition #12.C.1.))
 - b. Maintenance activities, with inspection schedules, repair actions, and replacements. (Special Condition #12.C.2.))
 - c. A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections. Either paper copy or electronic formats are acceptable. (Special Condition #12.C.3.))
- E. DDGS Cooling/Separating Cyclone (EU027)
- i. The permittee shall monitor and record the operating pressure drop across the DDGS Cooling/Separating Cyclone (EU027). at least once every 24 hours. (Special Condition #13.E)
 - a. The observation of pressure drop deviations for the DDGS Cooling/Separating Cyclone (EU027) outside of the range 3 to 8 in WC will be considered an excursion and corrective actions shall be implemented within a reasonable period. (Special Condition #13.E.1.))
 - b. An excursion does not necessarily indicate a violation of the applicable requirement. When the level of excursions exceed three percent of the of the total number of observations in a six month period and corrective actions fail to return the the DDGS Cooling/Separating Cyclone (EU027) to a no visible emission condition, then the permittee shall conduct source testing within 90 days of the last excursion to demonstrate compliance with 10 CSR 10-6.400. (Special Condition #13.E.2.))
 - c. If the test demonstrates noncompliance with the emission limitation, the permittee shall propose a schedule to implement further corrective actions to bring the source into compliance and demonstrate that compliance. (Special Condition #13.E.3.))
 - ii. The permittee shall maintain an operating and maintenance log for the the DDGS Cooling/Separating Cyclone (EU027) which shall include the following: (Special Condition #13.F.)
 - a. Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions; and (Special Condition #13.F.1.))

- b. Maintenance activities, with inspection schedule, repair actions, and replacements, etc. (Special Condition #13.F.2.))
 - c. A written record of regular inspection schedule, the date and results of all inspections including any actions or maintenance activities that result from that inspection. (Special Condition #13.F.3.))
- F. Regenerative Thermal Oxidizer (C008/EU038)
 - i. The permittee shall continuously monitor and record the operating temperature of the thermal oxidizer during operation. The permittee shall maintain the operating temperature of the thermal oxidizer at 1614 degrees Fahrenheit or greater based on a 1 hour averaging time. The acceptable temperature range may be reestablished by performing a new set of performance testing. (Special Condition #14.B.)
 - ii. The permittee shall maintain an operating and maintenance log for the thermal oxidizer which shall include the following: (Special Condition #14.C.)
 - a. Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions; and (Special Condition #14.C.1.))
 - b. Maintenance activities, with inspection schedule, repair actions, and replacements, etc. (Special Condition #14.C.2.))
 - c. A written record of regular inspection schedule, the date and results of all inspections including any actions or maintenance activities that result from that inspection. (Special Condition #14.C.3.))

Table 13: Summary of Monitoring and Recordkeeping Attachments for Add On Control Devices

Add On Control Device	The permittee shall monitor and record the...	The permittee shall use the attachments listed, or approved alternative recordkeeping
Baghouse #1 (C001) and Baghouse #2 (C002)	Visual indicator checks	Attachment Visual Indicator Checks
	Operating pressure drop	Attachment Pressure Drop
	Operating and maintenance log	Attachment Operating and Maintenance Log
DDGS Dump Pits/Augers (EU008), the DDGS Elevator (EU009), the DDGS Rail Load Spout (EU010), and the DDGS Truck Load Spout (EU012)	Method 22 visible emission observations	Attachment Method 22
Grain Storage Building (EU029) and all material handling and processing equipment located inside the building	Operating and maintenance log	Attachment Operating and Maintenance Log
Fermentation Scrubber (C003) and Distillation Scrubber (C004)	Operating pressure drop	Attachment Pressure Drop
	Operating and maintenance log	Attachment Operating and Maintenance Log
Flare (EU039)	Operating and maintenance log	Attachment Operating and Maintenance Log

Add On Control Device	The permittee shall monitor and record the...	The permittee shall use the attachments listed, or approved alternative recordkeeping
DDGS Cooling/Separating Cyclone (EU027)	Operating pressure drop	Attachment Pressure Drop
	Operating and maintenance log	Attachment Operating and Maintenance Log
Regenerative Thermal Oxidizer (C008/EU038)	Operating and maintenance log	Attachment Operating and Maintenance Log

4. The permittee shall comply with the following requirements to demonstrate compliance with the alternative operating scenario.
 - A. The permittee shall monitor and record the monthly and 12 consecutive month total number of hours the DDGS Dryer operated during Regenerative Thermal Oxidizer (C008/EU038) regeneration.
 - B. The permittee shall monitor and record the date and total number of hours of each Regenerative Thermal Oxidizer (C008/EU038) regeneration event.
 - C. The permittee shall use Attachment Regenerative Thermal Oxidizer Bypass, or an approved alternative recordkeeping method, to demonstrate compliance.
5. The permittee shall comply with all monitoring and recordkeeping detailed in the Performance Testing portion of this permit condition.
6. All records required by this permit shall be kept onsite for no less than 5 years and shall be made available to any Department of Natural Resources' personnel upon request. (Special Condition #19.)

Reporting:

1. The permittee shall report any exceedance of the emission limitations or operational limitations no later than ten days after an exceedance of the limitation.
2. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
3. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition Subpart A-Flare

10 CSR 10-6.070 New Source Performance Regulations
40 CFR part 60 Subpart A, General Provisions

Operational Limitations/Monitoring/Recordkeeping:

1. The permittee shall design for and operate the flare with no visible emissions as determined by the methods specified in §60.18(f), except for periods not to exceed a total of 5 minutes during any 2 consecutive hours. [§60.18(c)(1)]
2. The permittee shall operate the flare with a flame present at all times, as determined by the methods specified in §60.18(f). [§60.18(c)(2)]

3. The permittee has the choice of adhering to either the heat content specifications in §60.18(c)(3)(ii) and the maximum tip velocity specifications in §60.18(c)(4), or adhering to the requirements in §60.18(c)(3)(i). [§60.18(c)(3)]
- A. Flares shall be used that have a diameter of 3 inches or greater, are nonassisted, have a hydrogen content of 8.0 percent (by volume), or greater, and are designed for and operated with an exit velocity less than 37.2 m/sec (122 ft/sec) and less than the velocity, V_{max} , as determined by the following equation: [§60.18(c)(3)(i)(A)]

$$V_{max} = (X_{H_2} - K_1) * K_2$$

Where:

V_{max} = Maximum permitted velocity, m/sec.

K_1 = Constant, 6.0 volume-percent hydrogen.

K_2 = Constant, 3.9(m/sec)/volume-percent hydrogen.

X_{H_2} = The volume-percent of hydrogen, on a wet basis, as calculated by using the American Society for Testing and Materials (ASTM) Method D1946-77. (Incorporated by reference as specified in §60.17).

- B. The actual exit velocity of a flare shall be determined by the method specified in §60.18(f)(4). [§60.18(c)(3)(i)(B)]
4. The permittee shall use the flare only with the net heating value of the gas being combusted being 11.2 MJ/scm (300 Btu/scf) or greater if the flare is steam-assisted or air-assisted; or with the net heating value of the gas being combusted being 7.45 MJ/scm (200 Btu/scf) or greater if the flare is nonassisted. The net heating value of the gas being combusted shall be determined by the methods specified in §60.18(f)(3). [§60.18(c)(3)(ii)]
5. The permittee shall design and operate steam-assisted and nonassisted flares with an exit velocity, as determined by the methods specified in §60.18(f)(4), less than 18.3 m/sec (60 ft/sec), except as provided in §60.18(c)(4) (ii) and (iii). [§60.18(c)(4)(i)]
6. Steam-assisted and nonassisted flares designed for and operated with an exit velocity, as determined by the methods specified in §60.18(f)(4), equal to or greater than 18.3 m/sec (60 ft/sec) but less than 122 m/sec (400 ft/sec) are allowed if the net heating value of the gas being combusted is greater than 37.3 MJ/scm (1,000 Btu/scf). [§60.18(c)(4)(ii)]
7. Steam-assisted and nonassisted flares designed for and operated with an exit velocity, as determined by the methods specified in §60.18(f)(4), less than the velocity, V_{max} , as determined by the method specified in §60.18(f)(5), and less than 122 m/sec (400 ft/sec) are allowed. [§60.18(c)(4)(iii)]
8. The permittee shall design and operate air-assisted flares with an exit velocity less than the velocity, V_{max} , as determined by the method specified in §60.18(f)(6). [§60.18(c)(5)]
9. Flares used to comply with this section shall be steam-assisted, air-assisted, or nonassisted. [§60.18(c)(6)]
10. The permittee shall monitor these control devices to ensure that they are operated and maintained in conformance with their designs. [§60.18(d)]
11. The permittee shall operated the flares at all times when emissions may be vented to them. [§60.18(e)]
12. The permittee shall use Method 22 of appendix A to part 60 to determine the compliance of flares with the visible emission provisions of Subpart A of part 60. The observation period is 2 hours and shall be used according to Method 22 of appendix A to part 60. [§60.18(f)(1)]

13. The permittee shall monitor the presence of a flare pilot flame using a thermocouple or any other equivalent device to detect the presence of a flame. [§60.18(f)(2)]
14. The permittee shall calculate the net heating value of the gas being combusted in a flare using the following equation: [§60.18(f)(3)]

$$H_T = K * \sum_{i=1}^n C_i H_i$$

Where:

H_T = Net heating value of the sample, MJ/scm; where the net enthalpy per mole of offgas is based on combustion at 25 °C and 760 mm Hg, but the standard temperature for determining the volume corresponding to one mole is 20 °C;

K = constant $1.740 \times 10^{-7} \frac{g-mol*MJ}{ppm*scm*kcal}$, where the standard temperature for $\frac{g-mol}{scm}$ is 20°C

C_i = Concentration of sample component i in ppm on a wet basis, as measured for organics by Reference Method 18 and measured for hydrogen and carbon monoxide by ASTM D1946-77 or 90 (Reapproved 1994) (Incorporated by reference as specified in §60.17); and

H_i = Net heat of combustion of sample component i, kcal/g mole at 25 °C and 760 mm Hg. The heats of combustion may be determined using ASTM D2382-76 or 88 or D4809-95 (incorporated by reference as specified in §60.17) if published values are not available or cannot be calculated.

15. The actual exit velocity of a flare shall be determined by dividing the volumetric flowrate (in units of standard temperature and pressure), as determined by Reference Methods 2, 2A, 2C, or 2D as appropriate; by the unobstructed (free) cross sectional area of the flare tip. [§60.18(f)(4)]
16. The maximum permitted velocity, V_{max} , for flares complying with §60.18(c)(4)(iii) shall be determined by the following equation. [§60.18(f)(5)]

$$\log_{10} V_{max} = \frac{(H_T + 28.8)}{31.7}$$

Where:

V_{max} = Maximum permitted velocity, M/sec

28.8 = Constant

31.7 = Constant

H_T = The net heating value as determined in §60.18(f)(3).

17. The maximum permitted velocity, V_{max} , for air-assisted flares shall be determined by the following equation. [§60.18(f)(6)]

$$V_{max} = 8.706 + 0.7084(H_T)$$

Where:

V_{max} = Maximum permitted velocity, m/sec

8.706 = Constant

0.7084 = Constant

H_T = The net heating value as determined in §60.18(f)(3).

Reporting:

1. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
2. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition Subpart VV

10 CSR 10-6.070 New Source Performance Regulations

40 CFR part 60 Subpart A, General Provisions

40 CFR part 60 Subpart VV, Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006

Applicability:

1. This permit condition applies to all LDAR monitoring accounted for under emission point FS002-1.
2. If the permittee applies for one or more of the exemptions in §60.480(d), then the permittee shall maintain records as required in §60.486(i). [§60.480(d)(1)]
3. Any affected facility that produces beverage alcohol is exempt from §§60.482-1 through 60.482-10. [§60.480(d)(4)]
4. *Alternative means of compliance* [§60.480(e)]
 - A. *Option to comply with part 65.* [§60.480(e)(1)]
 - i. The permittee may choose to comply with the provisions of 40 CFR part 65, subpart F, to satisfy the requirements of §§60.482 through 60.487 for an affected facility. When choosing to comply with 40 CFR part 65, subpart F, the requirements of §§60.485(d), (e), and (f) and 60.486(i) and (j) still apply. Other provisions applying to the permittee choosing to comply with 40 CFR part 65 are provided in 40 CFR 65.1. [§60.480(e)(1)(i)]
 - ii. *Part 60, subpart A.* The permittee choosing to comply with 40 CFR part 65, subpart F must also comply with §§60.1, 60.2, 60.5, 60.6, 60.7(a)(1) and (4), 60.14, 60.15, and 60.16 for that equipment. All provisions of subpart A of part 60 that are not mentioned in §60.480(e)(1)(ii) do not apply to the permittee of equipment subject to this Subpart VV complying with 40 CFR part 65, subpart F, except that provisions required to be met prior to implementing 40 CFR part 65 still apply. The permittee who choose to comply with 40 CFR part 65, subpart F, must comply with 40 CFR part 65, subpart A. [§60.480(e)(1)(ii)]
 - B. *Subpart VVa.* The permittee may choose to comply with the provisions of 40 CFR part 60 Subpart VVa to satisfy the requirements of Subpart VV for an affected facility. [§60.480(e)(2)]
5. *Stay of standards.* The permittee is not required to comply with the definition of "process unit" in §60.481 and the requirements in §60.482-1(g) of Subpart VV until the EPA takes final action to require compliance and publishes a document in the Federal Register. While the definition of "process unit" is stayed, the permittee should use the following definition: [§60.480(f)]

Process unit means components assembled to produce, as intermediate or final products, one or more of the chemicals listed in §60.489 of part 60. A process unit can operate independently if supplied with sufficient feed or raw materials and sufficient storage facilities for the product.

General Standards:

1. Compliance with §§60.482-1 to 60.482-10 will be determined by review of records and reports, review of performance test results, and inspection using the methods and procedures specified in §60.485. [§60.482-1(b)]
2. The permittee may request a determination of equivalence of a means of emission limitation to the requirements of §§60.482-2, 60.482-3, 60.482-5, 60.482-6, 60.482-7, 60.482-8, and 60.482-10 as provided in §60.484. [§60.482-1(c)(1)]
3. If the Director makes a determination that a means of emission limitation is at least equivalent to the requirements of §60.482-2, §60.482-3, §60.482-5, §60.482-6, §60.482-7, §60.482-8, or §60.482-10, the permittee shall comply with the requirements of that determination. [§60.482-1(c)(2)]
4. Equipment that is in vacuum service is excluded from the requirements of §§60.482-2 to 60.482-10 if it is identified as required in §60.486(e)(5). [§60.482-1(d)]
5. Equipment that the permittee designates as being in VOC service less than 300 hours/year is excluded from the requirements of §§60.482-2 through 60.482-10 if it is identified as required in §60.486(e)(6) and it meets any of the conditions specified in §60.482-1(e)(1) through (3). [§60.482-1(e)]
 - A. The equipment is in VOC service only during startup and shutdown, excluding startup and shutdown between batches of the same campaign for a batch process. [§60.482-1(e)(1)]
 - B. The equipment is in VOC service only during process malfunctions or other emergencies. [§60.482-1(e)(2)]
 - C. The equipment is backup equipment that is in VOC service only when the primary equipment is out of service. [§60.482-1(e)(3)]
6. If a dedicated batch process unit operates less than 365 days during a year, the permittee may monitor to detect leaks from pumps and valves at the frequency specified in the following table instead of monitoring as specified in §§60.482-2, 60.482-7, and 60.483-2: [§60.482-1(f)(1)]

Table 14: Equivalent Monitoring Frequencies for Subpart VV

Operating Time (percent of hours during year)	Equivalent Monitoring Frequency Time in Use		
	Monthly	Quarterly	Semiannually
0 to <25	Quarterly	Annually	Annually.
25 to <50	Quarterly	Semiannually	Annually.
50 to <75	Bimonthly	Three quarters	Semiannually.
75 to 100	Monthly	Quarterly	Semiannually.

7. Pumps and valves that are shared among two or more batch process units that are subject to Subpart VV may be monitored at the frequencies specified in §60.482-1(f)(1), provided the operating time of all such process units is considered. [§60.482-1(f)(2)]
8. The monitoring frequencies specified in §60.482-1(f)(1) are not requirements for monitoring at specific intervals and can be adjusted to accommodate process operations. The permittee may monitor at any time during the specified monitoring period (e.g., month, quarter, year), provided the monitoring is conducted at a reasonable interval after completion of the last monitoring campaign. Reasonable intervals are defined in §60.482-1(f)(3)(i) through (iv). [§60.482-1(f)(3)]
 - A. When monitoring is conducted quarterly, monitoring events must be separated by at least 30 calendar days. [§60.482-1(f)(3)(i)]

- B. When monitoring is conducted semiannually (*i.e.*, once every 2 quarters), monitoring events must be separated by at least 60 calendar days. [§60.482-1(f)(3)(ii)]
- C. When monitoring is conducted in 3 quarters per year, monitoring events must be separated by at least 90 calendar days. [§60.482-1(f)(3)(iii)]
- D. When monitoring is conducted annually, monitoring events must be separated by at least 120 calendar days. [§60.482-1(f)(3)(iv)]

Standards for Pumps in light liquid service:

1. Each pump in light liquid service shall be monitored monthly to detect leaks by the methods specified in §60.485(b), except as provided in §60.482-1(c) and (f) and §60.482-2(d), (e), and (f). A pump that begins operation in light liquid service after the initial startup date for the process unit must be monitored for the first time within 30 days after the end of its startup period, except for a pump that replaces a leaking pump and except as provided in §60.482-1(c) and (f) and §60.482-2(d), (e), and (f). [§60.482-2(a)(1)]
2. Each pump in light liquid service shall be checked by visual inspection each calendar week for indications of liquids dripping from the pump seal, except as provided in §60.482-1(f). [§60.482-2(a)(2)]
3. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-2(b)(1)]
4. If there are indications of liquids dripping from the pump seal, the permittee shall follow the procedure specified in either §60.482-2(b)(2)(i) or (ii). This requirement does not apply to a pump that was monitored after a previous weekly inspection if the instrument reading for that monitoring event was less than 10,000 ppm and the pump was not repaired since that monitoring event. [§60.482-2(b)(2)]
 - A. Monitor the pump within 5 days as specified in §60.485(b). If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. The leak shall be repaired using the procedures in §60.482-2(c). [§60.482-2(b)(2)(i)]
 - B. Designate the visual indications of liquids dripping as a leak, and repair the leak within 15 days of detection by eliminating the visual indications of liquids dripping. [§60.482-2(b)(2)(ii)]
5. When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9. [§60.482-2(c)(1)]
6. A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. First attempts at repair include, but are not limited to, the practices described in §60.482-2(c)(2)(i) and (ii), where practicable. [§60.482-2(c)(2)]
 - A. Tightening the packing gland nuts; [§60.482-2(c)(2)(i)]
 - B. Ensuring that the seal flush is operating at design pressure and temperature. [§60.482-2(c)(2)(ii)]
7. Each pump equipped with a dual mechanical seal system that includes a barrier fluid system is exempt from the requirements of §60.482-2(a), provided the requirements specified in §60.482-2(d)(1) through (6) are met. [§60.482-2(d)]
 - A. Each dual mechanical seal system is: [§60.482-2(d)(1)]
 - i. Operated with the barrier fluid at a pressure that is at all times greater than the pump stuffing box pressure; or [§60.482-2(d)(1)(i)]
 - ii. Equipped with a barrier fluid degassing reservoir that is routed to a process or fuel gas system or connected by a closed vent system to a control device that complies with the requirements of §60.482-10; or [§60.482-2(d)(1)(ii)]
 - iii. Equipped with a system that purges the barrier fluid into a process stream with zero VOC emissions to the atmosphere. [§60.482-2(d)(1)(iii)]

- B. The barrier fluid system is in heavy liquid service or is not in VOC service. [§60.482-2(d)(2)]
- C. Each barrier fluid system is equipped with a sensor that will detect failure of the seal system, the barrier fluid system, or both. [§60.482-2(d)(3)]
- D. Each pump is checked by visual inspection, each calendar week, for indications of liquids dripping from the pump seals. [§60.482-2(d)(4)(i)]
- E. If there are indications of liquids dripping from the pump seal at the time of the weekly inspection, the permittee shall follow the procedure specified in either §60.482-2(d)(4)(ii)(A) or (B). [§60.482-2(d)(4)(ii)]
 - i. Monitor the pump within 5 days as specified in §60.485(b) to determine if there is a leak of VOC in the barrier fluid. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-2(d)(4)(ii)(A)]
 - ii. Designate the visual indications of liquids dripping as a leak. [§60.482-2(d)(4)(ii)(B)]
- F. Each sensor as described in §60.482-2(d)(3) is checked daily or is equipped with an audible alarm. [§60.482-2(d)(5)(i)]
- G. The permittee determines, based on design considerations and operating experience, a criterion that indicates failure of the seal system, the barrier fluid system, or both. [§60.482-2(d)(5)(ii)]
- H. If the sensor indicates failure of the seal system, the barrier fluid system, or both, based on the criterion established in §60.482-2(d)(5)(ii), a leak is detected. [§60.482-2(d)(5)(iii)]
- I. When a leak is detected pursuant to §60.482-2(d)(4)(ii)(A), it shall be repaired as specified in §60.482-2(c). [§60.482-2(d)(6)(i)]
- J. A leak detected pursuant to §60.482-2(d)(5)(iii) shall be repaired within 15 days of detection by eliminating the conditions that activated the sensor. [§60.482-2(d)(6)(ii)]
- K. A designated leak pursuant to §60.482-2(d)(4)(ii)(B) shall be repaired within 15 days of detection by eliminating visual indications of liquids dripping. [§60.482-2(d)(6)(iii)]
- 8. Any pump that is designated, as described in §60.486(e)(1) and (2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-2(a), (c), and (d) if the pump: [§60.482-2(e)]
 - A. Has no externally actuated shaft penetrating the pump housing, [§60.482-2(e)(1)]
 - B. Is demonstrated to be operating with no detectable emissions as indicated by an instrument reading of less than 500 ppm above background as measured by the methods specified in §60.485(c), and [§60.482-2(e)(2)]
 - C. Is tested for compliance with §60.482-2(e)(2) initially upon designation, annually, and at other times requested by the Director. [§60.482-2(e)(3)]
- 9. If any pump is equipped with a closed vent system capable of capturing and transporting any leakage from the seal or seals to a process or to a fuel gas system or to a control device that complies with the requirements of §60.482-10, it is exempt from §60.482-2(a) through (e). [§60.482-2(f)]
- 10. Any pump that is designated, as described in §60.486(f)(1), as an unsafe-to-monitor pump is exempt from the monitoring and inspection requirements of §60.482-2(a) and (d)(4) through (6) if: [§60.482-2(g)]
 - A. The permittee demonstrates that the pump is unsafe-to-monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-2(a); and [§60.482-2(g)(1)]
 - B. The permittee has a written plan that requires monitoring of the pump as frequently as practicable during safe-to-monitor times but not more frequently than the periodic monitoring schedule otherwise applicable, and repair of the equipment according to the procedures in §60.482-2(c) if a leak is detected. [§60.482-2(g)(2)]

11. Any pump that is located within the boundary of an unmanned plant site is exempt from the weekly visual inspection requirement of §60.482-2(a)(2) and (d)(4), and the daily requirements of §60.482-2(d)(5), provided that each pump is visually inspected as often as practicable and at least monthly. [§60.482-2(h)]

Standards for Compressors:

1. Each compressor shall be equipped with a seal system that includes a barrier fluid system and that prevents leakage of VOC to the atmosphere, except as provided in §60.482-1(c) and §60.482-3(h), (i), and (j). [§60.482-3(a)]
2. Each compressor seal system as required in §60.482-3(a) shall be: [§60.482-3(b)]
 - A. Operated with the barrier fluid at a pressure that is greater than the compressor stuffing box pressure; or [§60.482-3(b)(1)]
 - B. Equipped with a barrier fluid system degassing reservoir that is routed to a process or fuel gas system or connected by a closed vent system to a control device that complies with the requirements of §60.482-10; or [§60.482-3(b)(2)]
 - C. Equipped with a system that purges the barrier fluid into a process stream with zero VOC emissions to the atmosphere. [§60.482-3(b)(3)]
3. The barrier fluid system shall be in heavy liquid service or shall not be in VOC service. [§60.482-3(c)]
4. Each barrier fluid system as described in §60.482-3(a) shall be equipped with a sensor that will detect failure of the seal system, barrier fluid system, or both. [§60.482-3(d)]
5. Each sensor as required in §60.482-3(d) shall be checked daily or shall be equipped with an audible alarm. [§60.482-3(e)(1)]
6. The permittee shall determine, based on design considerations and operating experience, a criterion that indicates failure of the seal system, the barrier fluid system, or both. [§60.482-3(e)(2)]
7. If the sensor indicates failure of the seal system, the barrier system, or both based on the criterion determined under §60.482-3(e)(2), a leak is detected. [§60.482-3(f)]
8. When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9. [§60.482-3(g)(1)]
9. A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-3(g)(2)]
10. A compressor is exempt from the requirements of §60.482-3(a) and (b), if it is equipped with a closed vent system to capture and transport leakage from the compressor drive shaft back to a process or fuel gas system or to a control device that complies with the requirements of §60.482-10, except as provided in §60.482-3(i). [§60.482-3(h)]
11. Any compressor that is designated, as described in §60.486(e) (1) and (2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-3(a)-(h) if the compressor: [§60.482-3(i)]
 - A. Is demonstrated to be operating with no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as measured by the methods specified in §60.485(c); and [§60.482-3(i)(1)]
 - B. Is tested for compliance with §60.482-3(i)(1) initially upon designation, annually, and at other times requested by the Director. [§60.482-3(i)(2)]
12. Any existing reciprocating compressor in a process unit which becomes an affected facility under provisions of §60.14 or §60.15 is exempt from §60.482-3(a) through (e) and (h), provided the permittee demonstrates that recasting the distance piece or replacing the compressor are the only

options available to bring the compressor into compliance with the provisions of §60.482-3(a) through (e) and (h). [§60.482-3(j)]

Standards for pressure relief devices in gas/vapor service:

1. Except during pressure releases, each pressure relief device in gas/vapor service shall be operated with no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as determined by the methods specified in §60.485(c). [§60.482-4(a)]
2. After each pressure release, the pressure relief device shall be returned to a condition of no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as soon as practicable, but no later than 5 calendar days after the pressure release, except as provided in §60.482-9. [§60.482-4(b)(1)]
3. No later than 5 calendar days after the pressure release, the pressure relief device shall be monitored to confirm the conditions of no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, by the methods specified in §60.485(c). [§60.482-4(b)(2)]
4. Any pressure relief device that is routed to a process or fuel gas system or equipped with a closed vent system capable of capturing and transporting leakage through the pressure relief device to a control device as described in §60.482-10 is exempted from the requirements of §60.482-4(a) and (b). [§60.482-4(c)]
5. Any pressure relief device that is equipped with a rupture disk upstream of the pressure relief device is exempt from the requirements of §60.482-4(a) and (b), provided the permittee complies with the requirements in §60.482-4(d)(2). [§60.482-4(d)(1)]
6. After each pressure release, a new rupture disk shall be installed upstream of the pressure relief device as soon as practicable, but no later than 5 calendar days after each pressure release, except as provided in §60.482-9. [§60.482-4(d)(2)]

Standards for sampling connection systems:

1. Each sampling connection system shall be equipped with a closed-purge, closed-loop, or closed-vent system, except as provided in §60.482-1(c) and §60.482-5(c). [§60.482-5(a)]
2. Each closed-purge, closed-loop, or closed-vent system as required in §60.482-5(a) shall comply with the requirements specified in §60.482-5(b)(1) through (4). [§60.482-5(b)]
 - A. Gases displaced during filling of the sample container are not required to be collected or captured. [§60.482-5(b)(1)]
 - B. Containers that are part of a closed-purge system must be covered or closed when not being filled or emptied. [§60.482-5(b)(2)]
 - C. Gases remaining in the tubing or piping between the closed-purge system valve(s) and sample container valve(s) after the valves are closed and the sample container is disconnected are not required to be collected or captured. [§60.482-5(b)(3)]
 - D. Each closed-purge, closed-loop, or closed-vent system shall be designed and operated to meet requirements in either §60.482-5(b)(4)(i), (ii), (iii), or (iv). [§60.482-5(b)(4)]
 - i. Return the purged process fluid directly to the process line. [§60.482-5(b)(4)(i)]
 - ii. Collect and recycle the purged process fluid to a process. [§60.482-5(b)(4)(ii)]
 - iii. Capture and transport all the purged process fluid to a control device that complies with the requirements of §60.482-10. [§60.482-5(b)(4)(iii)]
 - iv. Collect, store, and transport the purged process fluid to any of the following systems or facilities: [§60.482-5(b)(4)(iv)]

- a. A waste management unit as defined in §63.111, if the waste management unit is subject to and operated in compliance with the provisions of 40 CFR part 63, subpart G, applicable to Group 1 wastewater streams; [§60.482-5(b)(4)(iv)(A)]
 - b. A treatment, storage, or disposal facility subject to regulation under 40 CFR part 262, 264, 265, or 266; [§60.482-5(b)(4)(iv)(B)]
 - c. A facility permitted, licensed, or registered by a state to manage municipal or industrial solid waste, if the process fluids are not hazardous waste as defined in 40 CFR part 261; [§60.482-5(b)(4)(iv)(C)]
 - d. A waste management unit subject to and operated in compliance with the treatment requirements of §61.348(a), provided all waste management units that collect, store, or transport the purged process fluid to the treatment unit are subject to and operated in compliance with the management requirements of §§61.343 through 61.347; or [§60.482-5(b)(4)(iv)(D)]
 - e. A device used to burn off-specification used oil for energy recovery in accordance with 40 CFR part 279, subpart G, provided the purged process fluid is not hazardous waste as defined in 40 CFR part 261. [§60.482-5(b)(4)(iv)(E)]
3. In situ sampling systems and sampling systems without purges are exempt from the requirements of §60.482-5(a) and (b). [§60.482-5(c)]

Standards for Open-ended valves or lines:

1. Each open-ended valve or line shall be equipped with a cap, blind flange, plug, or a second valve, except as provided in §60.482-1(c) and §60.482-6(d) and (e). [§60.482-6(a)(1)]
2. The cap, blind flange, plug, or second valve shall seal the open end at all times except during operations requiring process fluid flow through the open-ended valve or line. [§60.482-6(a)(2)]
3. Each open-ended valve or line equipped with a second valve shall be operated in a manner such that the valve on the process fluid end is closed before the second valve is closed. [§60.482-6(b)]
4. When a double block-and-bleed system is being used, the bleed valve or line may remain open during operations that require venting the line between the block valves but shall comply with §60.482-6(a) at all other times. [§60.482-6(c)]
5. Open-ended valves or lines in an emergency shutdown system which are designed to open automatically in the event of a process upset are exempt from the requirements of §60.482-6(a), (b) and (c). [§60.482-6(d)]
6. Open-ended valves or lines containing materials which would autocatalytically polymerize or would present an explosion, serious overpressure, or other safety hazard if capped or equipped with a double block and bleed system as specified in §60.482-6(a) through (c) are exempt from the requirements of §60.482-6(a) through (c). [§60.482-6(e)]

Standards for Valves in gas/vapor service and in light liquid service:

1. Each valve shall be monitored monthly to detect leaks by the methods specified in §60.485(b) and shall comply with §60.482-7(b) through (e), except as provided in §60.482-7(f), (g), and (h), §60.482-1(c) and (f), and §§60.483-1 and 60.483-2. [§60.482-7(a)(1)]
2. A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for the process unit must be monitored according to §60.482-7(a)(2)(i) or (ii), except for a valve that replaces a leaking valve and except as provided in §60.482-7(f), (g), and (h), §60.482-1(c), and §§60.483-1 and 60.483-2. [§60.482-7(a)(2)]

- A. Monitor the valve as in §60.482-7(a)(1). The valve must be monitored for the first time within 30 days after the end of its startup period to ensure proper installation. [§60.482-7(a)(2)(i)]
- B. If the valves on the process unit are monitored in accordance with §60.483-1 or §60.483-2, count the new valve as leaking when calculating the percentage of valves leaking as described in §60.483-2(b)(5). If less than 2.0 percent of the valves are leaking for that process unit, the valve must be monitored for the first time during the next scheduled monitoring event for existing valves in the process unit or within 90 days, whichever comes first. [§60.482-7(a)(2)(i)]
3. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-7(b)]
4. Any valve for which a leak is not detected for 2 successive months may be monitored the first month of every quarter, beginning with the next quarter, until a leak is detected. [§60.482-7(c)(1)(i)]
5. As an alternative to monitoring all of the valves in the first month of a quarter, the permittee may elect to subdivide the process unit into 2 or 3 subgroups of valves and monitor each subgroup in a different month during the quarter, provided each subgroup is monitored every 3 months. The permittee must keep records of the valves assigned to each subgroup. [§60.482-7(c)(1)(ii)]
6. If a leak is detected, the valve shall be monitored monthly until a leak is not detected for 2 successive months. [§60.482-7(c)(2)]
7. When a leak is detected, it shall be repaired as soon as practicable, but no later than 15 calendar days after the leak is detected, except as provided in §60.482-9. [§60.482-7(d)(1)]
8. A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-7(d)(2)]
9. First attempts at repair include, but are not limited to, the following best practices where practicable: [§60.482-7(e)]
 - A. Tightening of bonnet bolts; [§60.482-7(e)(1)]
 - B. Replacement of bonnet bolts; [§60.482-7(e)(2)]
 - C. Tightening of packing gland nuts; [§60.482-7(e)(3)]
 - D. Injection of lubricant into lubricated packing. [§60.482-7(e)(4)]
10. Any valve that is designated, as described in §60.486(e)(2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-7(a) if the valve: [§60.482-7(f)]
 - A. Has no external actuating mechanism in contact with the process fluid, [§60.482-7(f)(1)]
 - B. Is operated with emissions less than 500 ppm above background as determined by the method specified in §60.485(c), and [§60.482-7(f)(2)]
 - C. Is tested for compliance with §60.482-7(f)(2) initially upon designation, annually, and at other times requested by the Director. [§60.482-7(f)(3)]
11. Any valve that is designated, as described in §60.486(f)(1), as an unsafe-to-monitor valve is exempt from the requirements of §60.482-7(a) if: [§60.482-7(g)]
 - A. The permittee of the valve demonstrates that the valve is unsafe to monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-7(a), and [§60.482-7(g)(1)]
 - B. The permittee of the valve adheres to a written plan that requires monitoring of the valve as frequently as practicable during safe-to-monitor times. [§60.482-7(g)(2)]
12. Any valve that is designated, as described in §60.486(f)(2), as a difficult-to-monitor valve is exempt from the requirements of §60.482-7(a) if: [§60.482-7(h)]
 - A. The permittee of the valve demonstrates that the valve cannot be monitored without elevating the monitoring personnel more than 2 meters above a support surface. [§60.482-7(h)(1)]

- B. The process unit within which the valve is located either becomes an affected facility through §60.14 or §60.15 or the permittee designates less than 3.0 percent of the total number of valves as difficult-to-monitor, and [§60.482-7(h)(2)]
- C. The permittee of the valve follows a written plan that requires monitoring of the valve at least once per calendar year. [§60.482-7(h)(3)]

Standards for pumps and valves in heavy liquid service, pressure relief devices in light liquid or heavy liquid service, and connectors:

- 1. If evidence of a potential leak is found by visual, audible, olfactory, or any other detection method at pumps and valves in heavy liquid service, pressure relief devices in light liquid or heavy liquid service, and connectors, the permittee shall follow either one of the following procedures: [§60.482-8(a)]
 - A. The permittee shall monitor the equipment within 5 days by the method specified in §60.485(b) and shall comply with the requirements of §60.482-8(b) through (d). [§60.482-8(a)(1)]
 - B. The permittee shall eliminate the visual, audible, olfactory, or other indication of a potential leak within 5 calendar days of detection. [§60.482-8(a)(2)]
- 2. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-8(b)]
- 3. When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9. [§60.482-8(c)(1)]
- 4. The first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-8(c)(2)]
- 5. First attempts at repair include, but are not limited to, the best practices described under §§60.482-2(c)(2) and 60.482-7(e). [§60.482-8(d)]

Standards for Delay of repair:

- 1. Delay of repair of equipment for which leaks have been detected will be allowed if repair within 15 days is technically infeasible without a process unit shutdown. Repair of this equipment shall occur before the end of the next process unit shutdown. Monitoring to verify repair must occur within 15 days after startup of the process unit. [§60.482-9(a)]
- 2. Delay of repair of equipment will be allowed for equipment which is isolated from the process and which does not remain in VOC service. [§60.482-9(b)]
- 3. Delay of repair for valves will be allowed if: [§60.482-9(c)]
 - A. The permittee demonstrates that emissions of purged material resulting from immediate repair are greater than the fugitive emissions likely to result from delay of repair, and [§60.482-9(c)(1)]
 - B. When repair procedures are effected, the purged material is collected and destroyed or recovered in a control device complying with §60.482-10. [§60.482-9(c)(2)]
- 4. Delay of repair for pumps will be allowed if: [§60.482-9(d)]
 - A. Repair requires the use of a dual mechanical seal system that includes a barrier fluid system, and [§60.482-9(d)(1)]
 - B. Repair is completed as soon as practicable, but not later than 6 months after the leak was detected. [§60.482-9(d)(2)]
- 5. Delay of repair beyond a process unit shutdown will be allowed for a valve, if valve assembly replacement is necessary during the process unit shutdown, valve assembly supplies have been depleted, and valve assembly supplies had been sufficiently stocked before the supplies were depleted. Delay of repair beyond the next process unit shutdown will not be allowed unless the next

process unit shutdown occurs sooner than 6 months after the first process unit shutdown. [§60.482-9(e)]

6. When delay of repair is allowed for a leaking pump or valve that remains in service, the pump or valve may be considered to be repaired and no longer subject to delay of repair requirements if two consecutive monthly monitoring instrument readings are below the leak definition. [§60.482-9(f)]

Standards for closed vent systems and control devices:

1. The permittee of closed vent systems and control devices used to comply with Subpart VV shall comply with the provisions of §60.482-10. [§60.482-10(a)]
2. Vapor recovery systems (for example, condensers and absorbers) shall be designed and operated to recover the VOC emissions vented to them with an efficiency of 95 percent or greater, or to an exit concentration of 20 parts per million by volume, whichever is less stringent. [§60.482-10(b)]
3. Enclosed combustion devices shall be designed and operated to reduce the VOC emissions vented to them with an efficiency of 95 percent or greater, or to an exit concentration of 20 parts per million by volume, on a dry basis, corrected to 3 percent oxygen, whichever is less stringent or to provide a minimum residence time of 0.75 seconds at a minimum temperature of 816 °C. [§60.482-10(c)]
4. Flares used to comply with Subpart VV shall comply with the requirements of §60.18. [§60.482-10(d)]
5. The permittee of control devices used to comply with the provisions of Subpart VV shall monitor these control devices to ensure that they are operated and maintained in conformance with their designs. [§60.482-10(e)]
6. Except as provided in §60.482-10(i) through (k), each closed vent system shall be inspected according to the procedures and schedule specified in §60.482-10(f)(1) and (f)(2). [§60.482-10(f)]
 - A. If the vapor collection system or closed vent system is constructed of hard-piping, the permittee shall comply with the requirements specified in §60.482-10(f)(1)(i) and (f)(1)(ii): [§60.482-10(f)(1)]
 - i. Conduct an initial inspection according to the procedures in §60.485(b); and [§60.482-10(f)(1)(i)]
 - ii. Conduct annual visual inspections for visible, audible, or olfactory indications of leaks. [§60.482-10(f)(1)(ii)]
 - B. If the vapor collection system or closed vent system is constructed of ductwork, the permittee shall: [§60.482-10(f)(2)]
 - i. Conduct an initial inspection according to the procedures in §60.485(b); and [§60.482-10(f)(2)(i)]
 - ii. Conduct annual inspections according to the procedures in §60.485(b). [§60.482-10(f)(2)(ii)]
7. Leaks, as indicated by an instrument reading greater than 500 parts per million by volume above background or by visual inspections, shall be repaired as soon as practicable except as provided in §60.482-10(h). [§60.482-10(g)]
 - A. A first attempt at repair shall be made no later than 5 calendar days after the leak is detected. [§60.482-10(g)(1)]
 - B. Repair shall be completed no later than 15 calendar days after the leak is detected. [§60.482-10(g)(2)]
8. Delay of repair of a closed vent system for which leaks have been detected is allowed if the repair is technically infeasible without a process unit shutdown or if the permittee determines that emissions resulting from immediate repair would be greater than the fugitive emissions likely to result from

delay of repair. Repair of such equipment shall be complete by the end of the next process unit shutdown. [§60.482-10(h)]

9. If a vapor collection system or closed vent system is operated under a vacuum, it is exempt from the inspection requirements of §60.482-10(f)(1)(i) and (f)(2). [§60.482-10(i)]
10. Any parts of the closed vent system that are designated, as described in §60.482-10(l)(1), as unsafe to inspect are exempt from the inspection requirements of §60.482-10(f)(1)(i) and (f)(2) if they comply with the requirements specified in §60.482-10(j)(1) and (j)(2): [§60.482-10(j)]
 - A. The permittee determines that the equipment is unsafe to inspect because inspecting personnel would be exposed to an imminent or potential danger as a consequence of complying with §60.482-10(f)(1)(i) or (f)(2); and [§60.482-10(j)(1)]
 - B. The permittee has a written plan that requires inspection of the equipment as frequently as practicable during safe-to-inspect times. [§60.482-10(j)(2)]
11. Any parts of the closed vent system that are designated, as described in §60.482-10(l)(2), as difficult to inspect are exempt from the inspection requirements of §60.482-10(f)(1)(i) and (f)(2) if they comply with the requirements specified in §60.482-10(k)(1) through (k)(3): [§60.482-10(k)]
 - A. The permittee determines that the equipment cannot be inspected without elevating the inspecting personnel more than 2 meters above a support surface; and [§60.482-10(k)(1)]
 - B. The process unit within which the closed vent system is located becomes an affected facility through §§60.14 or 60.15, or the permittee designates less than 3.0 percent of the total number of closed vent system equipment as difficult to inspect; and [§60.482-10(k)(2)]
 - C. The permittee has a written plan that requires inspection of the equipment at least once every 5 years. A closed vent system is exempt from inspection if it is operated under a vacuum. [§60.482-10(k)(3)]
12. The permittee shall record the information specified in §60.482-10(l)(1) through (l)(5). [§60.482-10(l)]
 - A. Identification of all parts of the closed vent system that are designated as unsafe to inspect, an explanation of why the equipment is unsafe to inspect, and the plan for inspecting the equipment. [§60.482-10(l)(1)]
 - B. Identification of all parts of the closed vent system that are designated as difficult to inspect, an explanation of why the equipment is difficult to inspect, and the plan for inspecting the equipment. [§60.482-10(l)(2)]
 - C. For each inspection during which a leak is detected, a record of the information specified in §60.486(c). [§60.482-10(l)(3)]
 - D. For each inspection conducted in accordance with §60.485(b) during which no leaks are detected, a record that the inspection was performed, the date of the inspection, and a statement that no leaks were detected. [§60.482-10(l)(4)]
 - E. For each visual inspection conducted in accordance with §60.482-10(f)(1)(ii) during which no leaks are detected, a record that the inspection was performed, the date of the inspection, and a statement that no leaks were detected. [§60.482-10(l)(5)]
13. Closed vent systems and control devices used to comply with provisions of Subpart VV shall be operated at all times when emissions may be vented to them. [§60.482-10(m)]

Alternative standards for valves—allowable percentage of valves leaking:

1. The permittee may elect to comply with an allowable percentage of valves leaking of equal to or less than 2.0 percent. [§60.483-1(a)]

2. The following requirements shall be met if the permittee wishes to comply with an allowable percentage of valves leaking: [§60.483-1(b)]
 - A. The permittee must notify the Director that the permittee has elected to comply with the allowable percentage of valves leaking before implementing this alternative standard, as specified in §60.487(d). [§60.483-1(b)(1)]
 - B. A performance test as specified in §60.483-1(c) shall be conducted initially upon designation, annually, and at other times requested by the Director. [§60.483-1(b)(2)]
 - C. If a valve leak is detected, it shall be repaired in accordance with §60.482-7(d) and (e). [§60.483-1(b)(3)]
3. Performance tests shall be conducted in the following manner: [§60.483-1(c)]
 - A. All valves in gas/vapor and light liquid service within the affected facility shall be monitored within 1 week by the methods specified in §60.485(b). [§60.483-1(c)(1)]
 - B. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.483-1(c)(2)]
 - C. The leak percentage shall be determined by dividing the number of valves for which leaks are detected by the number of valves in gas/vapor and light liquid service within the affected facility. [§60.483-1(c)(3)]
4. The permittee who elects to comply with this alternative standard shall not have an affected facility with a leak percentage greater than 2.0 percent, determined as described in §60.485(h). [§60.483-1(d)]

Alternative standards for valves—skip period leak detection and repair:

1. The permittee may elect to comply with one of the alternative work practices specified in §60.483-2(b)(2) and (3). [§60.483-2(a)(1)]
2. The permittee must notify the Director before implementing one of the alternative work practices, as specified in §60.487(d). [§60.483-2(a)(2)]
3. The permittee shall comply initially with the requirements for valves in gas/vapor service and valves in light liquid service, as described in §60.482-7. [§60.483-2(b)(1)]
4. After 2 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 1 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2(b)(2)]
5. After 5 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 3 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2(b)(3)]
6. If the percent of valves leaking is greater than 2.0, the permittee shall comply with the requirements as described in §60.482-7 but can again elect to use §60.483-2. [§60.483-2(b)(4)]
7. The percent of valves leaking shall be determined as described in §60.485(h). [§60.483-2(b)(5)]
8. The permittee must keep a record of the percent of valves found leaking during each leak detection period. [§60.483-2(b)(6)]
9. A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for a process unit following one of the alternative standards in §60.483-2 must be monitored in accordance with §60.482-7(a)(2)(i) or (ii) before the provisions of §60.483-2 can be applied to that valve. [§60.483-2(b)(7)]

Equivalence of means of emission limitation:

1. The permittee may apply to the Administrator for determination of equivalence for any means of emission limitation that achieves a reduction in emissions of VOC at least equivalent to the reduction in emissions of VOC achieved by the controls required in Subpart VV. [§60.484(a)]
2. Determination of equivalence to the equipment, design, and operational requirements of Subpart V will be evaluated by the following guidelines: [§60.484(b)]
 - A. The permittee applying for an equivalence determination shall be responsible for collecting and verifying test data to demonstrate equivalence of means of emission limitation. [§60.484(b)(1)]
 - B. The Administrator will compare test data for demonstrating equivalence of the means of emission limitation to test data for the equipment, design, and operational requirements. [§60.484(b)(2)]
 - C. The Administrator may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the equipment, design, and operational requirements. [§60.484(b)(3)]
3. Determination of equivalence to the required work practices in Subpart VV will be evaluated by the following guidelines: [§60.484(c)]
 - A. The permittee applying for a determination of equivalence shall be responsible for collecting and verifying test data to demonstrate equivalence of an equivalent means of emission limitation. [§60.484(c)]
 - B. For each affected facility for which a determination of equivalence is requested, the emission reduction achieved by the required work practice shall be demonstrated. [§60.484(c)(2)]
 - C. For each affected facility, for which a determination of equivalence is requested, the emission reduction achieved by the equivalent means of emission limitation shall be demonstrated. [§60.484(c)(3)]
 - D. The permittee applying for a determination of equivalence shall commit in writing to work practice(s) that provide for emission reductions equal to or greater than the emission reductions achieved by the required work practice. [§60.484(c)(4)]
 - E. The Administrator will compare the demonstrated emission reduction for the equivalent means of emission limitation to the demonstrated emission reduction for the required work practices and will consider the commitment in §60.484(c)(4). [§60.484(c)(5)]
 - F. The Administrator may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the required work practice. [§60.484(c)(6)]
4. The permittee may offer a unique approach to demonstrate the equivalence of any equivalent means of emission limitation. [§60.484(d)]
5. After a request for determination of equivalence is received, the Administrator will publish a notice in the Federal Register and provide the opportunity for public hearing if the Administrator judges that the request may be approved. [§60.484(e)(1)]
6. After notice and opportunity for public hearing, the Administrator will determine the equivalence of a means of emission limitation and will publish the determination in the Federal Register. [§60.484(e)(2)]
7. Any equivalent means of emission limitations approved under this section shall constitute a required work practice, equipment, design, or operational standard within the meaning of section 111(h)(1) of the Clean Air Act. [§60.484(e)(3)]
8. Manufacturers of equipment used to control equipment leaks of VOC may apply to the Administrator for determination of equivalence for any equivalent means of emission limitation that

achieves a reduction in emissions of VOC achieved by the equipment, design, and operational requirements of this subpart. [§60.484(f)(1)]

9. The Administrator will make an equivalence determination according to the provisions of §60.484(b), (c), (d), and (e). [§60.484(f)(2)]

Test methods and procedures:

1. In conducting the performance tests required in §60.8, the permittee shall use as reference methods and procedures the test methods in appendix A of part 60 or other methods and procedures as specified in §60.485, except as provided in §60.8(b). [§60.485(a)]
2. The permittee shall determine compliance with the standards in §§60.482-1 through 60.482-10, 60.483, and 60.484 as follows: [§60.485(b)]
 - A. Method 21 shall be used to determine the presence of leaking sources. The instrument shall be calibrated before use each day of its use by the procedures specified in Method 21. The following calibration gases shall be used: [§60.485(b)(1)]
 - i. Zero air (less than 10 ppm of hydrocarbon in air); and [§60.485(b)(1)(i)]
 - ii. A mixture of methane or n-hexane and air at a concentration of about, but less than, 10,000 ppm methane or n-hexane. [§60.485(b)(1)(ii)]
3. The permittee shall determine compliance with the no detectable emission standards in §§60.482-2(e), 60.482-3(i), 60.482-4, 60.482-7(f), and 60.482-10(e) as follows: [§60.485(c)]
 - A. The requirements of §60.485(b). [§60.485(c)(1)]
 - B. Method 21 shall be used to determine the background level. All potential leak interfaces shall be traversed as close to the interface as possible. The arithmetic difference between the maximum concentration indicated by the instrument and the background level is compared with 500 ppm for determining compliance. [§60.485(c)(2)]
4. The permittee shall test each piece of equipment unless the permittee demonstrates that a process unit is not in VOC service, i.e., that the VOC content would never be reasonably expected to exceed 10 percent by weight. For purposes of this demonstration, the following methods and procedures shall be used: [§60.485(d)]
 - A. Procedures that conform to the general methods in ASTM E260-73, 91, or 96, E168-67, 77, or 92, E169-63, 77, or 93 (incorporated by reference—see §60.17) shall be used to determine the percent VOC content in the process fluid that is contained in or contacts a piece of equipment. [§60.485(d)(1)]
 - B. Organic compounds that are considered by the Director to have negligible photochemical reactivity may be excluded from the total quantity of organic compounds in determining the VOC content of the process fluid. [§60.485(d)(2)]
 - C. Engineering judgment may be used to estimate the VOC content, if a piece of equipment had not been shown previously to be in service. If the Director disagrees with the judgment, §60.485(d)(1) and (2) shall be used to resolve the disagreement. [§60.485(d)(3)]
5. The permittee shall demonstrate that a piece of equipment is in light liquid service by showing that all the following conditions apply: [§60.485(e)]
 - A. The vapor pressure of one or more of the organic components is greater than 0.3 kPa at 20 °C (1.2 in. H₂O at 68 °F). Standard reference texts or ASTM D2879-83, 96, or 97 (incorporated by reference—see §60.17) shall be used to determine the vapor pressures. [§60.485(e)(1)]
 - B. The total concentration of the pure organic components having a vapor pressure greater than 0.3 kPa at 20 °C (1.2 in. H₂O at 68 °F) is equal to or greater than 20 percent by weight. [§60.485(e)(2)]

- C. The fluid is a liquid at operating conditions. [§60.485(e)(3)]
6. Samples used in conjunction with §60.485(d), (e), and (g) shall be representative of the process fluid that is contained in or contacts the equipment or the gas being combusted in the flare. [§60.485(f)]
7. The permittee shall determine compliance with the standards of flares as follows: [§60.485(g)]
- A. Method 22 shall be used to determine visible emissions. [§60.485(g)(1)]
- B. A thermocouple or any other equivalent device shall be used to monitor the presence of a pilot flame in the flare. [§60.485(g)(2)]
- C. The maximum permitted velocity for air assisted flares shall be computed using the following equation: [§60.485(g)(3)]

$$V_{max} = K_1 + K_2 H_T$$

Where:

V_{max} = Maximum permitted velocity, m/sec (ft/sec)

H_T = Net heating value of the gas being combusted, MJ/scm (Btu/scf).

K_1 = 8.706 m/sec (metric units) , 28.56 ft/sec (English units)

K_2 = 0.7084 m⁴/(MJ-sec) (metric units), 0.087 ft⁴/(Btu-sec) (English units)

- D. The net heating value (H_T) of the gas being combusted in a flare shall be computed using the following equation: [§60.485(g)(4)]

$$H_T = K \sum_{i=1}^n C_i H_i$$

Where:

K = Conversion constant, $1.740 \times 10^{-7} \frac{g-mol*MJ}{ppm*scm*kcal}$ (metric units), $4.674 \times 10^{-6} \frac{g-mol*Btu}{ppm*scf*kcal}$ (English units)

C_i = Concentration of sample component “i,” ppm

H_i = Net heat of combustion of sample component “i” at 25 °C and 760 mm Hg (77 °F and 14.7 psi), kcal/g-mole

- E. Method 18 or ASTM D6420-99 (2004) (where the target compound(s) are those listed in Section 1.1 of ASTM D6420-99, and the target concentration is between 150 parts per billion by volume and 100 parts per million by volume) and ASTM D2504-67, 77 or 88 (Reapproved 1993) (incorporated by reference—see §60.17) shall be used to determine the concentration of sample component “i.” [§60.485(g)(5)]
- F. ASTM D2382-76 or 88 or D4809-95 (incorporated by reference—see §60.17) shall be used to determine the net heat of combustion of component “i” if published values are not available or cannot be calculated. [§60.485(g)(6)]
- G. Method 2, 2A, 2C, or 2D, as appropriate, shall be used to determine the actual exit velocity of a flare. If needed, the unobstructed (free) cross-sectional area of the flare tip shall be used. [§60.485(g)(7)]
8. The permittee shall determine compliance with §60.483-1 or §60.483-2 as follows: [§60.485(h)]
- A. The percent of valves leaking shall be determined using the following equation: [§60.485(h)(1)]

$$\%V_L = \left(\frac{V_L}{V_T} \right) * 100$$

Where:

$\%V_L$ = Percent leaking valves

V_L = Number of valves found leaking

V_T = The sum of the total number of valves monitored

- B. The total number of valves monitored shall include difficult-to-monitor and unsafe-to-monitor valves only during the monitoring period in which those valves are monitored. [§60.485(h)(2)]
- C. The number of valves leaking shall include valves for which repair has been delayed. [§60.485(h)(3)]
- D. Any new valve that is not monitored within 30 days of being placed in service shall be included in the number of valves leaking and the total number of valves monitored for the monitoring period in which the valve is placed in service. [§60.485(h)(4)]
- E. If the process unit has been subdivided in accordance with §60.482-7(c)(1)(ii), the sum of valves found leaking during a monitoring period includes all subgroups. [§60.485(h)(5)]
- F. The total number of valves monitored does not include a valve monitored to verify repair. [§60.485(h)(6)]

Recordkeeping:

- 1. The permittee shall comply with the recordkeeping requirements of §60.486. [§60.486(a)(1)]
- 2. The permittee of more than one affected facility subject to the provisions of Subpart VV may comply with the recordkeeping requirements for these facilities in one recordkeeping system if the system identifies each record by each facility. [§60.486(a)(2)]
- 3. When each leak is detected as specified in §§60.482-2, 60.482-3, 60.482-7, 60.482-8, and 60.483-2, the following requirements apply: [§60.486(b)]
 - A. A weatherproof and readily visible identification, marked with the equipment identification number, shall be attached to the leaking equipment. [§60.486(b)(1)]
 - B. The identification on a valve may be removed after it has been monitored for 2 successive months as specified in §60.482-7(c) and no leak has been detected during those 2 months. [§60.486(b)(2)]
 - C. The identification on equipment except on a valve, may be removed after it has been repaired. [§60.486(b)(3)]
- 4. When each leak is detected as specified in §§60.482-2, 60.482-3, 60.482-7, 60.482-8, and 60.483-2, the following information shall be recorded in a log and shall be kept for 2 years in a readily accessible location: [§60.486(c)]
 - A. The instrument and operator identification numbers and the equipment identification number. [§60.486(c)(1)]
 - B. The date the leak was detected and the dates of each attempt to repair the leak. [§60.486(c)(2)]
 - C. Repair methods applied in each attempt to repair the leak. [§60.486(c)(3)]
 - D. “Above 10,000” if the maximum instrument reading measured by the methods specified in §60.485(a) after each repair attempt is equal to or greater than 10,000 ppm. [§60.486(c)(4)]
 - E. “Repair delayed” and the reason for the delay if a leak is not repaired within 15 calendar days after discovery of the leak. [§60.486(c)(5)]

- F. The signature of the permittee (or designate) whose decision it was that repair could not be effected without a process shutdown. [§60.486(c)(6)]
- G. The expected date of successful repair of the leak if a leak is not repaired within 15 days. [§60.486(c)(7)]
- H. Dates of process unit shutdowns that occur while the equipment is unrepaired. [§60.486(c)(8)]
- I. The date of successful repair of the leak. [§60.486(c)(9)]
- 5. The following information pertaining to the design requirements for closed vent systems and control devices described in §60.482-10 shall be recorded and kept in a readily accessible location: [§60.486(d)]
 - A. Detailed schematics, design specifications, and piping and instrumentation diagrams. [§60.486(d)(1)]
 - B. The dates and descriptions of any changes in the design specifications. [§60.486(d)(2)]
 - C. A description of the parameter or parameters monitored, as required in §60.482-10(e), to ensure that control devices are operated and maintained in conformance with their design and an explanation of why that parameter (or parameters) was selected for the monitoring. [§60.486(d)(3)]
 - D. Periods when the closed vent systems and control devices required in §§60.482-2, 60.482-3, 60.482-4, and 60.482-5 are not operated as designed, including periods when a flare pilot light does not have a flame. [§60.486(d)(4)]
 - E. Dates of startups and shutdowns of the closed vent systems and control devices required in §§60.482-2, 60.482-3, 60.482-4, and 60.482-5. [§60.486(d)(5)]
- 6. The following information pertaining to all equipment subject to the requirements in §§60.482-1 to 60.482-10 shall be recorded in a log that is kept in a readily accessible location: [§60.486(e)]
 - A. A list of identification numbers for equipment subject to the requirements of Subpart VV. [§60.486(e)(1)]
 - B. A list of identification numbers for equipment that are designated for no detectable emissions under the provisions of §§60.482-2(e), 60.482-3(i) and 60.482-7(f). [§60.486(e)(2)(i)]
 - C. The designation of equipment as subject to the requirements of §60.482-2(e), §60.482-3(i), or §60.482-7(f) shall be signed by the permittee. Alternatively, the permittee may establish a mechanism with the Director that satisfies this requirement. [§60.486(e)(2)(ii)]
 - D. A list of equipment identification numbers for pressure relief devices required to comply with §60.482-4. [§60.486(e)(3)]
 - E. The dates of each compliance test as required in §§60.482-2(e), 60.482-3(i), 60.482-4, and 60.482-7(f). [§60.486(e)(4)(i)]
 - F. The background level measured during each compliance test. [§60.486(e)(4)(ii)]
 - G. The maximum instrument reading measured at the equipment during each compliance test. [§60.486(e)(4)(iii)]
 - H. A list of identification numbers for equipment in vacuum service. [§60.486(e)(5)]
 - I. A list of identification numbers for equipment that the permittee designates as operating in VOC service less than 300 hr/yr in accordance with §60.482-1(e), a description of the conditions under which the equipment is in VOC service, and rationale supporting the designation that it is in VOC service less than 300 hr/yr. [§60.486(e)(6)]
- 7. The following information pertaining to all valves subject to the requirements of §60.482-7(g) and (h) and to all pumps subject to the requirements of §60.482-2(g) shall be recorded in a log that is kept in a readily accessible location: [§60.486(f)]

- A. A list of identification numbers for valves and pumps that are designated as unsafe-to-monitor, an explanation for each valve or pump stating why the valve or pump is unsafe-to-monitor, and the plan for monitoring each valve or pump. [§60.486(f)(1)]
- B. A list of identification numbers for valves that are designated as difficult-to-monitor, an explanation for each valve stating why the valve is difficult-to-monitor, and the schedule for monitoring each valve. [§60.486(f)(2)]
8. The following information shall be recorded for valves complying with §60.483-2: [§60.486(g)]
 - A. A schedule of monitoring. [§60.486(g)(1)]
 - B. The percent of valves found leaking during each monitoring period. [§60.486(g)(2)]
9. The following information shall be recorded in a log that is kept in a readily accessible location: [§60.486(h)]
 - A. Design criterion required in §§60.482-2(d)(5) and 60.482-3(e)(2) and explanation of the design criterion; and [§60.486(h)(1)]
 - B. Any changes to this criterion and the reasons for the changes. [§60.486(h)(2)]
10. The following information shall be recorded in a log that is kept in a readily accessible location for use in determining exemptions as provided in §60.480(d): [§60.486(i)]
 - A. An analysis demonstrating the design capacity of the affected facility, [§60.486(i)(1)]
 - B. A statement listing the feed or raw materials and products from the affected facilities and an analysis demonstrating whether these chemicals are heavy liquids or beverage alcohol, and [§60.486(i)(2)]
 - C. An analysis demonstrating that equipment is not in VOC service. [§60.486(i)(3)]
11. Information and data used to demonstrate that a piece of equipment is not in VOC service shall be recorded in a log that is kept in a readily accessible location. [§60.486(j)]
12. The provisions of §60.7 (b) and (d) do not apply to affected facilities subject to this subpart. [§60.486(k)]

Reporting:

1. The permittee shall submit semiannual reports to the Director beginning six months after the initial startup date. [§60.487(a)]
2. The initial semiannual report to the Director shall include the following information: [§60.487(b)]
 - A. Process unit identification. [§60.487(b)(1)]
 - B. Number of valves subject to the requirements of §60.482-7, excluding those valves designated for no detectable emissions under the provisions of §60.482-7(f). [§60.487(b)(2)]
 - C. Number of pumps subject to the requirements of §60.482-2, excluding those pumps designated for no detectable emissions under the provisions of §60.482-2(e) and those pumps complying with §60.482-2(f). [§60.487(b)(3)]
 - D. Number of compressors subject to the requirements of §60.482-3, excluding those compressors designated for no detectable emissions under the provisions of §60.482-3(i) and those compressors complying with §60.482-3(h). [§60.487(b)(4)]
3. All semiannual reports to the Director shall include the following information, summarized from the information in §60.486: [§60.487(c)]
 - A. Process unit identification. [§60.487(c)(1)]
 - B. For each month during the semiannual reporting period, [§60.487(c)(2)]
 - i. Number of valves for which leaks were detected as described in §60.482-7(b) or §60.483-2, [§60.487(c)(2)(i)]

- ii. Number of valves for which leaks were not repaired as required in §60.482-7(d)(1), [§60.487(c)(2)(ii)]
- iii. Number of pumps for which leaks were detected as described in §60.482-2(b), (d)(4)(ii)(A) or (B), or (d)(5)(iii), [§60.487(c)(2)(iii)]
- iv. Number of pumps for which leaks were not repaired as required in §60.482-2(c)(1) and (d)(6), [§60.487(c)(2)(iv)]
- v. Number of compressors for which leaks were detected as described in §60.482-3(f), [§60.487(c)(2)(v)]
- vi. Number of compressors for which leaks were not repaired as required in §60.482-3(g)(1), and [§60.487(c)(2)(vi)]
- vii. The facts that explain each delay of repair and, where appropriate, why a process unit shutdown was technically infeasible. [§60.487(c)(2)(vii)]
- C. Dates of process unit shutdowns which occurred within the semiannual reporting period. [§60.487(c)(3)]
- D. Revisions to items reported according to §60.487(b) if changes have occurred since the initial report or subsequent revisions to the initial report. [§60.487(c)(4)]
- 4. The permittee electing to comply with the provisions of §60.483-1 or §60.483-2 shall notify the Director of the alternative standard selected 90 days before implementing either of the provisions. [§60.487(d)]
- 5. The permittee shall report the results of all performance tests in accordance with §60.8 of the General Provisions. The provisions of §60.8(d) do not apply to affected facilities subject to the provisions of Subpart VV except that a permittee must notify the Director of the schedule for the initial performance tests at least 30 days before the initial performance tests. [§60.487(e)]
- 6. The requirements of §60.487(a) through (c) remain in force until and unless EPA, in delegating enforcement authority to a State under section 111(c) of the Act, approves reporting requirements or an alternative means of compliance surveillance adopted by such State. In that event, affected sources within the State will be relieved of the obligation to comply with the requirements of §60.487(a) through (c), provided that they comply with the requirements established by the State. [§60.487(f)]
- 7. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
- 8. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition Subpart VVa

10 CSR 10-6.070 New Source Performance Regulations

40 CFR part 60 Subpart A, General Provisions

40 CFR part 60 Subpart VVa, Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After November 7, 2006

Applicability:

1. This permit condition applies to all LDAR monitoring accounted for under emission point FS002.
2. The provisions of Subpart VVa apply to affected facilities in the synthetic organic chemicals manufacturing industry. [§60.480a(a)(1)]
3. The group of all equipment (defined in §60.481a) within a process unit is an affected facility. [§60.480a(a)(2)]
4. Any affected facility under §60.480a(a) that commences construction, reconstruction, or modification after November 7, 2006, shall be subject to the requirements of Subpart VVa. [§60.480a(b)]
5. Addition or replacement of equipment for the purpose of process improvement which is accomplished without a capital expenditure shall not by itself be considered a modification under Subpart VVa. [§60.480a(c)]
6. If the permittee applies for one or more of the exemptions in §60.480a(d), then the permittee shall maintain records as required in §60.486a(i). [§60.480a(d)(1)]
7. Any affected facility that has the design capacity to produce less than 1,000 Mg/yr (1,102 ton/yr) of a chemical listed in §60.489 is exempt from §§60.482-1a through 60.482-11a. [§60.480a(d)(2)]
8. If an affected facility produces heavy liquid chemicals only from heavy liquid feed or raw materials, then it is exempt from §§60.482-1a through 60.482-11a. [§60.480a(d)(3)]
9. Any affected facility that produces beverage alcohol is exempt from §§60.482-1a through 60.482-11a. [§60.480a(d)(4)]
10. Any affected facility that has no equipment in volatile organic compounds (VOC) service is exempt from §§60.482-1a through 60.482-11a. [§60.480a(d)(5)]
11. *Alternative means of compliance* [§60.480a(e)]
 - A. *Option to comply with part 65.* [§60.480a(e)(1)]
 - i. The permittee may choose to comply with the provisions of 40 CFR part 65, subpart F, to satisfy the requirements of §§60.482-1a through 60.487a for an affected facility. When choosing to comply with 40 CFR part 65, subpart F, the requirements of §§60.485a(d), (e), and (f), and 60.486a(i) and (j) still apply. Other provisions applying to the permittee who chooses to comply with 40 CFR part 65 are provided in 40 CFR 65.1. [§60.480a(e)(1)(i)]
 - ii. *Part 60, subpart A.* The permittee who chooses to comply with 40 CFR part 65, subpart F must also comply with §§60.1, 60.2, 60.5, 60.6, 60.7(a)(1) and (4), 60.14, 60.15, and 60.16 for that equipment. All provisions of subpart A of part 60 that are not mentioned in §60.480a(e)(1)(ii) do not apply to the permittee of equipment subject to Subpart VVa complying with 40 CFR part 65, subpart F, except that provisions required to be met prior to implementing 40 CFR part 65 still apply. The permittee who chooses to comply with 40 CFR part 65, subpart F, must comply with 40 CFR part 65, subpart A. [§60.480a(e)(1)(ii)]
 - B. *Part 63, subpart H.* [§60.480a(e)(2)]
 - i. The permittee may choose to comply with the provisions of 40 CFR part 63, subpart H, to satisfy the requirements of §§60.482-1a through 60.487a for an affected facility. When

choosing to comply with 40 CFR part 63, subpart H, the requirements of §60.485a(d), (e), and (f), and §60.486a(i) and (j) still apply. [§60.480a(e)(2)(i)]

- ii. *Part 60, subpart A.* The permittee who chooses to comply with 40 CFR part 63, subpart H must also comply with §§60.1, 60.2, 60.5, 60.6, 60.7(a)(1) and (4), 60.14, 60.15, and 60.16 for that equipment. All provisions of subpart A of part 60 that are not mentioned in §60.480a(e)(2)(ii) do not apply to the permittee of equipment subject to Subpart VVa complying with 40 CFR part 63, subpart H, except that provisions required to be met prior to implementing 40 CFR part 63 still apply. The permittee who chooses to comply with 40 CFR part 63, subpart H, must comply with 40 CFR part 63, subpart A. [§60.480a(e)(2)(ii)]

12. *Stay of standards.* [§60.480a(f)]

- A. The permittee that starts a new, reconstructed, or modified affected source prior to November 16, 2007 is not required to comply with the requirements in §60.480a(f)(1) until EPA takes final action to require compliance and publishes a document in the Federal Register. [§60.480a(f)(1)]
 - i. The definition of “capital expenditure” in §60.481a of Subpart VVa. While the definition of “capital expenditure” is stayed, the permittee should use the definition found in §60.481 of Subpart VV of part 60. [§60.480a(f)(1)(i)]
- B. The permittee is not required to comply with the requirements in §60.480a(f)(1) until EPA takes final action to require compliance and publishes a document in the Federal Register. [§60.480a(f)(2)]
 - i. The definition of “process unit” in §60.481a of Subpart VVa. While the definition of “process unit” is stayed, the permittee should use the following definition: [§60.480a(f)(2)(i)]
Process unit means components assembled to produce, as intermediate or final products, one or more of the chemicals listed in §60.489 of part 60. A process unit can operate independently if supplied with sufficient feed or raw materials and sufficient storage facilities for the product.
 - ii. The method of allocation of shared storage vessels in §60.482-1a(g) of Subpart VVa. [§60.480a(f)(2)(ii)]
 - iii. The standards for connectors in gas/vapor service and in light liquid service in §60.482-11a of Subpart VVa. [§60.480a(f)(2)(iii)]

General Standards:

- 1. The permittee shall demonstrate compliance with the requirements of §§60.482-1a through 60.482-10a or §60.480a(e) for all equipment within 180 days of initial startup. [§60.482-1a(a)]
- 2. Compliance with §§60.482-1a to 60.482-10a will be determined by review of records and reports, review of performance test results, and inspection using the methods and procedures specified in §60.485a. [§60.482-1a(b)]
- 3. The permittee may request a determination of equivalence of a means of emission limitation to the requirements of §§60.482-2a, 60.482-3a, 60.482-5a, 60.482-6a, 60.482-7a, 60.482-8a, and 60.482-10a as provided in §60.484a. [§60.482-1a(c)(1)]
- 4. If the Administrator makes a determination that a means of emission limitation is at least equivalent to the requirements of §60.482-2a, §60.482-3a, §60.482-5a, §60.482-6a, §60.482-7a, §60.482-8a, or §60.482-10a, the permittee shall comply with the requirements of that determination. [§60.482-1a(c)(2)]
- 5. Equipment that is in vacuum service is excluded from the requirements of §§60.482-2a through 60.482-10a if it is identified as required in §60.486a(e)(5). [§60.482-1a(d)]

6. Equipment that the permittee designates as being in VOC service less than 300 hr/yr is excluded from the requirements of §§60.482-2a through 60.482-11a if it is identified as required in §60.486a(e)(6) and it meets any of the conditions specified in §60.482-1a(e)(1) through (3). [§60.482-1a(e)]
 - A. The equipment is in VOC service only during startup and shutdown, excluding startup and shutdown between batches of the same campaign for a batch process. [§60.482-1a(e)(1)]
 - B. The equipment is in VOC service only during process malfunctions or other emergencies. [§60.482-1a(e)(2)]
 - C. The equipment is backup equipment that is in VOC service only when the primary equipment is out of service. [§60.482-1a(e)(3)]
7. If a dedicated batch process unit operates less than 365 days during a year, the permittee may monitor to detect leaks from pumps, valves, and open-ended valves or lines at the frequency specified in the following table instead of monitoring as specified in §§60.482-2a, 60.482-7a, and 60.483.2a: [§60.482-1a(f)(1)]

Table 15: Monitoring frequency for Subpart VVa

Operating time (percent of hours during year)	Equivalent monitoring frequency time in use		
	Monthly	Quarterly	Semiannually
0 to <25	Quarterly	Annually	Annually.
25 to <50	Quarterly	Semiannually	Annually.
50 to <75	Bimonthly	Three quarters	Semiannually.
75 to 100	Monthly	Quarterly	Semiannually.

8. Pumps and valves that are shared among two or more batch process units that are subject to Subpart VVa may be monitored at the frequencies specified in §60.482-1a(f)(1), provided the operating time of all such process units is considered. [§60.482-1a(f)(2)]
 - A. The monitoring frequencies specified in §60.482-1a(f)(1) are not requirements for monitoring at specific intervals and can be adjusted to accommodate process operations. The permittee may monitor at any time during the specified monitoring period (e.g., month, quarter, year), provided the monitoring is conducted at a reasonable interval after completion of the last monitoring campaign. Reasonable intervals are defined in §60.482-1a(f)(3)(i) through (iv). [§60.482-1a(f)(3)]
 - B. When monitoring is conducted quarterly, monitoring events must be separated by at least 30 calendar days. [§60.482-1a(f)(3)(i)]
 - C. When monitoring is conducted semiannually (*i.e.*, once every 2 quarters), monitoring events must be separated by at least 60 calendar days. [§60.482-1a(f)(3)(ii)]
 - D. When monitoring is conducted in 3 quarters per year, monitoring events must be separated by at least 90 calendar days. [§60.482-1a(f)(3)(iii)]
 - E. When monitoring is conducted annually, monitoring events must be separated by at least 120 calendar days. [§60.482-1a(f)(3)(iv)]
9. If the storage vessel is shared with multiple process units, the process unit with the greatest annual amount of stored materials (predominant use) is the process unit the storage vessel is assigned to. If the storage vessel is shared equally among process units, and one of the process units has equipment subject to Subpart VVa, the storage vessel is assigned to that process unit. If the storage vessel is shared equally among process units, none of which have equipment subject to Subpart VVa, the

storage vessel is assigned to any process unit subject to Subpart VV. If the predominant use of the storage vessel varies from year to year, then the permittee must estimate the predominant use initially and reassess every 3 years. The permittee must keep records of the information and supporting calculations that show how predominant use is determined. All equipment on the storage vessel must be monitored when in VOC service. [§60.482-1a(g)]¹²

Standards for pumps in light liquid service:

1. Each pump in light liquid service shall be monitored monthly to detect leaks by the methods specified in §60.485a(b), except as provided in §60.482-1a(c) and (f) and §60.482-2a(d), (e), and (f). A pump that begins operation in light liquid service after the initial startup date for the process unit must be monitored for the first time within 30 days after the end of its startup period, except for a pump that replaces a leaking pump and except as provided in §60.482-1a(c) and §60.482-2a(d), (e), and (f). [§60.482-2a(a)(1)]
2. Each pump in light liquid service shall be checked by visual inspection each calendar week for indications of liquids dripping from the pump seal, except as provided in §60.482-1a(f). [§60.482-2a(a)(2)]
3. The instrument reading that defines a leak is specified in §60.482-2a(b)(1)(i) and (ii). [§60.482-2a(b)(1)]
 - A. 5,000 parts per million (ppm) or greater for pumps handling polymerizing monomers; [§60.482-2a(b)(1)(i)]
 - B. 2,000 ppm or greater for all other pumps. [§60.482-2a(b)(1)(ii)]
4. If there are indications of liquids dripping from the pump seal, the permittee shall follow the procedure specified in either §60.482-2a(b)(2)(i) or (ii). This requirement does not apply to a pump that was monitored after a previous weekly inspection and the instrument reading was less than the concentration specified in §60.482-2a(b)(1)(i) or (ii), whichever is applicable. [§60.482-2a(b)(2)]
 - A. Monitor the pump within 5 days as specified in §60.485a(b). A leak is detected if the instrument reading measured during monitoring indicates a leak as specified in §60.482-2a(b)(1)(i) or (ii), whichever is applicable. The leak shall be repaired using the procedures in §60.482-2a(c). [§60.482-2a(b)(2)(i)]
 - B. Designate the visual indications of liquids dripping as a leak, and repair the leak using either the procedures in §60.482-2a(c) or by eliminating the visual indications of liquids dripping. [§60.482-2a(b)(2)(ii)]
5. When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. [§60.482-2a(c)(1)]
6. A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. First attempts at repair include, but are not limited to, the practices described in §60.482-2a(c)(2)(i) and (ii), where practicable. [§60.482-2a(c)(2)]
 - A. Tightening the packing gland nuts; [§60.482-2a(c)(2)(i)]
 - B. Ensuring that the seal flush is operating at design pressure and temperature. [§60.482-2a(c)(2)(ii)]
7. Each pump equipped with a dual mechanical seal system that includes a barrier fluid system is exempt from the requirements of §60.482-2a(a), provided the requirements specified in §60.482-2a(d)(1) through (6) are met. [§60.482-2a(d)]
 - A. Each dual mechanical seal system is: [§60.482-2a(d)(1)]

¹² Effective Date Note: At 73 FR 31376, June 2, 2008, §60.482-1a(g) was stayed until further notice.

- i. Operated with the barrier fluid at a pressure that is at all times greater than the pump stuffing box pressure; or [§60.482-2a(d)(1)(i)]
 - ii. Equipped with a barrier fluid degassing reservoir that is routed to a process or fuel gas system or connected by a closed vent system to a control device that complies with the requirements of §60.482-10a; or [§60.482-2a(d)(1)(ii)]
 - iii. Equipped with a system that purges the barrier fluid into a process stream with zero VOC emissions to the atmosphere. [§60.482-2a(d)(1)(iii)]
- B. The barrier fluid system is in heavy liquid service or is not in VOC service. [§60.482-2a(d)(2)]
- C. Each barrier fluid system is equipped with a sensor that will detect failure of the seal system, the barrier fluid system, or both. [§60.482-2a(d)(3)]
- D. Each pump is checked by visual inspection, each calendar week, for indications of liquids dripping from the pump seals. [§60.482-2a(d)(4)(i)]
- E. If there are indications of liquids dripping from the pump seal at the time of the weekly inspection, the permittee shall follow the procedure specified in either §60.482-2a(d)(4)(ii)(A) or (B) prior to the next required inspection. [§60.482-2a(d)(4)(ii)]
 - i. Monitor the pump within 5 days as specified in §60.485a(b) to determine if there is a leak of VOC in the barrier fluid. If an instrument reading of 2,000 ppm or greater is measured, a leak is detected. [§60.482-2a(d)(4)(ii)(A)]
 - ii. Designate the visual indications of liquids dripping as a leak. [§60.482-2a(d)(4)(ii)(B)]
- F. Each sensor as described in §60.482-2a(d)(3) is checked daily or is equipped with an audible alarm. [§60.482-2a(d)(5)(i)]
- G. The permittee determines, based on design considerations and operating experience, a criterion that indicates failure of the seal system, the barrier fluid system, or both. [§60.482-2a(d)(5)(ii)]
- H. If the sensor indicates failure of the seal system, the barrier fluid system, or both, based on the criterion established in §60.482-2a(d)(5)(ii), a leak is detected. [§60.482-2a(d)(5)(iii)]
- I. When a leak is detected pursuant to §60.482-2a(d)(4)(ii)(A), it shall be repaired as specified in §60.482-2a(c). [§60.482-2a(d)(6)(i)]
- J. A leak detected pursuant to §60.482-2a(d)(5)(iii) shall be repaired within 15 days of detection by eliminating the conditions that activated the sensor. [§60.482-2a(d)(6)(ii)]
- K. A designated leak pursuant to §60.482-2a(d)(4)(ii)(B) shall be repaired within 15 days of detection by eliminating visual indications of liquids dripping. [§60.482-2a(d)(6)(iii)]
8. Any pump that is designated, as described in §60.486a(e)(1) and (2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-2a(a), (c), and (d) if the pump: [§60.482-2a(e)]
 - A. Has no externally actuated shaft penetrating the pump housing; [§60.482-2a(e)(1)]
 - B. Is demonstrated to be operating with no detectable emissions as indicated by an instrument reading of less than 500 ppm above background as measured by the methods specified in §60.485a(c); and [§60.482-2a(e)(2)]
 - C. Is tested for compliance with §60.482-2a(e)(2) initially upon designation, annually, and at other times requested by the Director. [§60.482-2a(e)(3)]
9. If any pump is equipped with a closed vent system capable of capturing and transporting any leakage from the seal or seals to a process or to a fuel gas system or to a control device that complies with the requirements of §60.482-10a, it is exempt from §60.482-2a(a) through (e). [§60.482-2a(f)]
10. Any pump that is designated, as described in §60.486a(f)(1), as an unsafe-to-monitor pump is exempt from the monitoring and inspection requirements of §60.482-2a(a) and (d)(4) through (6) if: [§60.482-2a(g)]

- A. The permittee demonstrates that the pump is unsafe-to-monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-2a(a); and [§60.482-2a(g)(1)]
 - B. The permittee has a written plan that requires monitoring of the pump as frequently as practicable during safe-to-monitor times, but not more frequently than the periodic monitoring schedule otherwise applicable, and repair of the equipment according to the procedures in §60.482-2a(c) if a leak is detected. [§60.482-2a(g)(2)]
11. Any pump that is located within the boundary of an unmanned plant site is exempt from the weekly visual inspection requirement of §60.482-2a(a)(2) and (d)(4), and the daily requirements of §60.482-2a(d)(5), provided that each pump is visually inspected as often as practicable and at least monthly. [§60.482-2a(h)]

Standards for compressors:

- 1. Each compressor shall be equipped with a seal system that includes a barrier fluid system and that prevents leakage of VOC to the atmosphere, except as provided in §60.482-1a(c) and §60.482-3a(h), (i), and (j). [§60.482-3a(a)]
- 2. Each compressor seal system as required in §60.482-3a(a) shall be: [§60.482-3a(b)]
 - A. Operated with the barrier fluid at a pressure that is greater than the compressor stuffing box pressure; or [§60.482-3a(b)(1)]
 - B. Equipped with a barrier fluid system degassing reservoir that is routed to a process or fuel gas system or connected by a closed vent system to a control device that complies with the requirements of §60.482-10a; or [§60.482-3a(b)(2)]
 - C. Equipped with a system that purges the barrier fluid into a process stream with zero VOC emissions to the atmosphere. [§60.482-3a(b)(3)]
- 3. The barrier fluid system shall be in heavy liquid service or shall not be in VOC service. [§60.482-3a(c)]
- 4. Each barrier fluid system as described in §60.482-3a(a) shall be equipped with a sensor that will detect failure of the seal system, barrier fluid system, or both. [§60.482-3a(d)]
- 5. Each sensor as required in §60.482-3a(d) shall be checked daily or shall be equipped with an audible alarm. [§60.482-3a(e)(1)]
- 6. The permittee shall determine, based on design considerations and operating experience, a criterion that indicates failure of the seal system, the barrier fluid system, or both. [§60.482-3a(e)(2)]
- 7. If the sensor indicates failure of the seal system, the barrier system, or both based on the criterion determined under §60.482-3a(e)(2), a leak is detected. [§60.482-3a(f)]
- 8. When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. [§60.482-3a(g)(1)]
- 9. A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-3a(g)(2)]
- 10. A compressor is exempt from the requirements of §60.482-3a(a) and (b), if it is equipped with a closed vent system to capture and transport leakage from the compressor drive shaft back to a process or fuel gas system or to a control device that complies with the requirements of §60.482-10a, except as provided in §60.482-3a(i). [§60.482-3a(h)]
- 11. Any compressor that is designated, as described in §60.486a(e)(1) and (2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-3a(a) through (h) if the compressor: [§60.482-3a(i)]

- A. Is demonstrated to be operating with no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as measured by the methods specified in §60.485a(c); and [§60.482-3a(i)(1)]
 - B. Is tested for compliance with §60.482-3a(i)(1) initially upon designation, annually, and at other times requested by the Director. [§60.482-3a(i)(2)]
12. Any existing reciprocating compressor in a process unit which becomes an affected facility under provisions of §60.14 or §60.15 is exempt from §60.482-3a(a) through (e) and (h), provided the permittee demonstrates that recasting the distance piece or replacing the compressor are the only options available to bring the compressor into compliance with the provisions of §60.482-3a(a) through (e) and (h). [§60.482-3a(j)]

Standards for pressure relief devices in gas/vapor service:

- 1. Except during pressure releases, each pressure relief device in gas/vapor service shall be operated with no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as determined by the methods specified in §60.485a(c). [§60.482-4a(a)]
- 2. After each pressure release, the pressure relief device shall be returned to a condition of no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as soon as practicable, but no later than 5 calendar days after the pressure release, except as provided in §60.482-9a. [§60.482-4a(b)(1)]
- 3. No later than 5 calendar days after the pressure release, the pressure relief device shall be monitored to confirm the conditions of no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, by the methods specified in §60.485a(c). [§60.482-4a(b)(2)]
- 4. Any pressure relief device that is routed to a process or fuel gas system or equipped with a closed vent system capable of capturing and transporting leakage through the pressure relief device to a control device as described in §60.482-10a is exempted from the requirements of §60.482-4a(a) and (b). [§60.482-4a(c)]
- 5. Any pressure relief device that is equipped with a rupture disk upstream of the pressure relief device is exempt from the requirements of §60.482-4a(a) and (b), provided the permittee complies with the requirements in §60.482-4a(d)(2). [§60.482-4a(d)(1)]
- 6. After each pressure release, a new rupture disk shall be installed upstream of the pressure relief device as soon as practicable, but no later than 5 calendar days after each pressure release, except as provided in §60.482-9a. [§60.482-4a(d)(2)]

Standards for sampling connection systems:

- 1. Each sampling connection system shall be equipped with a closed-purge, closed-loop, or closed-vent system, except as provided in §60.482-1a(c) and §60.482-5a(c). [§60.482-5a(a)]
- 2. Each closed-purge, closed-loop, or closed-vent system as required in §60.482-5a(a) shall comply with the requirements specified in §60.482-5a(b)(1) through (4). [§60.482-5a(b)]
 - A. Gases displaced during filling of the sample container are not required to be collected or captured. [§60.482-5a(b)(1)]
 - B. Containers that are part of a closed-purge system must be covered or closed when not being filled or emptied. [§60.482-5a(b)(2)]
 - C. Gases remaining in the tubing or piping between the closed-purge system valve(s) and sample container valve(s) after the valves are closed and the sample container is disconnected are not required to be collected or captured. [§60.482-5a(b)(3)]

- D. Each closed-purge, closed-loop, or closed-vent system shall be designed and operated to meet requirements in either §60.482-5a(b)(4)(i), (ii), (iii), or (iv). [§60.482-5a(b)(4)]
- i. Return the purged process fluid directly to the process line. [§60.482-5a(b)(4)(i)]
 - ii. Collect and recycle the purged process fluid to a process. [§60.482-5a(b)(4)(ii)]
 - iii. Capture and transport all the purged process fluid to a control device that complies with the requirements of §60.482-10a. [§60.482-5a(b)(4)(iii)]
 - iv. Collect, store, and transport the purged process fluid to any of the following systems or facilities: [§60.482-5a(b)(4)(iv)]
 - a. A waste management unit as defined in 40 CFR 63.111, if the waste management unit is subject to and operated in compliance with the provisions of 40 CFR part 63, subpart G, applicable to Group 1 wastewater streams; [§60.482-5a(b)(4)(iv)(A)]
 - b. A treatment, storage, or disposal facility subject to regulation under 40 CFR part 262, 264, 265, or 266; [§60.482-5a(b)(4)(iv)(B)]
 - c. A facility permitted, licensed, or registered by a state to manage municipal or industrial solid waste, if the process fluids are not hazardous waste as defined in 40 CFR part 261; [§60.482-5a(b)(4)(iv)(C)]
 - d. A waste management unit subject to and operated in compliance with the treatment requirements of 40 CFR 61.348(a), provided all waste management units that collect, store, or transport the purged process fluid to the treatment unit are subject to and operated in compliance with the management requirements of 40 CFR 61.343 through 40 CFR 61.347; or [§60.482-5a(b)(4)(iv)(D)]
 - e. A device used to burn off-specification used oil for energy recovery in accordance with 40 CFR part 279, subpart G, provided the purged process fluid is not hazardous waste as defined in 40 CFR part 261. [§60.482-5a(b)(4)(iv)(E)]
3. In-situ sampling systems and sampling systems without purges are exempt from the requirements of §60.482-5a(a) and (b). [§60.482-5a(c)]

Standards for open-ended valves or lines:

1. Each open-ended valve or line shall be equipped with a cap, blind flange, plug, or a second valve, except as provided in §60.482-1a(c) and §60.482-6a(d) and (e). [§60.482-6a(a)(1)]
2. The cap, blind flange, plug, or second valve shall seal the open end at all times except during operations requiring process fluid flow through the open-ended valve or line. [§60.482-6a(a)(2)]
3. Each open-ended valve or line equipped with a second valve shall be operated in a manner such that the valve on the process fluid end is closed before the second valve is closed. [§60.482-6a(b)]
4. When a double block-and-bleed system is being used, the bleed valve or line may remain open during operations that require venting the line between the block valves but shall comply with §60.482-6a(a) at all other times. [§60.482-6a(c)]
5. Open-ended valves or lines in an emergency shutdown system which are designed to open automatically in the event of a process upset are exempt from the requirements of §60.482-6a(a), (b), and (c). [§60.482-6a(d)]
6. Open-ended valves or lines containing materials which would autocatalytically polymerize or would present an explosion, serious overpressure, or other safety hazard if capped or equipped with a double block and bleed system as specified in §60.482-6a(a) through (c) are exempt from the requirements of §60.482-6a(a) through (c). [§60.482-6a(e)]

Standards for valves in gas/vapor service and in light liquid service:

1. Each valve shall be monitored monthly to detect leaks by the methods specified in §60.485a(b) and shall comply with §60.482-7a(b) through (e), except as provided in §60.482-7a(f), (g), and (h), §60.482-1a(c) and (f), and §§60.483-1a and 60.483-2a. [§60.482-7a(a)(1)]
2. A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for the process unit must be monitored according to §60.482-7a(a)(2)(i) or (ii), except for a valve that replaces a leaking valve and except as provided in §60.482-7a(f), (g), and (h), §60.482-1a(c), and §§60.483-1a and 60.483-2a. [§60.482-7a(a)(2)]
 - A. Monitor the valve as in §60.482-7a(a)(1). The valve must be monitored for the first time within 30 days after the end of its startup period to ensure proper installation. [§60.482-7a(a)(2)(i)]
 - B. If the existing valves in the process unit are monitored in accordance with §60.483-1a or §60.483-2a, count the new valve as leaking when calculating the percentage of valves leaking as described in §60.483-2a(b)(5). If less than 2.0 percent of the valves are leaking for that process unit, the valve must be monitored for the first time during the next scheduled monitoring event for existing valves in the process unit or within 90 days, whichever comes first. [§60.482-7a(a)(2)(ii)]
3. If an instrument reading of 500 ppm or greater is measured, a leak is detected. [§60.482-7a(b)]
4. Any valve for which a leak is not detected for 2 successive months may be monitored the first month of every quarter, beginning with the next quarter, until a leak is detected. [§60.482-7a(c)(1)(i)]
5. As an alternative to monitoring all of the valves in the first month of a quarter, the permittee may elect to subdivide the process unit into two or three subgroups of valves and monitor each subgroup in a different month during the quarter, provided each subgroup is monitored every 3 months. The permittee must keep records of the valves assigned to each subgroup. [§60.482-7a(c)(1)(ii)]
6. If a leak is detected, the valve shall be monitored monthly until a leak is not detected for 2 successive months. [§60.482-7a(c)(2)]
7. When a leak is detected, it shall be repaired as soon as practicable, but no later than 15 calendar days after the leak is detected, except as provided in §60.482-9a. [§60.482-7a(d)(1)]
8. A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-7a(d)(2)]
9. First attempts at repair include, but are not limited to, the following best practices where practicable: [§60.482-7a(e)]
 - A. Tightening of bonnet bolts; [§60.482-7a(e)(1)]
 - B. Replacement of bonnet bolts; [§60.482-7a(e)(2)]
 - C. Tightening of packing gland nuts; [§60.482-7a(e)(3)]
 - D. Injection of lubricant into lubricated packing. [§60.482-7a(e)(4)]
10. Any valve that is designated, as described in §60.486a(e)(2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-7a(a) if the valve: [§60.482-7a(f)]
 - A. Has no external actuating mechanism in contact with the process fluid, [§60.482-7a(f)(1)]
 - B. Is operated with emissions less than 500 ppm above background as determined by the method specified in §60.485a(c), and [§60.482-7a(f)(2)]
 - C. Is tested for compliance with §60.482-7a(f)(2) of this section initially upon designation, annually, and at other times requested by the Director. [§60.482-7a(f)(3)]
11. Any valve that is designated, as described in §60.486a(f)(1), as an unsafe-to-monitor valve is exempt from the requirements of §60.482-7a(a) if: [§60.482-7a(g)]

- A. The permittee demonstrates that the valve is unsafe to monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-7a(a), and [§60.482-7a(g)(1)]
 - B. The permittee adheres to a written plan that requires monitoring of the valve as frequently as practicable during safe-to-monitor times. [§60.482-7a(g)(2)]
12. Any valve that is designated, as described in §60.486a(f)(2), as a difficult-to-monitor valve is exempt from the requirements of §60.482-7a(a) if: [§60.482-7a(h)]
- A. The permittee demonstrates that the valve cannot be monitored without elevating the monitoring personnel more than 2 meters above a support surface. [§60.482-7a(h)(1)]
 - B. The process unit within which the valve: [§60.482-7a(h)(2)]
 - i. Has less than 3.0 percent of its total number of valves designated as difficult-to-monitor by the permittee. [§60.482-7a(h)(2)(ii)]
 - C. The permittee follows a written plan that requires monitoring of the valve at least once per calendar year. [§60.482-7a(h)(3)]

Standards for pumps, valves, and connectors in heavy liquid service and pressure relief devices in light liquid or heavy liquid service:

- 1. If evidence of a potential leak is found by visual, audible, olfactory, or any other detection method at pumps, valves, and connectors in heavy liquid service and pressure relief devices in light liquid or heavy liquid service, the permittee shall follow either one of the following procedures: [§60.482-8a(a)]
 - A. The permittee shall monitor the equipment within 5 days by the method specified in §60.485a(b) and shall comply with the requirements of §60.482-8a(b) through (d). [§60.482-8a(a)(1)]
 - B. The permittee shall eliminate the visual, audible, olfactory, or other indication of a potential leak within 5 calendar days of detection. [§60.482-8a(a)(2)]
- 2. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-8a(b)]
- 3. When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. [§60.482-8a(c)(1)]
- 4. The first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-8a(c)(2)]
- 5. First attempts at repair include, but are not limited to, the best practices described under §60.482-2a(c)(2) and 60.482-7a(e). [§60.482-8a(d)]

Standards for delay of repair:

- 1. Delay of repair of equipment for which leaks have been detected will be allowed if repair within 15 days is technically infeasible without a process unit shutdown. Repair of this equipment shall occur before the end of the next process unit shutdown. Monitoring to verify repair must occur within 15 days after startup of the process unit. [§60.482-9a(a)]
- 2. Delay of repair of equipment will be allowed for equipment which is isolated from the process and which does not remain in VOC service. [§60.482-9a(b)]
- 3. Delay of repair for valves and connectors will be allowed if: [§60.482-9a(c)]
 - A. The permittee demonstrates that emissions of purged material resulting from immediate repair are greater than the fugitive emissions likely to result from delay of repair, and [§60.482-9a(c)(1)]
 - B. When repair procedures are effected, the purged material is collected and destroyed or recovered in a control device complying with §60.482-10a. [§60.482-9a(c)(2)]

4. Delay of repair for pumps will be allowed if: [§60.482-9a(d)]
 - A. Repair requires the use of a dual mechanical seal system that includes a barrier fluid system, and [§60.482-9a(d)(1)]
 - B. Repair is completed as soon as practicable, but not later than 6 months after the leak was detected. [§60.482-9a(d)(2)]
5. Delay of repair beyond a process unit shutdown will be allowed for a valve, if valve assembly replacement is necessary during the process unit shutdown, valve assembly supplies have been depleted, and valve assembly supplies had been sufficiently stocked before the supplies were depleted. Delay of repair beyond the next process unit shutdown will not be allowed unless the next process unit shutdown occurs sooner than 6 months after the first process unit shutdown. [§60.482-9a(e)]
6. When delay of repair is allowed for a leaking pump, valve, or connector that remains in service, the pump, valve, or connector may be considered to be repaired and no longer subject to delay of repair requirements if two consecutive monthly monitoring instrument readings are below the leak definition. [§60.482-9a(f)]

Standards for closed vent systems and control devices:

1. The permittee of closed vent systems and control devices used to comply with provisions of Subpart VVa shall comply with the provisions of §60.482-10a. [§60.482-10a(a)]
2. Vapor recovery systems (for example, condensers and absorbers) shall be designed and operated to recover the VOC emissions vented to them with an efficiency of 95 percent or greater, or to an exit concentration of 20 parts per million by volume (ppmv), whichever is less stringent. [§60.482-10a(b)]
3. Enclosed combustion devices shall be designed and operated to reduce the VOC emissions vented to them with an efficiency of 95 percent or greater, or to an exit concentration of 20 ppmv, on a dry basis, corrected to 3 percent oxygen, whichever is less stringent or to provide a minimum residence time of 0.75 seconds at a minimum temperature of 816 °C. [§60.482-10a(c)]
4. Flares used to comply with this subpart shall comply with the requirements of §60.18. [§60.482-10a(d)]
5. The permittee of control devices used to comply with the provisions of Subpart VVa shall monitor these control devices to ensure that they are operated and maintained in conformance with their designs. [§60.482-10a(e)]
6. Except as provided in §60.482-10a(i) through (k), each closed vent system shall be inspected according to the procedures and schedule specified in §60.482-10a(f)(1) and (2). [§60.482-10a(f)]
 - A. If the vapor collection system or closed vent system is constructed of hard-piping, the permittee shall comply with the requirements specified in §60.482-10a(f)(1)(i) and (ii): [§60.482-10a(f)(1)]
 - i. Conduct an initial inspection according to the procedures in §60.485a(b); and [§60.482-10a(f)(1)(i)]
 - ii. Conduct annual visual inspections for visible, audible, or olfactory indications of leaks. [§60.482-10a(f)(1)(ii)]
 - B. If the vapor collection system or closed vent system is constructed of ductwork, the permittee shall: [§60.482-10a(f)(2)]
 - i. Conduct an initial inspection according to the procedures in §60.485a(b); and [§60.482-10a(f)(2)(i)]
 - ii. Conduct annual inspections according to the procedures in §60.485a(b). [§60.482-10a(f)(2)(ii)]

7. Leaks, as indicated by an instrument reading greater than 500 ppmv above background or by visual inspections, shall be repaired as soon as practicable except as provided in §60.482-10a(h). [§60.482-10a(g)]
 - A. A first attempt at repair shall be made no later than 5 calendar days after the leak is detected. [§60.482-10a(g)(1)]
 - B. Repair shall be completed no later than 15 calendar days after the leak is detected. [§60.482-10a(g)(2)]
8. Delay of repair of a closed vent system for which leaks have been detected is allowed if the repair is technically infeasible without a process unit shutdown or if the permittee determines that emissions resulting from immediate repair would be greater than the fugitive emissions likely to result from delay of repair. Repair of such equipment shall be complete by the end of the next process unit shutdown. [§60.482-10a(h)]
9. If a vapor collection system or closed vent system is operated under a vacuum, it is exempt from the inspection requirements of §60.482-10a(f)(1)(i) and (f)(2). [§60.482-10a(i)]
10. Any parts of the closed vent system that are designated, as described in §60.482-10a(l)(1), as unsafe to inspect are exempt from the inspection requirements of §60.482-10a(f)(1)(i) and (f)(2) if they comply with the requirements specified in §60.482-10a(j)(1) and (2): [§60.482-10a(j)]
 - A. The permittee determines that the equipment is unsafe to inspect because inspecting personnel would be exposed to an imminent or potential danger as a consequence of complying with §60.482-10a(f)(1)(i) or (f)(2); and [§60.482-10a(j)(1)]
 - B. The permittee has a written plan that requires inspection of the equipment as frequently as practicable during safe-to-inspect times. [§60.482-10a(j)(2)]
11. Any parts of the closed vent system that are designated, as described in §60.482-10a(l)(2), as difficult to inspect are exempt from the inspection requirements of §60.482-10a(f)(1)(i) and (f)(2) if they comply with the requirements specified in §60.482-10a(k)(1) through (3): [§60.482-10a(k)]
 - A. The permittee determines that the equipment cannot be inspected without elevating the inspecting personnel more than 2 meters above a support surface; and [§60.482-10a(k)(1)]
 - B. The process unit within which the closed vent system is located becomes an affected facility through §§60.14 or 60.15, or the permittee designates less than 3.0 percent of the total number of closed vent system equipment as difficult to inspect; and [§60.482-10a(k)(2)]
 - C. The permittee has a written plan that requires inspection of the equipment at least once every 5 years. A closed vent system is exempt from inspection if it is operated under a vacuum. [§60.482-10a(k)(3)]
12. The permittee shall record the information specified in §60.482-10a(l)(1) through (5). [§60.482-10a(l)]
 - A. Identification of all parts of the closed vent system that are designated as unsafe to inspect, an explanation of why the equipment is unsafe to inspect, and the plan for inspecting the equipment. [§60.482-10a(l)(1)]
 - B. Identification of all parts of the closed vent system that are designated as difficult to inspect, an explanation of why the equipment is difficult to inspect, and the plan for inspecting the equipment. [§60.482-10a(l)(2)]
 - C. For each inspection during which a leak is detected, a record of the information specified in §60.486a(c). [§60.482-10a(l)(3)]
 - D. For each inspection conducted in accordance with §60.485a(b) during which no leaks are detected, a record that the inspection was performed, the date of the inspection, and a statement that no leaks were detected. [§60.482-10a(l)(4)]

- E. For each visual inspection conducted in accordance with §60.482-10a(f)(1)(ii) during which no leaks are detected, a record that the inspection was performed, the date of the inspection, and a statement that no leaks were detected. [§60.482-10a(l)(5)]
13. Closed vent systems and control devices used to comply with provisions of Subpart VVa shall be operated at all times when emissions may be vented to them. [§60.482-10a(m)]

Standards for connectors in gas/vapor service and in light liquid service:¹³

1. The permittee shall initially monitor all connectors in the process unit for leaks by the later of either 12 months after the compliance date or 12 months after initial startup. If all connectors in the process unit have been monitored for leaks prior to the compliance date, no initial monitoring is required provided either no process changes have been made since the monitoring or the permittee can determine that the results of the monitoring, with or without adjustments, reliably demonstrate compliance despite process changes. If required to monitor because of a process change, the permittee is required to monitor only those connectors involved in the process change. [§60.482-11a(a)]
2. Except as allowed in §60.482-1a(c), §60.482-10a, or as specified in §60.482-11a(e), the permittee shall monitor all connectors in gas and vapor and light liquid service as specified in §60.482-11a(a) and (b)(3). [§60.482-11a(b)]
 - A. The connectors shall be monitored to detect leaks by the method specified in §60.485a(b) and, as applicable, §60.485a(c). [§60.482-11a(b)(1)]
 - B. If an instrument reading greater than or equal to 500 ppm is measured, a leak is detected. [§60.482-11a(b)(2)]
 - C. The permittee shall perform monitoring, subsequent to the initial monitoring required in §60.482-11a(a), as specified in §60.482-11a(b)(3)(i) through (iii), and shall comply with the requirements of §60.482-11a(b)(3)(iv) and (v). The required period in which monitoring must be conducted shall be determined from §60.482-11a(b)(3)(i) through (iii) using the monitoring results from the preceding monitoring period. The percent leaking connectors shall be calculated as specified in §60.482-11a(c). [§60.482-11a(b)(3)]
 - i. If the percent leaking connectors in the process unit was greater than or equal to 0.5 percent, then monitor within 12 months (1 year). [§60.482-11a(b)(3)(i)]
 - ii. If the percent leaking connectors in the process unit was greater than or equal to 0.25 percent but less than 0.5 percent, then monitor within 4 years. The permittee may comply with the requirements of §60.482-11a(b)(3)(ii) by monitoring at least 40 percent of the connectors within 2 years of the start of the monitoring period, provided all connectors have been monitored by the end of the 4-year monitoring period. [§60.482-11a(b)(3)(ii)]
 - iii. If the percent leaking connectors in the process unit was less than 0.25 percent, then monitor as provided in §60.482-11a(b)(3)(iii)(A) and either §60.482-11a(b)(3)(iii)(B) or (b)(3)(iii)(C), as appropriate. [§60.482-11a(b)(3)(iii)]
 - a. The permittee shall monitor at least 50 percent of the connectors within 4 years of the start of the monitoring period. [§60.482-11a(b)(3)(iii)(A)]
 - b. If the percent of leaking connectors calculated from the monitoring results in §60.482-11a(b)(3)(iii)(A) is greater than or equal to 0.35 percent of the monitored connectors, the permittee shall monitor as soon as practical, but within the next 6 months, all connectors that have not yet been monitored during the monitoring period. At the

¹³ Effective Date Note: At 73 FR 31376, June 2, 2008, §60.482-11a was stayed until further notice.

conclusion of monitoring, a new monitoring period shall be started pursuant to §60.482-11a(b)(3), based on the percent of leaking connectors within the total monitored connectors. [§60.482-11a(b)(3)(iii)(B)]

- c. If the percent of leaking connectors calculated from the monitoring results in §60.482-11a(b)(3)(iii)(A) is less than 0.35 percent of the monitored connectors, the permittee shall monitor all connectors that have not yet been monitored within 8 years of the start of the monitoring period. [§60.482-11a(b)(3)(iii)(C)]
 - iv. If, during the monitoring conducted pursuant to §60.482-11a(b)(3)(i) through (iii), a connector is found to be leaking, it shall be re-monitored once within 90 days after repair to confirm that it is not leaking. [§60.482-11a(b)(3)(iv)]
 - v. The permittee shall keep a record of the start date and end date of each monitoring period under this §60.482-11a(b) for each process unit. [§60.482-11a(b)(3)(v)]
3. For use in determining the monitoring frequency, as specified in §60.482-11a(a) and (b)(3), the percent leaking connectors as used in §60.482-11a(a) and (b)(3) shall be calculated by using the following equation: [§60.482-11a(c)]

$$\%C_L = \frac{C_L}{C_T} * 100$$

Where:

%C_L = Percent of leaking connectors as determined through periodic monitoring required in §60.482-11a(a) and (b)(3)(i) through (iii).

C_L = Number of connectors measured at 500 ppm or greater, by the method specified in §60.485a(b).

C_t = Total number of monitored connectors in the process unit or affected facility.

4. When a leak is detected pursuant to §60.482-11a(a) and (b), it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. A first attempt at repair as defined in Subpart VVa shall be made no later than 5 calendar days after the leak is detected. [§60.482-11a(d)]
5. Any connector that is designated, as described in §60.486a(f)(1), as an unsafe-to-monitor connector is exempt from the requirements of §60.482-11a(a) and (b) if: [§60.482-11a(e)]
 - A. The permittee of the connector demonstrates that the connector is unsafe-to-monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-11a(a) and (b); and [§60.482-11a(e)(1)]
 - B. The permittee of the connector has a written plan that requires monitoring of the connector as frequently as practicable during safe-to-monitor times but not more frequently than the periodic monitoring schedule otherwise applicable, and repair of the equipment according to the procedures in §60.482-11a(d) if a leak is detected. [§60.482-11a(e)(2)]
6. *Inaccessible, ceramic, or ceramic-lined connectors.* [§60.482-11a(f)]
 - A. Any connector that is inaccessible or that is ceramic or ceramic-lined (e.g., porcelain, glass, or glass-lined), is exempt from the monitoring requirements of §60.482-11a(a) and (b), from the leak repair requirements of §60.482-11a(d), and from the recordkeeping and reporting requirements of §§63.1038 and 63.1039. An inaccessible connector is one that meets any of the provisions specified in §60.482-11a(f)(1)(i) through (vi), as applicable: [§60.482-11a(f)(1)]
 - i. Buried; [§60.482-11a(f)(1)(i)]

- ii. Insulated in a manner that prevents access to the connector by a monitor probe; [§60.482-11a(f)(1)(ii)]
 - iii. Obstructed by equipment or piping that prevents access to the connector by a monitor probe; [§60.482-11a(f)(1)(iii)]
 - iv. Unable to be reached from a wheeled scissor-lift or hydraulic-type scaffold that would allow access to connectors up to 7.6 meters (25 feet) above the ground; [§60.482-11a(f)(1)(iv)]
 - v. Inaccessible because it would require elevating the monitoring personnel more than 2 meters (7 feet) above a permanent support surface or would require the erection of scaffold; or [§60.482-11a(f)(1)(v)]
 - vi. Not able to be accessed at any time in a safe manner to perform monitoring. Unsafe access includes, but is not limited to, the use of a wheeled scissor-lift on unstable or uneven terrain, the use of a motorized man-lift basket in areas where an ignition potential exists, or access would require near proximity to hazards such as electrical lines, or would risk damage to equipment. [§60.482-11a(f)(1)(vi)]
- B. If any inaccessible, ceramic, or ceramic-lined connector is observed by visual, audible, olfactory, or other means to be leaking, the visual, audible, olfactory, or other indications of a leak to the atmosphere shall be eliminated as soon as practical. [§60.482-11a(f)(2)]
7. Except for instrumentation systems and inaccessible, ceramic, or ceramic-lined connectors meeting the provisions of §60.482-11a(f), identify the connectors subject to the requirements of Subpart VVa. Connectors need not be individually identified if all connectors in a designated area or length of pipe subject to the provisions of Subpart VVa are identified as a group, and the number of connectors subject is indicated. [§60.482-11a(g)]

Alternative standards for valves—allowable percentage of valves leaking:

1. The permittee may elect to comply with an allowable percentage of valves leaking of equal to or less than 2.0 percent. [§60.483-1a(a)]
2. The following requirements shall be met if the permittee wishes to comply with an allowable percentage of valves leaking: [§60.483-1a(b)]
 - A. The permittee must notify the Administrator that the permittee has elected to comply with the allowable percentage of valves leaking before implementing this alternative standard, as specified in §60.487a(d). [§60.483-1a(b)(1)]
 - B. A performance test as specified in §60.483-1a(c) shall be conducted initially upon designation, annually, and at other times requested by the Director. [§60.483-1a(b)(2)]
 - C. If a valve leak is detected, it shall be repaired in accordance with §60.482-7a(d) and (e). [§60.483-1a(b)(3)]
3. Performance tests shall be conducted in the following manner: [§60.483-1a(c)]
 - A. All valves in gas/vapor and light liquid service within the affected facility shall be monitored within 1 week by the methods specified in §60.485a(b). [§60.483-1a(c)(1)]
 - B. If an instrument reading of 500 ppm or greater is measured, a leak is detected. [§60.483-1a(c)(2)]
 - C. The leak percentage shall be determined by dividing the number of valves for which leaks are detected by the number of valves in gas/vapor and light liquid service within the affected facility. [§60.483-1a(c)(3)]
4. The permittee who elects to comply with this alternative standard shall not have an affected facility with a leak percentage greater than 2.0 percent, determined as described in §60.485a(h). [§60.483-1a(d)]

Alternative standards for valves—skip period leak detection and repair:

1. The permittee may elect to comply with one of the alternative work practices specified in §60.483-2a(b)(2) and (3). [§60.483-2a(a)(1)]
2. The permittee must notify the Director before implementing one of the alternative work practices, as specified in §60.487(d)a. [§60.483-2a(a)(2)]
3. The permittee shall comply initially with the requirements for valves in gas/vapor service and valves in light liquid service, as described in §60.482-7a. [§60.483-2a(b)(1)]
4. After 2 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 1 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2a(b)(2)]
5. After 5 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 3 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2a(b)(3)]
6. If the percent of valves leaking is greater than 2.0, the permittee shall comply with the requirements as described in §60.482-7a but can again elect to use §60.483-2a. [§60.483-2a(b)(4)]
7. The percent of valves leaking shall be determined as described in §60.485a(h). [§60.483-2a(b)(5)]
8. The permittee must keep a record of the percent of valves found leaking during each leak detection period. [§60.483-2a(b)(6)]
9. A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for a process unit following one of the alternative standards in §60.483-2a must be monitored in accordance with §60.482-7a(a)(2)(i) or (ii) before the provisions of §60.483-2a can be applied to that valve. [§60.483-2a(b)(7)]

Equivalence of means of emission limitation:

1. The permittee may apply to the Administrator for determination of equivalence for any means of emission limitation that achieves a reduction in emissions of VOC at least equivalent to the reduction in emissions of VOC achieved by the controls required in Subpart VVa. [§60.484(a)]
2. Determination of equivalence to the equipment, design, and operational requirements of Subpart VVa will be evaluated by the following guidelines: [§60.484(b)]
 - A. The permittee applying for an equivalence determination shall be responsible for collecting and verifying test data to demonstrate equivalence of means of emission limitation. [§60.484(b)(1)]
 - B. The Administrator will compare test data for demonstrating equivalence of the means of emission limitation to test data for the equipment, design, and operational requirements. [§60.484(b)(2)]
 - C. The Administrator may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the equipment, design, and operational requirements. [§60.484(b)(3)]
3. Determination of equivalence to the required work practices in Subpart VVa will be evaluated by the following guidelines: [§60.484(c)]
 - A. The permittee applying for a determination of equivalence shall be responsible for collecting and verifying test data to demonstrate equivalence of an equivalent means of emission limitation. [§60.484(c)(1)]
 - B. For each affected facility for which a determination of equivalence is requested, the emission reduction achieved by the required work practice shall be demonstrated. [§60.484(c)(2)]

- C. For each affected facility, for which a determination of equivalence is requested, the emission reduction achieved by the equivalent means of emission limitation shall be demonstrated. [§60.484(c)(3)]
- D. The permittee applying for a determination of equivalence shall commit in writing to work practice(s) that provide for emission reductions equal to or greater than the emission reductions achieved by the required work practice. [§60.484(c)(4)]
- E. The Administrator will compare the demonstrated emission reduction for the equivalent means of emission limitation to the demonstrated emission reduction for the required work practices and will consider the commitment in §60.484(c)(4). [§60.484(c)(5)]
- F. The Administrator may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the required work practice. [§60.484(c)(6)]
- 4. The permittee may offer a unique approach to demonstrate the equivalence of any equivalent means of emission limitation. [§60.484(d)]
- 5. After a request for determination of equivalence is received, the Administrator will publish a notice in the Federal Register and provide the opportunity for public hearing if the Administrator judges that the request may be approved. [§60.484(e)(1)]
- 6. After notice and opportunity for public hearing, the Administrator will determine the equivalence of a means of emission limitation and will publish the determination in the Federal Register. [§60.484(e)(2)]
- 7. Any equivalent means of emission limitations approved under §60.484 shall constitute a required work practice, equipment, design, or operational standard within the meaning of section 111(h)(1) of the Clean Air Act. [§60.484(e)(3)]
- 8. Manufacturers of equipment used to control equipment leaks of VOC may apply to the Administrator for determination of equivalence for any equivalent means of emission limitation that achieves a reduction in emissions of VOC achieved by the equipment, design, and operational requirements of this subpart. [§60.484(f)(1)]
- 9. The Administrator will make an equivalence determination according to the provisions of §60.484(b), (c), (d), and (e). [§60.484(f)(2)]

Test methods and procedures:

- 1. In conducting the performance tests required in §60.8, the permittee shall use as reference methods and procedures the test methods in appendix A of part 60 or other methods and procedures as specified in §60.485a, except as provided in §60.8(b). [§60.485a(a)]
- 2. The permittee shall determine compliance with the standards in §§60.482-1a through 60.482-11a, 60.483a, and 60.484a as follows: [§60.485a(b)]
 - A. Method 21 shall be used to determine the presence of leaking sources. The instrument shall be calibrated before use each day of its use by the procedures specified in Method 21 of appendix A-7 of part 60. The following calibration gases shall be used: [§60.485a(b)(1)]
 - i. Zero air (less than 10 ppm of hydrocarbon in air); and [§60.485a(b)(1)(i)]
 - ii. A mixture of methane or n-hexane and air at a concentration no more than 2,000 ppm greater than the leak definition concentration of the equipment monitored. If the monitoring instrument's design allows for multiple calibration scales, then the lower scale shall be calibrated with a calibration gas that is no higher than 2,000 ppm above the concentration specified as a leak, and the highest scale shall be calibrated with a calibration gas that is approximately equal to 10,000 ppm. If only one scale on an instrument will be used during

monitoring, the permittee need not calibrate the scales that will not be used during that day's monitoring. [§60.485a(b)(1)(ii)]

- B. A calibration drift assessment shall be performed, at a minimum, at the end of each monitoring day. Check the instrument using the same calibration gas(es) that were used to calibrate the instrument before use. Follow the procedures specified in Method 21 of appendix A-7 of part 60, Section 10.1, except do not adjust the meter readout to correspond to the calibration gas value. Record the instrument reading for each scale used as specified in §60.486a(e)(7). Calculate the average algebraic difference between the three meter readings and the most recent calibration value. Divide this algebraic difference by the initial calibration value and multiply by 100 to express the calibration drift as a percentage. If any calibration drift assessment shows a negative drift of more than 10 percent from the initial calibration value, then all equipment monitored since the last calibration with instrument readings below the appropriate leak definition and above the leak definition multiplied by (100 minus the percent of negative drift/divided by 100) must be re-monitored. If any calibration drift assessment shows a positive drift of more than 10 percent from the initial calibration value, then, at the permittee's discretion, all equipment since the last calibration with instrument readings above the appropriate leak definition and below the leak definition multiplied by (100 plus the percent of positive drift/divided by 100) may be re-monitored. [§60.485a(b)(2)]
- 3. The permittee shall determine compliance with the no-detectable-emission standards in §§60.482-2a(e), 60.482-3a(i), 60.482-4a, 60.482-7a(f), and 60.482-10a(e) as follows: [§60.485a(c)]
 - A. The requirements of §60.485a(b) shall apply. [§60.485a(c)(1)]
 - B. Method 21 of appendix A-7 of part 60 shall be used to determine the background level. All potential leak interfaces shall be traversed as close to the interface as possible. The arithmetic difference between the maximum concentration indicated by the instrument and the background level is compared with 500 ppm for determining compliance. [§60.485a(c)(2)]
- 4. The permittee shall test each piece of equipment unless they demonstrate that a process unit is not in VOC service, i.e., that the VOC content would never be reasonably expected to exceed 10 percent by weight. For purposes of this demonstration, the following methods and procedures shall be used: [§60.485a(d)]
 - A. Procedures that conform to the general methods in ASTM E260-73, 91, or 96, E168-67, 77, or 92, E169-63, 77, or 93 (incorporated by reference—see §60.17) shall be used to determine the percent VOC content in the process fluid that is contained in or contacts a piece of equipment. [§60.485a(d)(1)]
 - B. Organic compounds that are considered by the Administrator to have negligible photochemical reactivity may be excluded from the total quantity of organic compounds in determining the VOC content of the process fluid. [§60.485a(d)(2)]
 - C. Engineering judgment may be used to estimate the VOC content, if a piece of equipment had not been shown previously to be in service. If the Administrator disagrees with the judgment, §60.485a(d)(1) and (2) shall be used to resolve the disagreement. [§60.485a(d)(3)]
- 5. The permittee shall demonstrate that a piece of equipment is in light liquid service by showing that all the following conditions apply: [§60.485a(e)]
 - A. The vapor pressure of one or more of the organic components is greater than 0.3 kPa at 20 °C (1.2 in. H₂O at 68 °F). Standard reference texts or ASTM D2879-83, 96, or 97 (incorporated by reference—see §60.17) shall be used to determine the vapor pressures. [§60.485a(e)(1)]

- B. The total concentration of the pure organic components having a vapor pressure greater than 0.3 kPa at 20 °C (1.2 in. H₂O at 68 °F) is equal to or greater than 20 percent by weight. [§60.485a(e)(2)]
- C. The fluid is a liquid at operating conditions. [§60.485a(e)(3)]
6. Samples used in conjunction with §60.485a(d), (e), and (g) shall be representative of the process fluid that is contained in or contacts the equipment or the gas being combusted in the flare. [§60.485a(f)]
7. The permittee shall determine compliance with the standards of flares as follows: [§60.485a(g)]
- A. Method 22 of appendix A-7 of part 60 shall be used to determine visible emissions. [§60.485a(g)(1)]
- B. A thermocouple or any other equivalent device shall be used to monitor the presence of a pilot flame in the flare. [§60.485a(g)(2)]
- C. The maximum permitted velocity for air assisted flares shall be computed using the following equation: [§60.485a(g)(3)]

$$V_{max} = K_1 + K_2 H_T$$

Where:

V_{max} = Maximum permitted velocity, m/sec (ft/sec).

H_T = Net heating value of the gas being combusted, MJ/scm (Btu/scf).

K_1 = 8.706 m/sec (metric units) , 28.56 ft/sec (English units).

K_2 = 0.7084 m⁴/(MJ-sec) (metric units), 0.087 ft⁴/(Btu-sec) (English units).

- D. The net heating value (HT) of the gas being combusted in a flare shall be computed using the following equation: [§60.485a(g)(4)]

$$H_T = K \sum_{i=1}^n C_i H_i$$

Where:

K = Conversion constant, $1.740 \times 10^{-7} \frac{g-mol*MJ}{ppm*scm*kcal}$ (metric units), $4.674 \times 10^{-6} \frac{g-mol*Btu}{ppm*scf*kcal}$ (English units).

C_i = Concentration of sample component “i,” ppm

H_i = net heat of combustion of sample component “i” at 25 °C and 760 mm Hg (77 °F and 14.7 psi), kcal/g-mole.

- E. Method 18 of appendix A-6 of part 60 or ASTM D6420-99 (2004) (where the target compound(s) are those listed in Section 1.1 of ASTM D6420-99, and the target concentration is between 150 parts per billion by volume and 100 ppmv) and ASTM D2504-67, 77, or 88 (Reapproved 1993) (incorporated by reference-see §60.17) shall be used to determine the concentration of sample component “i.” [§60.485a(g)(5)]
- F. ASTM D2382-76 or 88 or D4809-95 (incorporated by reference-see §60.17) shall be used to determine the net heat of combustion of component “i” if published values are not available or cannot be calculated. [§60.485a(g)(6)]

- G. Method 2, 2A, 2C, or 2D of appendix A-7 of part 60, as appropriate, shall be used to determine the actual exit velocity of a flare. If needed, the unobstructed (free) cross-sectional area of the flare tip shall be used. [§60.485a(g)(7)]
8. The permittee shall determine compliance with §60.483-1a or §60.483-2a as follows: [§60.485a(h)]
- A. The percent of valves leaking shall be determined using the following equation: [§60.485a(h)(1)]

$$\%V_L = \left(\frac{V_L}{V_T} \right) * 100$$

Where:

$\%V_L$ = Percent leaking valves.

V_L = Number of valves found leaking.

V_T = The sum of the total number of valves monitored.

- B. The total number of valves monitored shall include difficult-to-monitor and unsafe-to-monitor valves only during the monitoring period in which those valves are monitored. [§60.485a(h)(2)]
- C. The number of valves leaking shall include valves for which repair has been delayed. [§60.485a(h)(3)]
- D. Any new valve that is not monitored within 30 days of being placed in service shall be included in the number of valves leaking and the total number of valves monitored for the monitoring period in which the valve is placed in service. [§60.485a(h)(4)]
- E. If the process unit has been subdivided in accordance with §60.482-7a(c)(1)(ii), the sum of valves found leaking during a monitoring period includes all subgroups. [§60.485a(h)(5)]
- F. The total number of valves monitored does not include a valve monitored to verify repair. [§60.485a(h)(6)]

Recordkeeping:

1. The permittee shall comply with the recordkeeping requirements of §60.486a. [§60.486a(a)(1)]
2. The permittee of more than one affected facility subject to the provisions of Subpart VVa may comply with the recordkeeping requirements for these facilities in one recordkeeping system if the system identifies each record by each facility. [§60.486a(a)(2)]
3. The permittee shall record the information specified in §60.486a(a)(3)(i) through (v) for each monitoring event required by §§60.482-2a, 60.482-3a, 60.482-7a, 60.482-8a, 60.482-11a, and 60.483-2a. [§60.486a(a)(3)]
 - A. Monitoring instrument identification. [§60.486a(a)(3)(i)]
 - B. Operator identification. [§60.486a(a)(3)(ii)]
 - C. Equipment identification. [§60.486a(a)(3)(iii)]
 - D. Date of monitoring. [§60.486a(a)(3)(iv)]
 - E. Instrument reading. [§60.486a(a)(3)(v)]
4. When each leak is detected as specified in §§60.482-2a, 60.482-3a, 60.482-7a, 60.482-8a, 60.482-11a, and 60.483-2a, the following requirements apply: [§60.486a(b)]
 - A. A weatherproof and readily visible identification, marked with the equipment identification number, shall be attached to the leaking equipment. [§60.486a(b)(1)]
 - B. The identification on a valve may be removed after it has been monitored for 2 successive months as specified in §60.482-7a(c) and no leak has been detected during those 2 months. [§60.486a(b)(2)]

- C. The identification on a connector may be removed after it has been monitored as specified in §60.482-11a(b)(3)(iv) and no leak has been detected during that monitoring. [§60.486a(b)(3)]
- D. The identification on equipment, except on a valve or connector, may be removed after it has been repaired. [§60.486a(b)(4)]
- 5. When each leak is detected as specified in §§60.482-2a, 60.482-3a, 60.482-7a, 60.482-8a, 60.482-11a, and 60.483-2a, the following information shall be recorded in a log and shall be kept for 2 years in a readily accessible location: [§60.486a(c)]
 - A. The instrument and operator identification numbers and the equipment identification number, except when indications of liquids dripping from a pump are designated as a leak. [§60.486a(c)(1)]
 - B. The date the leak was detected and the dates of each attempt to repair the leak. [§60.486a(c)(2)]
 - C. Repair methods applied in each attempt to repair the leak. [§60.486a(c)(3)]
 - D. Maximum instrument reading measured by Method 21 of appendix A-7 of part 60 at the time the leak is successfully repaired or determined to be nonreparable, except when a pump is repaired by eliminating indications of liquids dripping. [§60.486a(c)(4)]
 - E. “Repair delayed” and the reason for the delay if a leak is not repaired within 15 calendar days after discovery of the leak. [§60.486a(c)(5)]
 - F. The signature of the permittee (or designate) whose decision it was that repair could not be effected without a process shutdown. [§60.486a(c)(6)]
 - G. The expected date of successful repair of the leak if a leak is not repaired within 15 days. [§60.486a(c)(7)]
 - H. Dates of process unit shutdowns that occur while the equipment is unrepaired. [§60.486a(c)(8)]
 - I. The date of successful repair of the leak. [§60.486a(c)(9)]
- 6. The following information pertaining to the design requirements for closed vent systems and control devices described in §60.482-10a shall be recorded and kept in a readily accessible location: [§60.486a(d)]
 - A. Detailed schematics, design specifications, and piping and instrumentation diagrams. [§60.486a(d)(1)]
 - B. The dates and descriptions of any changes in the design specifications. [§60.486a(d)(2)]
 - C. A description of the parameter or parameters monitored, as required in §60.482-10a(e), to ensure that control devices are operated and maintained in conformance with their design and an explanation of why that parameter (or parameters) was selected for the monitoring. [§60.486a(d)(3)]
 - D. Periods when the closed vent systems and control devices required in §§60.482-2a, 60.482-3a, 60.482-4a, and 60.482-5a are not operated as designed, including periods when a flare pilot light does not have a flame. [§60.486a(d)(4)]
 - E. Dates of startups and shutdowns of the closed vent systems and control devices required in §§60.482-2a, 60.482-3a, 60.482-4a, and 60.482-5a. [§60.486a(d)(5)]
- 7. The following information pertaining to all equipment subject to the requirements in §§60.482-1a to 60.482-11a shall be recorded in a log that is kept in a readily accessible location: [§60.486a(e)]
 - A. A list of identification numbers for equipment subject to the requirements of Subpart VVa. [§60.486a(e)(1)]
 - B. A list of identification numbers for equipment that are designated for no detectable emissions under the provisions of §§60.482-2a(e), 60.482-3a(i), and 60.482-7a(f). [§60.486a(e)(2)(i)]

- C. The designation of equipment as subject to the requirements of §60.482-2a(e), §60.482-3a(i), or §60.482-7a(f) shall be signed by the permittee. Alternatively, the permittee may establish a mechanism with the Director that satisfies this requirement. [§60.486a(e)(2)(ii)]
 - D. A list of equipment identification numbers for pressure relief devices required to comply with §60.482-4a. [§60.486a(e)(3)]
 - E. The dates of each compliance test as required in §§60.482-2a(e), 60.482-3a(i), 60.482-4a, and 60.482-7a(f). [§60.486a(e)(4)(i)]
 - F. The background level measured during each compliance test. [§60.486a(e)(4)(ii)]
 - G. The maximum instrument reading measured at the equipment during each compliance test. [§60.486a(e)(4)(iii)]
 - H. A list of identification numbers for equipment in vacuum service. [§60.486a(e)(5)]
 - I. A list of identification numbers for equipment that the permittee designates as operating in VOC service less than 300 hr/yr in accordance with §60.482-1a(e), a description of the conditions under which the equipment is in VOC service, and rationale supporting the designation that it is in VOC service less than 300 hr/yr. [§60.486a(e)(6)]
 - J. The date and results of the weekly visual inspection for indications of liquids dripping from pumps in light liquid service. [§60.486a(e)(7)]
 - K. Records of the information specified in §60.486a(e)(8)(i) through (vi) for monitoring instrument calibrations conducted according to sections 8.1.2 and 10 of Method 21 of appendix A-7 of part 60 and §60.485a(b). [§60.486a(e)(8)]
 - i. Date of calibration and initials of operator performing the calibration. [§60.486a(e)(8)(i)]
 - ii. Calibration gas cylinder identification, certification date, and certified concentration. [§60.486a(e)(8)(ii)]
 - iii. Instrument scale(s) used. [§60.486a(e)(8)(iii)]
 - iv. A description of any corrective action taken if the meter readout could not be adjusted to correspond to the calibration gas value in accordance with section 10.1 of Method 21 of appendix A-7 of part 60. [§60.486a(e)(8)(iv)]
 - v. Results of each calibration drift assessment required by §60.485a(b)(2) (i.e., instrument reading for calibration at end of monitoring day and the calculated percent difference from the initial calibration value). [§60.486a(e)(8)(v)]
 - vi. If the permittee makes their own calibration gas, a description of the procedure used. [§60.486a(e)(8)(vi)]
 - L. The connector monitoring schedule for each process unit as specified in §60.482-11a(b)(3)(v). [§60.486a(e)(9)]
 - M. Records of each release from a pressure relief device subject to §60.482-4a. [§60.486a(e)(10)]
8. The following information pertaining to all valves subject to the requirements of §60.482-7a(g) and (h), all pumps subject to the requirements of §60.482-2a(g), and all connectors subject to the requirements of §60.482-11a(e) shall be recorded in a log that is kept in a readily accessible location: [§60.486a(f)]
- A. A list of identification numbers for valves, pumps, and connectors that are designated as unsafe-to-monitor, an explanation for each valve, pump, or connector stating why the valve, pump, or connector is unsafe-to-monitor, and the plan for monitoring each valve, pump, or connector. [§60.486a(f)(1)]
 - B. A list of identification numbers for valves that are designated as difficult-to-monitor, an explanation for each valve stating why the valve is difficult-to-monitor, and the schedule for monitoring each valve. [§60.486a(f)(2)]

9. The following information shall be recorded for valves complying with §60.483-2a: [§60.486a(g)]
 - A. A schedule of monitoring. [§60.486a(g)(1)]
 - B. The percent of valves found leaking during each monitoring period. [§60.486a(g)(2)]
10. The following information shall be recorded in a log that is kept in a readily accessible location: [§60.486a(h)]
 - A. Design criterion required in §§60.482-2a(d)(5) and 60.482-3a(e)(2) and explanation of the design criterion; and [§60.486a(h)(1)]
 - B. Any changes to this criterion and the reasons for the changes. [§60.486a(h)(2)]
11. The following information shall be recorded in a log that is kept in a readily accessible location for use in determining exemptions as provided in §60.480a(d): [§60.486a(i)]
 - A. An analysis demonstrating the design capacity of the affected facility, [§60.486a(i)(1)]
 - B. A statement listing the feed or raw materials and products from the affected facilities and an analysis demonstrating whether these chemicals are heavy liquids or beverage alcohol, and [§60.486a(i)(2)]
 - C. An analysis demonstrating that equipment is not in VOC service. [§60.486a(i)(3)]
12. Information and data used to demonstrate that a piece of equipment is not in VOC service shall be recorded in a log that is kept in a readily accessible location. [§60.486a(j)]
13. The provisions of §60.7(b) and (d) do not apply to affected facilities subject to Subpart VVa. [§60.486a(k)]

Reporting:

1. The permittee shall submit semiannual reports to the Director beginning 6 months after the initial startup date. [§60.487a(a)]
2. The initial semiannual report to the Director shall include the following information: [§60.487a(b)]
 - A. Process unit identification. [§60.487a(b)(1)]
 - B. Number of valves subject to the requirements of §60.482-7a, excluding those valves designated for no detectable emissions under the provisions of §60.482-7a(f). [§60.487a(b)(2)]
 - C. Number of pumps subject to the requirements of §60.482-2a, excluding those pumps designated for no detectable emissions under the provisions of §60.482-2a(e) and those pumps complying with §60.482-2a(f). [§60.487a(b)(3)]
 - D. Number of compressors subject to the requirements of §60.482-3a, excluding those compressors designated for no detectable emissions under the provisions of §60.482-3a(i) and those compressors complying with §60.482-3a(h). [§60.487a(b)(4)]
 - E. Number of connectors subject to the requirements of §60.482-11a. [§60.487a(b)(5)]
3. All semiannual reports to the Director shall include the following information, summarized from the information in §60.486a: [§60.487a(c)]
 - A. Process unit identification. [§60.487a(c)(1)]
 - B. For each month during the semiannual reporting period, [§60.487a(c)(2)]
 - i. Number of valves for which leaks were detected as described in §60.482-7a(b) or §60.483-2a, [§60.487a(c)(2)(i)]
 - ii. Number of valves for which leaks were not repaired as required in §60.482-7a(d)(1), [§60.487a(c)(2)(ii)]
 - iii. Number of pumps for which leaks were detected as described in §60.482-2a(b), (d)(4)(ii)(A) or (B), or (d)(5)(iii), [§60.487a(c)(2)(iii)]
 - iv. Number of pumps for which leaks were not repaired as required in §60.482-2a(c)(1) and (d)(6), [§60.487a(c)(2)(iv)]

- v. Number of compressors for which leaks were detected as described in §60.482-3a(f), [§60.487a(c)(2)(v)]
 - vi. Number of compressors for which leaks were not repaired as required in §60.482-3a(g)(1), [§60.487a(c)(2)(vi)]
 - vii. Number of connectors for which leaks were detected as described in §60.482-11a(b) [§60.487a(c)(2)(vii)]
 - viii. Number of connectors for which leaks were not repaired as required in §60.482-11a(d), and [§60.487a(c)(2)(viii)]
 - ix. The facts that explain each delay of repair and, where appropriate, why a process unit shutdown was technically infeasible. [§60.487a(c)(2)(xi)]
- C. Dates of process unit shutdowns which occurred within the semiannual reporting period. [§60.487a(c)(3)]
- D. Revisions to items reported according to §60.487a(b) if changes have occurred since the initial report or subsequent revisions to the initial report. [§60.487a(c)(4)]
4. The permittee electing to comply with the provisions of §§60.483-1a or 60.483-2a shall notify the Administrator of the alternative standard selected 90 days before implementing either of the provisions. [§60.487a(d)]
5. The permittee shall report the results of all performance tests in accordance with §60.8 of the General Provisions. The provisions of §60.8(d) do not apply to affected facilities subject to Subpart VVa except that the permittee must notify the Director of the schedule for the initial performance tests at least 30 days before the initial performance tests. [§60.487a(e)]
6. The requirements of §60.487a(a) through (c) remain in force until and unless EPA, in delegating enforcement authority to a state under section 111(c) of the Clean Air Act, approves reporting requirements or an alternative means of compliance surveillance adopted by such state. In that event, affected sources within the state will be relieved of the obligation to comply with the requirements of §60.487a(a) through (c), provided that they comply with the requirements established by the state. [§60.487a(f)]
7. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
8. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition Subpart A-Alternative for Leaks

10 CSR 10-6.070 New Source Performance Regulations
40 CFR part 60 Subpart A, General Provisions

Alternative Work Practice Monitoring for Leaks:

1. Alternative work practice for monitoring equipment for leaks. §60.18(g), (h), and (i) apply to all equipment for which the applicable regulation requires monitoring with a 40 CFR part 60, appendix A-7, Method 21 monitor, except for closed vent systems, equipment designated as leakless, and equipment identified in the applicable regulation as having no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background. The permittee may use an optical gas imaging instrument instead of a 40 CFR part 60, appendix A-7, Method 21 monitor. Requirements in

the existing regulations that are specific to the Method 21 instrument do not apply under §60.18. All other requirements in the applicable regulation that are not addressed in §60.18(g), (h), and (i) apply to §60.18. For example, equipment specification requirements, and non-Method 21 instrument recordkeeping and reporting requirements in the applicable regulation continue to apply. The terms defined in §60.18(g)(1) through (5) have meanings that are specific to the alternative work practice standard in §60.18(g), (h), and (i). [§60.18(g)]

- A. Applicable regulation means the regulation in 40 CFR parts 60, 61, 63, or 65 that requires monitoring of equipment with a 40 CFR part 60, appendix A-7, Method 21 monitor. [§60.18(g)(1)]
 - B. Equipment means pumps, valves, pressure relief valves, compressors, open-ended lines, flanges, connectors, and other equipment covered by the applicable regulation that require monitoring with a 40 CFR part 60, appendix A-7, Method 21 monitor. [§60.18(g)(2)]
 - C. Imaging means making visible emissions that may otherwise be invisible to the naked eye. [§60.18(g)(3)]
 - D. Optical gas imaging instrument means an instrument that makes visible emissions that may otherwise be invisible to the naked eye. [§60.18(g)(4)]
 - E. Repair means that equipment is adjusted, or otherwise altered, in order to eliminate a leak. [§60.18(g)(5)]
 - F. Leak means: [§60.18(g)(6)]
 - i. Any emissions imaged by the optical gas instrument; [§60.18(g)(6)(i)]
 - ii. Indications of liquids dripping; [§60.18(g)(6)(ii)]
 - iii. Indications by a sensor that a seal or barrier fluid system has failed; or [§60.18(g)(6)(iii)]
 - iv. Screening results using a 40 CFR part 60, appendix A-7, Method 21 monitor that exceed the leak definition in the applicable regulation to which the equipment is subject. [§60.18(g)(6)(iv)]
2. The alternative work practice standard for monitoring equipment for leaks is available to all subparts in 40 CFR parts 60, 61, 63, and 65 that require monitoring of equipment with a 40 CFR part 60, appendix A-7, Method 21 monitor. [§60.18(h)]
- A. The permittee of an affected source subject to CFR parts 60, 61, 63, or 65 can choose to comply with the alternative work practice requirements in §60.18(i) instead of using the 40 CFR part 60, appendix A-7, Method 21 monitor to identify leaking equipment. The permittee must document the equipment, process units, and facilities for which the alternative work practice will be used to identify leaks. [§60.18(h)(1)]
 - B. Any leak detected when following the leak survey procedure in §60.18(i)(3) must be identified for repair as required in the applicable regulation. [§60.18(h)(2)]
 - C. If the alternative work practice is used to identify leaks, re-screening after an attempted repair of leaking equipment must be conducted using either the alternative work practice or the 40 CFR part 60, appendix A-7, Method 21 monitor at the leak definition required in the applicable regulation to which the equipment is subject. [§60.18(h)(4)]
 - D. The schedule for repair is as required in the applicable regulation. [§60.18(h)(4)]
 - E. When this alternative work practice is used for detecting leaking equipment, choose one of the monitoring frequencies listed in Table 1 to subpart A of this part in lieu of the monitoring frequency specified for regulated equipment in the applicable regulation. Reduced monitoring frequencies for good performance are not applicable when using the alternative work practice. [§60.18(h)(5)]

- F. When this alternative work practice is used for detecting leaking equipment the following are not applicable for the equipment being monitored: [§60.18(h)(6)]
- i. Skip period leak detection and repair; [§60.18(h)(6)(i)]
 - ii. Quality improvement plans; or [§60.18(h)(6)(ii)]
 - iii. Complying with standards for allowable percentage of valves and pumps to leak. [§60.18(h)(6)(iii)]
- G. When the alternative work practice is used to detect leaking equipment, the regulated equipment in §60.18(h)(1)(i) must also be monitored annually using a 40 CFR part 60, appendix A-7, Method 21 monitor at the leak definition required in the applicable regulation. The permittee may choose the specific monitoring period (for example, first quarter) to conduct the annual monitoring. Subsequent monitoring must be conducted every 12 months from the initial period. The permittee must keep records of the annual Method 21 screening results, as specified in §60.18(i)(4)(vii). [§60.18(h)(7)]
- i. The permittee of an affected source who chooses to use the alternative work practice must comply with the requirements of §60.18(i)(1) through (i)(5). [§60.18(h)(7)(i)]
 - a. Instrument Specifications. The optical gas imaging instrument must comply with the requirements in §60.18 (i)(1)(i) and (i)(1)(ii). [§60.18(h)(7)(i)(1)]
 - 1) Provide the operator with an image of the potential leak points for each piece of equipment at both the detection sensitivity level and within the distance used in the daily instrument check described in §60.18(i)(2). The detection sensitivity level depends upon the frequency at which leak monitoring is to be performed. [§60.18(h)(7)(i)(1)(i)]
 - 2) Provide a date and time stamp for video records of every monitoring event. [§60.18(h)(7)(i)(1)(ii)]
 - b. Daily Instrument Check. On a daily basis, and prior to beginning any leak monitoring work, test the optical gas imaging instrument at the mass flow rate determined in §60.18(i)(2)(i) in accordance with the procedure specified in §60.18(i)(2)(ii) through (i)(2)(iv) for each camera configuration used during monitoring (for example, different lenses used), unless an alternative method to demonstrate daily instrument checks has been approved in accordance with §60.18(i)(2)(v). [§60.18(h)(7)(i)(2)]
 - 1) Calculate the mass flow rate to be used in the daily instrument check by following the procedures in §60.18(i)(2)(i)(A) and (i)(2)(i)(B). [§60.18(h)(7)(i)(2)(i)]
 - i) For a specified population of equipment to be imaged by the instrument, determine the piece of equipment in contact with the lowest mass fraction of chemicals that are detectable, within the distance to be used in §60.18(i)(2)(iv)(B), at or below the standard detection sensitivity level. [§60.18(h)(7)(i)(2)(i)(A)]
 - ii) Multiply the standard detection sensitivity level, corresponding to the selected monitoring frequency in Table 1 of subpart A of this part, by the mass fraction of detectable chemicals from the stream identified in §60.18(i)(2)(i)(A) to determine the mass flow rate to be used in the daily instrument check, using the following equation. [§60.18(h)(7)(i)(2)(i)(B)]

$$E_{dic} = (E_{sds}) \sum_{i=1}^k x_i$$

Where:

E_{dic} = Mass flow rate for the daily instrument check, grams per hour

x_i = Mass fraction of detectable chemical(s) i seen by the optical gas imaging instrument, within the distance to be used in §60.18(i)(2)(iv)(B), at or below the standard detection sensitivity level, E_{sds} .

E_{sds} = Standard detection sensitivity level from Table 1 to subpart A of part 60, grams per hour

k = Total number of detectable chemicals emitted from the leaking equipment and seen by the optical gas imaging instrument.

- 2) Start the optical gas imaging instrument according to the manufacturer's instructions, ensuring that all appropriate settings conform to the manufacturer's instructions. [§60.18(h)(7)(i)(2)(ii)]
 - 3) Use any gas chosen by the user that can be viewed by the optical gas imaging instrument and that has a purity of no less than 98 percent. [§60.18(h)(7)(i)(2)(iii)]
 - 4) Establish a mass flow rate by using the following procedures:
[§60.18(h)(7)(i)(2)(iv)]
 - i) Provide a source of gas where it will be in the field of view of the optical gas imaging instrument. [§60.18(h)(7)(i)(2)(iv)(A)]
 - ii) Set up the optical gas imaging instrument at a recorded distance from the outlet or leak orifice of the flow meter that will not be exceeded in the actual performance of the leak survey. Do not exceed the operating parameters of the flow meter. [§60.18(h)(7)(i)(2)(iv)(B)]
 - iii) Open the valve on the flow meter to set a flow rate that will create a mass emission rate equal to the mass rate specified in §60.18(i)(2)(i) while observing the gas flow through the optical gas imaging instrument viewfinder. When an image of the gas emission is seen through the viewfinder at the required emission rate, make a record of the reading on the flow meter.
[§60.18(h)(7)(i)(2)(iv)(C)]
 - 5) Repeat the procedures specified in §60.18(i)(2)(ii) through (i)(2)(iv) for each configuration of the optical gas imaging instrument used during the leak survey.
[§60.18(h)(7)(i)(2)(v)]
 - 6) To use an alternative method to demonstrate daily instrument checks, apply to the Administrator for approval of the alternative under §60.13(i).
[§60.18(h)(7)(i)(2)(vi)]
- c. Leak Survey Procedure. Operate the optical gas imaging instrument to image every regulated piece of equipment selected for this work practice in accordance with the instrument manufacturer's operating parameters. All emissions imaged by the optical gas imaging instrument are considered to be leaks and are subject to repair. All emissions visible to the naked eye are also considered to be leaks and are subject to repair. [§60.18(h)(7)(i)(3)]

Recordkeeping:

1. The permittee must keep the records described in §60.18(i)(4)(i) through (i)(4)(vii):
[§60.18(h)(7)(i)(4)]
 - A. The equipment, processes, and facilities for which the permittee chooses to use the alternative work practice. [§60.18(h)(7)(i)(4)(i)]
 - B. The detection sensitivity level selected from Table 1 to subpart A of part 60 for the optical gas imaging instrument. [§60.18(h)(7)(i)(4)(ii)]
 - C. The analysis to determine the piece of equipment in contact with the lowest mass fraction of chemicals that are detectable, as specified in §60.18(i)(2)(i)(A). [§60.18(h)(7)(i)(4)(iii)]
 - D. The technical basis for the mass fraction of detectable chemicals used in the equation in §60.18(i)(2)(i)(B). [§60.18(h)(7)(i)(4)(iv)]
 - E. The daily instrument check. Record the distance, per §60.18(i)(2)(iv)(B), and the flow meter reading, per §60.18(i)(2)(iv)(C), at which the leak was imaged. Keep a video record of the daily instrument check for each configuration of the optical gas imaging instrument used during the leak survey (for example, the daily instrument check must be conducted for each lens used). The video record must include a time and date stamp for each daily instrument check. The video record must be kept for 5 years. [§60.18(h)(7)(i)(4)(v)]
 - F. Recordkeeping requirements in the applicable regulation. A video record must be used to document the leak survey results. The video record must include a time and date stamp for each monitoring event. A video record can be used to meet the recordkeeping requirements of the applicable regulations if each piece of regulated equipment selected for this work practice can be identified in the video record. The video record must be kept for 5 years. [§60.18(h)(7)(i)(4)(vi)]
 - G. The results of the annual Method 21 screening required in §60.18(h)(7). Records must be kept for all regulated equipment specified in §60.18(h)(1). Records must identify the equipment screened, the screening value measured by Method 21, the time and date of the screening, and calibration information required in the existing applicable regulation. [§60.18(h)(7)(i)(4)(vii)]

Reporting:

1. The permittee shall submit the records of the annual Method 21 screening required in §60.18(h)(7) to the Administrator via e-mail to CCG-AWP@EPA.GOV. [§60.18(h)(7)(i)(5)]
2. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
3. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition Subpart Dc

10 CSR 10-6.070 New Source Performance Regulations

40 CFR part 60 Subpart A, General Provisions

40 CFR part 60 Subpart Dc, Standards of Performance for Small Industrial-Commercial-Institutional
Steam Generating Units

Applicability:

This permit condition applies to EU028 Boiler #1 and EU034 Boiler #2.

Reporting and recordkeeping:

1. The permittee shall submit notification of the date of construction or reconstruction and actual startup, as provided by §60.7 of this part. This notification shall include: [§60.48c(a)]
 - A. The design heat input capacity of the affected facility and identification of fuels to be combusted in the affected facility. [§60.48c(a)(1)]
 - B. If applicable, a copy of any federally enforceable requirement that limits the annual capacity factor for any fuel or mixture of fuels under §60.42c, or §60.43c. [§60.48c(a)(2)]
 - C. The annual capacity factor at which the permittee anticipates operating the affected facility based on all fuels fired and based on each individual fuel fired. [§60.48c(a)(3)]
 - D. Notification if an emerging technology will be used for controlling SO₂ emissions. The Administrator will examine the description of the control device and will determine whether the technology qualifies as an emerging technology. In making this determination, the Administrator may require the permittee to submit additional information concerning the control device. The affected facility is subject to the provisions of §60.42c(a) or (b)(1), unless and until this determination is made by the Administrator. [§60.48c(a)(4)]
2. Except as provided under §60.48c(g)(2) and (g)(3), the permittee shall record and maintain records of the amount of each fuel combusted during each operating day. [§60.48c(g)(1)]
3. As an alternative to meeting the requirements of §60.48c(g)(1), the permittee may elect to record and maintain records of the amount of each fuel combusted during each calendar month. [§60.48c(g)(2)]
4. As an alternative to meeting the requirements of §60.48c(g)(1), the permittee may elect to record and maintain records of the total amount of each steam generating unit fuel delivered to that property during each calendar month. [§60.48c(g)(3)]
5. The permittee subject to a federally enforceable requirement limiting the annual capacity factor for any fuel or mixture of fuels under §60.42c or §60.43c shall calculate the annual capacity factor individually for each fuel combusted. The annual capacity factor is determined on a 12-month rolling average basis with a new annual capacity factor calculated at the end of the calendar month. [§60.48c(h)]
6. All records required under this section shall be maintained by the permittee for a period of five years following the date of such record. [§60.48c(i)]
7. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
8. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition Subpart Kb

10 CSR 10-6.070 New Source Performance Regulations
40 CFR part 60 Subpart A, General Provisions
40 CFR part 60 Subpart Kb, Standards of Performance for Volatile Organic Liquid Storage Vessels
(Including Petroleum Liquid Storage Vessels) for Which Construction, Reconstruction, or
Modification Commenced After July 23, 1984

Applicability:

This permit condition applies to T801, T802, T803, T804, T811, T815, T816, and T817.

Operational Limitations:

1. The permittee shall equip each storage vessel with a fixed roof in combination with an internal floating roof meeting the following specifications: [§60.112b(a)(1)]
 - A. The internal floating roof shall rest or float on the liquid surface (but not necessarily in complete contact with it) inside a storage vessel that has a fixed roof. The internal floating roof shall be floating on the liquid surface at all times, except during initial fill and during those intervals when the storage vessel is completely emptied or subsequently emptied and refilled. When the roof is resting on the leg supports, the process of filling, emptying, or refilling shall be continuous and shall be accomplished as rapidly as possible. [§60.112b(a)(1)(i)]
 - B. Each internal floating roof shall be equipped with one of the following closure devices between the wall of the storage vessel and the edge of the internal floating roof: [§60.112b(a)(1)(ii)]
 - i. A foam- or liquid-filled seal mounted in contact with the liquid (liquid-mounted seal). A liquid-mounted seal means a foam- or liquid-filled seal mounted in contact with the liquid between the wall of the storage vessel and the floating roof continuously around the circumference of the tank. [§60.112b(a)(1)(ii)(A)]
 - ii. Two seals mounted one above the other so that each forms a continuous closure that completely covers the space between the wall of the storage vessel and the edge of the internal floating roof. The lower seal may be vapor-mounted, but both must be continuous. [§60.112b(a)(1)(ii)(B)]
 - iii. A mechanical shoe seal. A mechanical shoe seal is a metal sheet held vertically against the wall of the storage vessel by springs or weighted levers and is connected by braces to the floating roof. A flexible coated fabric (envelope) spans the annular space between the metal sheet and the floating roof. [§60.112b(a)(1)(ii)(C)]
 - C. Each opening in a noncontact internal floating roof except for automatic bleeder vents (vacuum breaker vents) and the rim space vents is to provide a projection below the liquid surface. [§60.112b(a)(1)(iii)]
 - D. Each opening in the internal floating roof except for leg sleeves, automatic bleeder vents, rim space vents, column wells, ladder wells, sample wells, and stub drains is to be equipped with a cover or lid which is to be maintained in a closed position at all times (i.e., no visible gap) except when the device is in actual use. The cover or lid shall be equipped with a gasket. Covers on each access hatch and automatic gauge float well shall be bolted except when they are in use. [§60.112b(a)(1)(iv)]
 - E. Automatic bleeder vents shall be equipped with a gasket and are to be closed at all times when the roof is floating except when the roof is being floated off or is being landed on the roof leg supports. [§60.112b(a)(1)(v)]

- F. Rim space vents shall be equipped with a gasket and are to be set to open only when the internal floating roof is not floating or at the manufacturer's recommended setting. [§60.112b(a)(1)(vi)]
- G. Each penetration of the internal floating roof for the purpose of sampling shall be a sample well. The sample well shall have a slit fabric cover that covers at least 90 percent of the opening. [§60.112b(a)(1)(vii)]
- H. Each penetration of the internal floating roof that allows for passage of a column supporting the fixed roof shall have a flexible fabric sleeve seal or a gasketed sliding cover. [§60.112b(a)(1)(viii)]
- I. Each penetration of the internal floating roof that allows for passage of a ladder shall have a gasketed sliding cover. [§60.112b(a)(1)(ix)]

Monitoring:

- 1. After installing the control equipment required to meet §60.112b(a)(1) (permanently affixed roof and internal floating roof), the permittee shall: [§60.113b(a)]
 - A. Visually inspect the internal floating roof, the primary seal, and the secondary seal (if one is in service), prior to filling the storage vessel with VOL. If there are holes, tears, or other openings in the primary seal, the secondary seal, or the seal fabric or defects in the internal floating roof, or both, the permittee shall repair the items before filling the storage vessel. [§60.113b(a)(1)]
 - B. For vessels equipped with a liquid-mounted or mechanical shoe primary seal, visually inspect the internal floating roof and the primary seal or the secondary seal (if one is in service) through manholes and roof hatches on the fixed roof at least once every 12 months after initial fill. If the internal floating roof is not resting on the surface of the VOL inside the storage vessel, or there is liquid accumulated on the roof, or the seal is detached, or there are holes or tears in the seal fabric, the permittee shall repair the items or empty and remove the storage vessel from service within 45 days. If a failure that is detected during inspections required in §60.113b(a)(2) cannot be repaired within 45 days and if the vessel cannot be emptied within 45 days, a 30-day extension may be requested from the Director in the inspection report required in §60.115b(a)(3). Such a request for an extension must document that alternate storage capacity is unavailable and specify a schedule of actions the company will take that will assure that the control equipment will be repaired or the vessel will be emptied as soon as possible. [§60.113b(a)(2)]
 - C. For vessels equipped with a double-seal system as specified in §60.112b(a)(1)(ii)(B): [§60.113b(a)(3)]
 - i. Visually inspect the vessel as specified in §60.113b(a)(4) at least every 5 years; or [§60.113b(a)(3)(i)]
 - ii. Visually inspect the vessel as specified in §60.113b(a)(2). [§60.113b(a)(3)(ii)]
 - D. Visually inspect the internal floating roof, the primary seal, the secondary seal (if one is in service), gaskets, slotted membranes and sleeve seals (if any) each time the storage vessel is emptied and degassed. If the internal floating roof has defects, the primary seal has holes, tears, or other openings in the seal or the seal fabric, or the secondary seal has holes, tears, or other openings in the seal or the seal fabric, or the gaskets no longer close off the liquid surfaces from the atmosphere, or the slotted membrane has more than 10 percent open area, the permittee shall repair the items as necessary so that none of the conditions specified in §60.113b(a)(4) exist before refilling the storage vessel with VOL. In no event shall inspections conducted in accordance with this provision occur at intervals greater than 10 years in the case of vessels conducting the annual visual inspection as specified in §60.113b(a)(2) and (a)(3)(ii) and at

intervals no greater than 5 years in the case of vessels specified in §60.113b(a)(3)(i).
[§60.113b(a)(4)]

- E. Notify the Director in writing at least 30 days prior to the filling or refilling of each storage vessel for which an inspection is required by §60.113b(a)(1) and (a)(4) to afford the Director the opportunity to have an observer present. If the inspection required by §60.113b(a)(4) is not planned and the permittee could not have known about the inspection 30 days in advance or refilling the tank, the permittee shall notify the Director at least 7 days prior to the refilling of the storage vessel. Notification shall be made by telephone immediately followed by written documentation demonstrating why the inspection was unplanned. Alternatively, this notification including the written documentation may be made in writing and sent by express mail so that it is received by the Director at least 7 days prior to the refilling. [§60.113b(a)(5)]

Alternative means of emission limitation:

1. If, in the Administrator's judgment, an alternative means of emission limitation will achieve a reduction in emissions at least equivalent to the reduction in emissions achieved by any requirement in §60.112b, the Administrator will publish in the Federal Register a notice permitting the use of the alternative means for purposes of compliance with that requirement. [§60.114b(a)]
2. Any notice under §60.114b(a) will be published only after notice and an opportunity for a hearing. [§60.114b(b)]
3. Any person seeking permission under §60.114b shall submit to the Administrator a written application including: [§60.114b(c)]
 - A. An actual emissions test that uses a full-sized or scale-model storage vessel that accurately collects and measures all VOC emissions from a given control device and that accurately simulates wind and accounts for other emission variables such as temperature and barometric pressure. [§60.114b(c)(1)]
 - B. An engineering evaluation that the Administrator determines is an accurate method of determining equivalence. [§60.114b(c)(2)]
4. The Administrator may condition the permission on requirements that may be necessary to ensure operation and maintenance to achieve the same emissions reduction as specified in §60.112b. [§60.114b(d)]

Reporting and Recordkeeping:

1. The permittee shall keep copies of all reports and records required by §60.115b for at least five years.
2. After installing control equipment in accordance with §60.112b(a)(1) (fixed roof and internal floating roof), the permittee shall meet the following requirements. [§60.115b(a)]
 - A. Furnish the Director with a report that describes the control equipment and certifies that the control equipment meets the specifications of §60.112b(a)(1) and §60.113b(a)(1). This report shall be an attachment to the notification required by §60.7(a)(3). [§60.115b(a)(1)]
 - B. Keep a record of each inspection performed as required by §60.113b (a)(1), (a)(2), (a)(3), and (a)(4). Each record shall identify the storage vessel on which the inspection was performed and shall contain the date the vessel was inspected and the observed condition of each component of the control equipment (seals, internal floating roof, and fittings). [§60.115b(a)(2)]
 - C. If any of the conditions described in §60.113b(a)(2) are detected during the annual visual inspection required by §60.113b(a)(2), a report shall be furnished to the Director within 30 days of the inspection. Each report shall identify the storage vessel, the nature of the defects, and the

date the storage vessel was emptied or the nature of and date the repair was made.

[§60.115b(a)(3)]

- D. After each inspection required by §60.113b(a)(3) that finds holes or tears in the seal or seal fabric, or defects in the internal floating roof, or other control equipment defects listed in §60.113b(a)(3)(ii), a report shall be furnished to the Director within 30 days of the inspection. The report shall identify the storage vessel and the reason it did not meet the specifications of §61.112b(a)(1) or §60.113b(a)(3) and list each repair made. [§60.115b(a)(4)]
3. The permittee shall keep copies of all records required by §60.116b, except for the record required by §60.116b(b), for at least 5 years. The record required by §60.116b(b) will be kept for the life of the source. [§60.116b(a)]
 4. The permittee shall keep readily accessible records showing the dimension of the storage vessel and an analysis showing the capacity of the storage vessel. [§60.116b(b)]
 5. Except as provided in §60.116b(f) and (g), the permittee of each storage vessel either with a design capacity greater than or equal to 151 m³ storing a liquid with a maximum true vapor pressure greater than or equal to 3.5 kPa or with a design capacity greater than or equal to 75 m³ but less than 151 m³ storing a liquid with a maximum true vapor pressure greater than or equal to 15.0 kPa shall maintain a record of the VOL stored, the period of storage, and the maximum true vapor pressure of that VOL during the respective storage period. [§60.116b(c)]
 6. Except as provided in §60.116b(g), the permittee of each storage vessel either with a design capacity greater than or equal to 151 m³ storing a liquid with a maximum true vapor pressure that is normally less than 5.2 kPa or with a design capacity greater than or equal to 75 m³ but less than 151 m³ storing a liquid with a maximum true vapor pressure that is normally less than 27.6 kPa shall notify the Director within 30 days when the maximum true vapor pressure of the liquid exceeds the respective maximum true vapor pressure values for each volume range. [§60.116b(d)]
 7. Available data on the storage temperature may be used to determine the maximum true vapor pressure as determined below. [§60.116b(e)]
 - A. For vessels operated above or below ambient temperatures, the maximum true vapor pressure is calculated based upon the highest expected calendar-month average of the storage temperature. For vessels operated at ambient temperatures, the maximum true vapor pressure is calculated based upon the maximum local monthly average ambient temperature as reported by the National Weather Service. [§60.116b(e)(1)]
 - B. The vapor pressure: [§60.116b(e)(3)]
 - i. May be obtained from standard reference texts, or [§60.116b(e)(3)(i)]
 - ii. Determined by ASTM D2879-83, 96, or 97 (incorporated by reference—see §60.17); or [§60.116b(e)(3)(ii)]
 - iii. Measured by an appropriate method approved by the Administrator; or [§60.116b(e)(3)(iii)]
 - iv. Calculated by an appropriate method approved by the Administrator. [§60.116b(e)(3)(iv)]
 8. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
 9. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition 6.220¹⁴

10 CSR 10-6.220 Restriction of Emission of Visible Air Contaminants

Applicability:

This permit condition applies to the following:

1. All emission units controlled by Baghouse #1: EU001, EU002, EU041, EU004, EU005, EU006, EU007, EU042, EU003, EU043, EU044, EU008, EU011, EU009, and EU010.
2. All emission units controlled by Baghouse #2: EU013
3. Emission Unit EU027, and
4. The following natural gas fired combustion units: EU028, EU034, EU039, EU026, EU038.

Emission Limitation:

1. The permittee shall not cause or permit to be discharged into the atmosphere from these emission units any visible emissions with an opacity greater than 20 percent for any continuous six-minute period. [10 CSR 10-6.220(3)(A)1]
2. Exception: The permittee may discharge into the atmosphere from any emission unit visible emissions with an opacity up to 60 percent for one continuous six-minute period in any 60 minutes. [10 CSR 10-6.220(3)(A)2]
3. Failure to demonstrate compliance with 10 CSR 10-6.220(3)(A) solely because of the presences of uncombined water shall not be a violation. [10 CSR 10-6.220(3)(B)]

Monitoring:

1. Monitoring schedule:
 - A. The permittee shall conduct weekly observations for a minimum of eight consecutive weeks after permit issuance. Should no violation of this regulation be observed during this period then:
 - i. The permittee shall conduct observations once every two weeks for a period of eight weeks. If a violation is noted, the permittee shall revert to weekly monitoring. Should no violation of this regulation be observed during this period then:
 - a. The permittee shall conduct observations once per month. If a violation is noted, the permittee shall revert to weekly monitoring.
2. If the permittee reverts to weekly monitoring at any time, the monitoring schedule shall progress in an identical manner from the initial monitoring schedule.
3. Observations are only required when the emission units are operating and when the weather conditions allow.
4. Issuance of a new, amended, or modified operating permit does not restart the monitoring schedule.
5. The permittee shall conduct visible emissions observation on these emission units using the procedures contained in U.S. EPA Test Method 22. Each Method 22 observation shall be conducted for a minimum of six-minutes. If no visible emissions are observed from the emission unit using Method 22, then no Method 9 is required for the emission unit.

¹⁴ The version of this regulation contained in Missouri's State Implementation Plan (SIP) is different than the version currently contained in Missouri's Code of State Regulations (CSR). The CSR version exempts natural gas fired units (6.220(1)(L)). The current SIP version does not contain this exemption. Emission limitations and compliance methods are the same for both versions, therefore the requirements appear in one permit condition. When the current CSR version is adopted into the SIP, this permit condition will not apply to the units that meet the noted exemptions. A permit modification is not required to reflect this change.

6. For emission units with visible emissions, the permittee shall have a certified Method 9 observer conduct a U.S. EPA Test Method 9 opacity observation. The permittee may choose to forego Method 22 observations and instead begin with a Method 9 opacity observation. The certified Method 9 observer shall conduct each Method 9 opacity observation for a minimum of 30-minutes.

Recordkeeping:

1. The permittee shall maintain records of all observation results for each emission unit using Attachment Method 9 and Attachment Method 22, or equivalents, to demonstrate compliance.
2. The permittee shall maintain all records required by this permit for not less than five years and shall make them available immediately to any Missouri Department of Natural Resources' personnel upon request.

Reporting:

1. The permittee shall report any exceedance of the limitations no later than ten days after an exceedance of the emission limitation.
2. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification report required by Section IV of this permit.
3. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition 6.400

10 CSR 10-6.400, Restriction of Emission of Particulate Matter From Industrial Processes

Applicability:

This permit condition applies to the following:

1. All emission units controlled by Baghouse #1: EU001, EU002, EU041, EU004, EU005, EU006, EU007, EU042, EU003, EU043, EU044, EU008, EU011, EU009, and EU010.
2. All emission units controlled by Baghouse #2: EU013.
3. Emission Unit EU026.
4. Emission Unit EU027.

Emission Limitation:

The permittee shall not emit particulate matter in excess of the values shown in the following table in any one hour period: [6.440(3)(A)1.]

Table 16: 10 CSR 10-6.400 Emission Limitations

Stack ID	Control Device	Emission Unit #	PM Emission Limitation (lb/hr)
SV001	Baghouse #1 (C001)	EU001, EU002, EU041, EU004, EU005, EU006, EU007, EU042, EU003, EU043, EU044, EU008, EU011, EU009, and EU010	57.67
SV003	Baghouse #2 (C002)	EU013	62.22
SV012 (RTO) or SV006 (RTO bypass)	RTO (C008, EU038) or RTO Bypass	EU026	15.67
		EU027	

Operational Limitations:

The permittee shall comply with the following operational limitations:

Table 17: 10 CSR 10-6.400 Operational Limitations

Stack ID	Control Device	Emission Unit #	Monitoring Requirements
SV001	Baghouse #1 (C001)	EU001, EU002, EU041, EU004, EU005, EU006, EU007, EU042, EU003, EU043, EU044, EU008, EU011, EU009, and EU010, EU012	The permittee shall comply with the requirements of Permit Condition 072008-001. No additional operational limitations are required to demonstrate compliance with Permit Condition 6.400.
SV003	Baghouse #2 (C002)	EU013	
SV012 (RTO) or SV006 (RTO bypass)	RTO (C008, EU038) or RTO Bypass	EU026	1. The permittee shall operate the Multiclones at all times the DDGS Dryer is in operation. 2. The permittee shall equip the Multiclones with an air flow monitor to indicate the presence of air flow through the Multiclones. 3. The permittee shall equip the Multiclones with gauges or meters, which indicate the pressure drop across the Multiclones. These gauges or meters shall be located such that Department of Natural Resources' employees may easily observe them. 4. The permittee shall operate and maintain the Multiclones such that the pressure drop remains within the manufacturer's specifications.
SV007	None	EU027	The permittee shall comply with the requirements of Permit Condition 072008-001. No additional operational limitations are required to demonstrate compliance with Permit Condition 6.400.

Monitoring and Recordkeeping:

The permittee shall comply with the following requirements:

Table 18: 10 CSR 10-6.400 Monitoring and Recordkeeping Requirements

Stack ID	Control Device	Emission Unit #	Monitoring and Recordkeeping Requirements
SV001	Baghouse #1 (C001)	EU001, EU002, EU041, EU004, EU005, EU006, EU007, EU042, EU003, EU043, EU044, EU008, EU011, EU009, EU010, and EU012	1. The permittee shall monitor and record the particulate matter emissions on a pounds per hour basis. 2. The permittee shall comply with the monitoring and recordkeeping requirements in Permit Condition 072008-001.
SV003	Baghouse #2 (C002)	EU013	

Stack ID	Control Device	Emission Unit #	Monitoring and Recordkeeping Requirements
SV012 (RTO) or SV006 (RTO bypass)	RTO (C008, EU038) or RTO Bypass	EU026	1. The permittee shall monitor and record the air flow rate through the Multiclones once per working day. 2. The permittee shall monitor and record the pressure drop across the Multiclones once per working day. 3. The permittee shall keep the manufacturer's specifications on site. 4. The permittee shall inspect the solids discharge valve(s) for proper operation on a weekly basis. 5. The permittee shall inspect the structural components, including the Mutliclone ductwork and hoods for leaks and component failures on a monthly basis. 6. The permittee shall check for leaks in the system to ensure airflow from the dirty side does not infiltrate the clean side on a once every consecutive twelve month basis. 7. The permittee shall verify that the inlet and outlet ductwork on the Multiclones is in proper operating condition on a once every consecutive twelve month basis. 8. The permittee shall check the barrel and collecting tube(s) for deposit and/or excess wear on a once every consecutive twelve month basis. The permittee shall clean and repair the barrel and collecting tube(s) as needed. 9. The permittee shall maintain an operating and maintenance log for the Multiclones which shall include the following: A. A record of regular inspection schedule of Items 1-8 above, the date and results of all inspections, including any actions or maintenance activities that result from the inspections. B. Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions C. Maintenance activities, with inspection schedule, repair actions, and replacements, etc. D. The permittee shall use Attachment O/M Log, or equivalent, to demonstrate compliance.

Reporting:

1. The permittee shall report any exceedance of the limitations no later than ten days after an exceedance of the emission limitation.
2. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
3. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition 6.260¹⁵

10 CSR 10-6.260, Restriction of Emission of Sulfur Compounds

Applicability:

This permit condition applies to the following:

1. Emission Units controlled by the Fermentation Scrubber (C003): EU014, EU015, EU016, EU051, EU052, and EU0017.
2. Emission Units controlled by the Distillation Scrubber (C004): EU023, EU024, EU018, EU019, EU025, EU020, EU021, EU035, EU036, EU037, EU040, and EU022.
3. Centrifuge #4 (EU050) and Centrifuge #5 (EU053).

Emission Limitation:

The permittee shall not cause or permit the emission into the atmosphere gases containing more than five hundred parts per million by volume (500 ppmv) of sulfur dioxide or more than thirty-five milligrams per cubic meter (35 mg/m³) of sulfuric acid or sulfur trioxide or any combination of those gases averaged on any consecutive three hour time period from any of the emission units or control devices listed above.

Operational Limitation:

The permittee shall comply with Permit Condition 072008-001. No additional operational limitations are required to demonstrate compliance with Permit Condition 6.260.

Monitoring/Recordkeeping:

1. The permittee shall establish emission factors to quantify the emissions of sulfur dioxide, or sulfuric acid and sulfur trioxide, using the procedures of Permit Condition 072008-001. The permittee may petition the Air Pollution Control Program Compliance and Enforcement Section to use alternative procedures. The permittee must obtain concurrence from the Air Pollution Control Program Compliance and Enforcement Section prior to using any proposed alternative procedures.

¹⁵ As effective September 30, 2012. This regulation was rescinded by the State of Missouri on November 30, 2015. The regulation remains in this operating permit as it is contained in Missouri's SIP and remains an applicable federal requirement. This is a federal only requirement. This permit condition will no longer be applicable when EPA takes final action to incorporate 10 CSR 10-6.261 in Missouri's SIP in place of 10 CSR 10-6.260. No action is required on the part of the permittee to remove this permit condition from this operating permit upon incorporation of 10 CSR 10-6.261 into Missouri's SIP.

2. The permittee shall monitor and record the emissions of sulfur dioxide, or sulfuric acid and sulfur trioxide, in the same units as the emission limitations averaged on any consecutive three hour time period, to demonstrate compliance.

Reporting:

1. The permittee shall report any exceedance of any of the terms imposed by this permit condition, or any malfunction which could cause an exceedance of any of the terms imposed by this permit condition, no later than ten days after the exceedance or event causing the exceedance. The permittee shall submit these reports to EPA Region VII, 11201 Renner Blvd., Lenexa, KS 66219.
2. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit. The permittee shall submit these reports to both the EPA Region VII and Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition 6.261¹⁶

10 CSR 10-6.261, Control of Sulfur Dioxide Emissions

Applicability:

This permit condition applies to the following:

1. Emission Units EU028 Boiler #1 and EU034 Boiler #2,
2. Emission Unit EU039 Open Flare,
3. Emission Unit EU026 DDGS Dryer,
4. Emission Unit EU038, Regenerative Thermal Oxidizer, and
5. Emission Unit EU045, Emergency Generator

Operational Limitation:

The permittee shall comply with the record keeping requirements of 6.261(4). [6.261(1)(A)]

Monitoring/Recordkeeping:

1. The permittee shall maintain a record of data, calculations, results, records, and reports from all fuel deliveries. [6.261(4)(A)3.]
2. The permittee must maintain the fuel supplier certification information to certify all fuel deliveries. Bills of lading and/or other fuel delivery documentation containing the following information for all fuel purchases or deliveries are deemed acceptable: [6.261(4)(C)]
 - A. The name, address, and contact information of the fuel supplier; [6.261(4)(C)1.]
 - B. The type of fuel; [6.261(4)(C)2.]
 - C. The sulfur content or maximum sulfur content expressed in percent sulfur by weight or in ppm sulfur; and [6.261(4)(C)3.]
 - D. The heating value of the fuel. [6.261(4)(C)4.]

¹⁶ As effective March 30, 2019. This regulation has not yet been adopted into Missouri's SIP; therefore, this regulation is a state only requirement. Upon adoption into Missouri's SIP this regulation will be both a state and federal requirement.

3. The permittee must retain all required reports and records on-site for a minimum of five years and make available within five business days upon written or electronic request by the Director.
[6.261(4)(F)]
4. The permittee must furnish the Director all data necessary to determine compliance status.
[6.261(4)(G)]

Reporting:

1. The permittee shall report any deviations from the monitoring, recordkeeping, and reporting requirements of this permit condition in the annual compliance certification required by Section IV of this permit.
2. All reports and certifications shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

Permit Condition Compliance Plan

10 CSR 10-6.065, Operating Permits

1. At the time of operating permit issuance, compliance could not be confirmed for the following:
 - A. Plant wide and individual unit specific actual emissions less than the limitations contained in Permit Condition 072008-001.
 - B. The installation and operation of the following activities prior to determination of construction permit applicability:
 - i. Tote loading operations
 - ii. Ethanol receiving operations from outside suppliers
 - iii. Syrup loadout and shipping operations
 - iv. Corn oil extraction from the evaporator, corn oil storage, loadout, and shipping operations
 - v. Reboilers
 - vi. Evaporators, planned for 2019 construction
 - vii. Hot Slurry Tank, T204
 - C. Changes in the method of operation of the following activities:
 - i. Operation of the Fermentation Scrubber (C003) without the addition of sodium bisulfite as required in Permit Condition 072008-001.
 - ii. Disconnection of the Flare control device from Rail Loadout activities
 - iii. Installation of Boiler #2 without Low NOx Burners
2. Within 60 days of issuance of this operating permit, the permittee shall submit a compliance plan to address all concerns listed above. The compliance plan shall contain all of the following:
 - A. A description of the compliance status of the installation with respect to all applicable requirements;
 - B. A description as follows:
 - i. For applicable requirements with which the installation is in compliance, a statement that the installation will continue to comply with these requirements;
 - ii. For applicable requirements that will become effective during the permit term, a statement that the installation will comply with these requirements on a timely basis; and
 - iii. For any applicable requirements with which the installation is not in compliance at the time of permit issuance, a narrative description of how the installation will achieve compliance with these requirements.

C. A compliance schedule as follows:

- i. For applicable requirements with which the installation is in compliance, a statement that the installation will continue to comply with these requirements;
- ii. For applicable requirements that will become effective during the permit term, a statement that the installation will comply with these requirements on a timely basis. A statement that the installation will comply in a timely manner with applicable requirements that become effective during the permit term shall satisfy this provision, unless a more detailed schedule is expressly required by the applicable requirement; and
- iii. A schedule of compliance for all applicable requirements with which the installation is not in compliance at the time of permit issuance, including a schedule of remedial measures and an enforceable sequence of actions, with milestones, leading to compliance. This compliance schedule shall resemble and be equivalent in stringency to that contained in any judicial consent decree or administrative order to which the installation is subject.

D. For installations required to have a schedule of compliance, a schedule for the submission of certified progress reports no less frequently than every six months.

3. The permittee shall satisfy any additional requests in the timeframes prescribed by the Air Pollution Control Program Permits Section or the Air Pollution Control Program Compliance and Enforcement Section.
4. The permittee shall submit all information required by this permit condition to the Air Pollution Control Program's Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.

III. Core Permit Requirements

The installation shall comply with each of the following requirements. Consult the appropriate sections in the Code of Federal Regulations (CFR), Code of State Regulations (CSR), and local ordinances for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect on the date of permit issuance. The following are only excerpts from the regulation or code, and are provided for summary purposes only.

10 CSR 10-6.045 Open Burning Requirements

- 1) General Provisions. The open burning of tires, petroleum-based products, asbestos containing materials, and trade waste is prohibited, except as allowed below. Nothing in this rule may be construed as to allow open burning which causes or constitutes a public health hazard, nuisance, a hazard to vehicular or air traffic, nor which violates any other rule or statute.
- 2) Certain types of materials may be open burned provided an open burning permit is obtained from the director. The permit will specify the conditions and provisions of all open burning. The permit may be revoked if the owner or operator fails to comply with the conditions or any provisions of the permit.

10 CSR 10-6.050 Start-up, Shutdown and Malfunction Conditions

- 1) In the event of a malfunction, which results in excess emissions that exceed one hour, the permittee shall submit to the director within two business days, in writing, the following information:
 - a) Name and location of installation;
 - b) Name and telephone number of person responsible for the installation;
 - c) Name of the person who first discovered the malfunction and precise time and date that the malfunction was discovered.
 - d) Identity of the equipment causing the excess emissions;
 - e) Time and duration of the period of excess emissions;
 - f) Cause of the excess emissions;
 - g) Air pollutants involved;
 - h) Estimate of the magnitude of the excess emissions expressed in the units of the applicable requirement and the operating data and calculations used in estimating the magnitude;
 - i) Measures taken to mitigate the extent and duration of the excess emissions; and
 - j) Measures taken to remedy the situation that caused the excess emissions and the measures taken or planned to prevent the recurrence of these situations.
- 2) The permittee shall submit the paragraph 1 information to the director in writing at least ten days prior to any maintenance, start-up or shutdown activity which is expected to cause an excessive release of emissions that exceed one hour. If notice of the event cannot be given ten days prior to the planned occurrence, notice shall be given as soon as practicable prior to the activity.
- 3) Upon receipt of a notice of excess emissions issued by an agency holding a certificate of authority under section 643.140, RSMo, the permittee may provide information showing that the excess emissions were the consequence of a malfunction, start-up or shutdown. The information, at a minimum, should be the paragraph 1 list and shall be submitted not later than 15 days after receipt of the notice of excess emissions. Based upon information submitted by the permittee or any other pertinent information available, the director or the commission shall make a determination whether the excess emissions constitute a malfunction, start-up or shutdown and whether the nature, extent

and duration of the excess emissions warrant enforcement action under section 643.080 or 643.151, RSMo.

- 4) Nothing in this rule shall be construed to limit the authority of the director or commission to take appropriate action, under sections 643.080, 643.090 and 643.151, RSMo to enforce the provisions of the Air Conservation Law and the corresponding rule.
- 5) Compliance with this rule does not automatically absolve the permittee of liability for the excess emissions reported.

10 CSR 10-6.060 Construction Permits Required

The permittee shall not commence construction, modification, or major modification of any installation subject to this rule, begin operation after that construction, modification, or major modification, or begin operation of any installation which has been shut down longer than five years without first obtaining a permit from the permitting authority.

10 CSR 10-6.065 Operating Permits

The permittee shall file a complete application for renewal of this operating permit at least six months before the date of permit expiration. In no event shall this time be greater than eighteen months. The permittee shall retain the most current operating permit issued to this installation on-site. The permittee shall make such permit available within a reasonable period of time to any Missouri Department of Natural Resources personnel upon request.

10 CSR 10-6.080 Emission Standards for Hazardous Air Pollutants and 40 CFR Part 61 Subpart M National Emission Standard for Asbestos

The permittee shall follow the procedures and requirements of 40 CFR Part 61, Subpart M for any activities occurring at this installation which would be subject to provisions for 40 CFR Part 61, Subpart M, National Emission Standard for Asbestos.

10 CSR 10-6.110 Reporting of Emission Data, Emission Fees and Process Information

- 1) The permittee shall submit a Full Emissions Report either electronically via MoEIS, which requires Form 1.0 signed by an authorized company representative, or on Emission Inventory Questionnaire (EIQ) paper forms on the frequency specified in this rule and in accordance with the requirements outlined in this rule. Alternate methods of reporting the emissions, such as spreadsheet file, can be submitted for approval by the director.
- 2) Public Availability of Emission Data and Process Information. Any information obtained pursuant to the rule(s) of the Missouri Air Conservation Commission that would not be entitled to confidential treatment under 10 CSR 10-6.210 shall be made available to any member of the public upon request.
- 3) The permittee shall submit full EIQ's per the schedule in the rule. In the interim years the installation may submit a Reduced Reporting Form; however, if the installation's emissions increase or decrease by more than five tons when compared to their last submitted full EIQ, the installation shall submit a full EIQ rather than a Reduced Reporting Form.
- 4) In addition to the EIQ submittal schedule outlined above, any permit issued under 10 CSR 10-6.060 section (5) triggers a requirement that a full EIQ be submitted in the first full calendar year after the permitted equipment initially operates.

10 CSR 10-6.130 Controlling Emissions During Episodes of High Air Pollution Potential

This rule specifies the conditions that establish an air pollution alert (yellow/orange/red/purple), or emergency (maroon) and the associated procedures and emission reduction objectives for dealing with each. The permittee shall submit an appropriate emergency plan if required by the director.

10 CSR 10-6.150 Circumvention

The permittee shall not cause or permit the installation or use of any device or any other means which, without resulting in reduction in the total amount of air contaminant emitted, conceals or dilutes an emission or air contaminant which violates a rule of the Missouri Air Conservation Commission.

10 CSR 10-6.165 Restriction of Emission of Odors

This is a State Only permit requirement.

No person may cause, permit or allow the emission of odorous matter in concentrations and frequencies or for durations that odor can be perceived when one volume of odorous air is diluted with seven volumes of odor-free air for two separate trials not less than 15 minutes apart within the period of one hour.

10 CSR 10-6.170 Restriction of Particulate Matter to the Ambient Air Beyond the Premises of Origin

Emission Limitation:

- 1) The permittee shall not cause or allow to occur any handling, transporting or storing of any material; construction, repair, cleaning or demolition of a building or its appurtenances; construction or use of a road, driveway or open area; or operation of a commercial or industrial installation without applying reasonable measures as may be required to prevent, or in a manner which allows or may allow, fugitive particulate matter emissions to go beyond the premises of origin in quantities that the particulate matter may be found on surfaces beyond the property line of origin. The nature or origin of the particulate matter shall be determined to a reasonable degree of certainty by a technique proven to be accurate and approved by the director.
- 2) The permittee shall not cause nor allow to occur any fugitive particulate matter emissions to remain visible in the ambient air beyond the property line of origin.
- 3) Should it be determined that noncompliance has occurred, the director may require reasonable control measures as may be necessary. These measures may include, but are not limited to, the following:
 - a) Revision of procedures involving construction, repair, cleaning and demolition of buildings and their appurtenances that produce particulate matter emissions;
 - b) Paving or frequent cleaning of roads, driveways and parking lots;
 - c) Application of dust-free surfaces;
 - d) Application of water; and
 - e) Planting and maintenance of vegetative ground cover.

Monitoring:

The permittee shall conduct inspections of its facilities sufficient to determine compliance with this regulation. If the permittee discovers a violation, the permittee shall undertake corrective action to eliminate the violation.

The permittee shall maintain the following monitoring schedule:

- 1) The permittee shall conduct weekly observations for a minimum of eight (8) consecutive weeks after permit issuance.
- 2) Should no violation of this regulation be observed during this period then-
 - a) The permittee may observe once every two (2) weeks for a period of eight (8) weeks.
 - b) If a violation is noted, monitoring reverts to weekly.
 - c) Should no violation of this regulation be observed during this period then-
 - i) The permittee may observe once per month.
 - ii) If a violation is noted, monitoring reverts to weekly.
- 3) If the permittee reverts to weekly monitoring at any time, monitoring frequency will progress in an identical manner to the initial monitoring frequency.
- 4) Issuance of a new, amended, or modified operating permit does not restart the monitoring schedule.

Recordkeeping:

The permittee shall document all readings on Attachment 6.170, or its equivalent, noting the following:

- 1) Whether air emissions (except water vapor) remain visible in the ambient air beyond the property line of origin.
- 2) Whether equipment malfunctions contributed to an exceedance.
- 3) Any violations and any corrective actions undertaken to correct the violation.

10 CSR 10-6.180 Measurement of Emissions of Air Contaminants

- 1) The director may require any person responsible for the source of emission of air contaminants to make or have made tests to determine the quantity or nature, or both, of emission of air contaminants from the source. The director may specify testing methods to be used in accordance with good professional practice. The director may observe the testing. All tests shall be performed by qualified personnel.
- 2) The director may conduct tests of emissions of air contaminants from any source. Upon request of the director, the person responsible for the source to be tested shall provide necessary ports in stacks or ducts and other safe and proper sampling and testing facilities, exclusive of instruments and sensing devices as may be necessary for proper determination of the emission of air contaminants.
- 3) The director shall be given a copy of the test results in writing and signed by the person responsible for the tests.

10 CSR 10-6.250 Asbestos Abatement Projects

Certification, Accreditation, and Business Exemption Requirements

This is a State Only permit requirement.

The permittee shall conduct all asbestos abatement projects within the procedures established for certification and accreditation by 10 CSR 10-6.250. This rule requires individuals who work in asbestos abatement projects to be certified by the Missouri Department of Natural Resources Air Pollution Control Program. This rule requires training providers who offer training for asbestos abatement occupations to be accredited by the Missouri Department of Natural Resources Air Pollution Control Program. This rule requires persons who hold exemption status from certain requirements of this rule to allow the department to monitor training provided to employees.

10 CSR 10-6.280 Compliance Monitoring Usage

- 1) The permittee is not prohibited from using the following in addition to any specified compliance methods for the purpose of submission of compliance certificates:

- a) Monitoring methods outlined in 40 CFR Part 64;
 - b) Monitoring method(s) approved for the permittee pursuant to 10 CSR 10-6.065, "Operating Permits", and incorporated into an operating permit; and
 - c) Any other monitoring methods approved by the director.
- 2) Any credible evidence may be used for the purpose of establishing whether a permittee has violated or is in violation of any such plan or other applicable requirement. Information from the use of the following methods is presumptively credible evidence of whether a violation has occurred at an installation:
- a) Monitoring methods outlined in 40 CFR Part 64;
 - b) A monitoring method approved for the permittee pursuant to 10 CSR 10-6.065, "Operating Permits", and incorporated into an operating permit; and
 - c) Compliance test methods specified in the rule cited as the authority for the emission limitations.
- 3) The following testing, monitoring or information gathering methods are presumptively credible testing, monitoring, or information gathering methods:
- a) Applicable monitoring or testing methods, cited in:
 - i) 10 CSR 10-6.030, "Sampling Methods for Air Pollution Sources";
 - ii) 10 CSR 10-6.040, "Reference Methods";
 - iii) 10 CSR 10-6.070, "New Source Performance Standards";
 - iv) 10 CSR 10-6.080, "Emission Standards for Hazardous Air Pollutants"; or
 - b) Other testing, monitoring, or information gathering methods, if approved by the director, that produce information comparable to that produced by any method listed above.

40 CFR Part 82 Protection of Stratospheric Ozone (Title VI)

- 1) The permittee shall comply with the standards for labeling of products using ozone-depleting substances pursuant to 40 CFR Part 82, Subpart E:
- a) All containers in which a class I or class II substance is stored or transported, all products containing a class I substance, and all products directly manufactured with a class I substance must bear the required warning statement if it is being introduced into interstate commerce pursuant to 40 CFR §82.106.
 - b) The placement of the required warning statement must comply with the requirements of 40 CFR §82.108.
 - c) The form of the label bearing the required warning statement must comply with the requirements of 40 CFR §82.110.
 - d) No person may modify, remove, or interfere with the required warning statement except as described in 40 CFR §82.112.
- 2) The permittee shall comply with the standards for recycling and emissions reduction pursuant to 40 CFR Part 82, Subpart F, except as provided for motor vehicle air conditioners (MVACs) in Subpart B of 40 CFR Part 82:
- a) Persons opening appliances for maintenance, service, repair, or disposal must comply with the required practices described in 40 CFR §82.156.
 - b) Equipment used during the maintenance, service, repair, or disposal of appliances must comply with the standards for recycling and recovery equipment described in 40 CFR §82.158.
 - c) Persons performing maintenance, service, repair, or disposal of appliances must be certified by an approved technician certification program pursuant to 40 CFR §82.161.

- d) Persons disposing of small appliances, MVACs, and MVAC-like appliances must comply with the record keeping requirements of 40 CFR §82.166. ("MVAC-like" appliance as defined at 40 CFR §82.152).
 - e) Persons owning commercial or industrial process refrigeration equipment must comply with the leak repair requirements pursuant to 40 CFR §82.156.
 - f) Owners/operators of appliances normally containing 50 or more pounds of refrigerant must keep records of refrigerant purchased and added to such appliances pursuant to 40 CFR §82.166.
- 3) If the permittee manufactures, transforms, imports, or exports a class I or class II substance, the permittee is subject to all the requirements as specified in 40 CFR part 82, Subpart A, Production and Consumption Controls.
- 4) If the permittee performs a service on motor (fleet) vehicles when this service involves ozone-depleting substance refrigerant (or regulated substitute substance) in the motor vehicle air conditioner (MVAC), the permittee is subject to all the applicable requirements contained in 40 CFR part 82, Subpart B, Servicing of Motor Vehicle Air Conditioners. The term "motor vehicle" as used in Subpart B does not include a vehicle in which final assembly of the vehicle has not been completed. The term "MVAC" as used in Subpart B does not include the air-tight sealed refrigeration system used as refrigerated cargo, or system used on passenger buses using HCFC-22 refrigerant.
- 5) The permittee shall be allowed to switch from any ozone-depleting substance to any alternative that is listed in the Significant New Alternatives Program (SNAP) promulgated pursuant to 40 CFR part 82, Subpart G, Significant New Alternatives Policy Program. *Federal Only - 40 CFR Part 82.*

IV. General Permit Requirements

The installation shall comply with each of the following requirements. Consult the appropriate sections in the Code of Federal Regulations (CFR) and Code of State Regulations (CSR) for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect as of the date that this permit is issued.

Permit Duration

10 CSR 10-6.065, §(4)(C)1, §(5)(C)1.B, §(4)(E)2.C

This permit is issued for a term of five years, commencing on the date of issuance. This permit will expire at the end of this period unless renewed. If a timely and complete application for a permit renewal is submitted, but the Air Pollution Control Program fails to take final action to issue or deny the renewal permit before the end of the term of this permit, this permit shall not expire until the renewal permit is issued or denied.

General Record Keeping and Reporting Requirements

10 CSR 10-6.065, §(4)(C)1 and §(5)(C)1.C

1) Record Keeping

- a) All required monitoring data and support information shall be retained for a period of at least five years from the date of the monitoring sample, measurement, report or application.
- b) Copies of all current operating and construction permits issued to this installation shall be kept on-site for as long as the permits are in effect. Copies of these permits shall be made available within a reasonable period of time to any Missouri Department of Natural Resources' personnel upon request.

2) Reporting

- a) All reports shall be submitted to the Air Pollution Control Program, Compliance and Enforcement Section, P. O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov.
- b) The permittee shall submit a report of all required monitoring by:
 - i) April 1st for monitoring which covers the January through December time period.
 - ii) Exception. Monitoring requirements which require reporting more frequently than annually shall report no later than 30 days after the end of the calendar quarter in which the measurements were taken.
- c) Each report shall identify any deviations from emission limitations, monitoring, record keeping, reporting, or any other requirements of the permit.
- d) Submit supplemental reports as required or as needed. All reports of deviations shall identify the cause or probable cause of the deviations and any corrective actions or preventative measures taken.
 - i) Notice of any deviation resulting from an emergency (or upset) condition as defined in paragraph (5)(C)7 of 10 CSR 10-6.065 (Emergency Provisions) shall be submitted to the permitting authority either verbally or in writing within two working days after the date on which the emission limitation is exceeded due to the emergency, if the permittee wishes to assert an affirmative defense. The affirmative defense of emergency shall be demonstrated through properly signed, contemporaneous operating logs, or other relevant evidence that indicate an emergency occurred and the permittee can identify the cause(s) of the emergency. The permitted installation must show that it was operated properly at the time and that during

the period of the emergency the permittee took all reasonable steps to minimize levels of emissions that exceeded the emission standards or requirements in the permit. The notice must contain a description of the emergency, the steps taken to mitigate emissions, and the corrective actions taken.

- ii) Any deviation that poses an imminent and substantial danger to public health, safety or the environment shall be reported as soon as practicable.
- iii) Any other deviations identified in the permit as requiring more frequent reporting than the permittee's annual report shall be reported on the schedule specified in this permit.
- e) Every report submitted shall be certified by the responsible official, except that, if a report of a deviation must be submitted within ten days after the deviation, the report may be submitted without a certification if the report is resubmitted with an appropriate certification within ten days after that, together with any corrected or supplemental information required concerning the deviation.
- f) The permittee may request confidential treatment of information submitted in any report of deviation.

Risk Management Plan Under Section 112(r)

10 CSR 10-6.065 §(4)(C)1 and §(5)(C)1.D

If the installation is required to develop and register a risk management plan pursuant to Section 112(r) of the Act, the permittee will verify that it has complied with the requirement to register the plan.

General Requirements

10 CSR 10-6.065(4)(C)1.A

- 1) The permittee must comply with all of the terms and conditions of this permit. Any noncompliance with a permit condition constitutes a violation and is grounds for enforcement action, permit termination, permit revocation and re-issuance, permit modification or denial of a permit renewal application.
- 2) The permittee may not use as a defense in an enforcement action that it would have been necessary for the permittee to halt or reduce the permitted activity in order to maintain compliance with the conditions of the permit
- 3) The permit may be modified, revoked, reopened, reissued or terminated for cause. Except as provided for minor permit modifications, the filing of an application or request for a permit modification, revocation and reissuance, or termination, or the filing of a notification of planned changes or anticipated noncompliance, does not stay any permit condition.
- 4) This permit does not convey any property rights of any sort, nor grant any exclusive privilege.
- 5) The permittee shall furnish to the Air Pollution Control Program, upon receipt of a written request and within a reasonable time, any information that the Air Pollution Control Program reasonably may require to determine whether cause exists for modifying, reopening, reissuing or revoking the permit or to determine compliance with the permit. Upon request, the permittee also shall furnish to the Air Pollution Control Program copies of records required to be kept by the permittee. The permittee may make a claim of confidentiality for any information or records submitted under this rule.
- 6) Failure to comply with the limitations and conditions that qualify the installation for an Intermediate permit make the installation subject to the provisions of 10 CSR 10-6.065(5) and enforcement action for operating without a valid part 70 operating permit.

Reasonably Anticipated Operating Scenarios

10 CSR 10-6.065(4)(C)1.C

The installation does have reasonably anticipated operating scenarios as detailed in Permit Condition 072008-001.

Compliance Requirements

10 CSR 10-6.065, §(4)(B)4; §(4)(C)1, §(5)(C)3.B; and §(5)(C)3.D; and §(4)(C)3 and §(5)(C)3.E.(I) – (III) and (V) – (VI)

- 1) Any document (including reports) required to be submitted under this permit shall contain a certification signed by the responsible official.
- 2) Upon presentation of credentials and other documents as may be required by law, the permittee shall allow authorized officials of the Missouri Department of Natural Resources, or their authorized agents, to perform the following (subject to the installation's right to seek confidential treatment of information submitted to, or obtained by, the Air Pollution Control Program):
 - a) Enter upon the premises where a permitted installation is located or an emissions-related activity is conducted, or where records must be kept under the conditions of this permit;
 - b) Have access to and copy, at reasonable times, any records that must be kept under the conditions of this permit;
 - c) Inspect, at reasonable times and using reasonable safety practices, any facilities, equipment (including monitoring and air pollution control equipment), practices, or operations regulated or required under this permit; and
 - d) As authorized by the Missouri Air Conservation Law, Chapter 643, RSMo or the Act, sample or monitor, at reasonable times, substances or parameters for the purpose of assuring compliance with the terms of this permit, and all applicable requirements as outlined in this permit.
- 3) All progress reports required under an applicable schedule of compliance shall be submitted semiannually (or more frequently if specified in the applicable requirement). These progress reports shall contain the following:
 - a) Dates for achieving the activities, milestones or compliance required in the schedule of compliance, and dates when these activities, milestones or compliance were achieved, and
 - b) An explanation of why any dates in the schedule of compliance were not or will not be met, and any preventative or corrective measures adopted.
- 4) The permittee shall submit an annual certification that it is in compliance with all of the federally enforceable terms and conditions contained in this permit, including emissions limitations, standards, or work practices. These certifications shall be submitted annually by April 1st, unless the applicable requirement specifies more frequent submission. These certifications shall be submitted to the Air Pollution Control Program, Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov. All deviations and exceedances must be included in the compliance certifications. The compliance certification shall include the following:
 - a) The identification of each term or condition of the permit that is the basis of the certification;
 - b) The current compliance status, as shown by monitoring data and other information reasonably available to the installation;
 - c) Whether compliance was continuous or intermittent;
 - d) The method(s) used for determining the compliance status of the installation, both currently and over the reporting period; and
 - e) Such other facts as the Air Pollution Control Program will require in order to determine the compliance status of this installation.

Emergency Provisions

10 CSR 10-6.065, §(4)(C)1 and §(5)(C)7

- 1) An emergency or upset as defined in 10 CSR 10-6.065(5)(C)7.A shall constitute an affirmative defense to an enforcement action brought for noncompliance with technology-based emissions limitations. To establish an emergency- or upset-based defense, the permittee must demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence, the following:
 - a) That an emergency or upset occurred and that the permittee can identify the source of the emergency or upset,
 - b) That the installation was being operated properly,
 - c) That the permittee took all reasonable steps to minimize emissions that exceeded technology-based emissions limitations or requirements in this permit, and
 - d) That the permittee submitted notice of the emergency to the Air Pollution Control Program within two working days of the time when emission limitations were exceeded due to the emergency. This notice must contain a description of the emergency, any steps taken to mitigate emissions, and any corrective actions taken.
- 2) Be aware that an emergency or upset shall not include noncompliance caused by improperly designed equipment, lack of preventative maintenance, careless or improper operation, or operator error.

Off-Permit Changes

10 CSR 10-6.065(4)(C)5

- 1) Except as noted below, the permittee may make any change in its permitted installation's operations, activities or emissions that is not addressed in, constrained by or prohibited by this permit without obtaining a permit revision. Off-permit changes shall be subject to the following requirements and restrictions:
 - a) The change must meet all applicable requirements of the Act and may not violate any existing permit term or condition; the permittee may not change a permitted installation without a permit revision if this change is a Title I modification; Please Note: Changes at the installation which affect the emission limitation(s) classifying the installation as an intermediate source (add additional equipment to the record keeping requirements, increase the emissions above major source level) do not qualify for off-permit changes.
 - b) The permittee must provide contemporaneous written notice of the change to the Air Pollution Control Program, Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 or AirComplianceReporting@dnr.mo.gov, as well as Missouri Compliance Coordinator, Air Branch; Enforcement and Compliance Assurance Division; EPA Region VII; 11201 Renner Blvd., Lenexa, KS 66219. This written notice shall describe each change, including the date, any change in emissions, pollutants emitted and any applicable requirement that would apply as a result of the change; and
 - c) The permittee shall keep a record describing all changes made at the installation that result in emissions of a regulated air pollutant subject to an applicable requirement and the emissions resulting from these changes.

Responsible Official

10 CSR 10-6.020(2)(R)34

The application utilized in the preparation of this permit was signed by Roger Hill, General Manager. On May 10, 2019, the Air Pollution Control Program was informed that Mr. Kelly Bordewick now the responsible official. If this person terminates employment, or is reassigned different duties such that a different person becomes the responsible person to represent and bind the installation in environmental permitting affairs, the owner or operator of this air contaminant source shall notify the director of the Air Pollution Control Program of the change. Said notification shall be in writing and shall be submitted within 30 days of the change. The notification shall include the name and title of the new person assigned by the source owner or operator to represent and bind the installation in environmental permitting affairs. All representations, agreement to terms and conditions and covenants made by the former responsible person that were used in the establishment of limiting permit conditions on this permit will continue to be binding on the installation until such time that a revision to this permit is obtained that would change said representations, agreements and covenants.

Reopening-Permit for Cause

10 CSR 10-6.065 §(4)(E)4 and §(5)(E)6.A(III)(a)-(c)

This permit may be reopened for cause if:

- 1) The Missouri Department of Natural Resources (MoDNR) or EPA determines that the permit contains a material mistake or that inaccurate statements were made which resulted in establishing the emissions limitation standards or other terms of the permit,
- 2) Additional applicable requirements under the Act become applicable to the installation; however, reopening on this ground is not required if—:
 - a) The permit has a remaining term of less than three years;
 - b) The effective date of the requirement is later than the date on which the permit is due to expire;or
 - c) The additional applicable requirements are implemented in a general permit that is applicable to the installation and the installation receives authorization for coverage under that general permit,
- 3) MoDNR or EPA determines that the permit must be reopened and revised to assure compliance with applicable requirements.

Statement of Basis

10 CSR 10-6.065 §(4)(E)1.A and §(5)(E)1.C

This permit is accompanied by a statement setting forth the legal and factual basis for the permit conditions (including references to applicable statutory or regulatory provisions). This Statement of Basis, while referenced by the permit, is not an actual part of the permit.

V. Attachments

Attachments follow.

Attachment 6.170

Fugitive Emission Observations

[illegible]

Attachment Method 22

Visible Emission Observations

Method 22 Visible Emissions Observations					
Installation Name			Observer Name		
Location			Date		
Sky Conditions			Wind Direction		
Precipitation			Wind Speed		
Time			Emission unit		
Sketch emission unit: indicate observer position relative to emission unit; indicate potential emission points and/or actual emission points.					
Minute	Seconds				Comments
	0	15	30	45	
	Visible Emissions Yes (Y) or No (N)				
0					
1					
2					
3					
4					
5					
6					

If visible emissions are observed, the installation is not required to complete the entire six-minute observation. The installation shall note when the visible emissions were observed and shall conduct a Method 9 opacity observation.

Attachment Method 9
Method 9 Opacity Observations

Method 9 Opacity Observations		
Installation Name:	Sketch of the observer's position relative to the emission unit	
Emission Point:		
Emission Unit:		
Observer Name and Affiliation:		
Observer Certification Date:		
Method 9 Observation Date:		
Height of Emission Point:		
Time:	Start of observations	End of observations
Distance of Observer from Emission Point:		
Observer Direction from Emission Point:		
Approximate Wind Direction:		
Estimated Wind Speed:		
Ambient Temperature:		
Description of Sky Conditions (Presence and color of clouds):		
Plume Color:		
Approximate Distance Plume is Visible from Emission Point:		

Attachment Method 9 (Continued)

Method 9 Opacity Observations

Minute	Seconds				1-minute Avg. % Opacity ¹⁷	6-minute Avg. % Opacity ¹⁸	Steam Plume (check if applicable)		Comments
	0	15	30	45			Attached	Detached	
	Opacity Readings (% Opacity) ¹⁹								
0						N/A			
1						N/A			
2						N/A			
3						N/A			
4						N/A			
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									
22									
23									
24									
25									
26									
27									
28									
29									
30									

The emission unit is in compliance if each six-minute average opacity is less than or equal to 20%.
Exception: The emission unit is in compliance if one six-minute average opacity is greater than 20%, but less than 60%.

Was the emission unit in compliance at the time of evaluation (yes or no)?

Signature of Observer

¹⁷ 1-minute avg. % opacity is the average of the four 15 second opacity readings during the minute.

¹⁸ 6-minute avg. % opacity is the average of the six most recent 1-minute avg. % opacities.

¹⁹ Each 15 second opacity reading shall be recorded to the nearest 5% opacity as stated within Method 9.

Attachment Visual Indicator Check

Emission Point # and Unit Description: _____

[illegible]

²⁰ All openings must indicate the presence of negative pressure for compliance

Attachment Pressure Drop

Emission Point # and Unit Description: _____

[illegible]

Attachment Operating and Maintenance Log

Emission Point # and Unit Description: _____

[illegible]

Attachment Regenerative Thermal Oxidizer Bypass

RTO Bypass, Table 1: DDGS Dryer operating time during RTO Bypass

Date (month/day/year)	RTO Bypass Start Time (HH:MM)	RTO Bypass End Time (HH:MM)	Total RTO Bypass time this month (Hours)	Consecutive 12 month total RTO Bypass Time (Hours) ²¹

RTO Bypass, Table 2: RTO Regeneration Events

Date (month/day/year)	RTO Regeneration Event Start Time (HH:MM)	RTO Regeneration Event End Time (HH:MM)	Total RTO Regeneration Event time this month (Hours)	Consecutive 12 month total RTO Regeneration Event Time (Hours) ²²

²¹ Consecutive 12 month total is calculated by summing the Monthly DDGS Dryer operating time during RTO Bypass over the previous 12 month period. **Compliance is demonstrated when this value is less than or equal to 50 hours.**

²² Consecutive 12 month total is calculated by summing the Monthly RTO Regeneration Event Time over the previous 12 month period. **Compliance is demonstrated when this value is less than or equal to 12 hours.**

Attachment Cooling Towers

Cooling Towers, Table 1:TDS Sampling and Results

Cooling Tower # : _____

Date (Month/year)	TDS Concentration (ppm)	Is TDS Concentration less than or equal to 3,500 ppm? (yes/no) ²³

²³ Compliance is demonstrated when TDS Concentration is less than or equal to 3,500 ppm for each sample.

Attachment Ethanol Production Rate

[illegible]

²⁴ Ethanol production includes all types of ethanol produced at the installation.

²⁵ Consecutive 12 month total is calculated by summing the production over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 21,000,000 gallons.

Attachment Denaturant Usage Rate

[illegible]

²⁶ Denaturant usage includes all types of denaturants used at the installation. This includes natural gasoline, methanol, tertiary butyl alcohol, toluene, 3-chloropropyl triethoxysilane (CPTEO), ethyl acetate, isopropyl alcohol, ethyl 3-ethoxypropionate (3-EEP), methyl isobutyl ketone, and hexane. Usage of any other denaturant must be evaluated for construction permitting applicability prior to use.

²⁷ Consecutive 12 month total is calculated by summing the production over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 1,050,000 gallons.

Attachment Denatured Ethanol Production Rate

[illegible]

²⁸ Denatured ethanol production includes ethanol that is denatured with any denaturant(s), before or after loading operations.

²⁹ Consecutive 12 month total is calculated by summing the production over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 22,050,000 gallons.

Attachment Truck and Rail Grain Receiving

[illegible]

³⁰ Truck and Rail are the only grain receiving operations at the installation. The monthly total includes all grain received by Truck and Rail operations.

³¹ Consecutive 12 month total is calculated by summing the production over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 217,778 tons.

Attachment Truck and Rail DDGS Shipping

[illegible]

³² Truck and Rail are the only DDGS shipping operations at the installation. The monthly total includes all DDGS shipped by Truck and Rail operations.

³³ Consecutive 12 month total is calculated by summing the production over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 78,840 tons.

Attachment PM₁₀ Emissions

This attachment is organized in two sections. Section 1 contains the compliance calculation methods for the emission units that have individual PM₁₀ emission limitations. Section 2 contains the compliance calculation methods for the plant wide PM₁₀ emission limitation.

Section 1: Compliance calculations for emission units with individual PM₁₀ emission limitations:

PM₁₀ Emissions, Section 1, Table 1: Emissions from Units Controlled by Baghouse #2

EP #	Stack ID	Description	Monthly Throughput (tons)	Emission Factor (lb PM ₁₀ /ton)	Monthly Emissions (tons) ³⁴
EU013	SV003	Hammermill and Belt Scale ³⁵		0.0044	
SSM Emissions for this month ³⁶ (tons):					
Total Monthly PM ₁₀ Emissions of Baghouse #2 (tons):					
Sum of Consecutive 12 month PM₁₀ Emissions of Baghouse #2 (tons):³⁷					

³⁴ Monthly Emissions calculated as follows: $Monthly\ Throughput \times Emission\ Factor \times \frac{ton}{2000\ lb}$

³⁵ Emission factor sourced from 11/17/2008 Performance Test. Emission rate is 0.091 lbs PM₁₀/hour and production rate is 20.72 tons/hour. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

³⁶ Start-up, shutdown, and malfunction emissions as reported to the Air Pollution Control Program's Compliance and Enforcement Section during the current month must be included in the Total Monthly PM₁₀ Emissions of Baghouse #2.

³⁷ Consecutive 12 month total is calculated by summing the Total Monthly PM₁₀ Emissions of Baghouse #2 over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 1.314 tons.

PM₁₀ Emissions, Section 1, Table 2: Emissions from the DDGS Cooling/Separating Cyclone

EP #	Stack ID	Description	Total Beer Feed rate for this month (1000 gallons)	Emission Factor (lb PM ₁₀ /1000 gallon beer feed)	Monthly Emissions (tons) ³⁴
SV012 or SV006 for RTO bypass	EU027	DDGS Cooling/Separating Cyclone ³⁸		0.0227	
SSM Emissions for this month ³⁶ (tons):					
Total Monthly PM ₁₀ Emissions of DDGS Cooling/Separating Cyclone (tons):					
Sum of Consecutive 12 month PM₁₀ Emissions of DDGS Cooling/Separating Cyclone (tons):³⁹					

³⁸ Emission factor sourced from 11/17/2016 Performance Test. Emission rate is 0.313 lbs PM₁₀/hour at a beer feed rate of 230 gallons/minute. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

³⁹ Consecutive 12 month total is calculated by summing the Total Monthly PM₁₀ Emissions of DDGS Cooling/Separating Cyclone over the previous 12 month period. **Compliance is demonstrated when this value is less than or equal to 1.34 tons.**

PM₁₀ Emissions, Section 1, Table 3: Emissions from units controlled by Baghouse #1

EP #	Stack ID	Description	Control Device	Monthly Throughput (tons)	Emission Factor (lb PM ₁₀ /ton)	Monthly Emissions (tons) ³⁴
EU001	SV001	Corn Dump Pit with Auger	⁴⁰ Baghouse #1		2.68x10 ⁻³	
EU002	SV001	Corn Elevator				
EU041	SV001	Corn Bin Distribution Conveyor				
EU004	SV001	Corn Bin-Surge Bin				
EU005	SV001	Corn Bin #1				
EU006	SV001	Corn Bin #2				
EU007	SV001	Corn Bin #3				
EU042	SV001	Corn Bin Unloading Conveyor				
EU003	SV001	Corn Scalper				
EU043	SV001	DDGS Conveyor				
EU044	SV001	DDGS Loadout Conveyor				
EU008	SV001	DDGS Pit A				
EU008	SV001	DDGS Pit B				
EU008	SV001	DDGS Pit C				
EU008	SV001	Screw Auger A				
EU008	SV001	Screw Auger B				
EU008	SV001	Screw Auger C				
EU011	SV001	DDGS Screw Conveyor				
EU009	SV001	DDGS Elevator				
EU010	SV001	DDGS Rail Load Spout				
EU012	SV001	DDGS Truck Load Spout				
SSM Emissions for this month ³⁶ (tons):						
Total Monthly PM ₁₀ Emissions of Baghouse #1 (tons):						
Sum of Consecutive 12 month PM ₁₀ Emissions of Baghouse #1 (tons): ⁴¹						

⁴⁰ Emission factor sourced from 11/18/2008 Performance Test. Emission rate is 0.156 lbs PM₁₀/hour at a production rate of 58.319 tons/hour. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁴¹ Consecutive 12 month total is calculated by summing the Total Monthly PM₁₀ Emissions of Baghouse #1 over the previous 12 month period.
Compliance is demonstrated when this value is less than or equal to 0.272 tons.

PM₁₀ Emissions, Section 1, Table 4: Emissions from Boiler #1

EP #	Stack ID	Description	Monthly Throughput (MMCF)	Emission Factor (lb PM ₁₀ /MMCF) ⁴²	Monthly Emissions (tons) ³⁴
EU028	SV008	Boiler #1		7.6	
SSM Emissions for this month ³⁶ (tons):					
Total Monthly PM ₁₀ Emissions of Boiler #1 (tons):					
Sum of Consecutive 12 month PM ₁₀ Emissions of Boiler #1 (tons): ⁴³					

PM₁₀ Emissions, Section 1, Table 5: Emissions from Boiler #2

EP #	Stack ID	Description	Monthly Throughput (MMCF)	Emission Factor (lb PM ₁₀ /MMCF) ⁴²	Monthly Emissions (tons) ³⁴
EU034	SV011	Boiler #2		7.6	
SSM Emissions for this month ³⁶ (tons):					
Total Monthly PM ₁₀ Emissions of Boiler #2 (tons):					
Sum of Consecutive 12 month PM ₁₀ Emissions of Boiler #2 (tons): ⁴⁴					

⁴² Emission factor sourced from webFIRE SCC 10200602.

⁴³ Consecutive 12 month total is calculated by summing the Total Monthly PM₁₀ Emissions of Boiler #1 over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 2.01 tons.

⁴⁴ Consecutive 12 month total is calculated by summing the Total Monthly PM₁₀ Emissions of Boiler #2 over the previous 12 month period.

Compliance is demonstrated when this value is less than or equal to 2.01 tons.

PM₁₀ Emissions, Section 1, Table 6: Emissions from the DDGS Dryer and RTO

EP #	Stack ID	Description	Total Beer Feed rate for this month (1000 gallons/hr)	Emission Factor (lb PM ₁₀ /1000 gallon beer feed) ⁴⁵	Monthly Emissions (tons) ³⁴
EU026 and EU038	SV012	DDGS Dryer and RTO		0.328	
SSM Emissions (SV006) for this month ³⁶ (tons):					
Total Monthly PM ₁₀ Emissions of DDGS Dryer and RTO (tons):					
Sum of Consecutive 12 month PM₁₀ Emissions of DDGS Dryer and RTO (tons):⁴⁶					

⁴⁵ Emission factor sourced from 11/17/2016 Performance Test. Emission rate is 4.586 lbs PM₁₀/hour at a beer feed rate of 233.24 gallons/minute. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁴⁶ Consecutive 12 month total is calculated by summing the Total Monthly PM₁₀ Emissions of DDGS Dryer and RTO over the previous 12 month period. **Compliance is demonstrated when this value is less than or equal to 8.67 tons.**

Section 2: Compliance calculations for the plant wide PM₁₀ emission limitation:

PM₁₀ Emissions, Section 2, Table 1: Emissions from Grain Operations⁴⁷:

EP #	Stack ID	Description	Monthly Throughput (tons)	Emission Factor (lb PM ₁₀ /ton) ⁴⁸	Monthly Emissions (tons) ³⁴
FS003-1	FS003	Uncaptured Grains ⁴⁹			
		Hopper Trucks receiving		0.0078	
		Straight Truck receiving		0.059	
		Rail receiving		0.0078	
Total Monthly PM ₁₀ Emissions from the sources listed in this table: (tons):					
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 1, Table 3: Emissions from units controlled by Baghouse #1 (tons):					
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 1, Table 1: Emissions from Units Controlled by Baghouse #2 (tons):					
Total Monthly PM ₁₀ Emissions of Grain Operations (tons) ⁵⁰ :					

⁴⁷ EU029, FS003, Grain/DDGS Storage Building is not included in the table as an emissions source. No capture efficiency is provided for the building. All grain operations are either listed in the table above, controlled by Baghouse #1, or controlled by Baghouse #2.

⁴⁸ Sourced from AP-42, Section 9.9.1, Table 9.9.1-1 for grain receiving.

⁴⁹ FS003-1, FS003, Uncaptured Grains represents the emissions from the corn dump pits that are not captured by Baghouse #1. Baghouse #1 is assigned 80% capture efficiency for corn dump pit emissions. The throughput value is calculated by (Total amount received at corn dump pits)*(0.20). Throughput must be recorded for each type of receiving operation, hopper trucks and rail.

⁵⁰ Calculated as follows: Total Monthly PM₁₀ Emissions from the sources listed in this table + Total Monthly PM₁₀ Emissions of Baghouse #1 + Total Monthly PM₁₀ Emissions of Baghouse #2.

PM₁₀ Emissions, Section 2, Table 2: Emissions from Fermentation Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput	Emission Factor (lb PM ₁₀ /1000 bushels ⁵¹	Monthly Emissions (tons) ³⁴
EU014	SV004	Fermentation Tank #1 (T301)	Fermentation Scrubber (C003)	1000 bushels	0.0587	
EU015	SV004	Fermentation Tank #2 (T302)				
EU016	SV004	Fermentation Tank #3 (T303)				
EU051	SV004	Fermentation Tank #4 (T314)				
EU052	SV004	Fermentation Tank #5 (T315)				
EU017	SV004	Beer Well (T304)				
SSM Emissions for this month ³⁶ (tons):						
Total Monthly PM ₁₀ Emissions of Fermentation Operations (tons) ⁵² :						

⁵¹ Emission factor sourced from 9/2008 Performance Test. Emission rate is 0.055 lbs PM₁₀/hour at a production rate of 0.937 1000 bushels/hr. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁵² Calculated as follows: Total Monthly PM₁₀ Emissions from the sources listed in this table + SSM Emissions for this month.

PM₁₀ Emissions, Section 2, Table 3: Emissions from Distillation Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput	Emission Factor	Monthly Emissions (tons) ³⁴
EU023	SV005	Slurry Tank (T203)	Distillation Scrubber (C004)	1000 gallons	0.0017 lb PM ₁₀ /1000 gallons ⁵³	
EU018	SV005	Beer Stripper Distillation Column (T401)				
EU019	SV005	Rectifier Distillation Column (T402)				
EU025	SV005	Rundown Tank (T406)				
EU020	SV005	Side Stripper Distillation Column (T403)				
EU021	SV005	Molecular Sieve Dehydration Vacuum Pump Vent (T421, T422, T423)				
EU035	SV005	Industrial Extractive Distillation Column #1				
EU036	SV005	Industrial Rectifier Distillation Column #2				
EU037	SV005	Industrial Demthylization Distillation Column #3				
EU040	SV005	Industrial Distillation Recycle Rectifier Column #4 (T524)				
EU022	SV005	Evaporator				
FS005		Cooling Tower #1	None	MM gallons	245.98 lb PM ₁₀ /MMGal ⁵⁴	
FS006		Cooling Tower #2	None	MM gallons	245.98 lb PM ₁₀ /MMGal ⁵⁵	
SSM Emissions for this month ³⁶ (tons):						
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 1, Table 4: Emissions from Boiler #1 (tons):						
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 1, Table 5: Emissions from Boiler #2 (tons):						
Total Monthly PM ₁₀ Emissions of Distillation Operations (tons) ⁵⁶ :						

⁵³ Emission factor sourced from 9/2008 Performance Test. Emission rate is 0.004 lbs PM₁₀/hour at a production rate of 2.4 1000 gallons/hr. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁵⁴ Emission factor derived using calculation methodology in "Calculating TSP, PM-10, and PM-2.5 from Cooling Towers", New Mexico, September 9, 2013; as follows for TDS=3,500 ppm, 1.4% drift loss, 7,700 gal/min recirculation rate, and a 60.16% particle size distribution for PM₁₀:

$$\frac{3,500 \text{ mg}}{L} \times \frac{1 \text{ lb}}{453,600 \text{ mg}} \times \frac{3.785 \text{ L}}{\text{gal}} \times \frac{7,700 \text{ gal}}{\text{min}} \times \frac{1.4\%}{100\%} \times \frac{60 \text{ min}}{\text{hr}} \times \frac{60.16\%}{100\%} \div \left(\frac{7,700 \text{ gal}}{\text{min}} \times \frac{60 \text{ min}}{\text{hr}} \times \frac{1 \text{ MM gal}}{1 \times 10^6 \text{ gal}} \right) = 245.98 \frac{\text{lb PM}_{10}}{\text{MMgal}}$$

⁵⁵ Emission factor derived using same methodology as for Cooling Tower #1, except with a 9,600 gal/min recirculation rate.

⁵⁶ Calculated as follows: Total Monthly PM₁₀ Emissions from the sources listed in this table + SSM Emissions + Boiler #1 Emissions + Boiler #2 Emissions.

PM₁₀ Emissions, Section 2, Table 4: Emissions from Ethanol Loading Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput	Emission Factor (Lb PM ₁₀ /MMCF) ⁵⁷	Monthly Emissions (tons) ³⁴
EU039		Open Flare, Natural gas fired	N/A	MMCF	7.6	
SSM Emissions for this month ³⁶ (tons):						
Total Monthly PM ₁₀ Emissions of Ethanol Loading Operations (tons) ⁵⁸ :						

⁵⁷ Emission factor sourced from webFIRE SCC 10200602.

⁵⁸ Calculated as follows: Total Monthly PM₁₀ Emissions from the sources listed in this table + SSM Emissions.

PM₁₀ Emissions, Section 2, Table 5: Emissions from WDGS and DDGS Operations:

EP #	Stack ID	Description	Monthly Throughput	Emission Factor (lb PM ₁₀ /ton) ⁵⁹	Monthly Emissions (tons) ³⁴
FS020		Wetcake Conveyor	Tons wetcake	0.034	
FS021		Wetcake Pad	Tons wetcake	0.034	
FS022		Wetcake Truck Loading	Tons wetcake	0.029	
EU026		Dryer Feed Conveyor	Tons DDGS	0.034	
EU026		Dried Solids Conveyor	Tons DDGS	0.034	
EU008		DDGS Dump Auger	Tons DDGS	0.034	
EU029		DDGS Stockpile	Tons DDGS	0.034	
P802-A		Bulk Scale	Tons DDGS	0.034	
C800		Railcar Grate	Tons DDGS	0.034	
C800		Railcar Loading Pit	Tons DDGS	0.034	
C801		Truck Loading Grate	Tons DDGS	0.034	
C801		Truck Loading Pit	Tons DDGS	0.034	
FS003		Uncaptured DDGS ⁶⁰			
		Truck Loadout (not aspirated)	Tons DDGS	0.029	
		Rail Loadout (not aspirated)	Tons DDGS	0.0022	
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 1, Table 2: Emissions from the DDGS Cooling/Separating Cyclone (tons):					
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 1, Table 6: Emissions from the DDGS Dryer and RTO (tons):					
Total Monthly PM ₁₀ Emissions of WDGS and DDGS Operations (tons) ⁶¹ :					

⁵⁹ Sourced from AP-42, Section 9.9.1, Table 9.9.1-1 for grain handling, truck loading, and rail loading .

⁶⁰ FS003, Uncaptured DDGS represents the emissions from the DDGS handling and loading operations not captured by Baghouse #1. Baghouse #1 is assigned 80% capture efficiency for DDGS emission units.. The throughput value is calculated by (Total amount DDGS produced)*(0.20) and must be recorded by each loading type.

⁶¹ Calculated as follows: Total Monthly PM₁₀ Emissions from the sources listed in this table + SSM Emissions + DDGS Cooling/Separating Cyclone Emissions+DDGS Dryer and RTO Emissions+RTO Emissions.

PM₁₀ Emissions, Section 2, Table 6: Emissions from Miscellaneous Operations

EP #	Stack ID	Description	Monthly Throughput	Emission Factor (lb PM ₁₀ /MMCF) ⁶²	Monthly Emissions (tons) ³⁴
EU045		Emergency Generator	MMCF	0.079	
Total Monthly PM ₁₀ Emissions of Miscellaneous Operations (tons):					

PM₁₀ Emissions, Section 2, Table 7: Emissions from FS004 Haul Roads

EP #	Stack ID	Type of Haul Road Activity	Monthly Throughput (tons)	Monthly Emissions (tons) ³⁴
FS004	FS004	Denaturant receiving (all types)		
		Enzyme receiving (all types)		
		Corrosion inhibitor receiving		
		Caustic receiving		
		Anhydrous Ammonia receiving		
		Sulfuric acid receiving		
		Grain receiving		
		Ethanol truck shipping ⁶³		
		DDGS truck shipping		
		WDGS truck shipping		
Sum of all Haul Road Activities for the current month (tons):				
Emission Factor for all Haul Road Activity (lb PM ₁₀ /ton) ⁶⁴				0.0259
Total Monthly PM ₁₀ Emissions of Truck Haul Roads (tons) ⁶⁵				

⁶² Emission factor sourced from AP-42, Section 3.2, Table 3.2-2 for 4 stroke lean burn engines. Converted using heating value of 1020 MMBtu/MMCF.

⁶³ This value shall include all types of ethanol product that leaves the site in any type of container in any type of truck.

⁶⁴ Calculated according to methodology in AP-42, Section 13.2.1, Paved Roads. Round trip road length = 0.6 miles, 200 tons/hr and 20 tons/truck activity rate to encompass all activities, silt loading of 8.2 g/m², and 100 rain days.

⁶⁵ Calculated as follows: Sum of all Haul Road Activities for the current month x Emission Factor for all Haul Road Activity x 0.0005

PM₁₀ Emissions, Section 2, Table 8: Summation of Plant Wide Monthly and Consecutive 12 Month Total PM₁₀ Emissionse

Values Carried Forward From the Following PM₁₀ Emissions Calculation Tables	Monthly Emissions (tons)
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 2, Table 1: Emissions from Grain Operations (tons):	
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 2, Table 2: Emissions from Fermentation Operations (tons):	
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 2, Table 3: Emissions from Distillation Operations (tons):	
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 2, Table 4: Emissions from Ethanol Loading Operations (tons):	
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 2, Table 5: Emissions from WDGS and DDGS Operations (tons):	
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 2, Table 6: Emissions from Miscellaneous Operations (tons):	
Total Monthly PM ₁₀ Emissions from PM ₁₀ Emissions, Section 2, Table 7: Emissions from FS004 Haul Roads	
Total Monthly PM ₁₀ Emissions of all plant wide sources for this month (tons) ⁶⁶ :	
Sum of Consecutive 12 month PM₁₀ Emissions of all plant wide sources (tons):⁶⁷	

⁶⁶ Sum the monthly emissions of all sources listed in the table.

⁶⁷ Consecutive 12 month total is calculated by summing the Total Monthly PM₁₀ Emissions of all plant wide sources for this month (tons) over the previous consecutive 12 month period. **Compliance is demonstrated when this value is less than 95.0 tons.**

Attachment NO_x Emissions

This attachment is organized in two sections. Section 1 contains the compliance calculation methods for the emission units that have group NO_x emission limitations. Section 2 contains the compliance calculation methods for the plant wide NO_x emission limitation.

Section 1: Compliance calculations for emission units with individual NO_x emission limitations:

NO_x Emissions, Section 1, Table 1: Boiler #1 and DDGS Dryer Outlet

EP #	Stack ID	Description	Monthly Throughput (units)	Emission Factor (lb NO _x /units) ⁶⁸	Monthly Emissions (tons) ³⁴
EU028	SV008	Boiler #1	MMBtu	0.03	
EU026	SV012	DDGS dryer outlet	MMBtu	0.04	
SSM Emissions for this month from the sources listed in this table (tons) ³⁶ :					
Total Monthly NO _x Emissions from the sources listed in this table (tons) ⁶⁹ :					
Sum of Consecutive 12 month NO_x Emissions of Boiler #1 and DDGS Dryer (tons):⁷⁰					

⁶⁸ Emission factor sourced from Permit Condition 072008-001. A construction permit amendment is required to change this value.

⁶⁹ Calculated as follows: Monthly NO_x Emissions of Boiler #1 + Monthly NO_x Emissions of DDGS Dryer + SSM Emissions.

⁷⁰ Consecutive 12 month total is calculated by summing the Total Monthly NO_x Emissions over the previous 12 month period. **Compliance is demonstrated when this value is less than or equal to 20.24 tons.**

Section 2: Compliance calculations for the plant wide NO_x emission limitation:

NO_x Emissions, Section 2, Table 1: Calculation of Plant Wide Monthly and Consecutive 12 month total NO_x Emissions

EP #	Stack ID	Description	Monthly Throughput (units)	Emission Factor (lb NO _x /unit)	Monthly Emissions (tons) ³⁴
EU028		⁷¹ Boiler #1	MMBtu	0.03	
EU034		⁷² Boiler #2	MMBtu	0.032	
EU039		⁷³ Open Flare	MMCF	69.36	
EU026	SV012 or SV006 for RTO bypass	⁷⁴ DDGS Dryer outlet	MMBtu	0.04	
EU038	SV012	⁷⁴ DDGS Dryer Regenerative Thermal Oxidizer	1000 gallons beer feed	0.393	
EU045		⁷⁵ Emergency Generator	MMCF	4,161.6	
SSM Emissions for this month from the sources listed in this table: ³⁶ (tons):					
Total Monthly NO _x Emissions of all plant wide sources for this month (tons) ⁷⁶ :					
Sum of Consecutive 12 month NO_x Emissions of all plant wide sources (tons):⁷⁷					

⁷¹ Emission factor sourced from 11/17/2016 Performance Test. Emission rate is 1.583 lb/hr at a firing rate of 71.4% of MHDR. Converted using a heating value of 1020 MMBtu/MMCF. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁷² A Boiler #2 specific NO_x emission factor has not been established through testing. Until such time that one is established, the permittee shall use the value shown, sourced from AP-42, Section 4.1, Table 1.4-1.

⁷³ Emission factor sourced from AP-42, Section 13.5, Table 13.5-1, converted using 1020 MMBtu/MMCF..

⁷⁴ Emission factor sourced from 11/17/2016 Performance Test. Emission rate is 5.503 lbs NO_x/hour at a beer feed rate of 233.24 gallons/minute. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁷⁵ Emission factor sourced from AP-42, Section 3.2, Table 3.2-2 for 4 stroke lean burn engines. Converted using heating value of 1020 MMBtu/MMCF.

⁷⁶ Calculated as the sum of all the Monthly NO_x Emissions for all units in the table + SSM Emissions.

⁷⁷ Consecutive 12 month total is calculated by summing the Total Monthly NO_x Emissions over the previous 12 month period. **Compliance is demonstrated when this value is less than 95.0 tons.**

Attachment CO Emissions

This attachment contains the compliance calculation methods for the plant wide CO emission limitation.

CO Emissions, Table 1: Calculation of Plant Wide Monthly and Consecutive 12 month total CO Emissions

EP #	Stack ID	Description	Monthly Throughput (units)	Emission Factor (lb CO /unit)	Monthly Emissions (tons) ³⁴
EU028		⁷⁸ Boiler #1	MMCF	0.929	
EU034		⁷⁹ Boiler #2	MMCF	4.937	
EU039		⁸⁰ Open Flare	MMCF	84	
EU026	SV012 or SV006 for RTO bypass	⁸¹ DDGS Dryer	1000 gallons beer feed	0.201	
EU038	SV012	⁸¹ DDGS Dryer Regenerative Thermal Oxidizer	1000 gallons beer feed	0.201	
EU045		⁸² Emergency Generator	MMCF	568.14	
SSM Emissions for this month from the sources listed in this table (tons) ³⁶ :					
Total Monthly CO Emissions of all plant wide sources for this month (tons) ⁸³ :					
Sum of Consecutive 12 month CO Emissions of all plant wide sources (tons):⁸⁴					

⁷⁸ Emission factor sourced from 11/17/2016 Performance Test. Emission rate is 0.039 lb/hr at a firing rate of 71.4% of MHDR, converting using heating value of 1020 MMBtu/MMCF. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁷⁹ Emission factor sourced from 11/17/2016 Performance Test. Emission rate is 0.2736 lb/hr at a firing rate of 93.33% of MHDR. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved.

⁸⁰ Emission factor sourced from AP-42, Section 4.1, Table 1.4-1.

⁸¹ Emission factor sourced from 11/17/2016 Performance Test. Emission rate is 2.813 lbs CO/hour at a beer feed rate of 233.24 gallons/minute. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date the most recent test results are approved. Future testing shall be conducted such that the emissions from each source are quantified separately.

⁸² Emission factor sourced from AP-42, Section 3.2, Table 3.2-2 for 4 stroke lean burn engines. Converted using heating value of 1020 MMBtu/MMCF.

⁸³ Calculated as the sum of all the Monthly CO Emissions for all units in the table + SSM Emissions.

⁸⁴ Consecutive 12 month total is calculated by summing the Total Monthly CO Emissions over the previous 12 month period. **Compliance is demonstrated when this value is less than 95.0 tons.**

Attachment SO_x Emissions

This attachment contains the compliance calculation methods for the plant wide SO_x emission limitation.

SO_x Emissions, Table 1: Calculation of Plant Wide Monthly and Consecutive 12 month total SO_x Emissions

EP #	Stack ID	Description	Monthly Throughput (units)	Emission Factor (lb SO _x /unit)	Monthly Emissions (tons) ³⁴
EU050		⁸⁵ Centrifuge #4	1000 gallons whole stillage	2.87x10 ⁻⁴	
EU053		⁸⁵ Centrifuge #5	1000 gallons whole stillage	2.87x10 ⁻⁴	
SV004		⁸⁶ Fermentation Scrubber (C003)	1000 gallons beer feed	1.42x10 ⁻³	
SV005		⁸⁷ Distillation Scrubber (C004)	1000 gallons beer feed	7.09x10 ⁻⁴	
EU028		⁸⁸ Boiler #1	MMCF	0.6	
EU034		Boiler #2	MMCF	0.6	
EU039		Open Flare	MMCF	0.6	
EU026	SV012 or SV006 for RTO bypass	DDGS Dryer	MMCF	0.6	
EU038	SV012	DDGS Dryer Regenerative Thermal Oxidizer	MMCF	0.6	
EU045		⁸⁹ Emergency Generator	MMCF	0.6	
SSM Emissions for this month from the sources listed in this table (tons) ³⁶ :					
Total Monthly SO _x Emissions of all plant wide sources for this month (tons) ⁹⁰ :					
Sum of Consecutive 12 month SO_x Emissions of all plant wide sources (tons):⁹¹					

⁸⁵ Emission factor sourced from March 2017 Performance Test on Centrifuge #4. Test results were 0.0039 lb/hr at a production rate of 226.46 gal/min whole stillage. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

⁸⁶ Emission factor sourced from March 2017 Performance Test on Fermentation Scrubber. Test results were 0.01 lb/hr at a production rate of 235 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

⁸⁷ Emission factor sourced from March 2017 Performance Test on Distillation Scrubber. Test results were 0.6673 lb/hr at a production rate of 235 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

⁸⁸ Emission factor sourced from AP-42, Section 1.4-2 for Natural Gas Combustion.

⁸⁹ Emission factor sourced from AP-42, Section 3.2, Table 3.2-2 for 4 stroke lean burn engines. Converted using heating value of 1020 MMBtu/MMCF.

⁹⁰ Calculated as the sum of all the Monthly SO_x Emissions for all units in the table + SSM Emissions.

⁹¹ Consecutive 12 month total is calculated by summing the Total Monthly SO_x Emissions over the previous 12 month period. **Compliance is demonstrated when this value is less than 95.0 tons.**

Attachment VOC Emissions

This attachment contains the compliance calculation methods for the plant wide VOC emission limitation.

VOC Emissions, Table 1: Emissions from Fermentation Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb VOC/unit)	Monthly Emissions (tons) ³⁴
EU014	SV004	Fermentation Tank #1 (T301)	⁹² Fermentation Scrubber (C003)	1000 gallons beer feed	0.066	
EU015	SV004	Fermentation Tank #2 (T302)				
EU016	SV004	Fermentation Tank #3 (T303)				
EU051	SV004	Fermentation Tank #4 (T314)				
EU052	SV004	Fermentation Tank #5 (T315)				
EU017	SV004	Beer Well (T304)				
T231		⁹³ Alpha Amylase Enzyme Tank				
TA001		⁹³ Anhydrous Ammonia Tank				
T232		⁹³ Gluc Amylase Enzyme Tank				
Monthly VOC emissions from the sources in this table (tons)						
SSM Emissions from the sources listed in this table for this month ³⁶ (tons):						
Total Monthly VOC Emissions of Fermentation Operations (tons) ⁹⁴ :						

⁹² Emission factor sourced from March 2017 Performance Test on Fermentation Scrubber. Test results were 0.9354 lb total VOC/hr at a production rate of 235 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

⁹³ The permittee shall use the calculation methodology contained in AP-42, Section 7.1 and the SDS for the material to calculate monthly emissions from all tanks. The calculations result in estimated emissions on a lb/month basis. The permittee shall divide this value by 2000 to convert to tons and shall record the monthly tons of emissions rounded to a minimum of four decimal places. The permittee shall document all calculations.

⁹⁴ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 2: Emissions from Distillation Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb VOC/unit)	Monthly Emissions (tons) ³⁴	
EU023	SV005	Slurry Tank (T203)	⁹⁵ Distillation Scrubber (C004)	1000 gallons beer feed	0.047		
EU018	SV005	Beer Stripper Distillation Column (T401)					
EU019	SV005	Rectifier Distillation Column (T402)					
EU025	SV005	Rundown Tank (T406)					
EU020	SV005	Side Stripper Distillation Column (T403)					
EU021	SV005	Molecular Sieve Dehydration Vacuum Pump Vent (T421, T422, T423)					
EU024	SV005	Yeast Tank (T307)					
EU035	SV005	Industrial Extractive Distillation Column #1					
EU036	SV005	Industrial Rectifier Distillation Column #2					
EU037	SV005	Industrial Demthylization Distillation Column #3					
EU040	SV005	Industrial Distillation Recycle Rectifier Column #4 (T524)					
EU022	SV005	Evaporator					
T801		190 Proof Ethanol Storage Tank ⁹³					
T802		200 Proof Ethanol Storage Tank ⁹³					
T830		Impurities Storage Tank ⁹³					
T807		HQ 200 Ethanol Tank ⁹³					
T808		HQ 190 Ethanol Tank ⁹³					
T809		HQ 190 Ethanol Tank ⁹³					
EU046		HQ Heat Exchanger					
T802		200 Proof Ethanol Storage Tank ⁹³					
EU054		HQ Molecular Sieve Dehydration					
T802		200 Industrial Alcohol Tank ⁹³					
T806		HQ 200 Ethanol Tank ⁹³					
EU028		Boiler #1, Natural gas fired ⁹⁶					
EU034		Boiler #2, Natural gas fired					
FS005		¹²⁷ Cooling Tower #1					
FS006		Cooling Tower #2					
Monthly VOC emissions from the sources in this table (tons)							
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :							
Total Monthly VOC Emissions of Distillation Operations (tons) ⁹⁷ :							

⁹⁵ Emission factor sourced from March 2017 Performance Test on Fermentation Scrubber. Test results were 0.6673 lb total VOC/hr at a production rate of 235 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

⁹⁶ Emission factor sourced from AP-42, Section 1.4-2 for Natural Gas Combustion.

⁹⁷ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 3: Emissions from Ethanol Product Loading Operations, Part 1: Storage Vessel Emissions

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb VOC/unit)	Monthly Emissions (tons)
T802		Fuel Shift Tank ⁹³	None			
T421		Fuel Sieves Storage Tank ⁹³	Distillation Scrubber (C004)			
T422		Fuel Sieves Storage Tank ⁹³	Distillation Scrubber (C004)			
T423		Fuel Sieves Storage Tank ⁹³	Distillation Scrubber (C004)			
T803		Denatured Ethanol Fuel Grade Storage Tank ⁹³				
T805		Natural Gasoline Storage Tank ⁹³				
T810		Corrosion Inhibitor Storage Tank ⁹³				
T815		HQ 190 Ethanol Tank ⁹³				
T816		HQ 190 Ethanol Tank ⁹³				
T817		HQ 190 Ethanol Tank ⁹³				
T804		HQ 200 Ethanol Tank ⁹³				
T811		HQ 200 Ethanol Tank ⁹³				
FS008		Methanol Storage Tank ⁹³				
FS009		Tertiary Butyl Alcohol Storage Tank ⁹³				
FS010		Toluene Storage Tank ⁹³				
FS011		3-Chloropropyl Triethoxysilane, CPTEO, Storage Tank ⁹³				
FS012		Ethyl Acetate Storage Tank ⁹³				
FS013		Isopropyl Alcohol Storage Tote ⁹³				
FS014		Ethyl 3-Ethoxypropionate, 3-EEO, Storage Tote ⁹³				
FS016		Hexane Storage Barrel ⁹³				
Monthly VOC emissions from the sources in this table (tons)						
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :						
Total Monthly VOC Emissions of Ethanol Product Loading Operations, Part 1 (tons) ⁹⁸ :						

⁹⁸ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

Ethanol Product Loading Operations, Part 2: The following tables reflect the various loading operations and the emissions associated with each loading scenario.

VOC Emissions, Table 4: Truck and Railcar Loading of Denaturants at the Denaturant and Railcar Loading Racks

Material Name	L _L (lb VOC/1000 gallons) ⁹⁹	Amount of Material Loaded this Month (1000 gallons)	Monthly VOC Emissions (tons) ¹⁰⁰
3-chloro triethoxysilane (CPTEO)	0.0104		
Ethyl 3-Ethoxypropionate (EEP)	0.0104		
Ethyl Acetate	1.2331		
Hexane	2.0752		
Isopropyl Alcohol	0.3533		
Methyl Alcohol	0.5610		
tert-butyl alcohol	0.3837		
Toluene	0.3482		
Total Truck and Railcar Denaturant Loading VOC Emissions this Month (tons) ¹⁰¹ :			
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :			
Total Monthly Truck and Railcar Denaturant Loading VOC Emissions this Month (tons) ¹⁰² :			

⁹⁹ Calculated using methodology found in AP-42 Section 5.2 and Section 7.1. Assumes S=0.6 for submerged loading of dedicated normal service trucks. Trucks are either dedicated normal service or clean cargo trucks, and all are submerged loaded. For CPTEO and EEP, compound specific data was not available. CPTE and EEP emissions used Distillate #2 as surrogates.

¹⁰⁰ Calculated as follows: $L_L \times \text{Amount of Material Loaded this Month} \times \frac{1 \text{ ton}}{2000 \text{ lbs}}$

¹⁰¹ Sum all monthly emissions in the table.

¹⁰² Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 5: Truck and Railcar Loading of Ethanol (no denaturants in Truck, Railcar, Tanks, or Stingers) at Truck and Railcar Loading Racks

Material Name	L _L (lb VOC/1000 gallons) ¹⁰³	Amount of Material Loaded this Month (1000 gallons)	Monthly VOC Emissions (tons) ³⁴
Ethanol	0.3419		
Total Truck and Railcar Ethanol Loading (no denaturants) VOC Emissions this Month (tons) ¹⁰⁴ :			
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :			
Total Monthly Truck and Railcar Ethanol Loading (no denaturants) VOC Emissions this Month (tons) ¹⁰⁵ :			

¹⁰³ This table represents ethanol loading when there are no denaturants present in the truck, the loading tank, or the loading stinger. Calculated using methodology found in AP-42 Section 5.2 and Section 7.1. Assumes S=0.6 for submerged loading of dedicated normal service trucks. Trucks are either dedicated normal service or clean cargo trucks, and all are submerged loaded. .

¹⁰⁴ Sum all monthly emissions in the table.

¹⁰⁵ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 6: Truck and Railcar Loading of Ethanol after Denaturant Loading in Truck or Railcar¹⁰⁶

Material Name	L _L (lb VOC/1000 gallons) ¹⁰⁷	Amount of Material Loaded this Month (1000 gallons)	Monthly VOC Emissions (tons) ¹⁰⁰
Ethanol	0.2849		
3-chloro triethoxysilane (CPTEO)	0.0086		
Ethyl 3-Ethoxypropionate (EEP)	0.0086		
Ethyl Acetate	1.0276		
Hexane	1.7293		
Isopropyl Alcohol	0.2944		
Methyl Alcohol	0.4675		
tert-butyl alcohol	0.3198		
Toluene	0.2902		
Total Truck and Railcar Denaturant then Ethanol Loading VOC Emissions this Month (tons) ¹⁰⁸ :			
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :			
Total Monthly Truck and Railcar Denaturant then Ethanol Loading VOC Emissions this Month (tons) ¹⁰⁹ :			

¹⁰⁶ When using denaturants, the denaturants are loaded into the trucks and railcars at the denaturant and railcar loading racks, then ethanol is loaded at the truck and railcar loading racks. To quantify emissions from this process, the permittee must determine the emissions from the loading of denaturants (see VOC Emissions, Table 4) and also the emissions from the subsequent loading of ethanol (see VOC Emissions, Table 6). Emissions from denaturants are expected at both loading operations.

¹⁰⁷ Calculated using methodology found in AP-42 Section 5.2 and Section 7.1. Assumes S=0.5 for all materials to account for headspace saturated with denaturant vapors prior to ethanol loading. .

¹⁰⁸ Sum all monthly emissions in the table.

¹⁰⁹ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 7: Controlled Truck Loading of Fuel Grade Ethanol at Truck Loading Rack¹¹⁰

Material Name	Emission Factor (units)	Amount of Material throughput this Month (units)	Monthly VOC Emissions (tons) ³⁴
Fuel Grade Ethanol Loading (Controlled)	0.0097 lb/1000 gallons	1000 gallons	
Flare	5.5 lb/MMCF	MMCF	
Total Controlled Fuel Grade Ethanol Truck Loading VOC Emissions this Month (tons) ¹¹¹ :			
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :			
Total Monthly Controlled Fuel Grade Ethanol Truck Loading VOC Emissions this Month (tons) ¹¹² :			

¹¹⁰ Loading losses reflect a mixture of Ethanol, 2% by volume Natural Gasoline, 2% by volume Corrosion Inhibitor, and flare controls. Calculated using methodology found in AP-42 Section 5.2 and Section 7.1. Assumes S=0.6 for submerged loading of dedicated normal service trucks. Trucks are either dedicated normal service or clean cargo trucks, and all are submerged loaded.

¹¹¹ Sum all monthly emissions in the table.

¹¹² Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 8: Emissions from Wet Distillers Grains with Syrup (WDGS) and Dried Distillers Grains with Syrup (DDGS) Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb VOC/unit)	Monthly Emissions (tons) 34
		Centrifuge Deck ¹¹³	None	Tons whole stillage	0.089	
EU050		Centrifuge #4		Tons whole stillage	0.089	
EU053		Centrifuge #5		Tons whole stillage	0.089	
T604		Concentrate Tank ⁹³				
FS020		Wetcake Conveyor		Tons wetcake	0.089	
FS021		Wetcake Pad		Tons wetcake	0.089	
FS022		Wetcake Truck Loading		Tons wetcake	0.089	
EU026		Dryer Feed Conveyor		Tons wetcake	0.089	
T501		Thin Stillage Tank ⁹³				
T505		Syrup Storage Tank ⁹³				
T505A		Syrup Storage Tank ⁹³				
EU026	SV012 or SV006 for RTO bypass	DDGS Dryer, Natural gas fired ¹¹⁴	RTO (EU038, SV012) or	1000 gallons beer feed	0.119	

¹¹³ Emission factor sourced from March 2017 Performance Test on Centrifuge Vent #4. Test results were 1.2092 lb total VOC/hr at a production rate of 226.46 gal/min whole stillage. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test. The permittee requested to use this value for all units as shown in the table.

¹¹⁴ Emission factor sourced from November 2016 Performance Test on RTO. Test results were 1.6627 lb total VOC/hr at a production rate of 233.24 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb VOC/unit)	Monthly Emissions (tons) ³⁴
EU027	SV012 or SV006 for RTO bypass	DDGS Cooling/Separating Cyclone ¹¹⁵	uncontrolled RTO Bypass (SV006)	1000 gallons beer feed	0.089	
EU038	SV012	DDGS Dryer Regenerative Thermal Oxidizer, Natural gas fired ¹¹⁶	None	1000 gallons beer feed	0.119	
SV006	SV006	DDGS Dryer Regenerative Thermal Oxidizer Bypass ¹¹⁶	None	1000 gallons beer feed	0.119	
EU026	SV007	Dried Solids Conveyor ¹¹⁷		Tons DDGS	0.595	
EU043	SV001	DDGS Conveyor		Tons DDGS	0.595	
EU044	SV001	DDGS Loadout Conveyor		Tons DDGS	0.595	
EU008	SV001	DDGS Dump Auger		Tons DDGS	0.595	
EU029	FS003	DDGS Stockpile		Tons DDGS	0.595	
EU008	SV001	DDGS Pit A		Tons DDGS	0.595	
EU008	SV001	DDGS Pit B		Tons DDGS	0.595	
EU008	SV001	DDGS Pit C		Tons DDGS	0.595	
EU008	SV001	Screw Auger A		Tons DDGS	0.595	

¹¹⁵ Emission factor sourced from November 2016 Performance Test on Cooling cyclone. Test results were 1.2227 lb total VOC/hr at a production rate of 230 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

¹¹⁶ Emission factor sourced from November 2016 Performance Test on RTO. Test results were 1.6627 lb total VOC/hr at a production rate of 233.24 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

¹¹⁷ Emission factor back calculated from thermal oxidizer emission factor, assuming a conservative 80% overall VOC control from thermal oxidizer.

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb VOC/unit)	Monthly Emissions (tons) ³⁴
EU008	SV001	Screw Auger B		Tons DDGS	0.595	
EU008	SV001	Screw Auger C		Tons DDGS	0.595	
EU043	SV001	DDGS Pit Conveyor		Tons DDGS	0.595	
EU011	SV001	DDGS Screw Conveyor		Tons DDGS	0.595	
EU009	SV001	DDGS Elevator		Tons DDGS	0.595	
P802-A		Bulk Scale		Tons DDGS	0.595	
EU010	SV001	DDGS Rail Load Spout		Tons DDGS	0.595	
C800		Railcar Grate		Tons DDGS	0.595	
C800		Railcar Loading Pit		Tons DDGS	0.595	
EU012	SV001	DDGS Truck Load Spout		Tons DDGS	0.595	
C801		Truck Loading Grate		Tons DDGS	0.595	
C801		Truck Loading Pit		Tons DDGS	0.595	
FS003		Uncaptured DDGS		Tons DDGS	0.595	
Monthly VOC emissions from the sources in this table (tons)						
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :						
Total Monthly VOC Emissions of DDGS and WDGS Operations (tons) ¹¹⁸ :						

¹¹⁸ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 9: Emissions from Miscellaneous Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb VOC/unit)	Monthly Emissions (tons) ³⁴
EU045		Emergency Generator ¹¹⁹	None	MMCF	120.36	
FS002-1		LDAR Monitoring for Fugitive Emissions for units subject to Subpart VV		ppm		
FS002-2		LDAR Monitoring for Fugitive Emissions for units subject to Subpart VVa		ppm		
Monthly VOC emissions from the sources in this table (tons)						
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :						
Total Monthly VOC Emissions of Miscellaneous Operations (tons) ¹²⁰ :						

¹¹⁹ Emission factor sourced from AP-42, Section 3.2, Table 3.2-2 for 4 stroke lean burn engines. Converted using heating value of 1020 MMBtu/MMCF.

¹²⁰ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

VOC Emissions, Table 10: Summation of Plant Wide Monthly and Consecutive 12 Month Total VOC Emissions

Values Carried Forward From the Following VOC Emissions Calculation Tables	Monthly Emissions (tons)
Total Monthly VOC Emissions from VOC Emissions, Table 1: Emissions from Fermentation Operations (tons):	
Total Monthly VOC Emissions from VOC Emissions, Table 2: Emissions from Distillation Operations (tons):	
Total Monthly VOC Emissions from VOC Emissions, Table 3: Emissions from Ethanol Product Loading Operations, Part 1 (tons):	
Total Monthly VOC Emissions, Table 4: Truck and Railcar Loading of Denaturants at the Denaturant and Railcar Loading Racks	
Total Monthly VOC Emissions, Table 5: Truck and Railcar Loading of Ethanol (no denaturants in Truck, Railcar, Tanks, or Stingers) at Truck and Railcar Loading Racks	
Total Monthly VOC Emissions, Table 6: Truck and Railcar Loading of Ethanol after Denaturant Loading in Truck or Railcar	
Total Monthly VOC Emissions, Table 7: Controlled Truck Loading of Fuel Grade Ethanol at Truck Loading Rack	
Total Monthly VOC Emissions Table 8: Emissions from Wet Distillers Grains with Syrup (WDGS) and Dried Distillers Grains with Syrup (DDGS) Operations (tons):	
Total Monthly VOC Emissions Table 9: Emissions from Miscellaneous Operations (tons):	
Total Monthly VOC Emissions of all plant wide sources for this month (tons) ¹²¹ :	
Sum of Consecutive 12 month VOC Emissions of all plant wide sources (tons):¹²²	

¹²¹ Sum the monthly emissions of all sources listed in the table.

¹²² Consecutive 12 month total is calculated by summing the Total Monthly VOC Emissions of all plant wide sources for this month (tons) over the previous consecutive 12 month period. **Compliance is demonstrated when this value is less than 95.0 tons.**

Attachment Total HAPs Emissions

This attachment contains the compliance calculation methods for the plant wide HAPs emission limitation.

Total HAP Emissions, Table 1: Emissions from Fermentation Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb Total HAPs/unit)	Monthly Emissions (tons) ³⁴
EU014	SV004	Fermentation Tank #1 (T301)	¹²³ Fermentation Scrubber (C003)	1000 gallons beer feed	0.03	
EU015	SV004	Fermentation Tank #2 (T302)				
EU016	SV004	Fermentation Tank #3 (T303)				
EU051	SV004	Fermentation Tank #4 (T314)				
EU052	SV004	Fermentation Tank #5 (T315)				
EU017	SV004	Beer Well (T304)				
T231		Alpha Amylase Enzyme Tank ⁹³				
TA001		Anhydrous Ammonia Tank ⁹³				
T232		Gluc Amylase Enzyme Tank ⁹³				
Monthly Total HAP emissions from the sources in this table (tons)						
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :						
Total Monthly Total HAP Emissions of Fermentation Operations (tons) ¹²⁴ :						

¹²³ Emission factor sourced from March 2017 Performance Test on Fermentation Scrubber. Test results were 0.4640 lb total HAPs/hr at a production rate of 235 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

¹²⁴ Calculated as follows: Total Monthly Total HAPs Emissions from the sources listed in this table + SSM Emissions for this month.

Total HAP Emissions, Table 2: Emissions from Distillation Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb Total HAPs/unit)	Monthly Emissions (tons) ³⁴
EU023	SV005	Slurry Tank (T203)	¹²⁵ Distillation Scrubber (C004)	1000 gallons beer feed	0.045	
EU018	SV005	Beer Stripper Distillation Column (T401)				
EU019	SV005	Rectifier Distillation Column (T402)				
EU025	SV005	Rundown Tank (T406)				
EU020	SV005	Side Stripper Distillation Column (T403)				
EU021	SV005	Molecular Sieve Dehydration Vacuum Pump Vent (T421, T422, T423)				
EU024	SV005	Yeast Tank (T307)				
EU035	SV005	Industrial Extractive Distillation Column #1				
EU036	SV005	Industrial Rectifier Distillation Column #2				
EU037	SV005	Industrial Demthylization Distillation Column #3				
EU040	SV005	Industrial Distillation Recycle Rectifier Column #4 (T524)				
EU022	SV005	Evaporator				
T801		190 Proof Ethanol Storage Tank ⁹³	None			
T802		200 Proof Ethanol Storage Tank ⁹³				
T830		Impurities Storage Tank ⁹³	Distillation Scrubber (C004)			
T807		HQ 200 Ethanol Tank ⁹³				
T808		HQ 190 Ethanol Tank ⁹³				
T809		HQ 190 Ethanol Tank ⁹³				
EU046		HQ Heat Exchanger				

¹²⁵ Emission factor sourced from March 2017 Performance Test on Fermentation Scrubber. Test results were 0.6280 lb total HAPs/hr at a production rate of 235 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb Total HAPs/unit)	Monthly Emissions (tons) ³⁴
T802		200 Proof Ethanol Storage Tank ⁹³				
EU054		HQ Molecular Sieve Dehydration	Distillation Scrubber (C004)			
T802		200 Industrial Alcohol Tank ⁹³				
T806		HQ 200 Ethanol Tank ⁹³	None			
EU028		Boiler #1, Natural gas fired ¹²⁶		MMCF	1.89	
EU034		Boiler #2, Natural gas fired		MMCF	1.89	
FS005		¹²⁷ Cooling Tower #1		Lb/MMgal	1.2	
FS006		Cooling Tower #2		Lb/MMgal	1.2	
Monthly Total HAPs emissions from the sources in this table (tons)						
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :						
Total Monthly Total HAPs Emissions of Distillation Operations (tons) ¹²⁸ :						

¹²⁶ Emission factor sourced from AP-42, Section 1.4-2 for Natural Gas Combustion.

¹²⁷ Emission factor sourced from AP-42, Section 5.1., with 20% of volatiles also meeting the definition of HAPs.

¹²⁸ Calculated as follows: Total Monthly Total HAPs Emissions from the sources listed in this table + SSM Emissions for this month.

Total HAP Emissions, Table 3: Emissions from Ethanol Product Loading Operations, Part 1: Storage Vessels

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb Total HAPs/unit)	Monthly Emissions (tons) ³⁴
T802		Fuel Shift Tank ⁹³	None			
T421		Fuel Sieves Storage Tank ⁹³	Distillation Scrubber (C004)			
T422		Fuel Sieves Storage Tank ⁹³	Distillation Scrubber (C004)			
T423		Fuel Sieves Storage Tank ⁹³	Distillation Scrubber (C004)			
T803		Denatured Ethanol Fuel Grade Storage Tank ⁹³				
T805		Natural Gasoline Storage Tank ⁹³				
T810		Corrosion Inhibitor Storage Tank ⁹³				
T815		HQ 190 Ethanol Tank ⁹³				
T816		HQ 190 Ethanol Tank ⁹³				
T817		HQ 190 Ethanol Tank ⁹³				
T804		HQ 200 Ethanol Tank ⁹³				
T811		HQ 200 Ethanol Tank ⁹³				
FS008		Methanol Storage Tank ⁹³				
FS009		Tertiary Butyl Alcohol Storage Tank ⁹³				
FS010		Toluene Storage Tank ⁹³				
FS011		3-Chloropropyl Triethoxysilane, CPTEO, Storage Tank ⁹³				
FS012		Ethyl Acetate Storage Tank ⁹³				
FS013		Isopropyl Alcohol Storage Tote ⁹³				
FS014		Ethyl 3-Ethoxypropionate, 3-EOP, Storage Tote ⁹³				
FS016		Hexane Storage Barrel ⁹³				
Monthly Total HAPs emissions from the sources in this table (tons):						
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :						
Total Monthly Total HAPs Emissions of Ethanol Product Loading Operations, Part 1 (tons) ¹²⁹ :						

¹²⁹ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

Ethanol Product Loading Operations, Part 2: The following tables reflect the various loading operations and the emissions associated with each loading scenario.

Total HAP Emissions, Table 4: Truck and Railcar Loading of Denaturants at the Denaturant and Railcar Truck Loading Racks

Material Name	L _L (lb total HAPs/1000 gallons) ¹³⁰	Amount of Material Loaded this Month (1000 gallons)	Monthly Emissions (tons) ¹⁰⁰
Hexane	1.245		
Methyl Alcohol	0.5610		
Toluene	0.3482		
Total Truck and Railcar Denaturant Loading Total HAPs Emissions this Month (tons) ¹³¹ :			
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :			
Total Monthly Truck and Railcar Denaturant Loading Total HAPs Emissions this Month (tons) ¹³² :			

¹³⁰ Calculated using methodology found in AP-42 Section 5.2 and Section 7.1. Assumes S=0.6 for submerged loading of dedicated normal service trucks. Trucks are either dedicated normal service or clean cargo trucks, and all are submerged loaded. For CPTEO and EEP, compound specific data was not available. CPTE and EEP emissions used Distillate #2 as surrogates. Materials not listed in this table do not contain any HAPs according to the SDS submitted by the permittee.

¹³¹ Sum all monthly emissions in the table.

¹³² Calculated as follows: Total Monthly Total HAPs Emissions from the sources listed in this table + SSM Emissions for this month.

Total HAP Emissions, Table 5: Truck and Railcar Loading of Ethanol after Denaturant Loading in Truck or Railcar¹³³

Material Name	L _L (lb total HAPs/1000 gallons) ¹³⁴	Amount of Material Loaded this Month (1000 gallons)	Monthly Emissions (tons) ¹⁰⁰
Hexane	1.245		
Methyl Alcohol	0.5610		
Toluene	0.3482		
Total Truck and Railcar Denaturant then Ethanol Loading Total HAPs Emissions this Month (tons) ¹³⁵ :			
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :			
Total Monthly Truck and Railcar Denaturant then Ethanol Loading Total HAPs Emissions this Month (tons) ¹³⁶ :			

¹³³ When using denaturants, the denaturants are loaded into the trucks and railcars at the denaturant and railcar loading racks, then ethanol is loaded at the truck and railcar loading racks. To quantify emissions from this process, the permittee must determine the emissions from the loading of denaturants (see Total HAPs Emissions, Table 4) and also the emissions from the subsequent loading of ethanol (see Total HAPs Emissions, Table 5). Emissions from denaturants are expected at both loading operations. Materials not listed in this table do not contain any HAPs according to the SDS submitted by the permittee.

¹³⁴ Calculated using methodology found in AP-42 Section 5.2 and Section 7.1. Assumes S=0.5 for all materials to account for headspace saturated with denaturant vapors prior to ethanol loading. .

¹³⁵ Sum all monthly emissions in the table.

¹³⁶ Calculated as follows: Total Monthly Total HAPs Emissions from the sources listed in this table + SSM Emissions for this month.

Total HAP Emissions, Table 6: Controlled Truck Loading of Fuel Grade Ethanol at Truck Loading Rack¹³⁷

Material Name	Emission Factor (units)	Amount of Material throughput this Month (units)	Monthly Emissions (tons) ³⁴
Fuel Grade Ethanol Loading (Controlled)	0.0001 lb/1000 gallons	1000 gallons	
Flare	1.89 lb/MMCF	MMCF	
Total Controlled Fuel Grade Ethanol Truck Loading Total HAPs Emissions this Month (tons) ¹³⁸ :			
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :			
Total Monthly Controlled Fuel Grade Ethanol Truck Loading Total HAPs Emissions this Month (tons) ¹³⁹ :			

¹³⁷ Loading losses reflect a mixture of Ethanol, 2% by volume Natural Gasoline, 2% by volume Corrosion Inhibitor, and flare controls. Calculated using methodology found in AP-42 Section 5.2 and Section 7.1. Assumes S=0.6 for submerged loading of dedicated normal service trucks. Trucks are either dedicated normal service or clean cargo trucks, and all are submerged loaded.

¹³⁸ Sum all monthly emissions in the table.

¹³⁹ Calculated as follows: Total Monthly VOC Emissions from the sources listed in this table + SSM Emissions for this month.

Total HAP Emissions, Table 7: Emissions from Wet Distillers Grains with Syrup (WDGS) and Dried Distillers Grains with Syrup (DDGS) Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb Total HAPs/unit)	Monthly Emissions (tons) 34
		¹⁴⁰ Centrifuge Deck	None	Tons whole stillage	0.004	
EU050		Centrifuge #4		Tons whole stillage	0.004	
EU053		Centrifuge #5		Tons whole stillage	0.004	
T604		Concentrate Tank ⁹³				
FS020		Wetcake Conveyor		Tons wetcake	0.004	
FS021		Wetcake Pad		Tons wetcake	0.004	
FS022		Wetcake Truck Loading		Tons wetcake	0.004	
EU026		Dryer Feed Conveyor		Tons wetcake	0.004	
T501		Thin Stillage Tank ⁹³				
T505		Syrup Storage Tank ⁹³				
T505A		Syrup Storage Tank ⁹³				
EU026	SV012 or SV006 for RTO bypass	DDGS Dryer, Natural gas fired ¹⁴¹	RTO (EU038, SV012) or uncontrolled RTO Bypass (SV006)	1000 gallons beer feed	0.036	
EU027	SV012 or SV006 for RTO bypass	DDGS Cooling/Separating Cyclone		1000 gallons beer feed	0.036	
EU038	SV012	DDGS Dryer Regenerative Thermal Oxidizer, Natural gas fired	N/A	1000 gallons beer feed	0.036	
SV006	SV006	DDGS Dryer Regenerative Thermal Oxidizer Bypass	None	1000 gallons beer feed	0.036	

¹⁴⁰ Emission factor sourced from March 2017 Performance Test on Centrifuge Vent #4. Test results were 0.0495 lb total HAPs/hr at a production rate of 226.46 gal/min whole stillage. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test. The permittee requested to use this value for all units as shown in the table.

¹⁴¹ Emission factor sourced from November 2016 Performance Test on RTO. Test results were 0.5027 lb total HAPs/hr at a production rate of 233.24 gal/min beer feed. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test. Future testing shall quantify emissions from each emission unit separately.

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb Total HAPs/unit)	Monthly Emissions (tons) ³⁴	
EU026		Dried Solids Conveyor ¹⁴²		Tons DDGS	0.18		
EU043	SV001	DDGS Conveyor		Tons DDGS	0.18		
EU044	SV001	DDGS Loadout Conveyor		Tons DDGS	0.18		
EU008		DDGS Dump Auger		Tons DDGS	0.18		
EU029		DDGS Stockpile		Tons DDGS	0.18		
EU008	SV001	DDGS Pit A		Tons DDGS	0.18		
EU008	SV001	DDGS Pit B		Tons DDGS	0.18		
EU008	SV001	DDGS Pit C		Tons DDGS	0.18		
EU008	SV001	Screw Auger A		Tons DDGS	0.18		
EU008	SV001	Screw Auger B		Tons DDGS	0.18		
EU008	SV001	Screw Auger C		Tons DDGS	0.18		
EU043	SV001	DDGS Pit Conveyor		Tons DDGS	0.18		
EU011	SV001	DDGS Screw Conveyor		Tons DDGS	0.18		
EU009	SV001	DDGS Elevator		Tons DDGS	0.18		
P802-A		Bulk Scale		Tons DDGS	0.18		
EU010	SV001	DDGS Rail Load Spout		Tons DDGS	0.18		
C800		Railcar Grate		Tons DDGS	0.18		
C800		Railcar Loading Pit		Tons DDGS	0.18		
EU012	SV001	DDGS Truck Load Spout		Tons DDGS	0.18		
C801		Truck Loading Grate		Tons DDGS	0.18		
C801		Truck Loading Pit		Tons DDGS	0.18		
FS003		Uncaptured DDGS		Tons DDGS	0.18		
Monthly Total HAPs emissions from the sources in this table (tons)							
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :							
Total Monthly Total HAPs Emissions of DDGS and WDGs Operations (tons) ¹⁴³ :							

¹⁴² Emission factor back calculated from thermal oxidizer emission factor, assuming a conservative 80% overall HAPs control from thermal oxidizer.

¹⁴³ Calculated as follows: Total Monthly Total HAPs Emissions from the sources listed in this table + SSM Emissions for this month.

Total HAP Emissions, Table 8: Miscellaneous Operations

EP #	Stack ID	Description	Control Device	Monthly Throughput (units)	Emission Factor (lb Total HAPs/unit)	Monthly Emissions (tons) ³⁴
EU045		Emergency Generator ¹⁴⁴	None	MMCF	73.64	
FS002-1		LDAR Monitoring for Fugitive Emissions for Emission Units subject to Supbart VV		ppm		
FS002-2		LDAR Monitoring for Fugitive Emissions for Emission Units subject to Supbart VVa		ppm		
Monthly Total HAPs emissions from the sources in this table (tons)						
SSM Emissions from the sources listed in this table for this month (tons) ³⁶ :						
Total Monthly Total HAPs Emissions of Miscellaneous Operations (tons) ¹⁴⁵ :						

¹⁴⁴ Emission factor sourced from AP-42, Section 3.2, Table 3.2-2 for 4 stroke lean burn engines. Converted using heating value of 1020 MMBTU/MMCF.

¹⁴⁵ Calculated as follows: Total Monthly Total HAPs Emissions from the sources listed in this table + SSM Emissions for this month.

Total HAP Emissions, Section 2, Table 9: Summation of Plant Wide Monthly and Consecutive 12 Month Total HAPs Emissions

Values Carried Forward From the Following Total HAPs Emissions Calculation Tables	Monthly Emissions (tons)
Total Monthly Total HAPs Emissions from Table 1: Emissions from Fermentation Operations (tons):	
Total Monthly Total HAPs Emissions from Table 2: Emissions from Distillation Operations (tons):	
Total Monthly Total HAPs Emissions from Table 3: Emissions from Ethanol Product Loading Operations, Part 1 (tons):	
Total Monthly Total HAPs Emissions from Table 4: Truck and Railcar Loading of Denaturants at the Denaturant and Railcar Truck Loading Racks (tons):	
Total Monthly Total HAPs Emissions from Table 5: Truck and Railcar Loading of Ethanol after Denaturant Loading in Truck or Railcar (tons):	
Total Monthly Total HAPs Emissions from Table 6: Controlled Truck Loading of Fuel Grade Ethanol at Truck Loading Rack (tons):	
Total Monthly Total HAPs Emissions from Table 7: Emissions from Wet Distillers Grains with Syrup (WDGS) and Dried Distillers Grains with Syrup (DDGS) Operations (tons):	
Total Monthly Total HAPs Emissions from Table 8: Miscellaneous Operations (tons):	
Total Monthly Total HAPs Emissions of all plant wide sources for this month (tons) ¹⁴⁶ :	
Sum of Consecutive 12 month Total HAPs Emissions of all plant wide sources (tons):¹⁴⁷	

¹⁴⁶ Sum the monthly emissions of all sources listed in the table.

¹⁴⁷ Consecutive 12 month total is calculated by summing the Total Monthly Total HAPs Emissions of all plant wide sources for this month (tons) over the previous consecutive 12 month period. **Compliance is demonstrated when this value is less than 24.0 tons.**

Attachment Individual HAPs Emissions

This attachment is organized in two sections. Section 1 contains the compliance calculation methods for the emission units that emit HAPs. Section 2 contains the individual HAP emission factors for each emissions unit.

Section 1: Compliance calculation methods for emission units that emit HAPs:

Individual HAP Emissions, Section 1, Table 1: Calculation Method for Individual HAPs¹⁴⁸

Calculations for Individual HAP: _____ (HAP name)						
EP #	Stack ID	Description	HAP Control Device ¹⁴⁹	Monthly Throughput (units)	Emission Factor (lb/unit)	Monthly Emissions (tons) ³⁴
Sum of the values listed above (tons):						
SSM Emissions for this month for this individual HAPs from the sources listed in this table (tons) ³⁶ :						
Total Monthly Emissions of all plant wide sources for this month for this individual HAP (tons) ¹⁵⁰ :						
Sum of Consecutive 12 month emissions of all plant wide sources for this individual HAP (tons):¹⁵¹						

¹⁴⁸ The permittee shall create a table for each individual HAP. Each table shall contain all emission units for each individual HAP.

¹⁴⁹ Fuel Grade Ethanol Truck Loading operations are controlled by the Flare (EU039). The flare controls 98% of HAPs from the Denatured Ethanol Fuel Grade Storage Tank (T803). The flare creates combustion HAPs which do not receive a control reduction. All other control devices do not receive a control reduction in this table. All other HAP control devices were performance tested downstream of the control device and the emission factor already includes emission reductions from the control device.

¹⁵⁰ Calculated as the sum of all the Monthly Emissions for all units in the table + SSM Emissions.

¹⁵¹ Consecutive 12 month total is calculated by summing the Total Monthly Emissions over the previous 12 month period. **Compliance is demonstrated when this value is less than 9.0 tons.**

Section 2: Individual HAP Emission Factors: The permittee shall use the emission factors listed in this section to calculate individual HAP emissions in Section 1.

Natural Gas Combustion Units:

EP#:	Stack ID	Description
EU028		Boiler #1, Natural gas fired
EU034		Boiler #2, Natural gas fired
EU039		Open Flare, Natural gas fired
EU026	SV012 or SV006 for RTO bypass	DDGS Dryer, Natural gas fired
EU038	SV012	DDGS Dryer Regenerative Thermal Oxidizer, Natural gas fired

Individual HAP Name	Emission Factor (lb/MMSCF) ¹⁵²
Lead	0.0005
2-Methylnaphthalene	2.4E-5
3-Methylcholanthrene	1.8E-6
7,12-Dimethylbenz(a)anthracene	1.6E-5
Acenaphthene	1.8E-6
Acenaphthylene	1.8E-6
Anthracene	2.4E-6
Benz(a)anthracene	1.8E-6
Benzene	2.1E-3
Benzo(a)pyrene	1.2E-6
Benzo(b)fluoranthene	1.8E-6
Benzo(g,h,i)perylene	1.2E-6
Benzo(k)fluoranthene	1.8E-6
Chrysene	1.8E-6
Dibenzo(a,h)anthracene	1.2E-6
Dichlorobenzene	1.2E-3
Fluoranthene	3.0E-6
Fluorene	2.8E-6
Formaldehyde	7.5E-2
Hexane	1.8
Indeno(1,2,3-cd)pyrene	1.8E-6
Naphthalene	6.1E-4
Phenanthrene	1.7E-5
Pyrene	5.0E-6
Toluene	3.4E-3
Arsenic	2.0E-4
Beryllium	1.2E-5

¹⁵² Emission factors sources from AP-42, Tables 1.4-1, 1.4-3, and 1.4-4. To convert to lb pollutant/MMBtu, divide by 1020.

Individual HAP Name	Emission Factor (lb/MMSCF) ¹⁵²
Cadmium	1.1E-3
Chromium	1.4E-3
Cobalt	8.4E-5
Manganese	3.8E-4
Mercury	2.6E-4
Nickel	2.1E-3
Selenium	2.4E-5

EP#:	Stack ID	Description
EU045		Emergency Generator, Natural gas fired, spark ignition

Individual HAP Name	Emission Factor (lb/MMSCF) ¹⁵³
1,1,2,2-Tetrachloroethane	4.00E-5
1,1,2-Trichloroethane	3.18E-5
1,3-Butadiene	2.67E-4
1,3-Dichloropropene	2.64E-5
2-Methylnaphthalene	3.32E-5
2,2,4-Trimethylpentane	2.50E-4
Acenaphthene	1.25E-6
Acenaphthylene	5.53E-6
Acetaldehyde	8.36E-3
Acrolein	5.14E-3
Benzene	4.40E-4
Benzo(b)fluoranthene	1.66E-7
Benzo(e)pyrene	4.15E-7
Benzo(g,h,i)perylene	4.14E-7
Biphenyl	2.12E-4
Carbon Tetrachloride	3.67E-5
Chlorobenzene	3.04E-5
Chloroform	2.85E-5
Chrysene	6.93E-7
Ethylbenzene	3.97E-5
Ethylene Dibromide	4.43E-5
Fluoranthene	1.11E-6
Fluorene	5.67E-6
Formaldehyde	5.28E-2
Methanol	2.50E-3
Methylene Chloride	2.00E-5
n-Hexane	1.11E-3
Naphthalene	7.44E-5
PAH	2.69E-5

¹⁵³ Emission factors sources from AP-42, Table 3.2-2. To convert to lb pollutant/MMCF, multiply by 1020.

Individual HAP Name	Emission Factor (lb/MMSCF) ¹⁵³
Phenanthrene	1.04E-5
Phenol	2.40E-5
Pyrene	1.36E-6
Styrene	2.36E-5
Tetrachloroethane	2.48E-6
Toluene	4.08E-4
Vinyl Chloride	1.49E-5
Xylene	1.84E-4

Natural Gas Combustion Units that also have process emissions:

EP#:	Stack ID	Description
EU026	SV012 or SV006 for RTO bypass	DDGS Dryer, Natural gas fired
EU038	SV012	DDGS Dryer Regenerative Thermal Oxidizer, Natural gas fired
SV006	SV006	DDGS Dryer Regenerative Thermal Oxidizer Bypass

Individual HAP Name	Emission Factor (lb/1000 gallons beer feed) ¹⁵⁴
Acetaldehyde	0.0174
Acrolein	0.0059
Formaldehyde	0.0095
Methanol	0.0031

Units Controlled by Fermentation Scrubber:

EP#:	Stack ID	Description	Control Device
EU014	SV004	Fermentation Tank #1 (T301)	Fermentation Scrubber (C003)
EU015	SV004	Fermentation Tank #2 (T302)	
EU016	SV004	Fermentation Tank #3 (T303)	
EU051	SV004	Fermentation Tank #4 (T314)	
EU052	SV004	Fermentation Tank #5 (T315)	
EU017	SV004	Beer Well (T304)	

¹⁵⁴ Emission factors sourced from November 2016 Performance Test. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

Individual HAP Name	Emission Factor (lb/1000 gallons beer feed) ¹⁵⁵
Acetaldehyde	0.0325
Acrolein	0.0001
Formaldehyde	0.0002
Methanol	0.0001

Units Controlled by Distillation Scrubber:

EP#:	Stack ID	Description	Control Device
EU023	SV005	Slurry Tank (T203)	Distillation Scrubber (C004)
EU018	SV005	Beer Stripper Distillation Column (T401)	
EU019	SV005	Rectifier Distillation Column (T402)	
EU025	SV005	Rundown Tank (T406)	
EU020	SV005	Side Stripper Distillation Column (T403)	
EU021	SV005	Molecular Sieve Dehydration Vacuum Pump Vent (T421, T422, T423)	
EU024	SV005	Yeast Tank (T307)	
EU035	SV005	Industrial Extractive Distillation Column #1	
EU036	SV005	Industrial Rectifier Distillation Column #2	
EU037	SV005	Industrial Demthylization Distillation Column #3	
EU040	SV005	Industrial Distillation Recycle Rectifier Column #4 (T524)	
EU022	SV005	Evaporator	

Individual HAP Name	Emission Factor (lb/1000 gallons beer feed) ¹⁵⁶
Acetaldehyde	0.0444
Acrolein	0.00002
Formaldehyde	0.0001
Methanol	0.00001

¹⁵⁵ Emission factors sourced from May 2017 Performance Test. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

¹⁵⁶ Emission factors sourced from May 2017 Performance Test. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

Centrifuge Units:

EP#:	Stack ID	Description
		Centrifuge Deck
EU050		Centrifuge #4
EU053		Centrifuge #5

Individual HAP Name	Emission Factor (lb/1000 gallons beer feed) ¹⁵⁷
Acetaldehyde	0.003
Acrolein	0.0001
Formaldehyde	0.0001
Methanol	0.001

WDGS Handling Units:

EP#:	Stack ID	Description
FS020		Wetcake Conveyor
FS021		Wetcake Pad
FS022		Wetcake Truck Loading
EU026		Dryer Feed Conveyor

Individual HAP Name	Emission Factor (lb/tons wetcake)
Acetaldehyde	0.003
Acrolein	0.0001
Formaldehyde	0.0001
Methanol	0.001

¹⁵⁷ Emission factors sourced from May 2017 Performance Test. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test. Same values used for all wetcake sources.

DDGS Handling Units

EP#:	Stack ID	Description
EU027	SV012 or SV006 for RTO bypass	DDGS Cooling/Separating Cyclone

Individual HAP Name	Emission Factor (lb/1000 gallons beer feed) ¹⁵⁸
Acetaldehyde	0.0053
Acrolein	0.0007
Formaldehyde	0.0011
Methanol	0.0019

EP#:	Stack ID	Description
EU026		Dried Solids Conveyor
EU043	SV001 ¹⁵⁹	DDGS Conveyor
EU044	SV001	DDGS Loadout Conveyor
EU008		DDGS Dump Auger
EU029		DDGS Stockpile
EU008	SV001	DDGS Pit A
EU008	SV001	DDGS Pit B
EU008	SV001	DDGS Pit C
EU008	SV001	Screw Auger A
EU008	SV001	Screw Auger B
EU008	SV001	Screw Auger C
EU043	SV001	DDGS Pit Conveyor
EU011	SV001	DDGS Screw Conveyor
EU009	SV001	DDGS Elevator
P802-A		Bulk Scale
EU010	SV001	DDGS Rail Load Spout
C800		Railcar Grate
C800		Railcar Loading Pit
EU012	SV001	DDGS Truck Load Spout
C801		Truck Loading Grate
C801		Truck Loading Pit
FS003		Uncaptured DDGS

¹⁵⁸ Emission factors sourced from November 2016 Performance Test. If this unit undergoes future performance testing, the permittee shall use the value derived from the most recent approved performance test in all compliance calculations starting on the date of the most recent approved test.

¹⁵⁹ All emission units with Stack ID SV001 are controlled by Baghouse #1 (C001). The baghouse is not expected to provide emissions control of HAPs, therefore it is not listed as a control device in this attachment.

Individual HAP Name	Emission Factor (lb/ton DDGS ¹⁶⁰)
Acetaldehyde	0.087
Acrolein	0.0295
Formaldehyde	0.0475
Methanol	0.0155

Storage Tanks¹⁶¹:

EP#:	Stack ID	Description
T803		Denatured Ethanol Fuel Grade Storage Tank ¹⁶²

Individual HAP Name	Emission Factor (units)
N-Hexane	
Xylenes	
Toluene	
Benzene	
Cumene	

EP#:	Stack ID	Description
T805		Natural Gasoline Storage Tank

Individual HAP Name	Emission Factor (units)
N-Hexane	
Xylenes	
Toluene	
Benzene	

EP#:	Stack ID	Description
T810		Corrosion Inhibitor Storage Tank

Individual HAP Name	Emission Factor
Xylene	
Cumene	

¹⁶⁰ Back calculated from dryer emissions, assuming an overall 80% control.

¹⁶¹ The permittee shall use the SDS of the material and the calculation methodology contained in AP-42, Section 7.1 to calculate monthly emissions from all tanks. The permittee shall complete the data shown in the tables and use these values in compliance calculations. The permittee shall keep copies of all documentation (such as SDS), and all calculations and supporting documentation used to derive emission factors. If an SDS presents a range, the permittee shall use the largest value of the range.

¹⁶² This tank contains a mixture of ethanol, natural gasoline, and corrosion inhibitor.

EP#:	Stack ID	Description
FS008		Methanol Storage Tank

Individual HAP Name	Emission Factor (units)
Methanol	

EP#:	Stack ID	Description
FS010		Toluene Storage Tank

Individual HAP Name	Emission Factor (units)
Toluene	

EP#:	Stack ID	Description
FS016		Hexane Storage Barrel

Individual HAP Name	Emission Factor (units)
Hexane	

Other Units:

EP#:	Stack ID	Description
T231		Alpha Amylase Enzyme Tank
TA001		Anhydrous Ammonia Tank
T232		Glucose Amylase Enzyme Tank
P407-2		Heat Exchanger Pump
T801		190 Proof Ethanol Storage Tank
T802		200 Proof Ethanol Storage Tank
T830		Impurities Storage Tank
T807		HQ 200 Ethanol Tank
T808		HQ 190 Ethanol Tank
T809		HQ 190 Ethanol Tank
EU046		HQ Heat Exchanger
T802		200 Proof Ethanol Storage Tank
EU054		HQ Molecular Sieve Dehydration
T802		200 Industrial Alcohol Tank
T806		HQ 200 Ethanol Tank
FS005		Cooling Tower #1
FS006		Cooling Tower #2
T802		Fuel Shift Tank
T421		Fuel Sieves Storage Tank
T422		Fuel Sieves Storage Tank
T423		Fuel Sieves Storage Tank
T815		HQ 190 Ethanol Tank

T816		HQ 190 Ethanol Tank
T817		HQ 190 Ethanol Tank
T804		HQ 200 Ethanol Tank
T811		HQ 200 Ethanol Tank
T604		Concentrate Tank
T501		Thin Stillage Tank
EU022	SV005	Evaporator
T505		Syrup Storage Tank
T505A		Syrup Storage Tank
FS002		LDAR Monitoring for Fugitive Emissions

The permittee shall create an individual HAP tracking table (as below) for each unit shown in the table above. The permittee shall complete the data shown in the tables and use these values in compliance calculations. The permittee shall keep copies of all documentation (such as SDS), and all calculations used to derive emission factors. If an SDS presents a range, the permittee shall use the largest value of the range

Individual HAP Name	Emission Factor (units)

STATEMENT OF BASIS

Voluntary Limitations

In order to qualify for this Intermediate State Operating Permit, the permittee has accepted voluntary, federally enforceable emission limitations. Per 10 CSR 10-6.065(4)(C)1.A.(VI), if these limitations are exceeded, the installation becomes subject to 10 CSR 10-6.065(5) and enforcement action for operating without a valid part 70 operating permit. It is the permittee's responsibility to monitor emission levels and apply for a part 70 operating permit far enough in advance to avoid this situation. This may mean applying more than eighteen months in advance of the exceedance, since it can take that long or longer to obtain a part 70 operating permit.

Installation Description

Golden Triangle Energy is a denatured ethanol, beverage ethanol, and denatured industrial ethanol production facility. Current permits limit production to 21 million gallons of ethanol, maximum amount of denaturant of 1,050,000 gallons, with a total denatured ethanol production limit of 22,050,000 gallons per consecutive twelve month period. The installation has emission limitations on the following criteria pollutants to qualify for an Intermediate operating permit: PM₁₀, NO_x, VOC, CO, SO_x, and both individual and total HAPs.

Grain Operations:

Trucks arrive using the Truck Haul Road (FS004) and travel to the Grain Storage Building (EU029). The Grain Storage Building (EU029) is a multipurpose building that encloses corn unloading, DDGS storage, and DDGS truck loading operations. The building is a four walled structure with multiple bay doors for trucks, personnel doors, and building vent fans that exhaust to the atmosphere. Inside, the building is divided into halves by a low concrete wall separating the DDGS storage from the corn unloading area.

Straight and Hopper trucks enter the building and unload the corn over one of two grates that feed the Corn Dump Pit with Auger (EU001). The corn is transferred via an auger at the bottom of the pit to the Corn Elevator (EU002) and then to the Corn Bin Distribution Conveyor (EU041), which distributes the corn to one of the four corn storage bins, Corn Bin EU004, EU005, EU006, or EU007. The Corn Bin Unloading Conveyor (EU042) is located under the bins and transfers the corn back to the Corn Elevator (EU002), which then sends the corn to a Scalper (EU003) to remove debris such as pieces of cob or stalk that may be in the grain. The corn exists the Scalper (EU003) and is transferred via the Corn Bin Distribution Conveyor (EU041) to the Corn Bin (EU005). From the Corn Bin (EU005), the corn passes over a magnet to remove metal and is gravity fed through enclosed piping to the Hammermill (EU013). The hammermill grinds the corn into a flour or meal, which then passes through a sizing screen. The resulting corn flour is pneumatically conveyed to a filter receiver where the fabric captures the flour and the conveying air is discharged into the atmosphere.

Additionally, solids collected by Baghouse C001 exit the bottom of the baghouse onto the Dust Conveyor, which feeds the solids into the Corn Elevator (EU002), where it is combined with the corn from the Corn Dump Pit with Auger (EU001) to recycle the baghouse solids into the grain receiving process.

Fermentation Operations:

Then corn flour drops into the Slurry Tank (EU023) where water and enzymes are added to form a slurry. Much of the water is from recycled sources within the process including thin stillage from centrifuges, scrubber water, and evaporator condensate. Steam is added if necessary to bring the temperature up to 140 to 155°F. Alpha amylase enzyme is pumped from the Enzyme Tank T231 and added to convert the starch to dextrose. Ammonia is pumped from the Ammonia Tank TA001 and is added to raise the pH and as a nutrient to the yeast. The slurry is now called a mash mixture.

The mash passes through a steam jet cooker which rapidly raises the temperature to 230°F. The hot mash flows through the cook tube under pressure for about ten minutes. The time and temperature sterilizes the mash slurry. The cook tube discharges to the flash tank that has a vacuum pulled on it. The steam that is flashed out is reused in the side stripper distillation column. The mash cools to 190 to 195°F. Cooled mash then drops into the liquefaction tank.

Liquefaction is a series of three tanks that are designed to hold the mash slurry for two hours at 190°F. Additional alpha amylase enzyme is pumped from the Enzyme Tank T231 to liquefy and reduce the viscosity of the slurry.

Mash cooling occurs after liquefaction to lower the temperature for fermentation. The first cooling heat exchanger preheats the beer prior to distillation in order to recover energy and reduce steam use. The subsequent heat exchangers are cooled by cooling tower water. The final mash temperature target is 85 to 90°F for fermentation.

Fermentation consists of five fermentation tanks and one beer well. The mash from liquefaction is pumped to one of the five Fermenters (EU014, EU015, EU016, EU051 and EU052) where yeast is added from the Yeast Tank (EU024) and gluco amylase enzyme is added from the Gluco Amylase Enzyme Tank T232. It takes about 48 to 50 hours for the enzyme to convert the remaining sugars to glucose and for the yeast to convert the glucose to ethanol and carbon dioxide. For each pound of ethanol produced, about one pound of carbon dioxide is also produced. The carbon dioxide along with some ethanol vapors collect in the headspace of the fermenter where it is evacuated to the water scrubber for ethanol recovery and pollution control. Carbon dioxide passes through the scrubber and is discharged into the atmosphere. Excess heat generated in the fermenters is removed by heat exchanger tubes with recirculating cooling tower water. The ethanol content at the end of fermentation is approximately 14 to 18% by volume.

After fermentation, the resulting “beer” is pumped to a storage tank called the Beer Well (EU017). The purpose of this tank is to allow the batch fermentation process and the continuous distillation process to occur simultaneously without interruption. Before the next batch fill, the fermenter is cleaned with a spray ball chemical solution to prevent infections.

Distillation Operations:

Distillation is used to separate the ethanol from water and remaining spent mash solids. From the Beer Well (EU017), the product is pumped to the distillation process. The first distillation process is the Beer Stripper Distillation Column (EU018). The beer enters the top of the Beer Stripper Distillation Column (EU018) and drops over a series of splash trays as it drops to the bottom. At the bottom the ethanol has been boiled off and the remaining material is called whole stillage, which will eventually be turned into

DDGS. Exiting the top of the Beer Stripper Distillation Column (EU018) is a 50/50 mixture of ethanol and water vapors that flow to the Rectifier Distillation Column (EU019).

The Rectifier Distillation Column (EU019) concentrates the ethanol to 95% with 5% water remaining. This is called 190 proof ethanol. The 190 proof vapors exit the top of the Rectifier Distillation Column (EU019) and are condensed in a heat exchanger. Vapors from the heat exchanger are removed by the vacuum pump P426 and then sent to the Distillation Scrubber C004. The liquid 190 proof ethanol is pumped to the Rundown Tank (EU025). The liquid in the Rundown Tank (EU025) is pumped to fuel sieves T421, T422, T423 or 190 Proof Ethanol Storage Tank (T801). A small amount of fusel oil is drawn off the column at its mid-point. The fusel oil is added back into the 190 proof ethanol prior to dehydration when the fuel grade product is made.

The Side Stripper Distillation Column (EU020) is used to remove water from the bottoms of the Rectifier Distillation Column (EU019), which contain about 50% water. Ethanol is boiled off to the top of the Side Stripper Distillation Column (EU020) and the water is collected at the bottom of the column. The ethanol is returned to the Rectifier Distillation Column (EU019) and the water is recycled back into the Slurry Tank (EU023).

After passing through these three distillation columns, the 190 proof ethanol is pumped from Rectifier Distillation Column (EU019) to 190 Proof Ethanol Storage Tank (T801). Molecular sieve dehydration is used to remove the 5% water from the 190 proof ethanol for purification. Liquid 190 proof is pumped from the 190 Proof Ethanol Storage Tank (T801) to a steam driven vaporizer. The superheated vapor passes a bed of molecular sieve beads where the beads selectively absorb water due to the smaller size of water molecules compared to ethanol. The dehydrated ethanol is 200 proof or 100% ethanol. The 200 proof vapors are condensed in a heat exchanger and stored in the 200 proof tank T802. A molecular sieve bed is regenerated by passing steam through it in a counter flow direction and pulling a vacuum on the bed (EU021). The gases produced during regeneration are condensed to a 50% ethanol/water liquid. This liquid is pumped to the bottom of the Rectifier Distillation Column (EU019) for reprocessing.

At this point, the fuel grade ethanol process is complete. The finished fuel grade ethanol products is 200 proof Ethanol T802.

The industrial and beverage ethanol products require further purification, consisting of four additional distillation columns and additional molecular sieve dehydration.

The crude 190 Proof Ethanol is pumped from 190 Proof Ethanol Storage Tank (T801) to the Industrial Extractive Distillation Column #1 (EU035). The ethanol is distilled under a high dilution condition with water. The pressurized column collects the light boiling impurities such as propanols and fusel oils at the top of the column. The ethanol discharges from the bottom of the column. The top impurities are condensed to liquid and pumped to Impurities Storage Tank (T830), then to the Fuel Sieve Storage Tanks T803 to be incorporated into the fuel grade product during loading operations.

The Industrial Rectifier Distillation Column #2 (EU036) receives the ethanol bottoms from Industrial Extractive Distillation Column #1 (EU035). The atmospheric pressure column distills the approximately 12% ethanol near its azeotrope point near the top of the column. This ethanol is condensed and pumped to the Industrial Demethylization Distillation Column #3 (EU037). Impurities of fusel oil and propanols are removed through side draw ports on the column. These impurities are pumped to Impurities Storage

Tank (T830) then pumped to the fuel sieves T421, T422, T423 to be incorporated into the fuel grade product during loading operations.

The Industrial Demthylization Distillation Column #3 (EU037) receives the condensed ethanol from Industrial Rectifier Distillation Column #2 (EU036). The demethylizer column operates at atmospheric pressure to remove methanol. The resulting high quality ethanol is pumped from the bottom of the column to the Industrial Distillation Recycle Rectifier Column #4 (EU040) for additional purification. The top of the column discharges the methanol which is condensed in a series of condensers. Methanol impurities are blended into the fuel grade product.

Industrial Distillation Recycle Rectifier Column #4 (EU040) receives the ethanol bottoms from the Industrial Demthylization Distillation Column #3 (EU037) and acts as a vaporizer prior to molecular sieve dehydration. The ethanol leaving Industrial Distillation Recycle Rectifier Column #4 (EU040) is 95% ethanol (190 proof) and the impurities are sent to Impurities Storage Tank (T830) then fed to the fuel sieves T421, T422, T423. The ethanol product is stored in 190 Industrial Alcohol Tanks (T807, T808, T809, T815, T816 or T817) to await shipment or proceeds to Molecular Sieve Dehydration for additional purification.

After passing through these four distillation columns, the 190 proof ethanol is pumped from 190 Industrial Alcohol Tanks (T807, T808, T809, T815, T816 or T817). Molecular sieve dehydration is used to remove the 5% water from the 190 proof ethanol for purification. Liquid 190 proof is pumped from the 190 Industrial Alcohol Tanks (T807, T808, T809, T815, T816 or T817) to a steam driven vaporizer. The superheated vapor passes a bed of molecular sieve beads where the beads selectively absorb water due to the smaller size of water molecules compared to ethanol. The dehydrated ethanol is 200 proof or 100% ethanol. The 200 proof vapors are condensed in the HQ Heat Exchanger (EU046) and stored in the 200 Proof Ethanol Storage Tank T802. A molecular sieve bed is regenerated by passing steam through it in a counter flow direction and pulling a vacuum on the bed (EU054). The gases produced during regeneration are condensed to a 50% ethanol/water liquid. This liquid is pumped to the bottom of the Industrial Distillation Recycle Rectifier Column #4 (EU040) for reprocessing. This ethanol product is stored in 200 Industrial Alcohol Tanks (T802, T806 or T811) to await shipment.

The distillation processes require steam, which is provided by two boilers (EU028 and EU034). Boiler #1 (EU028) provides steam to Beer Stripper Distillation Column (EU018), the Rectifier Distillation Column (EU019), and the Side Stripper Distillation Column (EU020), with Cooling Tower #1 (FS005) receiving the steam from these distillation columns; and Boiler #2 (EU034) provides steam to the four industrial ethanol production columns: Industrial Extractive Distillation Column #1 (EU035), Industrial Rectifier Distillation Column #2 (EU036), Industrial Demthylization Distillation Column #3 (EU037), and Industrial Distillation Recycle Rectifier Column #4 (EU040), with Cooling Tower #2 (FS006) receiving the steam from these distillation columns..

Ethanol Product Loading Operations:

This installation produces three different types of ethanol: denatured fuel grade, High Quality (HQ) 190 ethanol, and HQ 200 ethanol. Denatured fuel grade ethanol is denatured in the storage tank and loaded into trucks only. Denatured HQ ethanol is denatured in the truck or railcar at the time of loading. HQ 190 GNS (grain neutral spirits) ethanol and HQ 200 GNS ethanol does not contain any denaturants. The loading operations consists of four distinct areas: ethanol staging tanks, denaturant tanks, truck loading operations, and railcar loading operations.

Ethanol Staging Tanks

Ethanol product from the Fuel Sieves Storage Tanks (T421, T422, T423) is pumped to the Fuel Shift Tank (T802). This is then pumped to the Denatured Ethanol Fuel Grade Storage Tank (T803), where it is combined with a denaturant, natural gasoline (pumped from Natural Gasoline Storage Tank (T805)), and corrosion inhibitor (pumped from Corrosion Inhibitor Storage Tank (T810)). Denatured Ethanol Fuel Grade Storage Tank (T803) holds this denatured fuel grade ethanol and stages it for loading.

Ethanol product from the HQ 190 Ethanol Tank (T809) is pumped to HQ 190 Ethanol Tanks (T815, T816, or T817), where it is staged for loading.

Ethanol product from the HQ 200 Ethanol Tank (T807) is pumped to HQ 200 Ethanol Tank (T804), where it is staged for loading.

Ethanol Product from the HQ 200 Ethanol Tank (T806) is pumped to HQ 200 Ethanol Tank (T811) where it is staged for loading.

Denaturants

For Denatured HQ ethanol, this installation adds varying recipes of nine different denaturants to the truck or railcar. These recipes are regulated by 27 CFR part 21 and require the denaturants to be added at specific ratios for specific product types. The denaturants are loaded at the denaturants loading skid located at the HQ Ethanol Rail Loading Rack (FS001-1).

The trucks are staged by the denaturant loading skid and the hose for the specific denaturant to be loaded is connected to the camlock connector at the bottom of the truck and loaded into the truck before ethanol is added. The denaturants are loaded first to allow mixing with the ethanol when the ethanol is loaded at the Truck Bottom Load Rack. They are loaded into trucks that are clean and contain no existing vapors.

The rail cars also receive denaturant at the denaturant loading skid. The denaturants are loaded by connecting the hose to a camlock connector on a pipe with a ball valve that enters the rail loading stinger above a check valve. Once the denaturant has been added, the ball valve on the denaturant pipe is closed and the ethanol is loaded and flushes the stinger in the process. The installation uses dedicated rail cars to control railcar emissions.

All denaturants for the Denatured HQ ethanol product are stored in tanks or totes located in the denaturant storage area. Denaturants include:

- Methanol Storage Tank (FS008),
- Tertiary Butyl Alcohol Storage Tank (FS009),
- Toluene Storage Tank (FS010),
- 3-Chloropropyl Triethoxysilane, CPTEO, Storage Tank (FS011),
- Ethyl Acetate Storage Tank (FS012),
- Isopropyl Alcohol Storage Tote (FS013),
- Ethyl 3-Ethoxypropionate, 3-EEP, Storage Tote (FS014), and
- Hexane Storage Barrel (FS016)

Truck Bottom Load Rack and Tote Loading

Truck Bottom Load Rack

Emissions from the Truck Bottom Load Rack are accounted for using two emission point due to different denaturants, and therefore different pollutants and emissions, are expected from the fuel grade and HQ ethanol product loading operations:

- Emission Point FS007-1 is used to account for the loading of denatured fuel grade ethanol into trucks from Tank T803.
- Emission Point FS007-2 is used to account for the loading of HQ ethanol products into trucks: HQ 190 Ethanol from Tanks T815, T816, or T817, and HQ 200 Ethanol from Tanks T804 and T811.

The Truck Bottom Load Rack consists of four different stingers, each with a capacity of 500 gallons/minute. However, due to equipment layout, only one truck can be loaded at any given time. There is one stinger for fuel grade and three stingers for HQ ethanol.

Fuel grade products (FS007-1) are loaded from a stinger fed by the 200 Denatured Ethanol Fuel Grade Storage tank (T803). All trucks loaded from this stinger have a vent line with a camlock fitting designed to capture vapors. These vapors are sent to an Open Flare (C007, EU039).

HQ ethanol products (FS007-2) are loaded from the following tanks and stingers:

- HQ 200 ethanol Tank (T804) feeds a stinger,
- HQ 190 ethanol Tanks (T-815, T-816, T-817) feed a single stinger, and
- HQ 200 ethanol Tank (T-811) feeds a stinger

The stinger is connected to the truck and product is pumped into the truck. Trucks exit the installation using the Truck Haul Road (FS004).

Railcar Loading

The HQ Ethanol Rail Loading Rack (FS001-1) controls emissions by only loading product into dedicated rail cars. The HQ Ethanol Rail Loading Rack (FS001-1) has one stinger with a loading capacity of 500 gallons per minute. This single stinger receives product from Tanks (T804, T811, T-815, T-816, and T-817). Due to equipment layout, only one rail car can be loaded at any given time. The HQ Ethanol Rail Loading Rack (FS001-1) is used to load HQ 190 GNS and HQ 200 GNS ethanol.

Dried Distillers Grain with Solubles (DDGS) Operations:

Thin Stillage:

As mentioned above, the Beer Stripper Distillation Column (EU018) step separates the ethanol from the water and solids in the beer stream. The remaining water and solids exit the bottom of the Beer Stripper Distillation Column (EU018), are called whole stillage, and will eventually be turned into DDGS. This whole stillage stream is sent to the Centrifuge Deck and flows into one of two centrifuges: Centrifuge #4 (EU050) or Centrifuge #5 (EU053). The horizontal decanting style centrifuges separate spent distillers wet grains called wet cake (WDGS) from the thin stillage liquid. Approximately 50% of the thin stillage liquid is recycled back to the Slurry Tank (EU023). The remaining 50% of the thin stillage liquid is pumped to the Concentrate Tank T604 for further processing.

Wet Cake:

The wet cake from centrifuges (EU050 and EU053) is transferred via Wetcake Conveyor FS020 to a concrete pad outside, Wetcake Pad FS021, to be sold as wet distillers grains (WDGS), or transferred via Dryer Feed Conveyor C602 to the DDGS Dryer (EU026).

Syrup:

The Concentrate Tank T604 feeds the Thin Stillage Tank (T501). The thin stillage is then pumped to the Evaporator (EU022). Evaporators use steam to boil off water vapor in the thin stillage. There are three stages of evaporation to concentrate the organic slurry with corn oil remaining after the water boils off. This product is called Condensed Distillers Solubles (CDS) or syrup. The syrup is sent to Syrup Storage Tanks (T505 and T505A) and is then sent to the DDGS Dryer (EU026) to be combined with the wetcake and become part of the DDGS product.

DDGS Dryer:

The natural gas, direct fired, rotary DDGS Dryer (EU026) receives wetcake from the Dryer Feed Conveyor C602 and syrup from the Syrup Storage Tanks (T505 and T505A) and removes moisture from these two products simultaneously. When dried, this product is called dried distillers grains with syrup (DDGS). The dryers directly exhaust to a group of four multiclone cyclones, DDGS Cooling/Separating Cyclone (EU027), which separate the spent grain solids from the hot moist air. The air with some organic vapors is treated in a natural gas fired DDGS Dryer Regenerative Thermal Oxidizer (EU038) prior to discharge to the atmosphere. As an alternative operating scenario, the RTO may be bypassed up to 50 hours per consecutive 12 month period, DDGS Dryer Regenerative Thermal Oxidizer Bypass EU026. The dried solids from the DDGS Dryer Multiclone Cyclones EU026 are transferred via Dried Solids Conveyor EU026 into the Grain Storage Building (EU029).

From the DDGS Cooling Cyclone (EU027), the DDGS is gravity fed through a pipe into the Grain Storage Building (EU029), and drops onto the DDGS Conveyor (EU043) where it passes through the DDGS Dump Auger EU008 and then free falls onto the DDGS Stockpile EU029. Underneath the DDGS Stockpile EU029 are three floor grates which feed three separate DDGS Pits (Pit A (EU029), Pit B (EU029), and Pit C (EU029)). Each pit has a screw auger (Screw Auger A (EU008), Screw Auger B (EU008), and Screw Auger C ((EU008))) which transfers the DDGS from the corresponding Pit onto the DDGS Pit Conveyor, (EU043), followed by DDGS Screw Conveyor (EU011). The DDGS Screw Conveyor (EU011) feeds the DDGS Elevator (EU009). The DDGS Elevator (EU009) feeds the Bulk Scale P802-A. From the Bulk Scale P802-A, the DDGS can either be directly dropped into a railcar by the DDGS Rail Load Spout (EU010); or can drop onto DDGS Truck Loading Conveyor (EU044) to transfer to the DDGS Truck Load Spout (EU012) and fall into a truck.

DDGS Railcar Loading:

Overflow from the DDGS Rail Load Spout (EU010) can pass into the Railcar Grate C801 that is located underneath the railcar. From the Railcar Grate C801, the DDGS falls onto Railcar Loading Pit C801, which drops the DDGS onto the DDGS Screw Conveyor (EU011) to be recycled into the loading process.

DDGS Truck Loading:

Overflow from the DDGS Truck Load Spout (EU012) can pass into the Truck Loading Grate C800 into the Truck Loading Pit C800 located under the grate. From the Truck Loading Pit C800, the DDGS free falls onto the DDGS Pit Conveyor, (EU043) to be recycled back into the loading process. Additionally,

the DDGS Truck Load Spout (EU012) can also divert product from loading into the truck and deposit the DDGS onto the DDGS Stockpile EU029 at Pit C EU029.

Loaded trucks exit the installation using the Truck Haul Road (FS004). Emissions of any uncaptured DDGS are accounted for using Uncaptured DDGS (FS003).

Miscellaneous Emission Sources:

The installation also has an emergency generator, EU045. Additionally, the installation is subject to LDAR monitoring under Subparts VV and VVa. The installation uses emission point FS002 to account for LDAR emissions.

Equipment Updates:

The following equipment changes have occurred since previous permits:

Table 19: Equipment changes since issuance of previous operating permit

Emission Unit #	Emission Point Description	Change
EU033	Hominy System	Removed from site. Baghouse C002 remains and now controls emissions from the Hammermill (EU013)
EU051	Fermenter #4 (Tank 314)	Constructed in 2016, not included in previous operating permit.
EU052	Fermenter #5 (Tank 315)	Constructed in 2016, not included in previous operating permit.
EU032	Beer Well #2	Permitted but never installed. The installation must undergo a construction permitting evaluation prior to installing.
EU047	Centrifuge #1	Removed from site in 2018.
EU048	Centrifuge #2	Removed from site in 2018.
EU049	Centrifuge #3	Removed from site in 2018.
EU050	Centrifuge #4	Constructed 2015, not included in previous operating permit.
EU053	Centrifuge #5	Constructed in 2018, not included in previous operating permit.
T812	Storage Tank	Permitted but never installed. The installation must undergo a construction permitting evaluation prior to installing.
T813	Storage Tank	Permitted but never installed. The installation must undergo a construction permitting evaluation prior to installing.
T814	Storage Tank	Permitted but never installed. The installation must undergo a construction permitting evaluation prior to installing.
EU045	Emergency Generator	Not included in previous operating permit.

Emission Unit #	Emission Point Description	Change
FS001-1	Railcar Loading Rack	This rack contains two stingers. One stinger is operational as explained in the Railcar Loading section above. There is another stinger, which received fuel grade product from Tank 803. This stinger is decommissioned in place because it does not vent to the flare. If the permittee wishes to use this stinger, a construction permit may be required.

Table 20: Emissions Profile, tons per year

Pollutants	Reported Emissions					Potential Emissions
	2014	2015	2016	2017	2018	
Particulate Matter ≤ Ten Microns (PM ₁₀)	20.13	18.60	20.99	19.63	15.57	<95.0
Particulate Matter ≤ 2.5 Microns (PM _{2.5})	3.70	4.30	16.59	14.48	11.35	<95.0 ¹⁶³
Sulfur Oxides (SO _x)	0.55	0.45	0.47	1.93	1.61	<95.0
Nitrogen Oxides (NO _x)	31.43	44.76	39.70	42.40	24.35	<95.0
Volatile Organic Compounds (VOC)	48.85	56.34	51.02	66.41	83.14	<95.0
Carbon Monoxide (CO)	6.14	6.01	4.67	8.83	6.99	<95.0
Hazardous Air Pollutants (HAPs)	6.52	6.51	7.53	2.94	8.63	<9.0 individual and <24.0 total

Permit Reference Documents

These documents were relied upon in the preparation of the operating permit. Because they are not incorporated by reference, they are not an official part of the operating permit.

1. Intermediate Operating Permit Application, received May 30, 2014; revised September 26, 2019;
2. 2018 Emissions Inventory Questionnaire, received March 25, 2019;
3. U.S. EPA document AP-42, *Compilation of Air Pollutant Emission Factors*; Volume I, Stationary Point and Area Sources, Fifth Edition;
4. webFIRE;
5. All documents listed in Construction Permit History; and
6. Consent Decree for Civil Action # 05-6032-CV-SJ-SOW.

¹⁶³ The PM_{2.5} potential emissions are indirectly limited by the PM₁₀ emission limitation .

Applicable Requirements Included in the Operating Permit but Not in the Application or Previous Operating Permits

In the operating permit application, the installation indicated they were not subject to the following regulation(s). However, in the review of the application, the agency has determined that the installation is subject to the following regulation(s) for the reasons stated.

See Other Regulatory Determinations

Other Air Regulations Determined Not to Apply to the Operating Permit

The Air Pollution Control Program has determined the following requirements to not be applicable to this installation at this time for the reasons stated.

See Other Regulatory Determinations

Construction Permit History

The following construction permits were issued to this installation:

1. Construction Permit #022000-004
This permit was issued January 28, 2000 to authorize the construction of the initial installation of an 18 million gallon per year denatured ethanol plant. This permit contains 15 special conditions. This permit was superseded by Construction Permit #072008-001.
2. Construction Permit #052003-014
This permit was issued April 29, 2003 for the installation of 2-34,500 gallon capacity ethanol storage tanks. Each tank is 14 ft diameter, 30 feet tall, and are equipped with a fixed cone roof. This permit was superseded by Construction Permit #072008-001.
3. No Permit Required, Project #2004-04-056
This determination was for the installation of a new receiver for corn flour. Corn flour will be loaded into this receiver from rail and then feed into an existing line. The maximum hourly design rate of this receiver is 7.3 tons/hour. The emissions from this equipment will be collected by a baghouse rated at 99% control efficiency. Using an emission factor for a similar process (SCC 30200817), the maximum controlled emission rate is estimated at 0.000876 lbs PM10/hr. The installation of this receiver does not increase the production of the plant. This value is less than the permitting threshold, therefore no construction permit is required.
4. Construction Permit #052003-014A
This construction permit amendment, completed June 13, 2005, is to process an extension request and a proposed change in the tank sizes of Tank T806 and T807 originally permitted in Construction Permit #052003-014. This amendment states the extension request could not be granted because of the proposed changes to the tanks; and explains a permit modification is required. In lieu of a modification of Construction Permit #052003-014, the installation obtained a new permit, Construction Permit #072005-033. This permit amendment was superseded by Construction Permit #072008-001.

5. Construction Permit #072005-033
This permit was issued July 18, 2005 for the installation of 4-39,725 gallon capacity ethanol tanks (T806, T807, T808, and T809). This permit was superseded by Construction Permit #072008-001.
6. Construction Permit #122005-004
This permit was issued December 6, 2005 for the addition of Industrial Alcohol Production equipment, including three distillation columns, a boiler, a cooling tower, four storage tanks, and control equipment. This permit also increases the annual throughput to 21 million gallons per year of ethanol and 1.05 million gallons per year of denaturant. This permit incorporates the conditions of the Consent Decree for Civil Action # 05-6032-CV-SJ-SOW. This permit was superseded by Construction Permit #072008-001.
7. Construction Permit #072008-001
This permit was issued July 2, 2008 for the addition of equipment; revisions to the NOx cap emission factors; as well as revisions to particulate emissions from the haul roads, cooling tower, grain receiving, and DDGS operations. This permit also incorporates the conditions of the Consent Decree for Civil Action # 05-6032-CV-SJ-SOW

This permit contains 20 Special Conditions, as detailed below:

- A. Special Condition #1: This condition states this permit supersedes all special conditions found in the following permits. This Special Condition is administrative and not included in this operating permit.
 - i. Construction Permit #022000-004.
 - ii. Construction Permit #052003-014.
 - iii. Construction Permit #052003-014A.
 - iv. Construction Permit #072005-033, and
 - v. Construction Permit #122005-004.
- B. Special Condition #2: This permit condition requires stack testing to determine acrolein emission rates; and modeling for all installation wide sources of acrolein to demonstrate compliance with the Risk Assessment Levels (RAL). The requirements of this permit condition have been satisfied, therefore it does not appear in the operating permit.

In the modeling analysis document titled “Ambient Air Quality Impact Analysis (AAQIA) for Golden Triangle Energy, LLC”, dated April 1, 2009, the Program explains the project and modeling methods in depth. The recommendations of the report are as follows:

1. The point source emission rates contained in the following table should not be exceeded:

Table 21: Acrolein emission rates

Stack ID	Description	24 hr Acrolein Emission Rate (lbs/hr)
SV007	Cooling Cyclone	6.60×10^{-3}
SV012	Dryer/RTO Stack	1.19×10^{-2}

Stack ID	Description	24 hr Acrolein Emission Rate (lbs/hr)
SV005	Distillation Scrubber	1.1397×10^{-3}
SV004	Fermentation Scrubber	4.4303×10^{-3}

2. If significant alterations are made to the facility design as proposed in the construction permit application, the installation is required to submit an updated air quality analysis in order to ensure continued compliance with the air quality standards.
- C. Special Conditions #3 and 4: This permit condition applies to the paved haul roads, and requires partial paving, maintenance of pavement, pavement washing/cleaning, and application of dust suppressant to unpaved sections to control fugitive emissions. In August 2018, the Program received Construction Permit Amendment Application asserting that all haul roads are paved, which is currently under review. Until these special conditions are modified, they are considered applicable requirements and must appear in the operating permit.
- D. Special Condition #5: This permit condition establishes a grain receiving throughput limit of 217,778 tons of grain per year by truck and rail receiving operations.
- E. Special Condition #6: This permit condition establishes a DDGS shipment throughput limit of 78,840 tons of DDGS shipped per year by truck and rail.
- F. Special Condition #7: This permit condition establishes a denatured ethanol production rate of 22,050,000 gallons per consecutive 12 month period. This condition states total ethanol production remains at 21,000,000 gallons, with a maximum amount of denaturant of 1,050,000 gallons.
- G. Special Condition #8: This permit condition establishes various emission limitations as detailed below:
- i. Subpart A establishes PM₁₀ emission limitations on a stack, equipment, and plant wide basis.
 - ii. Subpart B establishes NO_x emission limitations on a stack, equipment and plant wide wide basis. It also establishes an alternate operating scenario for the thermal oxidizer.
 - iii. Subpart C establishes HAP emissions limitations on a plant wide basis for total HAPs, individual HAPs, and a specific acrolein limitation. Subpart C.3. establishes a plant wide limit on acrolein emissions and has been superceded in Construction Permit Amendment #072008-001A. All other provisions of Subpart C still apply.
 - iv. Subpart D establishes VOC emission limitations on an equipment and plant wide wide basis.
 - v. Subpart E establishes CO emission limitations on a stack and plant wide basis.
 - vi. Subpart F establishes SO_x emission limitations on a plant wide basis and restricts the installation to only combusting natural gas.
- H. Special Condition #9: This permit condition establishes capture efficiency requirements on the grain handling and process equipment (EU001 to EU012), the grain milling

- equipment (EU013), and the Grain Storage Building which contains the DDGS Storage Pile, the DDGS Truck Loading, and Corn Dump Pits.
- I. Special Condition #10: This permit condition establishes operational requirements on Baghouse #1 (C001) and Baghouse #2 (C002).
 - J. Special Condition #11: This permit condition establishes operational requirements on the Fermentation Scrubber (C003) and the Distillation Scrubber (C004).
 - K. Special Condition #12: This permit condition establishes operational requirements on the Flare (FS001).
 - L. Special Condition #13: This permit condition establishes operational requirements on the Cooling/Separating Cyclone (EU027). This condition also establishes HAP and VOC emission limitations on the Cooling/Separating Cyclone (EU027). Special Condition #13.B. contains duplicative performance testing requirements as Special Condition #15.A.2.); therefore only Special Condition #15.A.2.) appears in the operating permit as a streamlining measure.
 - M. Special Condition #14: This permit condition establishes operational requirements on the Regenerative Thermal Oxidizer. This condition also establishes an alternative operating scenario.
 - N. Special Condition #15: This permit condition establishes performance testing requirements for the various limits in the permit. Subpart A.7) establishes SO₂ testing requirements on the Fermentation Scrubber (C003), Distillation Scrubber (C004), and the RTO for the DDGS Dryer (EU026 and EU038) and has been superseded in Construction Permit Amendment #072008-001B. The amendment restates the testing requirements for the RTO for the DDGS Dryer (EU026 and EU038). All other provisions of this Special Condition still apply.
 - O. Special Condition #16: This permit condition establishes proposed test plan and final test plan procedures for the performance testing.
 - P. Special Condition #17: This permit condition establishes operational requirements on the Cooling Tower #1 (FS005) and Cooling Tower #2 (FS006).
 - Q. Special Condition #18: This permit condition requires the installation to apply for an Intermediate operating permit.
 - R. Special Condition #19: This permit condition contains recordkeeping requirements.
 - S. Special Condition #20: This permit condition contains reporting requirements.
8. Construction Permit Amendment #072008-001A
- This amendment was issued on June 12, 2009. Construction Permit #072008-001, Special Condition #2 required the installation to conduct stack testing and ambient air quality modeling to revise the acrolein emission factor. Special Condition #8.C.3. limiting the amount of acrolein emissions, was subject to revision based on the outcome of the stack testing.

Using the results of the stack testing, annual potential emissions of acrolein are estimated as shown in the following table:

Table 22: Acrolein emissions table for Construction Permit #072008-001A

EU#	Description	Emission Rate (lb/hr)	PTE (tons/yr)
SV-007	Cooling Cyclone	0.00276	0.0120
SV-012	Dryer/RTO Stack	0.004987	0.0218
SV-005	Distillation Stack	0.004776	0.0209
SV-004	Fermenter Scrubber	0.001856	0.0081
Total Plantwide Acrolein:			0.0628

Because the potential emissions are greater than the Screen Model Action Level (SMAL), modeling was required and the ambient air quality analysis was revised using refined modeling. The following table shows the revised modeling results:

Table 23: Acrolein modeling Results for Construction Permit #072008-001A

Pollutant	Modeled Impact ($\mu\text{g}/\text{m}^3$)	RAL ($\mu\text{g}/\text{m}^3$)	Time Period
Acrolein	0.1199	0.55	24-hour
	0.014	0.02	Annual

The potential emissions are much less than the limit established in the construction permit, and the revised refined modeling demonstrates compliance with the RAL standards. Therefore, the limit established in Special Condition #8.C.3. is no longer required to prevent the installation from exceeding the RAL limits and is superceded.

9. No Permit Required, Project #2010-04-022

This determination was for the construction of a new 270,000 gallon capacity storage tank (T815) for the storage of 190 proof beverage grade ethyl alcohol and the use of four new denaturants:

- A. Storage Tote, Industrial Denaturant -Isopropyl Alcohol, received and dispensed by 320 gallon totes FS013
- B. Storage Tote, Industrial Denaturant -ethyl 3-ethoxypropionate, 3-EEP, received and dispensed by 320 gallon totes, FS014
- C. Storage Tote, Industrial Denaturant-Methyl Isobutyl Ketone, received and dispensed by 320 gallon totes, FS015 (subsequently removed)
- D. Storage Tote, Industrial Denaturant-Hexane, received and dispensed by 320 gallon totes, FS016 (subsequently changed to Hexane Storage Barrel)

All these denaturants are considered volatile organic compounds, with methyl isobutyl ketone and hexane also listed as hazardous air pollutants. The denatured ethyl alcohol is loaded out by either truck or railcar. Railcar is considered the worst case for potential emission purposes.

The potential uncontrolled VOC emissions for the new storage tank and railcar loading of denaturants are estimated to be 0.78 tons/year; methyl isobutyl ketone emissions are estimated to be 0.28 tons/year; and hexane emissions are estimated to be 0.03 tons/year. These values are less than the permitting thresholds for VOC and HAPs, therefore no construction permit is required.

10. Construction Permit Amendment #072008-001B

This amendment was issued on September 12, 2013. Construction Permit #072008-001, Special Condition #15.A.7. required stack testing to determine the sulfur dioxide (SO₂) emissions from the stacks of the Fermentation Scrubber (SV004, C003) and the Distillation Scrubber (SV005, C004). According to the permit, the test is performed once every five years and the results used to track SO₂ emissions for compliance with the plant wide SO_x emissions limitation established in Construction Permit #072008-001, Special Condition #8.F.1). The initial testing performed in 2008 indicated the following results:

Table 24: SO_x emission Rates for Construction Permit 072008-001B

EU #	Description	Emission Rate (lb/hr)	Emission Rate (tons/year)
SV-005	Distillation Scrubber	0.002	0.009
SV-004	Fermentation Scrubber	0.02	0.088

While future testing may show some variations, at such low levels, it is unlikely that these variations would increase the actual SO_x emissions above the plant wide limit. The results from the 2008 testing, as shown above, shall be used to track emissions for SO_x limit compliance. This amendment only eliminates the SO₂ testing requirement for the Distillation Scrubber (SV005) and the Fermentation Scrubber (SV004). Therefore, this amendment supersedes Construction Permit #072008-001, Special Condition #15.A.7). and restates the requirement for SO₂ testing of the Regenerative Thermal Oxidizer (RTO) used to control emission from the Dried Distillers Grain with Solubles (DDGS) Dryer. The testing condition for the RTO is not eliminated but is restated as part of this amendment.

This amendment contains two Special Conditions as detailed below:

- A. Special Condition #1: This permit condition supercedes Construction Permit #072008-001, Special Condition #15.A.7).
- B. Special Condition #2: This permit condition requires SO₂ testing of the Regenerative Thermal Oxidizer (RTO) used to control emission from the Dried Distillers Grain with Solubles (DDGS) Dryer (EU026, EU038) in accordance with Construction Permit #072008-001. This condition appears in the Operating Permit within Permit Condition 072008-001.

11. Construction Permit #092015-013

This permit was issued September 22, 2015 for the installation of a new centrifuge (EU050) and three backup centrifuges (EU047, EU048, EU049). The centrifuges are used to separate thin stillage and corn oil from the spent corn mash. The corn mash is then sent to the dryer for processing into DDGS or sent to an outdoor concrete pad to be sold as WDGS. The thin stillage flows to the thin stillage storage tank, and is then pumped to the evaporators. The evaporators remove water, leaving a condensed syrup product. The syrup is mixed with WDGS for animal feed.

The new larger centrifuge (EU050) has a MHDR of 13,800 gallons per hour. The three smaller centrifuges are used only when the larger unit is undergoing maintenance. The installation continues to operation under the overall production limit of 22.05 million gallons per year, established in Construction Permit #072008-001. The emissions from the centrifuges are uncontrolled.

This permit contains two special conditions as detailed below:

- A. Special Condition #1: This permit condition requires stack testing on centrifuge (EU050) by performing EPA test method 320, or other preapproved test method, on the new centrifuge system while the plant is fully operational. A pre-survey of the centrifuge using method 18, method 207, or other preapproved test method, will be required in order to ensure the ratio of the known VOC peak area to the total VOC peak area is at least 95%. Formaldehyde, acetaldehyde, acrolein, and methanol must be considered in the EPA method 320 performance test. This test shall be conducted within 10% of the maximum hourly production rate. The results from the stack tests shall be used to calculate actual and potential HAP and VOC emissions as mass.. This permit condition also contains requirements concerning testing timelines, submission of proposed testing plans, and submissions of performance test results.
- B. Special Condition #2: This permit condition contains requirements that the installation must follow if the stack testing results indicate any of the following:
 - i. A maximum hourly VOC emission rate equal to or greater than 2.75 lb VOC per hour;
 - ii. A maximum hourly individual HAP emission rate equal to or greater than 0.5 lb HAP per hour; or
 - iii. A maximum annual individual HAP emission rate greater than or equal to the individual HAP SMAL value

The installation performed the required testing March 28-29, 2017. The test results are detailed in the following table:

Table 25: March 2017 Performance Test Data for Centrifuge EU050

Pollutant	Tested Emission Rate at 226.46 gpm whole stillage (lb/hr)	Emission Rate Scaled to 257 gpm whole stillage (lb/hr) and (ton/year) for HAPs	Regulatory Thresholds (lb/hr, HAP SMAL ton/yr)
SOx	0.0039	0.004	2.75 lb/hr
VOC	1.21	1.37	2.75 lb/hr
acetaldehyde	0.037	0.04 lb/hr, 0.18 tons/yr	0.5 lb/hr, 9 tons/yr
acrolein	0.0009	0.001 lb/hr, 0.004 tons/yr	0.5 lb/hr, 0.04 tons/yr
methanol	0.0102	0.01 lb/hr, 0.04 tons/yr	0.5 lb/hr, 10 tons/yr
formaldehyde	0.0013	0.001 lbs/hr, 0.004 tons/yr	0.5 lb/hr, 2 tons/yr
Total HAPs	0.0494	0.06	N/A

Because the potential emission rates at the scaled processing rate of 257 gallons/minute of whole stillage are less than the regulatory thresholds, no additional action is required by this construction permit. This construction permit contained one time testing requirements which have been satisfied, therefore this construction permit does not appear as a permit condition in the operating permit.

12. No Permit Required, Project #2015-08-019

This determination was for the installation of a new submerged top railcar loading system (FS001) and two ethanol storage tanks. The new submerged top ethanol loading system will replace the existing railcar loading system. The maximum railcar ethanol loading rate of 13,000,000 gallons/year and production limit of 21,000,000 gallons are as established in Construction Permit #072008-001. The remainder is loaded into trucks using a separate loading system. Potential emissions are estimated to be 0.86 lbs VOC/hour, which is less than the permitting thresholds. Therefore, no construction permit is required.

13. Construction Permit #072016-004

This permit was issued July 14, 2016 to authorize the construction of two new fermentation tanks, (T314 and T315). These tanks are in addition to the three existing permitted fermentation tanks (T301, T302, and T303), and will work in parallel with them to produce existing product types. All five tanks are permitted to operate simultaneously. Fermentation is bottlenecked by the downstream distillation column to approximately 257 gallons per minute. The two additional fermentation tanks will not cause an increase in potential annual production, nor regain lost capacity. The tanks are being installed to maintain capacity while the existing tanks undergo maintenance. The additional tanks may allow a longer fermentation time and allow more alcohol to be produced from the same amount of grain, or conversely allow the same amount of alcohol to be produced from a lesser amount of grain. As the fermentation tanks are bottlenecked, it is believed the same amount of alcohol will be produced from a lesser amount of grain. The existing and new fermentation tanks are controlled by the Fermentation Scrubber, C003. Due to the bottleneck, this project does not increase the potential emissions of the installation.

This permit contains two special conditions.

- A. Special Condition 1.A. requires controlling the emissions from the two new fermentation tanks with the existing Fermentation Scrubber (C003).
- B. Special Condition 1.B. requires the installation to comply with all existing special conditions pertaining to the Fermentation Scrubber (C003) contained in Construction Permit 072008-001.

14. No Permit Required, Project #2018-02-050

This determination, dated May 11, 2018 is for the replacement of three existing centrifuges (EU047 Centrifuge #1, SV013; EU048 Centrifuge #2, SV014; and EU049, Centrifuge #3, SV015) with one new larger centrifuge (EU053 Centrifuge #5, SV016). The three existing centrifuges were installed in 1999 without a construction permit. The three existing passive non-powered 8 inch vent stacks (SV013, SV014, and SV015) will be retained as back-up for maintenance down times. The new centrifuge will vent to an existing 12 inch diameter stack (SV-16) that is powered by a fan.

Prior to this project, whole stillage leaving the beer stripper entered EU047 Centrifuge #1, EU048 Centrifuge #2, EU049 Centrifuge #3, or EU050 Centrifuge #4. Centrifuge #4 (EU050) was constructed in 2015 under Construction Permit 092015-013. After this project, whole stillage leaving the beer stripper will enter EU050 Centrifuge #4 and/or EU053 Centrifuge #5. The centrifuges separate the whole stillage into thin stillage and wet cake. Both EU050 Centrifuge #4 and EU053 Centrifuge #5 vent to SV016. The maximum design rate of EU053 Centrifuge #5 is 300 gallons per minute of whole stillage; however, use of EU050 Centrifuge #4

and EU053 Centrifuge #5 is bottlenecked to 257 gallons per minute as the beer stripper cannot process more than 257 gallons per minute.

Emissions from EU053 Centrifuge #5 were determined using stack testing results conducted on the similar EU050 Centrifuge #4 in March 2017. The stack testing was required by Construction Permit 092015-013. To determine the potential emissions from EU053 Centrifuge #5, the stack tested emission rates from EU050 Centrifuge #4 were scaled up to the bottlenecked capacity of 257 gpm. Potential emissions are estimated to be 0.86 lbs VOC/hour, which is less than the permitting thresholds. Therefore, no construction permit is required.

The March 2017 stack testing event was not coordinated with the Air Program, and Air Program staff did not have the opportunity to review testing protocols or attend the testing event. The March 2017 testing report was submitted as part of this permit determination project. On October 17, 2018, the Air Program mailed a Letter of Warning to communicate concerns regarding the testing data and results. The Air Program is continuing to address concerns with the permittee. If retesting is deemed necessary, the determination made in this permit project may require further review.

New Source Performance Standards (NSPS) Applicability

40 CFR part 60 Subpart A, General Provisions

The installation is required to comply with the provisions for flares contained in this regulation. These provisions are included in the operating permit.

40 CFR part 60, Subpart Dc, Standards of Performance for Small Industrial-Commercial-Institutional Steam Generating Units

This regulation applies to Boiler #1 and Boiler #2. Both boilers combust solely natural gas, therefore only the provisions for natural gas combustion are included in the permit.

40 CFR part 60 Subpart DD, Standards of Performance for Grain Elevators

This regulation does not apply because the installation has permanent grain storage capacity less than 88,100 m³ (2.5 million U.S. bushels).

40 CF part 60 Subpart III, Standards of Performance for VOC Emissions from SOCMi Air Oxidation Unit Processes,

40 CFR part 60 Subpart NNN, Standards of Performance for VOC Emissions from SOCMi Distillation Operations, and

40 CFR part 60, Subpart RRR, Standards of Performance for VOC Emissions from SOCMi Reactor Processes

These regulations do not apply to manufacturing facilities that produce ethanol (fuel or beverage grade) through fermentation (biological synthesis).

40 CFR part 60 Subpart VV, Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006

This regulation applies and is included in the operating permit.

40 CFR part 60 Subpart VVa, Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for which Construction, Reconstruction, or Modification Commenced After November 7, 2006

This regulation applies and is included in the operating permit.

40 CFR part 60 Subpart Kb, Standards of Performance for Volatile Organic Liquid Storage Vessels (Including Petroleum Liquid Storage Vessels) for Which Construction, Reconstruction, or Modification Commenced After July 23, 1984

This regulation applies to the following tanks: T801, T802, T803, T804, T805, and T811; and has been applied in the operating permit.

Maximum Achievable Control Technology (MACT) Applicability

40 CFR part 63 Subpart Q, National Emission Standards for Hazardous Air Pollutants for Industrial Process Cooling Towers

This regulation applies to industrial process cooling towers that have been operated with chromium-based water treatment chemicals. The cooling towers at this installation have never used these materials, therefore this regulation does not apply.

40 CFR part 63 Subpart ZZZZ, National Emission Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines

This regulation applies to the emergency generator and is included in the operating permit.

National Emission Standards for Hazardous Air Pollutants (NESHAP) Applicability

40 CFR Part 61 Subpart M National Emission Standard for Asbestos

This regulation applies and has been included in the operating permit.

Greenhouse Gas Emissions

Note that this source may be subject to the Greenhouse Gas Reporting Rule. However, the preamble of the GHG Reporting Rule clarifies that Part 98 requirements do not have to be incorporated in operating permits at this time. In addition, Missouri regulations do not require the installation to report CO₂ emissions in their Missouri Emissions Inventory Questionnaire; therefore, the installation's CO₂ emissions were not included within this permit. If required to report, the applicant is required to report the data directly to EPA. The public may obtain CO₂ emissions data by visiting <http://epa.gov/ghgreporting/ghgdata/reportingdatasets.html>.

Other Regulatory Determinations

10 CSR 10-6.220 Restriction of Emission of Visible Air Contaminants

This regulation applies and is included in the operating permit.

10 CSR 10-6.400, Restriction of Emission of Particulate Matter From Industrial Processes

This regulation applies and is included in the operating permit.

10 CSR 10-6.260, Restriction of Emission of Sulfur Compounds

This regulation was rescinded from the code of state regulations (CSR). However, this regulation is still contained in Missouri's State Implementation Plan (SIP). This regulation is a federally enforceable requirement until it is removed from the SIP, therefore it must appear in this Operating Permit.

Units that combust solely natural gas are exempt from this regulation. This regulation applies to the process emissions from the Fermentation Scrubber, Distillation Scrubber, and Centrifuges; and is included in the operating permit.

10 CSR 10-6.261, Control of Sulfur Dioxide Emissions

This regulation applies to the natural gas combustion units and is included in the operating permit.

10 CSR 10-6.065, Operating Permits, Compliance Plan

At the time of issuance of this operating permit, the installation is addressing various compliance concerns with the Air Pollution Control Program. Until the concerns are resolved, a compliance plan is included in the operating permit.

Other Regulations Not Cited in the Operating Permit or the Above Statement of Basis

Any regulation which is not specifically listed in either the Operating Permit or in the above Statement of Basis does not appear, based on this review, to be an applicable requirement for this installation for one or more of the following reasons.

1. The specific pollutant regulated by that rule is not emitted by the installation.
2. The installation is not in the source category regulated by that rule.
3. The installation is not in the county or specific area that is regulated under the authority of that rule.
4. The installation does not contain the type of emission unit which is regulated by that rule.
5. The rule is only for administrative purposes.

Should a later determination conclude that the installation is subject to one or more of the regulations cited in this Statement of Basis or other regulations which were not cited, the installation shall determine and demonstrate, to the Air Pollution Control Program's satisfaction, the installation's compliance with that regulation(s). If the installation is not in compliance with a regulation which was not previously cited, the installation shall submit to the Air Pollution Control Program a schedule for achieving compliance for that regulation(s).

Response to Public Comments

The draft Intermediate Operating Permit for Golden Triangle Energy, LLC was placed on public notice December 13, 2019 for a 30-day comment period. The public notice was published on the Department of Natural Resources' Air Pollution Control Program's web page at: <https://dnr.mo.gov/env/apcp/permit-public-notice.htm>. Public comments were received from Mr. Mark Wilson, Golden Triangle Energy. The significant comments are addressed below. Some comments are summarized for brevity.

Comment #1: The installation requests an evaluation of the applicability of New Source Performance Standard 40 CFR part 60 Subpart JJJJ, Standards of Performance for Stationary Spark Ignition Internal Combustion Engines to the emergency generator. The engine was manufactured in January 2005, ordered from the manufacturer in December 2007, delivered to the installation in January 2008, and commissioned into service in June 2008. The unit is a spark ignition engine combusting natural gas, with a maximum hourly design rate of 74 HP.

Response to Comment #1: This response will begin with a discussion of the applicability of 40 CFR part 63 Subpart ZZZZ, National Emission Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines, MACT ZZZZ, then discuss the applicability of 40 CFR part 60 Subpart JJJJ, Standards of Performance for Stationary Spark Ignition Internal Combustion Engines, NSPS JJJJ.

MACT ZZZZ

MACT ZZZZ applicability is detailed in §63.6585 and §63.6590. An affected source is any existing, new, or reconstructed stationary reciprocating internal combustion engine, RICE, located at a major or area source of HAP. For area HAP sources, the following apply:

1. An existing RICE is one which commenced construction before June 12, 2006.
2. A new RICE is one which commenced construction after June 12, 2006

Additional definitions of new and existing are found in 40 CFR part 63, Subpart A, General Provisions. §63.2 contains the pertinent definitions of commenced and construction, as summarized below:

1. Commenced means the undertaking of a continuous program of construction or the entering into a contractual obligation to undertake and complete a continuous program of construction.
2. Construction means the installation of an affected source.

Based on these concepts, it is concluded the engine at this installation is a new affected source located at an area HAP source for the purposes of MACT ZZZZ. Therefore, the provisions of §63.6590(c)(1) apply. These state that new affected sources located at an area HAP source meet the requirements of MACT ZZZZ by meeting the requirements of NSPS JJJJ and specifies that no other requirements of MACT ZZZZ apply. Therefore, the engine at this installation does not have any applicable requirements in MACT ZZZZ and must comply with the applicable requirements of NSPS JJJJ.

NSPS JJJJ

NSPS JJJJ applicability is detailed in §60.4230, with the pertinent concepts summarized below. For this regulation, the date of commencing construction is the date the engine is ordered by the installation.

1. The RICE must commence construction after June 12, 2006, and
2. The emergency RICE, greater than 25 HP, must be manufactured on or after January 1, 2009

Based on these concepts, the emergency engine at this installation does not meet the applicability of NSPS JJJJ. Although the engine was ordered after June 12, 2006, it was manufactured prior to January 1, 2009. The applicability specifically states both conditions must be satisfied. Therefore, the emergency engine at this installation is not subject to either MACT ZZZZ or NSPS JJJJ. Accordingly, Permit Condition MACT ZZZZ has been removed from the operating permit and language has been added to the Statement of Basis.

Comment #2: The permittee requests the following for the denaturants:

1. The permittee states MIBK is not currently on site, but wants this denaturant included in the operating permit to provide flexibility.
2. The permittee requests the description of FS016 Hexane Storage Tote be updated to FS016, Hexane Storage Barrel. There is no storage tote for hexane, barrels are ordered as needed.
3. The permittee requests the addition of Heptane and Brucine Sulfate denaturants to the operating permit. There is none at the facility at this time but may be needed for future business.

Response to Comment #2:

1. The operating permit must contain all equipment at the time of permit issuance. Although MIBK usage is addressed in historical construction permit actions, the permittee does not currently have or use MIBK. Therefore it is not appropriate for the operating permit to include MIBK. If the permittee wishes to use MIBK in the future, a construction permit applicability evaluation is required prior to bringing MIBK on site.
2. The description is changed as requested for FS016 Hexane Storage Barrel.
3. The permittee does not currently have or use Heptane or Brucine Sulfate, nor have these denaturants been evaluated in historical permitting actions. As explained above for MIBK, it is not appropriate for the operating permit to include these denaturants. If the permittee wishes to use these, or any other new denaturants in the future, a construction permit applicability evaluation is required prior to bringing them on site. The permittee is encouraged to contact the Air Pollution Control Program Permits Section for additional guidance.

Comment #3: The permittee indicated T204, Hot Slurry Tank, was added at some point in the past; but was not part of the original permitted installation. The Slurry Tank, T203, that was part of the original permitted installation is still in place and operational. The installation requests including T204 Hot Slurry Tank in the operating permit.

Response to Comment #3: At the time of operating permit issuance, insufficient information was provided to demonstrate compliance with 10 CSR 10-6.060, Construction Permits Required, for the construction of T204, Hot Slurry Tank.

Installation of a second slurry tank could de-bottleneck the fermentation process. Therefore, this tank is not included in this operating permit. Usage of this tank is not currently a permitted activity. This compliance demonstration is required as part of Permit Condition Compliance Plan.

Comment #4: All VOC storage tanks either have a floating roof or press/vac conservation vent as a control device. The emission calculations take the control devices into account. MOEIS does not need to show a control device for these tanks because it would introduce needless complication. However for other future rule applicability determinations, it should be noted the tanks have control devices.

Response to Comment #4: The operating permit contains the federally enforceable control requirements established in issued construction permits and 40 CFR part 60 Subpart Kb, as applicable. Storage tanks for which control devices are not listed in the operating permit do not have federally enforceable requirements for usage or maintenance of the control devices. Therefore, these unlisted control devices are considered voluntary at this time. Voluntary controls cannot be used in compliance demonstrations for emissions or operational limitations. The permittee is encouraged to contact the Air Pollution Control Program Permits Section for additional guidance.

Comment #5: The permittee indicates Boiler #2, EU034, Low NO_x Burners should be removed from the permit as a control device and flue gas recirculation (FGR) should be added as the control device. The permittee requests usage of an emission factor of 32 lb NO_x/MMCF, from SCC 10200602, for this unit.

Response to Comment #5: Construction Permit 072008-001, Table 4, indicates Boiler #2, EU034, was permitted with Low NO_x Burners. Confirmation of the existence or absence of Low NO_x Burners on Boiler #2, EU034, is required as part of Permit Condition Compliance Plan. The operating permit shows Low NO_x Burners as the control device for Boiler #2, EU034, to satisfy the requirements of Construction Permit 072008-001. The NO_x emission factor for Boiler #2, EU034, is established in Construction Permit 072008-001; changing the emission factor requires a construction permit amendment. The permittee is encouraged to contact the Air Pollution Control Program Permits Section for additional guidance.

Comment #6: The permittee indicates the emission point number for HQ Molecular Sieve Dehydration should be changed from EU045 to EU054.

Response to Comment #6: The emission point number is changed as requested.

Comment #7: The permittee indicates changes to the construction dates and maximum hourly design capacities for various equipment.

Response to Comment #7: According to Air Pollution Control Program records, the installation commenced construction on a greenfield site in the year 2000. As referenced in the operating permit, the construction date indicates the date the equipment was installed at the site unless otherwise noted, see NSPS JJJJ applicability above. The proposed dates that pre-date 2000 have not been incorporated into the operating permit. The maximum hourly design rates in the operating permit reflect the rates as permitted in construction permits or those rates with documented process bottlenecks. The proposed capacities that are greater than these documented rates are not incorporated in the

operating permit. Additionally, the permittee proposed revisions to design rates based on stack testing. Testing does not re-establish a new maximum hourly design rate with each subsequent test. Because the rates are documented and formed the basis of construction permitting actions, changes the maximum hourly design rates may require construction permit amendments. The permittee is encouraged to contact the Air Pollution Control Program Permits Section for additional guidance.

Comment #8: The permittee requests the addition of the following control devices:

1. Grain Building, EU029, as a control device for the Dried Solids Conveyor,
2. LDAR as a control device for LDAR Monitoring, FS002-1 and FS002-2,
3. EU026 DDGS Multiclones as a control device to EU026 DDGS Dryer Regenerative Thermal Oxidizer Bypass

Response to Comment #8:

1. The Grain Building, EU029 is not considered a control device. As stated in Construction Permit 072008-001, no capture efficiency is claimed for the building.
2. LDAR is not considered a control device for the listed emission points. LDAR is a work practice designed to identify leaking equipment so that emissions can be reduced through repairs or replacement.
3. EU026 DDGS Multiclones are located prior to the DDGS Dryer and the DDGS Dryer Bypass. Although emissions pass through the DDGS Multiclones prior to the bypass, it is not accurate to state they control bypass emissions.

Comment #9: The permittee requests the following changes to Permit Condition 072008-001, Table 2: Summary of NO_x Emission Limitations:

1. Removal of emission limitations for the following equipment: EU026 DDGS Dryer; EU028, Boiler #1; EU034, Boiler #2, and the combined emission limitation for EU026, DDGS Dryer and EU038, RTO; on the basis that the values shown are emission factors, not limits per the Consent Decree.
2. Add language indicating the DDGS Dryer outlet is also the RTO inlet.

Response to Comment #9:

1. The cited emission limitations are established in Construction Permit 082007-001. A construction permit amendment is required to change or eliminate these limits. The permittee is encouraged to contact the Air Pollution Control Program Permits Section for additional guidance.
2. The emission limit applies to the sum of NO_x emissions from Boiler #1, EU028, and the DDGS Dryer outlet, EU026. The demonstrate compliance with the limit, NO_x emissions must be quantified at the dryer outlet.

Comment #10: The permittee requests the removal of the Open Flare (EU039) as a control device for FS001-1 HQ Ethanol Rail Loading Rack.

Response to Comment #10: Usage of the flare is required by Construction Permit 072008-001, Special Condition 12.A. Changing this requirement requires a construction permit amendment. The permittee is encouraged to contact the Air Pollution Control Program Permits Section for additional guidance.

Comment #11: The permittee requests to modify the requirements of Construction Permit 072008-001 Special Condition 11.C. and D. requiring sodium bisulfite usage in the Fermentation

Scurbber (C003). Additionally, the permittee requests the ability to use chilled water without any additives (bisulfate) in the Fermentation Scrubber (C003).

Response to Comment #11: Usage of sodium bisulfite in the Fermentation Scrubber (C003) is required by Construction Permit 082007-001, Special Conditions 11C. and 11.D. A construction permit amendment is required to change these requirements. The permittee is encouraged to contact the Air Pollution Control Program Permits Section for additional guidance.

Comment #12: The permittee requests that performance testing demonstrate emissions in units of pounds per hour only, and not include any production data. The permittee also requests to change emission factors in the attachments to units of pounds per hour.

Response to Comment #12: Emission rates are expected to vary with the material throughput, which varies over time for each emission unit. Therefore, calculating emissions on a operational hours basis does not accurately reflect actual emissions. The attachments present emission factors to use when calculating actual emissions to demonstrate compliance with applicable emission limitations. The performance testing requirements are designed establish emission factors to support the attachments. As required in the operating permit, the permittee shall use the attachments, or approved alternatives, to demonstrate compliance.

Comment #13: The permittee requests clarification regarding Permit Condition 6.400, Operational Limitations, for the installation of an air flow monitor on the DDGS Multiclones. The permittee requests confirmation that the air flow monitor is the same as the pressure drop monitor.

Response to Comment #13: The requirements to install an air flow monitor and a pressure drop monitor are separate monitoring requirements. Inlet gas velocity, as measured by an air flow monitor, is a primary indicator of cyclone performance, while pressure drop, as measured by pressure drop gauges, is a secondary indicator. The permit condition requires both an air flow monitor and pressure drop gauges.

Comment #14: The permittee requests clarifications regarding semi-annual reporting requirements.

Response to Comment #14: Permittee's that hold Intermediate Operating Permits must submit annual reports unless otherwise required by specific regulations. 10 CSR 10-6.065, Operating Permits, does not require semi-annual reports for Intermediate Operating Permits. The references to semi-annual reporting, under the authority of 10 CSR 10-6.065, are removed from the operating permit. Semi-annual reporting requirements by other underlying requirements remain in the permit.

Comment #15: The permittee requests clarification on the applicability of Permit Condition 6.260.

Response to Comment #15: The underlying requirement for this permit condition is 10 CSR 10-6.260, Restriction of Emission of Sulfur Compounds. This rule applies to all sources of sulfur compound emissions, with specific exemptions. The emission units detailed in the applicability section of this permit condition meet the applicability of the rule and do not meet any of the exemptions. Therefore the rule applies and must appear in the operating permit. The limits established in this rule are different than those

established in Construction Permit 072008-001. Compliance with Permit Condition 6.260 is demonstrated using the performance testing procedures of Permit Condition 072008-001, as published emission factors are not available. Permit Condition 6.260 does not require installation of a continuous emissions monitoring system (CEMS).

Comment #16: The permittee requests clarification on the Monitoring and Recordkeeping provisions of Permit Condition 6.261.

Response to Comment #16: The underlying regulation for Permit Condition 6.261 is 10 CSR 10-6.261, Control of Sulfur Dioxide Emissions. This regulation contains an exception for natural gas fired units. To qualify for the exception, the rule requires monitoring and recordkeeping. These requirements must appear in the permit condition.

Comment #17: For Permit Condition Compliance Plan, the permittee requests a 90 day timeline for submission of the required compliance plan instead of the stated 30 day timeline.

Response to Comment #17: Discussion of possible compliance concerns began approximately May 2019, and have occurred frequently since then between various Air Program staff and the permittee. Due to the complex nature of the possible compliance concerns, the Air Program agrees to a 60 day timeline for permitting purposes. This timeline may be re-evaluated by the Air Pollution Control Program Compliance and Enforcement Section.

Comment #18: The permittee requests clarification of the Monitoring requirements for 10 CSR 10-6.170.

Response to Comment #18: The Monitoring section requires weekly observations for a minimum of eight consecutive weeks after permit issuance. Issuance of a renewal operating permit does not restart the schedule.

Comment #19: The permittee references various information for inclusion in the operating permit. These references includes John Zink manufacturer guarantees, DENCO emission factors, various process flow diagrams, various spreadsheets and logs, derived emission factors, and other information.

Response to Comment #19: At the time of operating permit issuance, no supporting documentation was received to facilitate evaluation of the information. Therefore, the requested changes are not incorporated into the operating permit.

Comment #20: The permittee indicates straight truck receiving operations occur in the Grains Receiving Building.

Response to Comment #20: Straight truck receiving is added to the installation description and to Attachment PM₁₀, Section 2, Table 1.

Comment #21: The permittee requests the removal of references to the Truck Loading Grate and the Truck Loading Pit. The truck loading system is a top load system and does not contain a loading grate or loading pit.

Response to Comment #21: Process flow diagrams provided in Construction Permit 72008-001 indicate truck loading occurs with a top loading system while the truck is positioned over a grate to catch overflow materials. These materials pass through the grate and fall into a pit that feeds DDGS Pit Conveyor EU043. This grate and pit are

considered emission sources and must be accounted for in the operating permit.
Therefore, there is no change to the operating permit in response to this comment.

Comment #22: The permittee requests the removal of Tank 805 from Permit Condition Subpart Kb because the tank capacity is less than the applicability threshold.

Response to Comment #22: The capacity of Tank 805 is 18,000 gallons, which is less than the applicability threshold of 19,813 gallons. Therefore Tank 805 has been removed from the permit condition as requested.

Comment #23: The permittee requests removal of EU027 from Table 19: 10 CSR 10-6.400 Emission Limitations.

Response to Comment #23: Emission Unit EU027 is the DDGS Cooling/Separating Cyclone. Particulate emissions are expected from DDGS handling operations. Based on the information available at the time of operating permit issuance, it appears 10 CSR 10-6.400 is an applicable requirement. Therefore, there is no change to the operating permit in response to this comment.

Comment #24: The permittee requests removal of the Belt Scale from the operating permit. The unit has been removed from site.

Response to Comment #24: The Belt Scale has been removed as requested.

Comment #25: The permittee requests the addition of T423 to the description of EU021 as follows: Molecular Sieve Dehydration Vacuum Pump Vent (T421, T422, T423)

Response to Comment #25: The operating permit has been changed as requested.

Comment #26: For Attachment VOC Emissions, Table 2, the permittee provides the following comments:

1. Tank 830, Impurities Storage Tank emissions are controlled by the Distillation Scrubber.
2. EU046 HQ Heat Exchanger is a closed loop. No emissions, delete from permit.
3. EU054, Molecular Sieve Dehydration emissions are controlled by the Distillation Scrubber.
4. Tank T802 is on the list multiple times.

Response to Comment #26: The intent of the table is to identify the Distillation Operation emission units that have the potential to emit VOC, and to clarify the emission calculation methods for these units. All tanks are accompanied by footnote 93, which requires the permittee to calculate emissions using the methodology contained in AP-42, Section 7.1. This AP-42 section will detail how to include the various tank characteristics into the emission calculations. Tank controls are also discussed in Response to Comment #4. The Distillation Scrubber has been added as a control device to EU054, Molecular Sieve Dehydration throughout the permit. Tank T802 seems to identify multiple tanks, not duplicative identification of the same tank. The permit and attachments must identify each tank. Based on information available at the time of operating permit issuance, it appears the EU046 HQ Heat Exchanger is an emission unit and must remain in the operating permit.

Comment #27: For Attachment VOC Emissions, Table 3, the installation provides the following comments:

1. Fuel Sieve Tanks 421, 422, and 423 control device is the Distillation Scrubber. The Fuel Sieve Tanks are flow through process vessels, not storage tanks for the point of view of permitting.

Response to Comment #27: The intent of this attachment is to demonstrate compliance with the VOC emission limitations established in Construction Permit 072008-001, issued under the authority of 10 CSR 10-6.060, Construction Permits Required. This regulation does not differentiate between process vessels or storage tanks; they are equivalent under provisions of this regulation. The emissions from these units must be included in the compliance demonstrations for the applicable emission limitations. The Distillation Scrubber has been added as the control device for the Fuel Sieve Tanks 421, 422, and 423 throughout the operating permit as requested.

Comment #28: For Attachment HAP Emissions, Tables 4 and 5, the permittee questions if Tertiary Butyl, CPTEO, Ethyl Acetate, Isopropyl, EEP, Heptane, and Brucine Sulfate need to be added to the table.

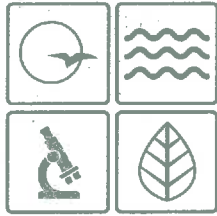
Response to Comment #28: All the listed denaturants, except Heptane and Brucine Sulfate, were evaluated for inclusion into this table. Heptane and Brucine Sulfate are discussed in Response to Comment #2. This table only lists the denaturants that are believed to emit hazardous air pollutants (HAPs). For denaturants not listed on this table, except Heptane and Brucine Sulfate, an analysis of the Safety Data Sheets (SDS) did not reveal the presence of HAPs. Therefore, there is no change to the operating permit in response to this comment.

Comment #29: In the Statement of Basis, Construction Permit History, Item #11, Construction Permit 092015-013, the installation requests removal of references to backup centrifuges EU047, EU048, and EU049 as they have been removed.

Response to Comment #29: The purpose of this section is to detail the issued construction permits and all no construction permit required determinations to show the development of the installation over time and to ensure historical compliance with permitting requirements. These three centrifuges were installed and operated under the authority of the referenced construction permit. The removal of these three centrifuges is detailed in this same section, Item #14, No Permit Required, Project #2018-02-050. Therefore, there is no change to the operating permit in response to this comment.

Comment #30: The permittee requests to clarify the description of the Emergency Generator, EU045, as a 4 stroke, spark ignition, natural gas fired unit. The permittee submitted supporting documentation for this request.

Response to Comment #30: The description for the Emergency Generator, EU045, is updated and corresponding emission factors are included in the attachments.



Missouri Department of dnr.mo.gov
NATURAL RESOURCES

Michael L. Parson, Governor

Carol S. Comer, Director

MAR 16 2020

Kelly Bordewick
Golden Triangle Energy, LLC
15053 Highway 111
Craig, MO 64437

Re: Golden Triangle Energy, LLC, 087-0016
Permit Number: OP2020-009

Dear Kelly Bordewick:

Enclosed with this letter is your Intermediate operating permit. Please review this document carefully. Operation of your installation in accordance with the rules and regulations cited in this document is necessary for continued compliance. It is very important that you read and understand the requirements contained in your permit.

This permit may include requirements with which you may not be familiar. If you would like the department to meet with you to discuss how to understand and satisfy the requirements contained in this permit, an appointment referred to as a Compliance Assistance Visit (CAV) can be set up with you. To request a CAV, please contact your local regional office or fill out an online request. The regional office contact information can be found at <http://dnr.mo.gov/regions/>. The online CAV request can be found at <http://dnr.mo.gov/cav/compliance.htm>.

You may appeal this permit to the Administrative Hearing Commission (AHC), P.O. Box 1557, Jefferson City, MO 65102, as provided in RSMo 643.078.16 and 621.250.3. If you choose to appeal, you must file a petition with the AHC within thirty days after the date this decision was mailed or the date it was delivered, whichever date was earlier. If any such petition is sent by registered mail or certified mail, it will be deemed filed on the date it is mailed. If it is sent by any method other than registered mail or certified mail, it will be deemed filed on the date it is received by the AHC.

If you have any questions or need additional information regarding this permit, please contact the Air Pollution Control Program (APCP) at (573) 751-4817, or you may write to the Department of Natural Resources, Air Pollution Control Program, P.O. Box 176, Jefferson City, MO 65102.

Sincerely,

AIR POLLUTION CONTROL PROGRAM

Michael J. Stansfield, P.E.
Operating Permit Unit Chief

MJS:NWA

Enclosures

c: PAMS File: 2014-05-089

